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(54) **CTLA4-BINDING PROTEINS AND METHODS OF TREATING CANCER**

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C07K 2317/56 (2013.01); **C07K 2317/626**

(2013.01); **C07K 2317/71** (2013.01); **C07K**

2317/76 (2013.01); **C07K 2319/21** (2013.01)

(57)

ABSTRACT

The present disclosure provides engineered antigen-binding proteins that bind to cytotoxic T-lymphocyte-associated antigen-4 (CTLA4). Nucleic acids, vectors, host cells, and conjugates are also provided herein. Further provided are kits and pharmaceutical compositions comprising said entities as well as methods of making the engineered antigen-binding proteins and methods of treating a subject in need thereof.

Specification includes a Sequence Listing.

Related U.S. Application Data

(60) Provisional application No. 63/139,510, filed on Jan. 20, 2021.

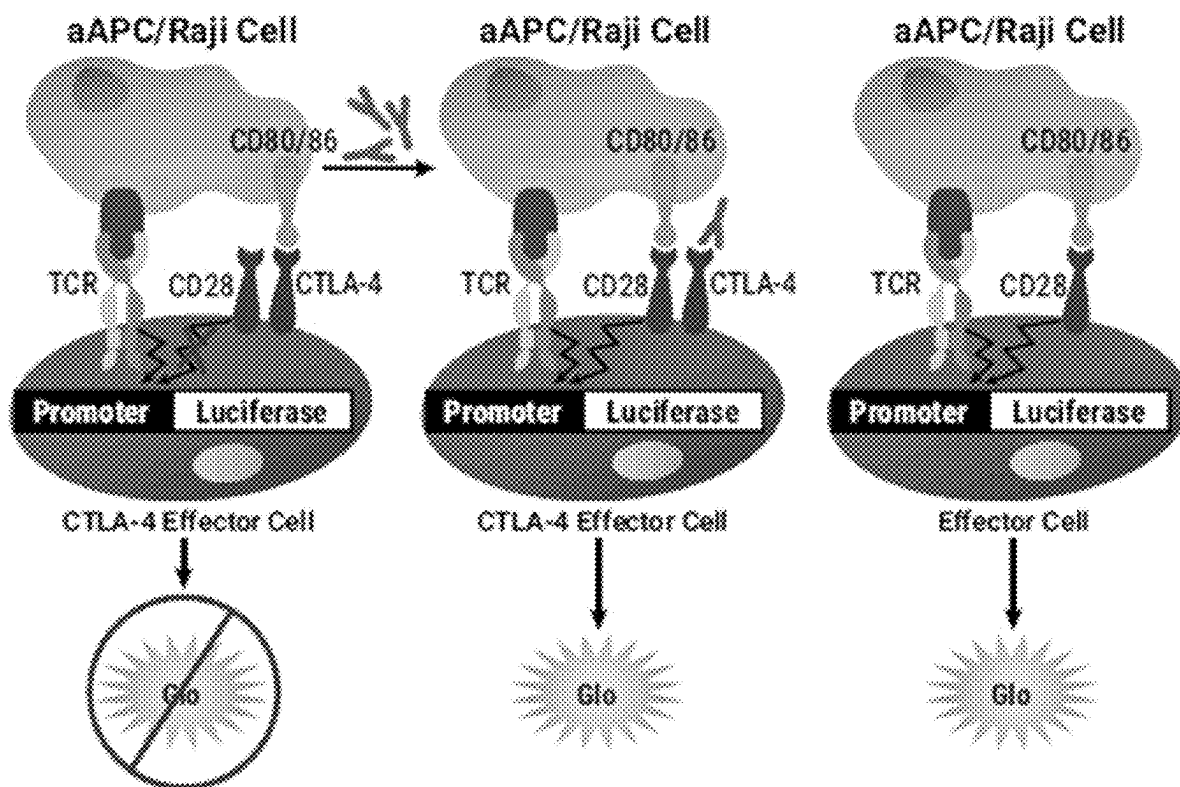


Fig. 1

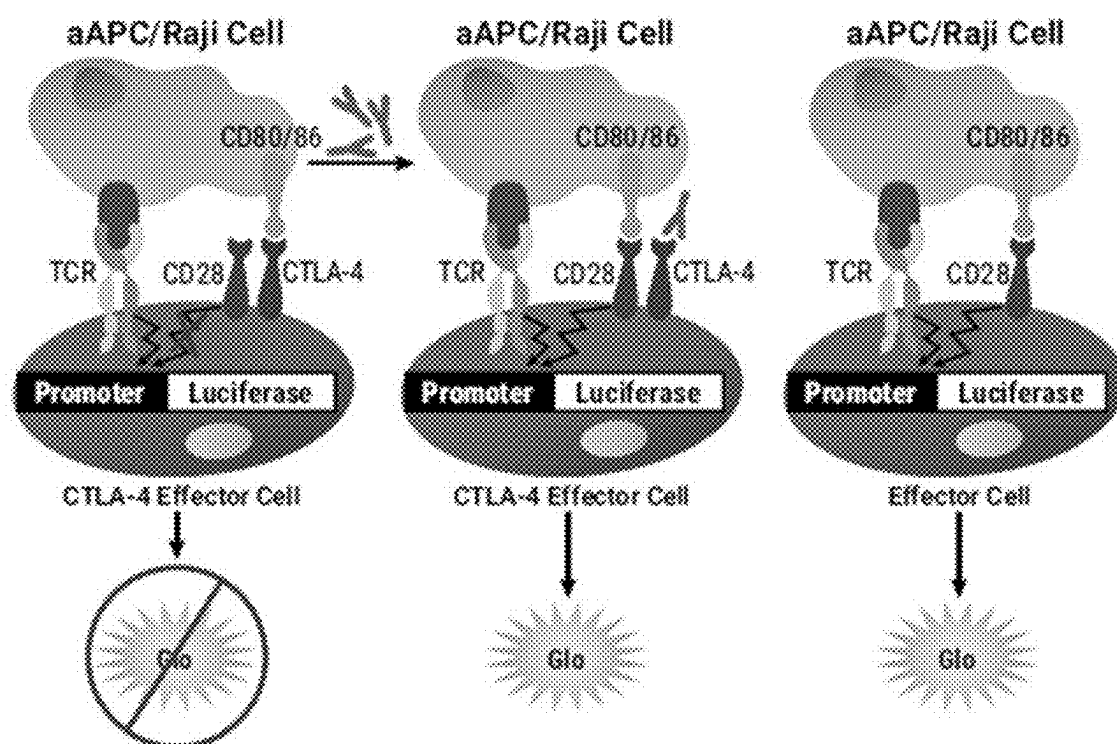


Fig. 2A

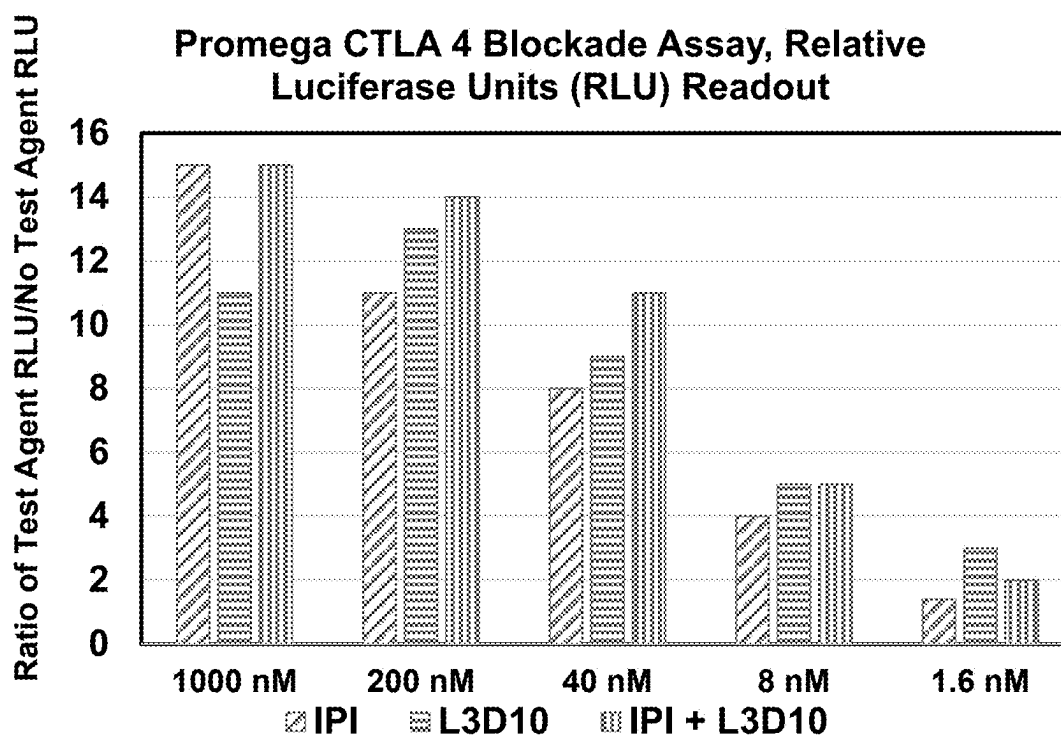


Fig. 2B

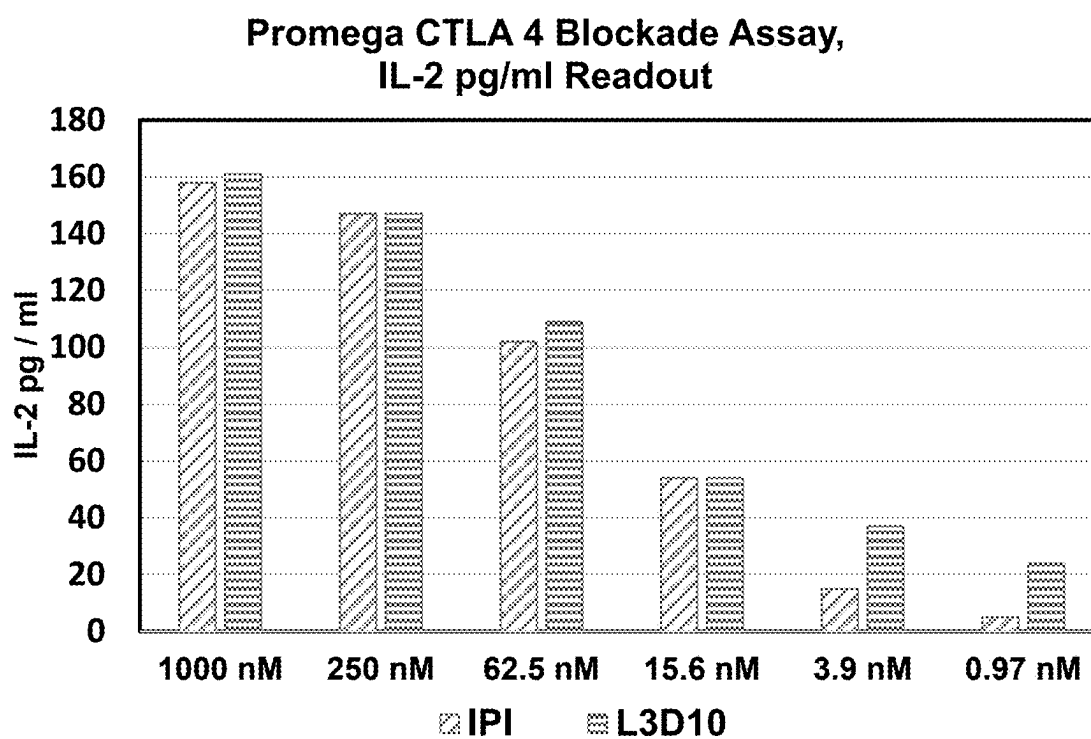


Fig. 3

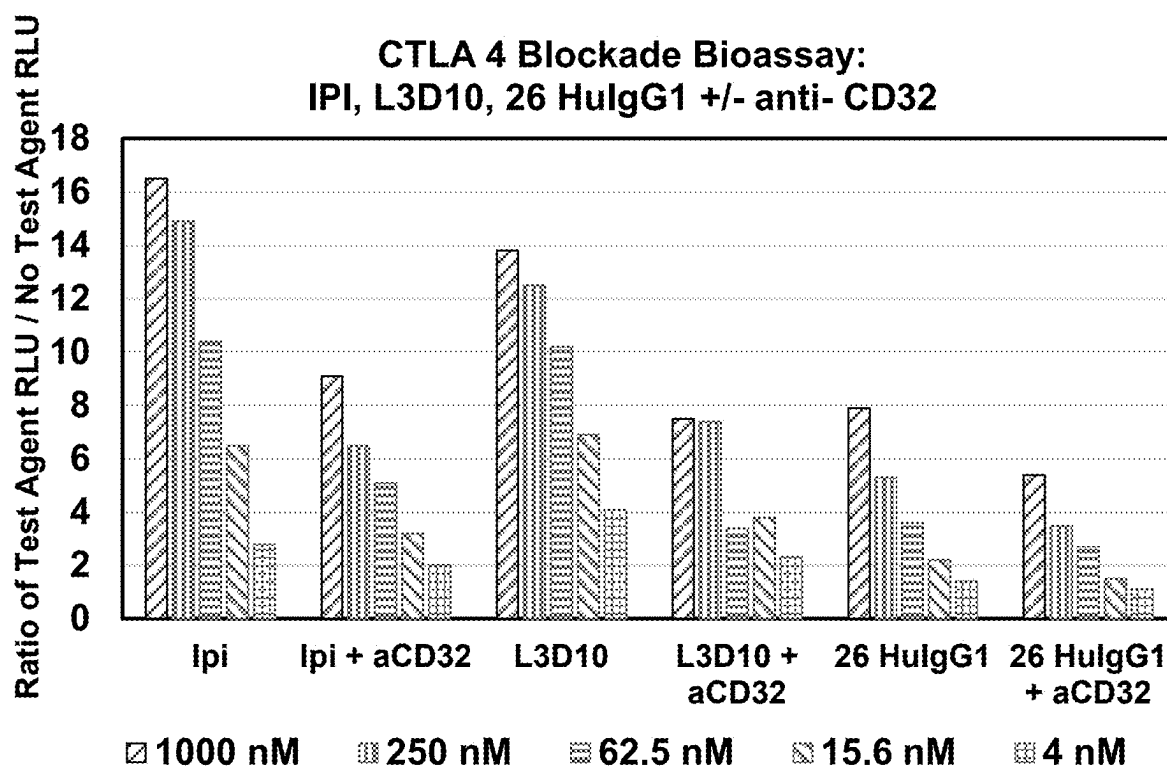


Fig. 4

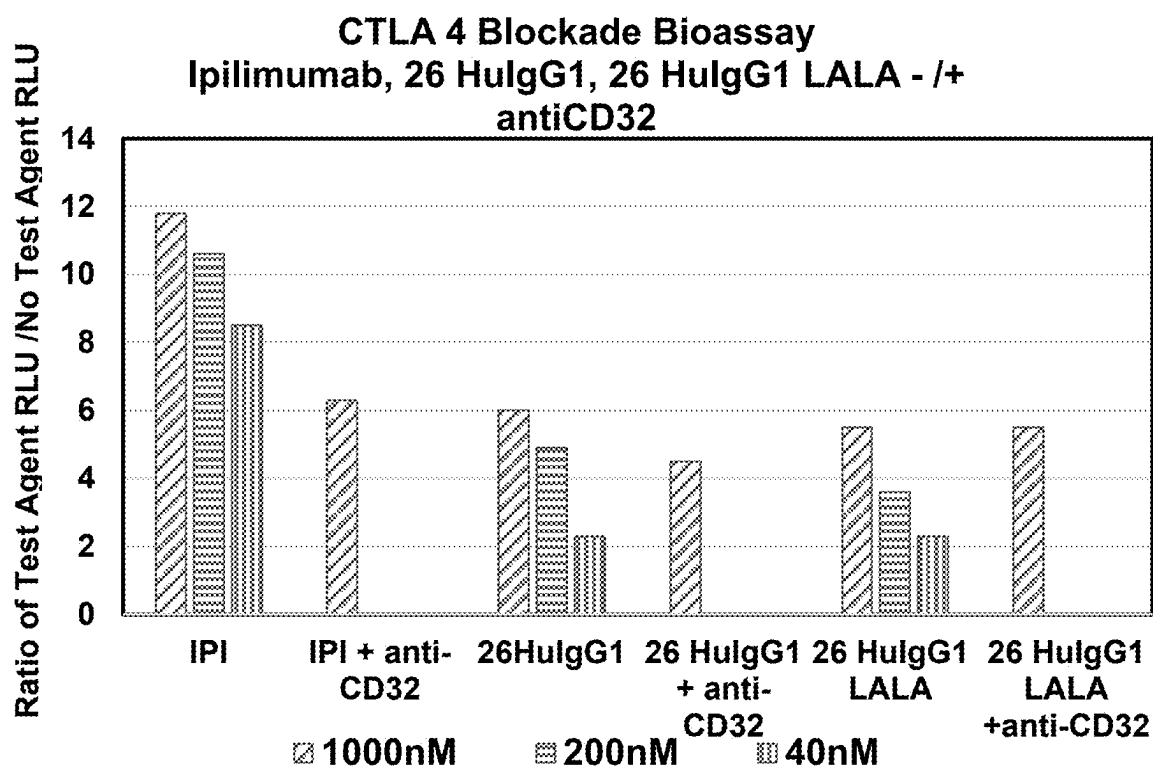


Fig. 5

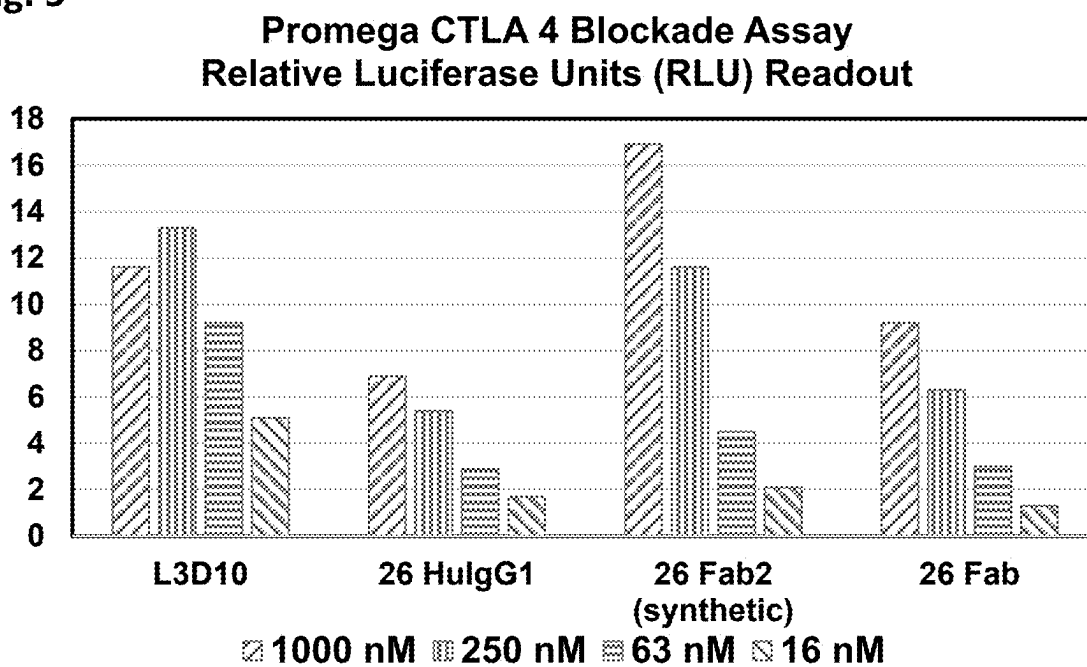


Fig. 6

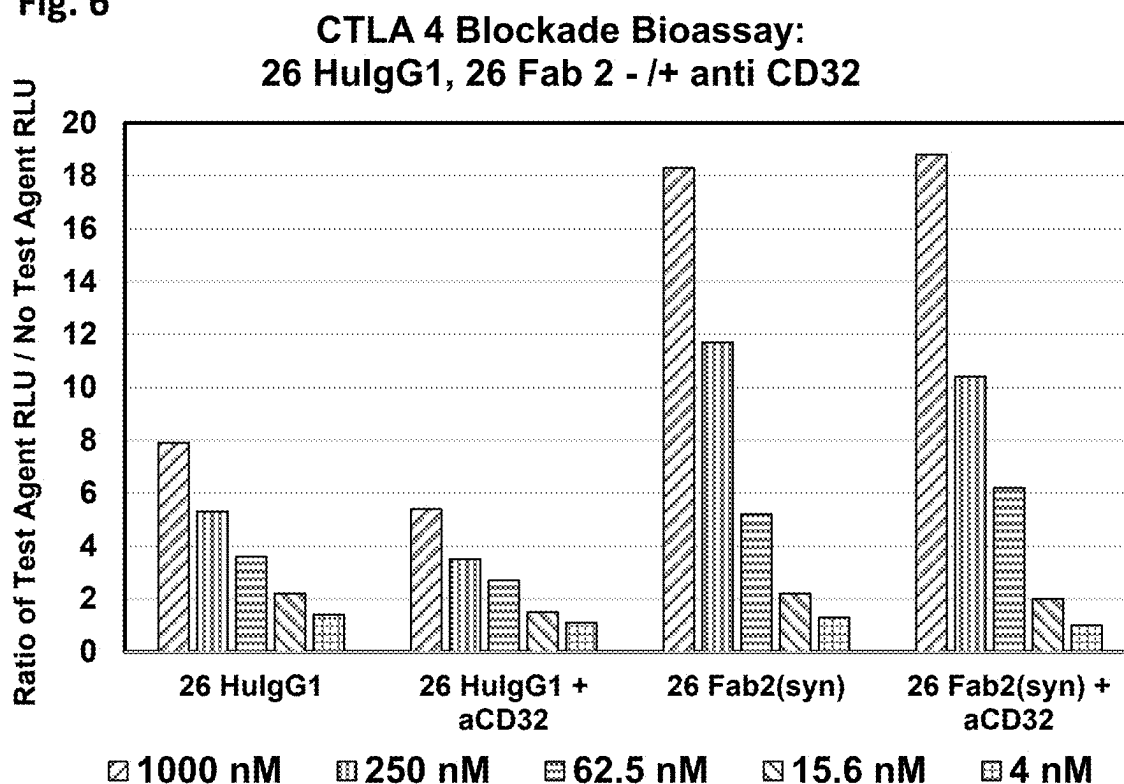


Fig. 7

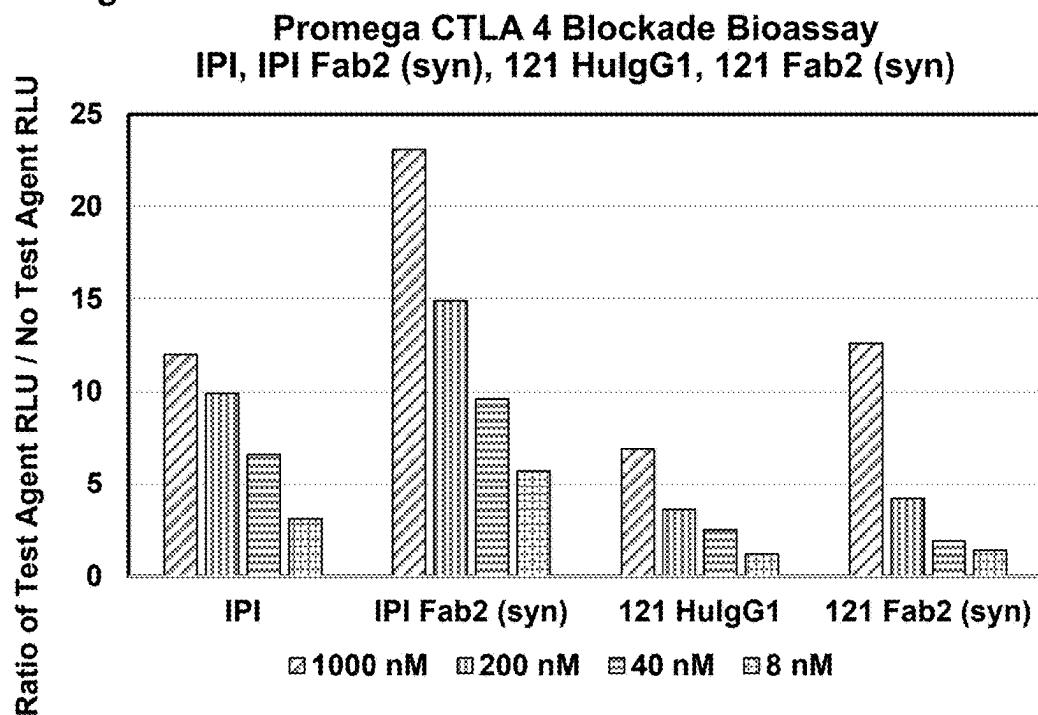


Fig. 8

Promega CTLA 4 Blockade Bioassay
IPI, IPI ScFv, 121 HulgG1, 121 ScFv

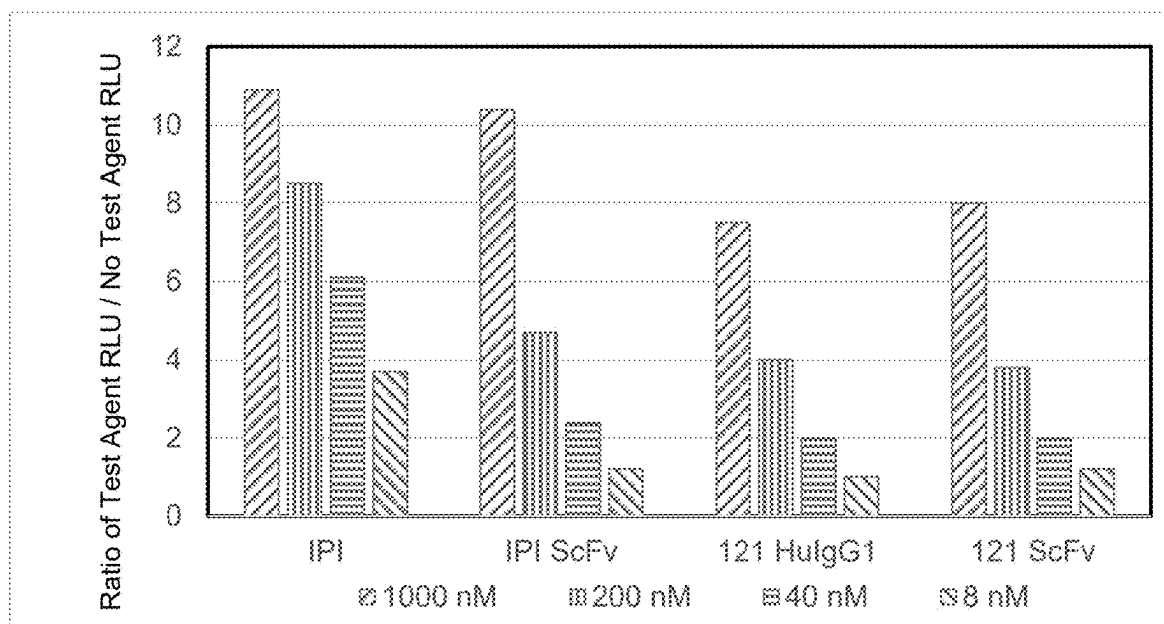


Fig. 9A

Diabody: Fv Antibody A + Fv Antibody B

V _H A	Linker A	V _L B	Linker C	V _H B	Linker B	V _L A	His ₆ Tag
V _H B	Linker A	V _L A	Linker C	V _H A	Linker B	V _L B	His ₆ Tag
V _L A	Linker A	V _H B	Linker C	V _L B	Linker B	V _H A	His ₆ Tag
V _L B	Linker A	V _H A	Linker C	V _L A	Linker B	V _H B	His ₆ Tag

Linker A = Linker B, (GlyGlySer)_N For N < 3

Linker C ~ (GlyGlySer)_N for N > 4, Single Chain Diabody

Linker C ~ (GlyGlySer)_N for N < 4, Homodimeric Diabody

Fig. 9B

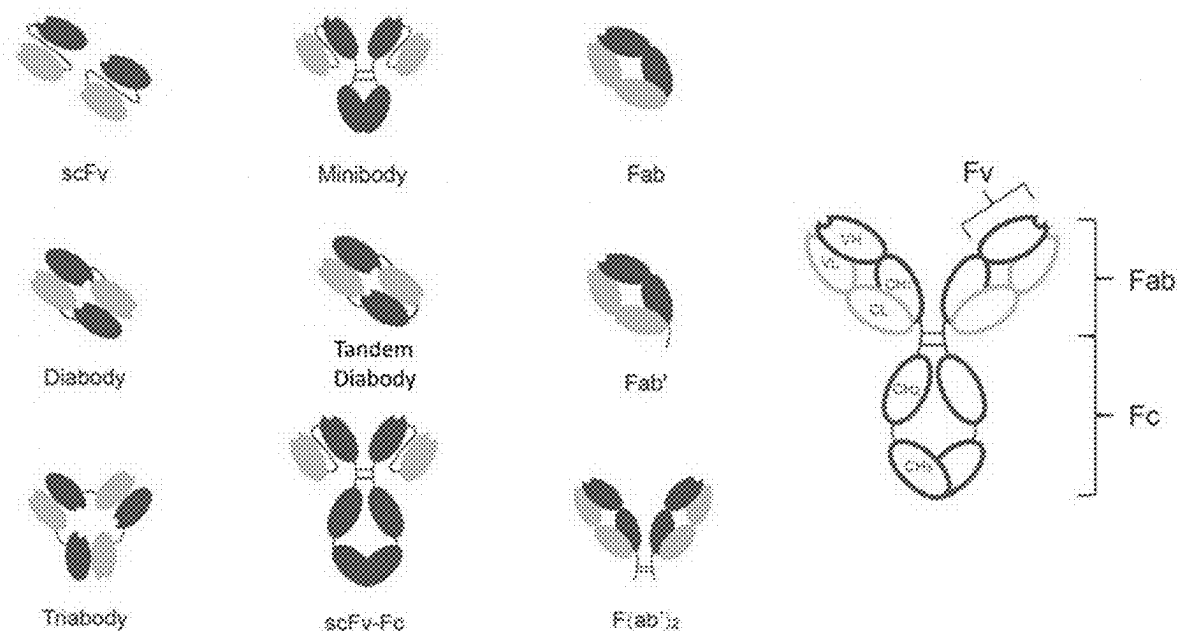
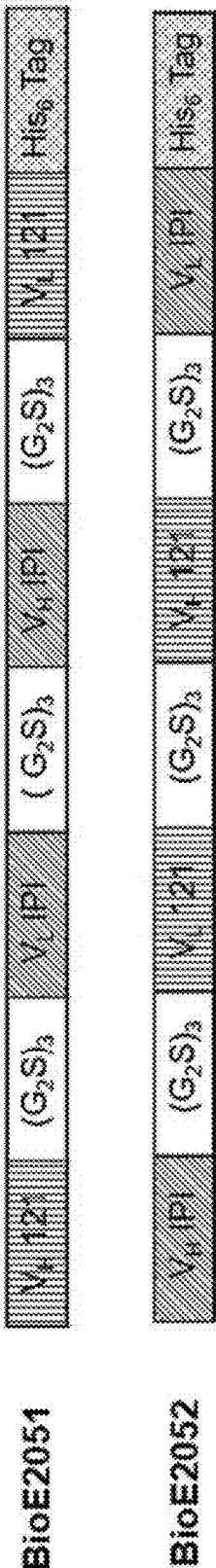


Fig. 10A

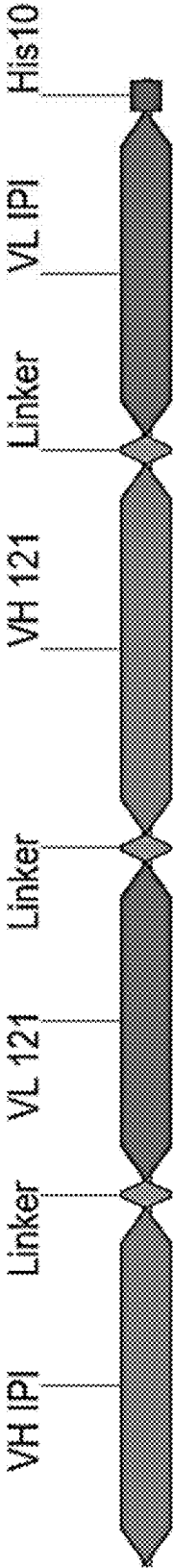
Domain Structure of Diabodies, BioE2051 & BioE2052



Fv of antibody 121: V_H 121, V_L 121

Fv of antibody Ipilimumab: V_H IPI, V_L IPI

Fig. 10B



BioE2052 GID016379 492 aa



Tandem
Diabody

1 CQQLVSSGG WQPSRL SCASQTFE STYMPWQA KKKLEWTFE ISIDENHY ADVSRFTI SRRNSFTY LQMSLRAD TALLVCARTG
Linker
VH-IPI
101 MLEPDYMQ GTLVTVSSGG SGGSSGGIQ MTPSPSLA SVERWTTC SATSLTMS WQPSWAP KLLIDTSHL ASQPSFSG SGGSTYTLT
Linker
VL 121
201 ISSQEDPA YTYQWSSY PLTYGNTV KLRSSSGG GSKVLQSG FGLWKSQTL SLTCTSSGES LYSIDWNR QFFQRLKWL GTWAGDTTN
Linker
VL 121
301 TNSALMERLT ISRDTSENYV SLRSTYTA DTVVTCAG FHMAMST AMYNAQTL VTVSSSGSG SGGSDVYQ SPQSLSG KRAVLSGAS
Linker
VL IPI
401 QWGSSTLAW YQRPQAPR LLTGAFSA TLHSPSG GSGDTFTI SREEDPAV YTYQWSSP WTYQWTFE INHHHHHH HH
His10

Fig. 10D

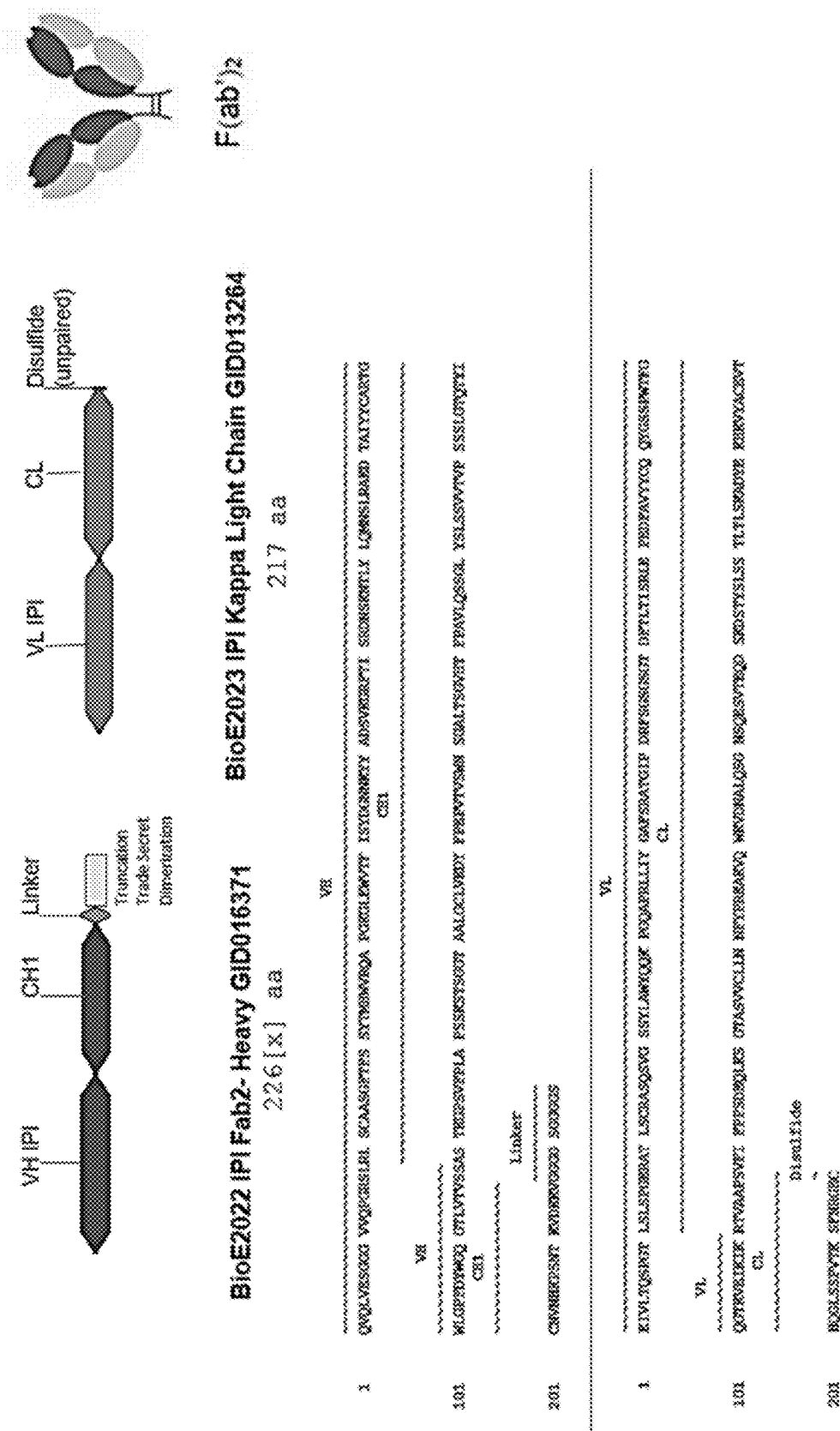


Fig. 10F

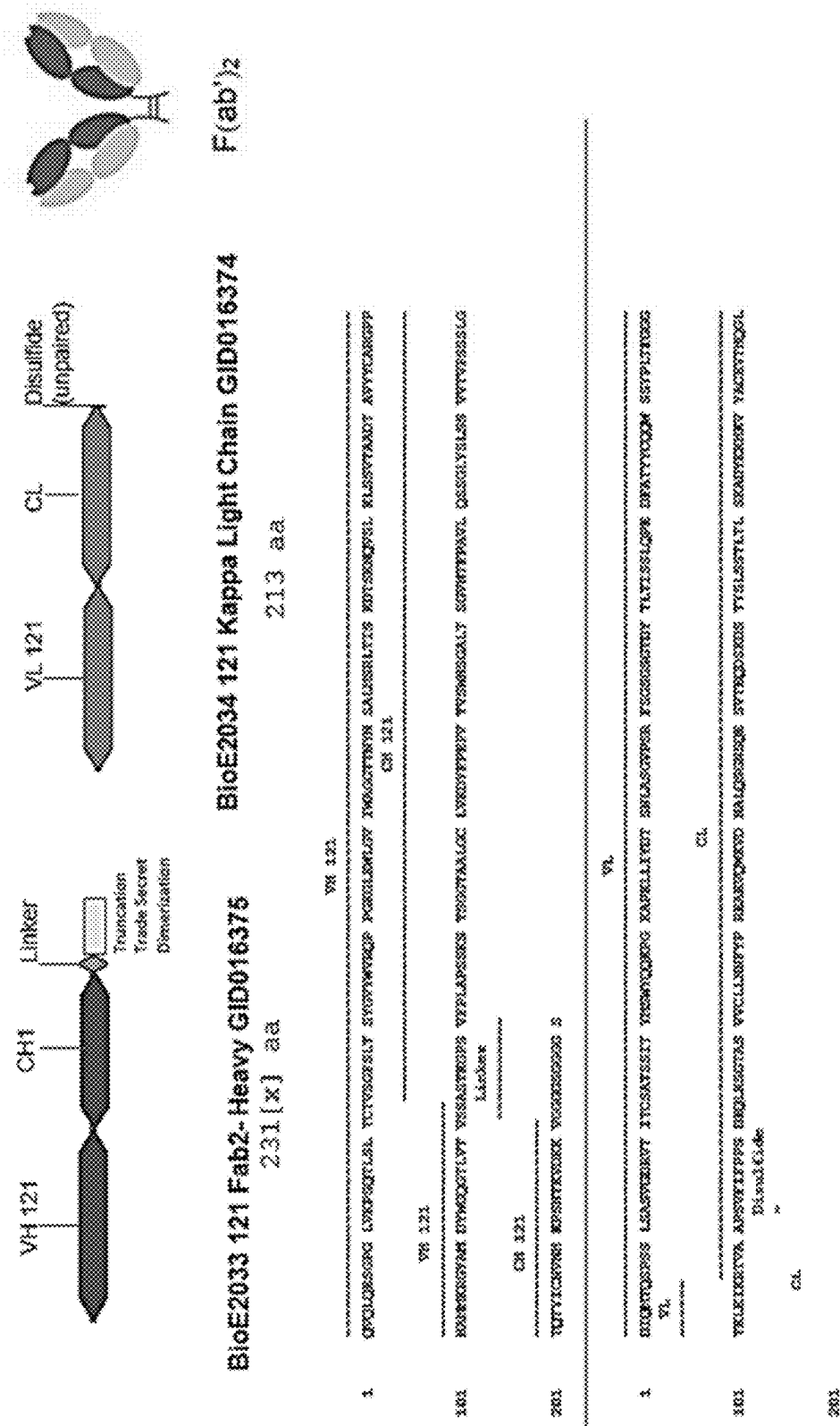
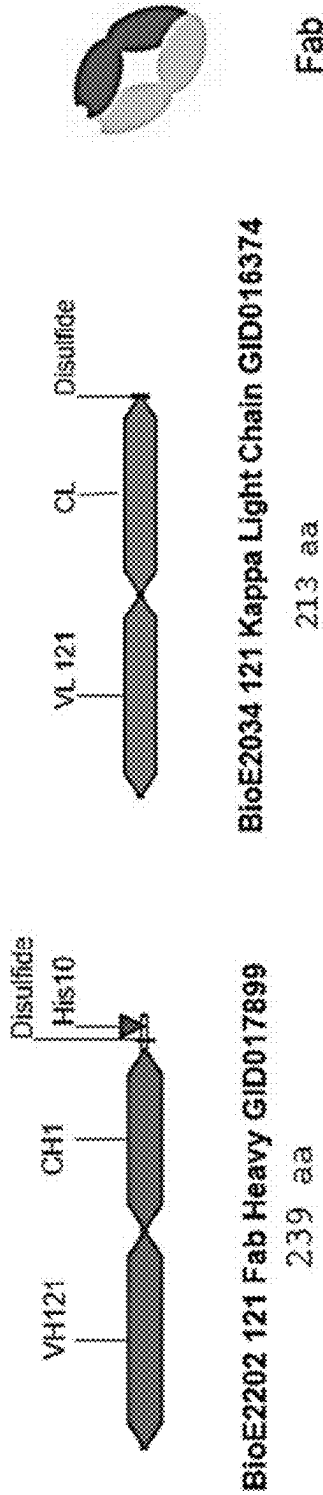


Fig. 10G



BioE2202 121 Fab Heavy GID017899 BioE2034 121 Kappa Light Chain GID016374

1		VH	
		CH	
101		Disulfide	
		CH	
201		His10	
		Disulfide	
		VL	
1		CL	
		Disulfide	
101		CL	
		Disulfide	
201		Disulfide	

Fig. 10H

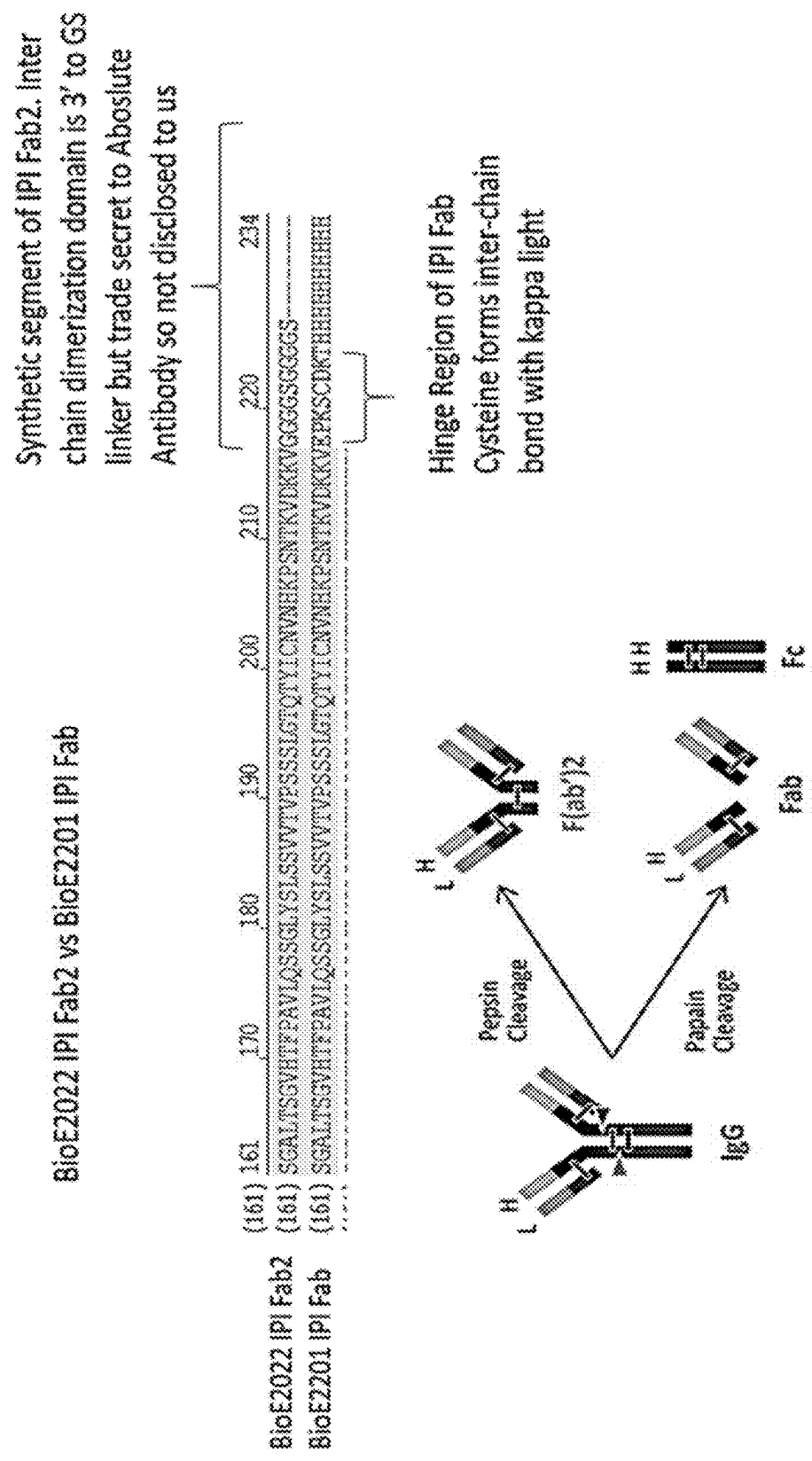


Fig. 10I

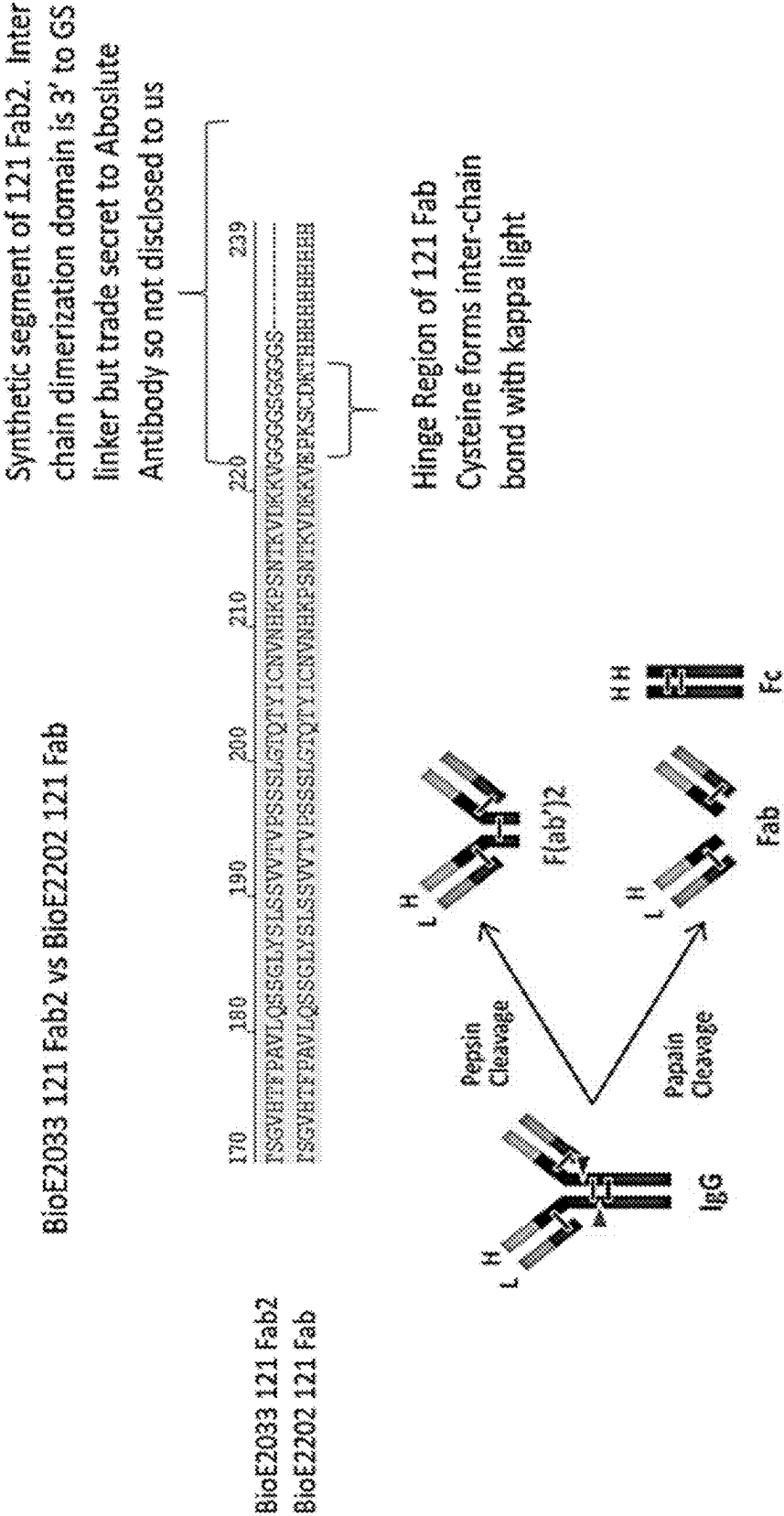


Fig. 11

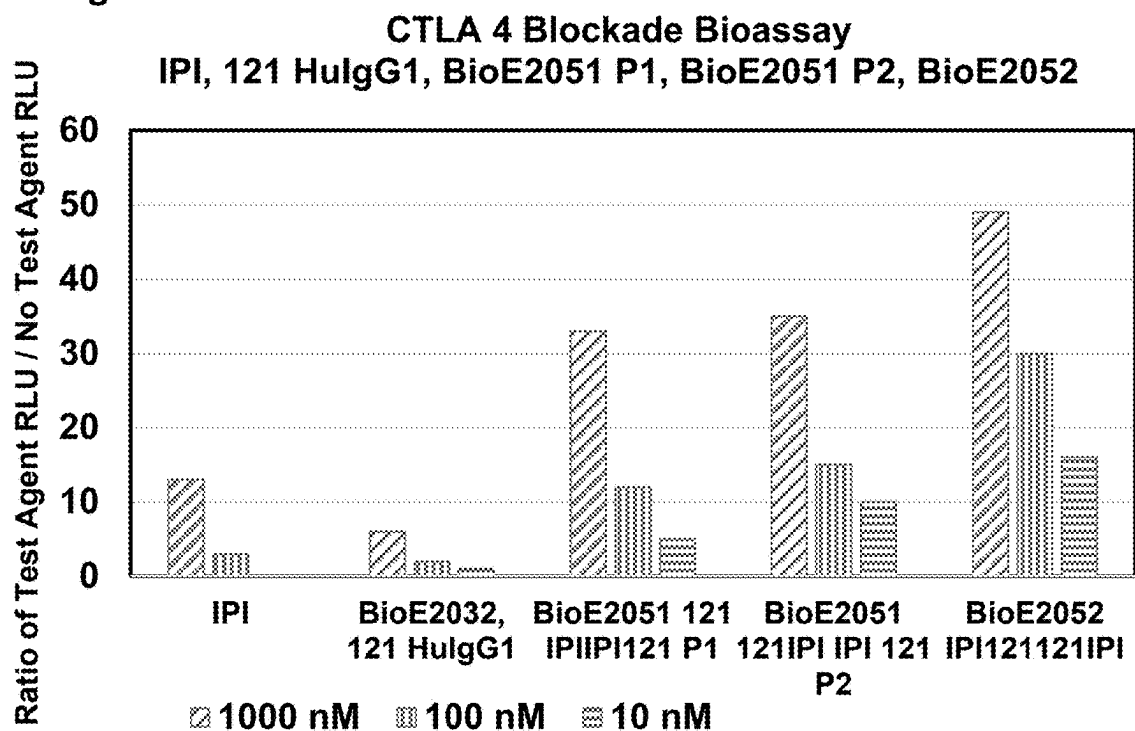


Fig. 12

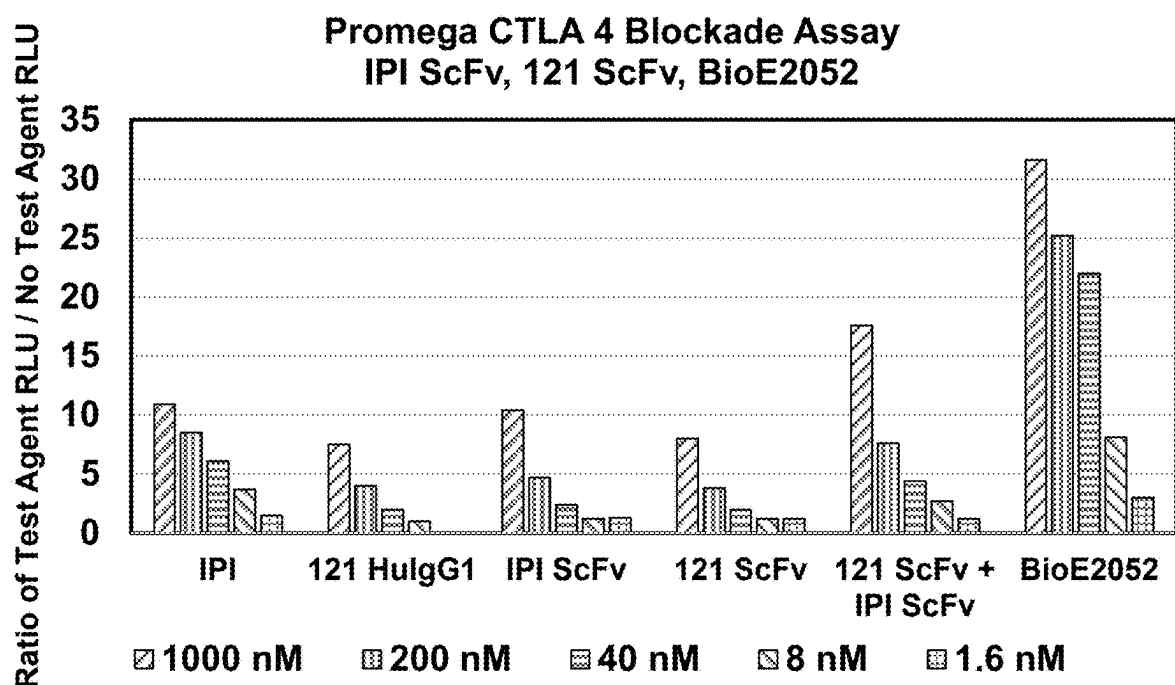


Fig. 13

**Promega CTLA 4 Blockade Bioassay:
Ipilimumab, BioE2052, 121 HulgG1**

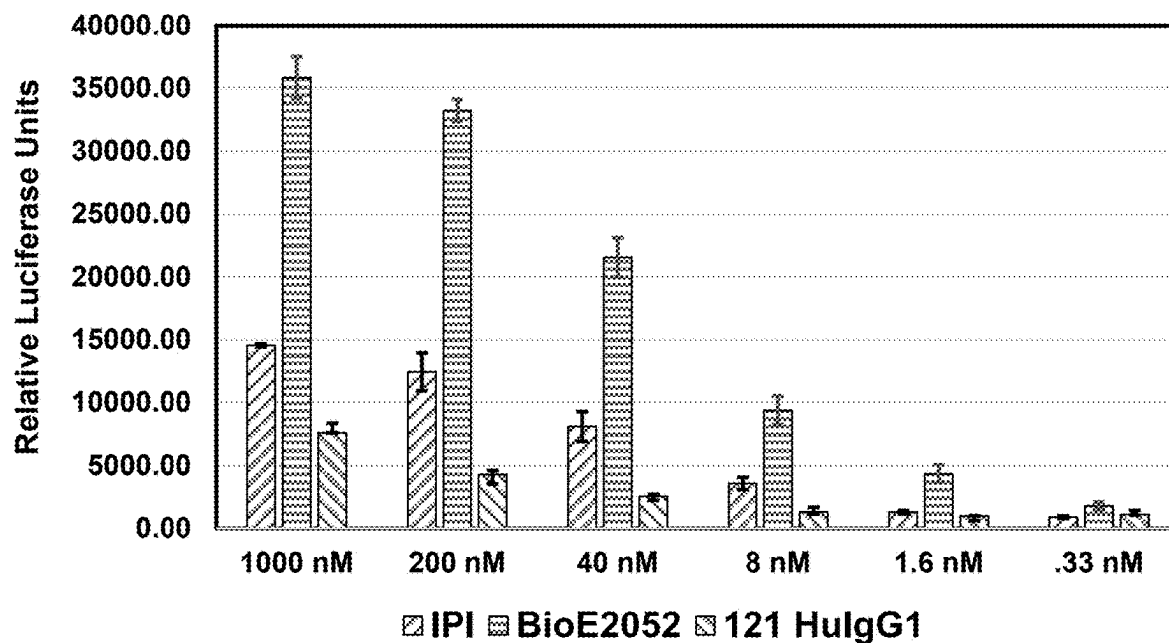
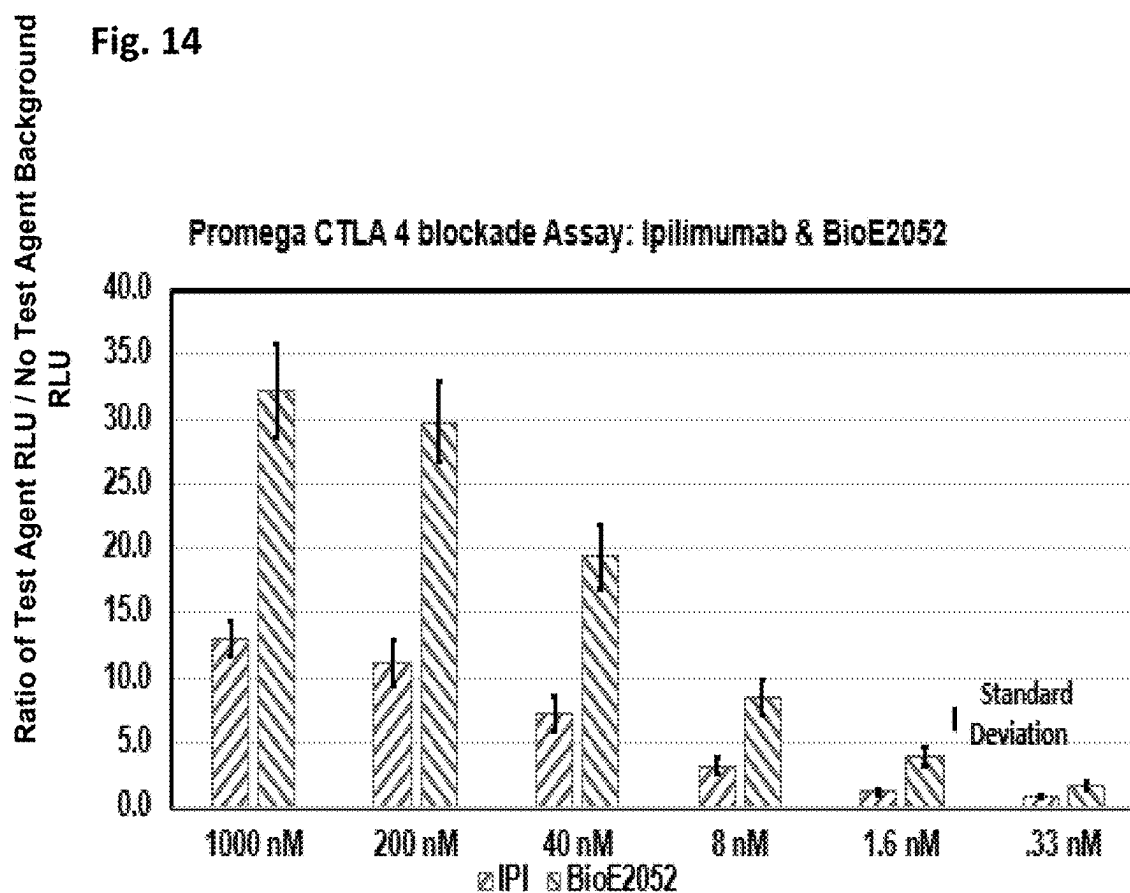


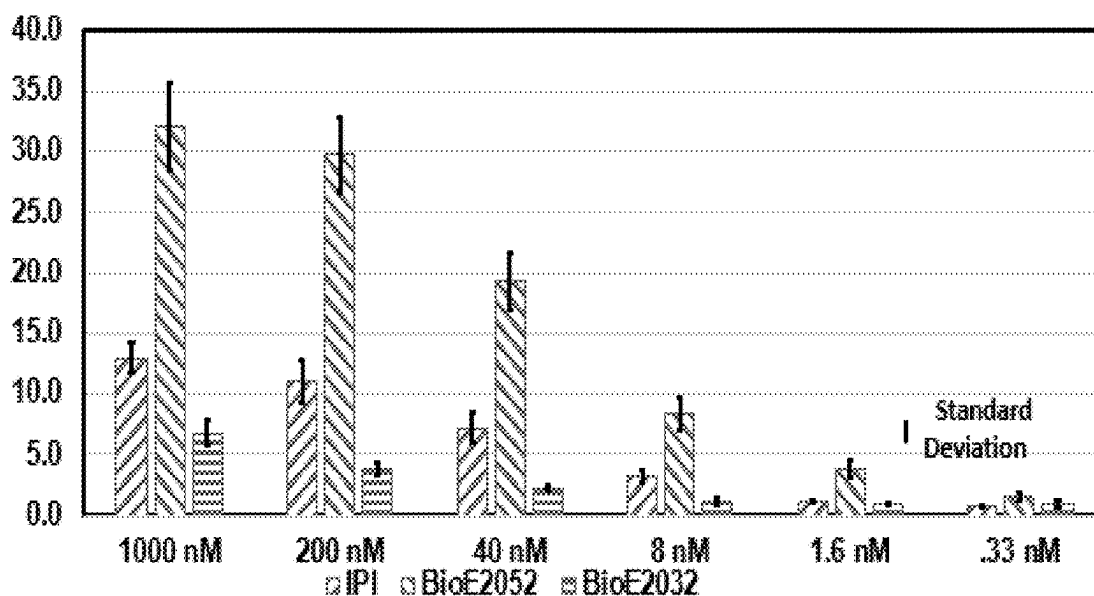
Fig. 14



Ratio of Test Agent RLU / No Test Agent Background RLU

Fig. 15

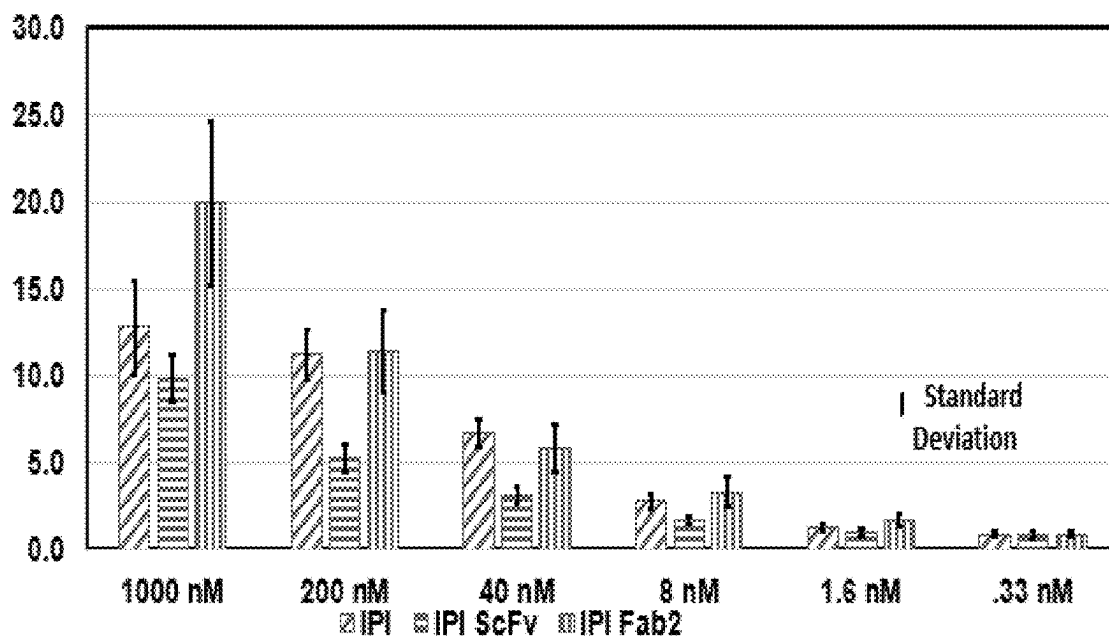
Promega CTLA 4 blockade Assay: Ipilimumab, BioE2052, BioE2032



Ratio of Test Agent RLU / No Test Agent Background RLU

Fig. 16

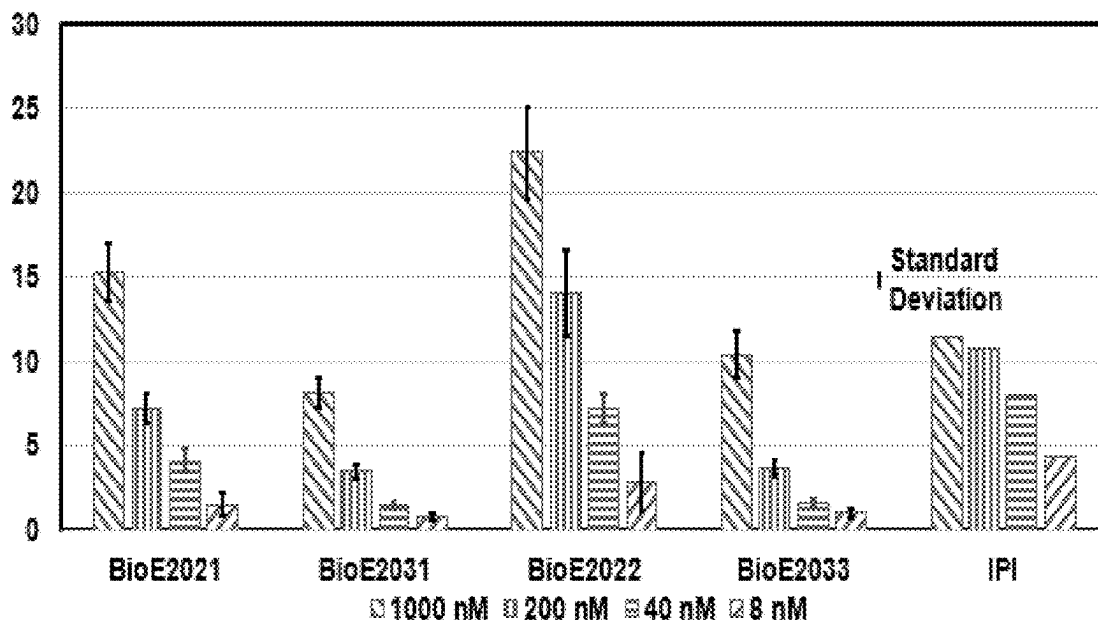
CTLA 4 Blockade Assay: Ipilimumab, IPI ScFv & IPI Fab2



Ratio of Test Agent RLU / No Test Agent Background RLU

Fig. 17

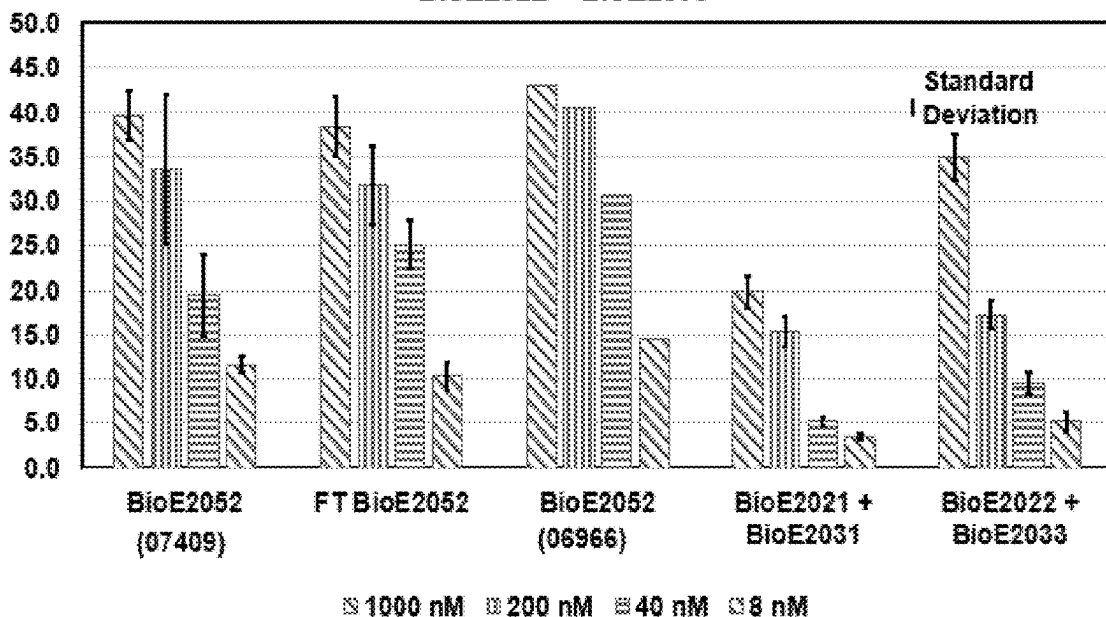
Promega CTLA 4 Assay: BioE2021, BioE2022, BioE2031, BioE2033 & IPI



Ratio of Test Agent RLU / No Test Agent Background RLU

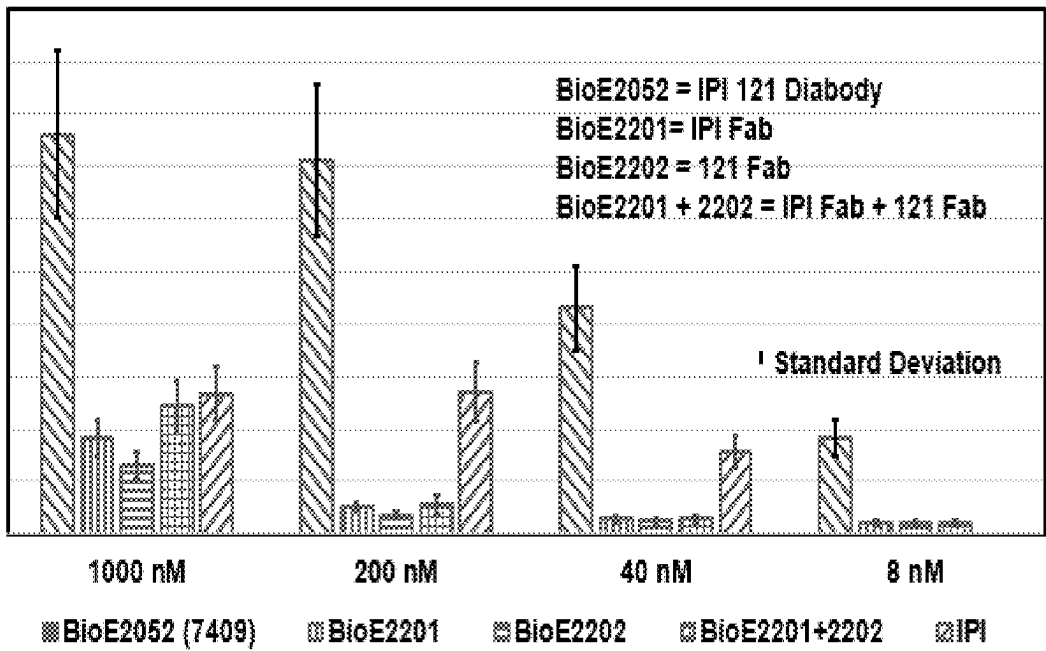
Fig. 18

Promega CTLA 4 Blockade Assay: BioE2052, BioE2021 + BioE2031, BioE2022 + BioE2033



Ratio of Test Agent RLU / No Test Agent Background RLU

Fig. 19



Ratio of Test Agent RLU / No Test Agent Background RLU

Fig. 20

Promega CTLA 4 Blockade Assay: BioE2052, BioE2022, BioE2033 & IPI

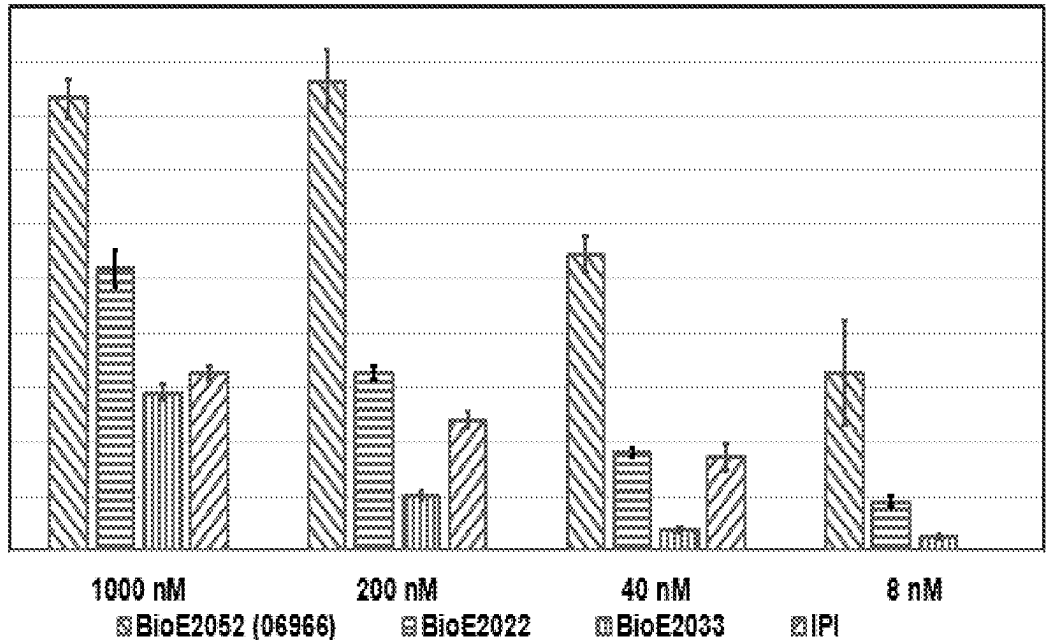


Fig. 21

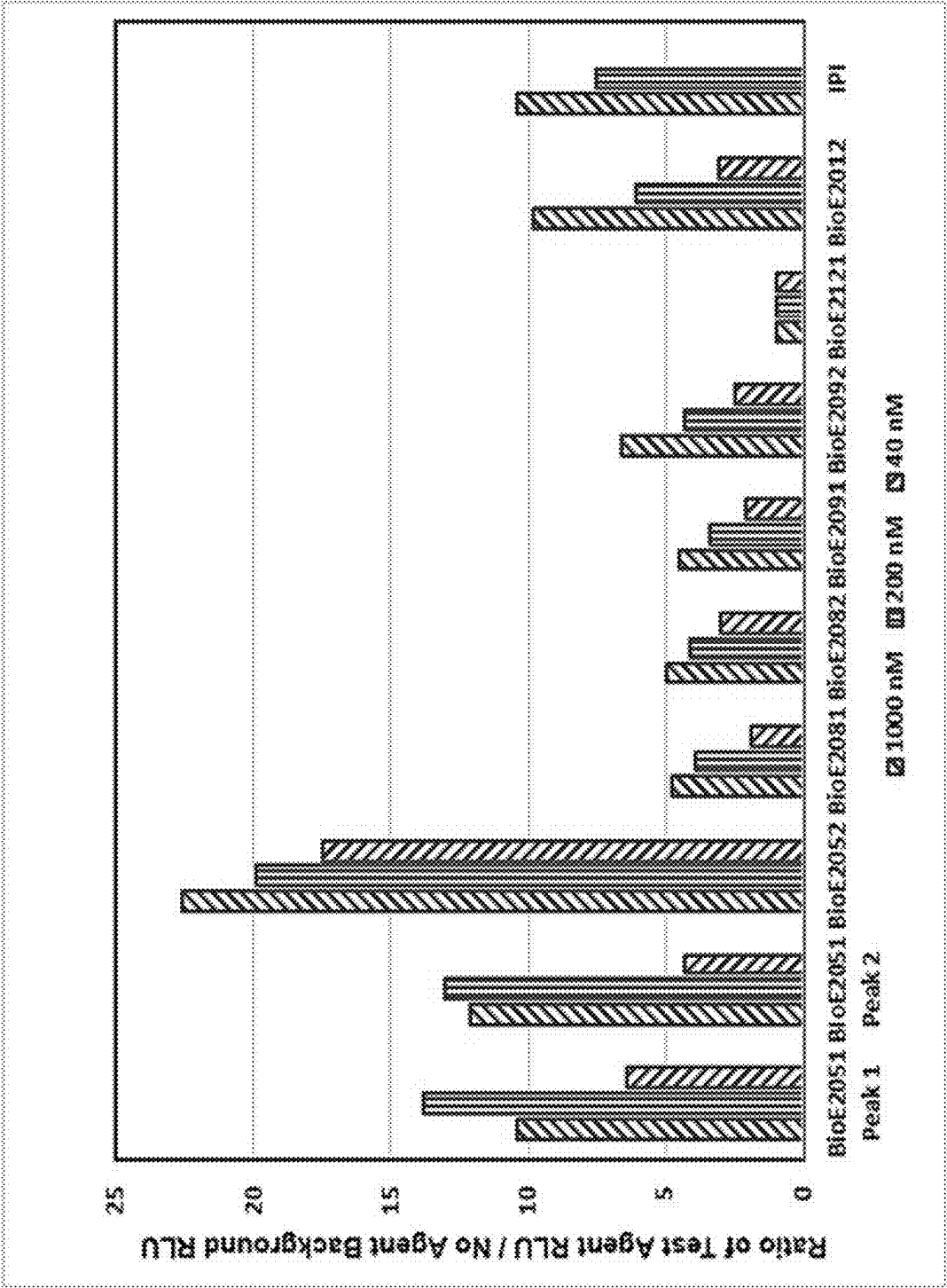


Fig. 22

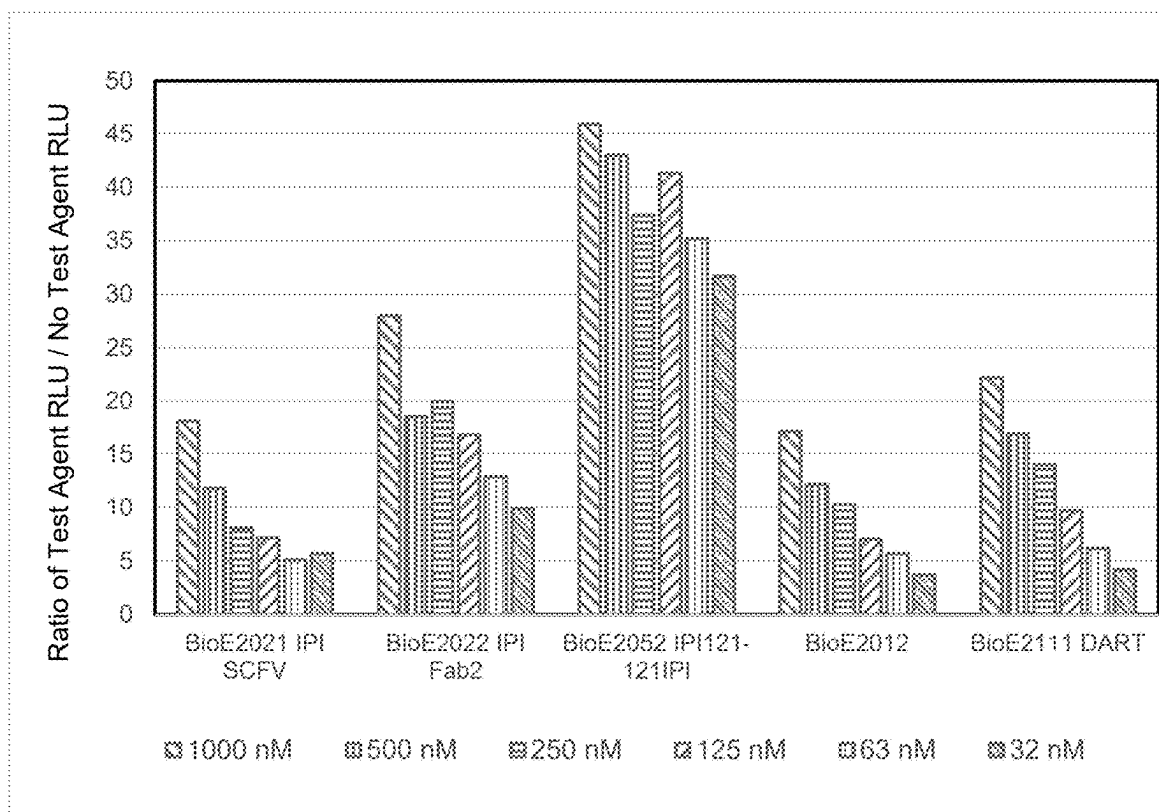


Fig. 23A

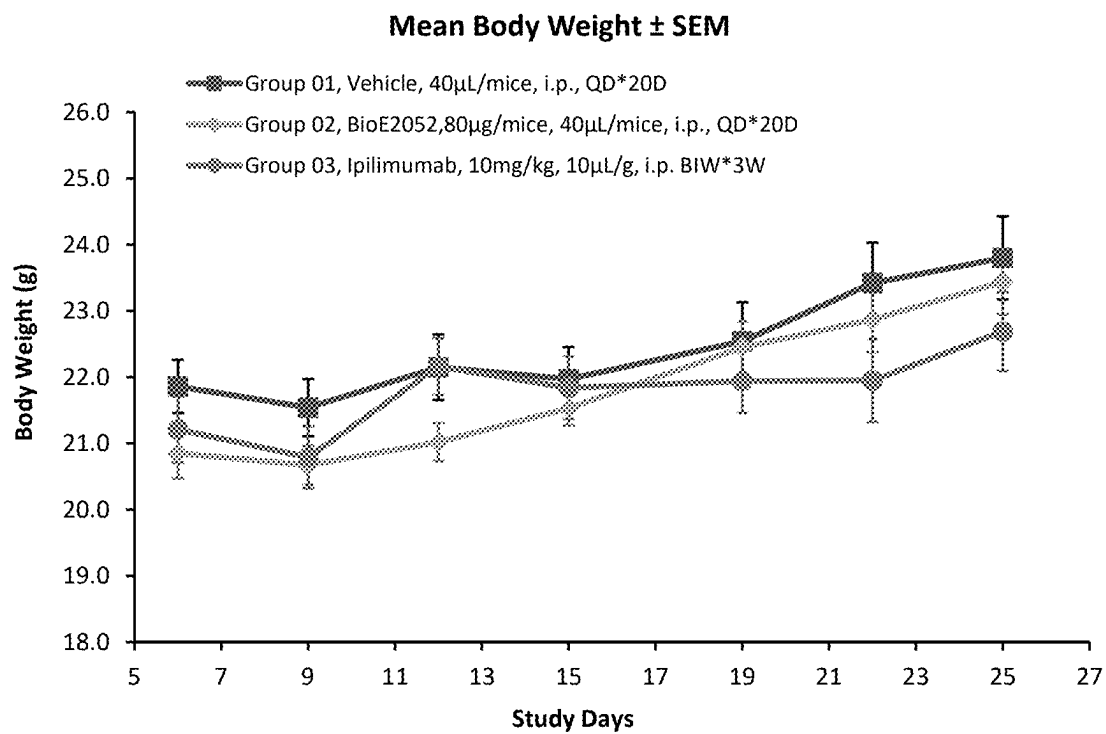


Fig. 23B

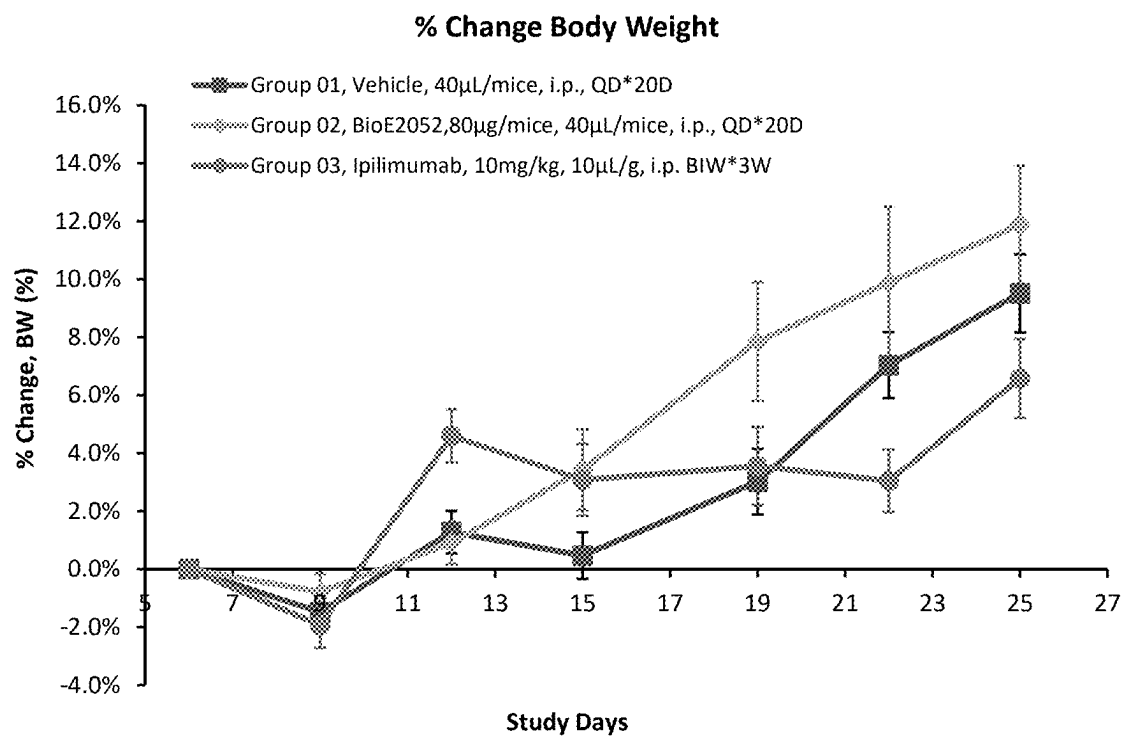


Fig. 24

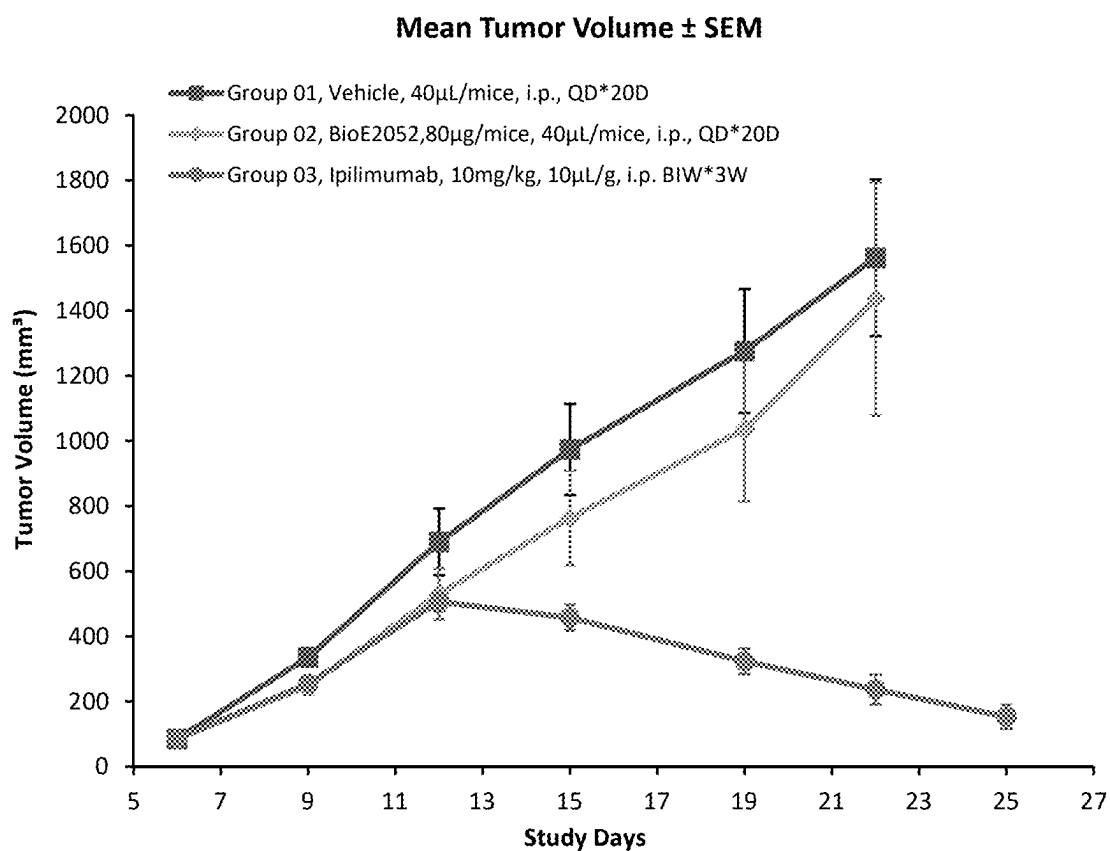
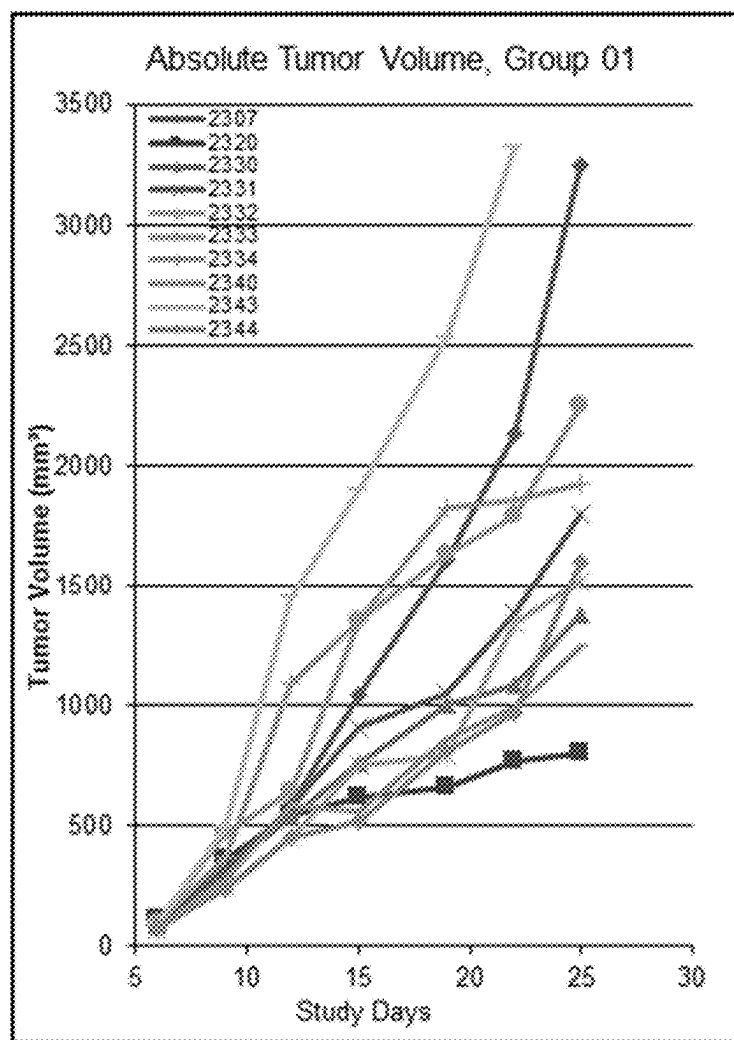
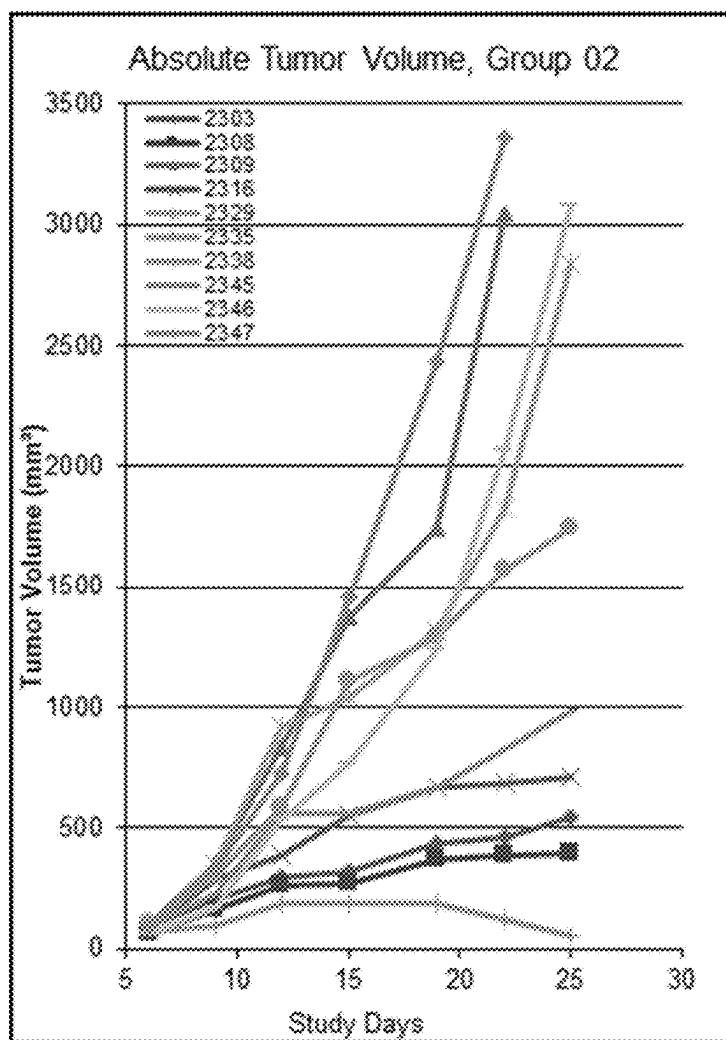


Fig. 25A



Saline 40 ul IP QD X 20D

Fig. 25B



BioE2052 4 mg/Kg IP QD X 20 D

Fig. 25C

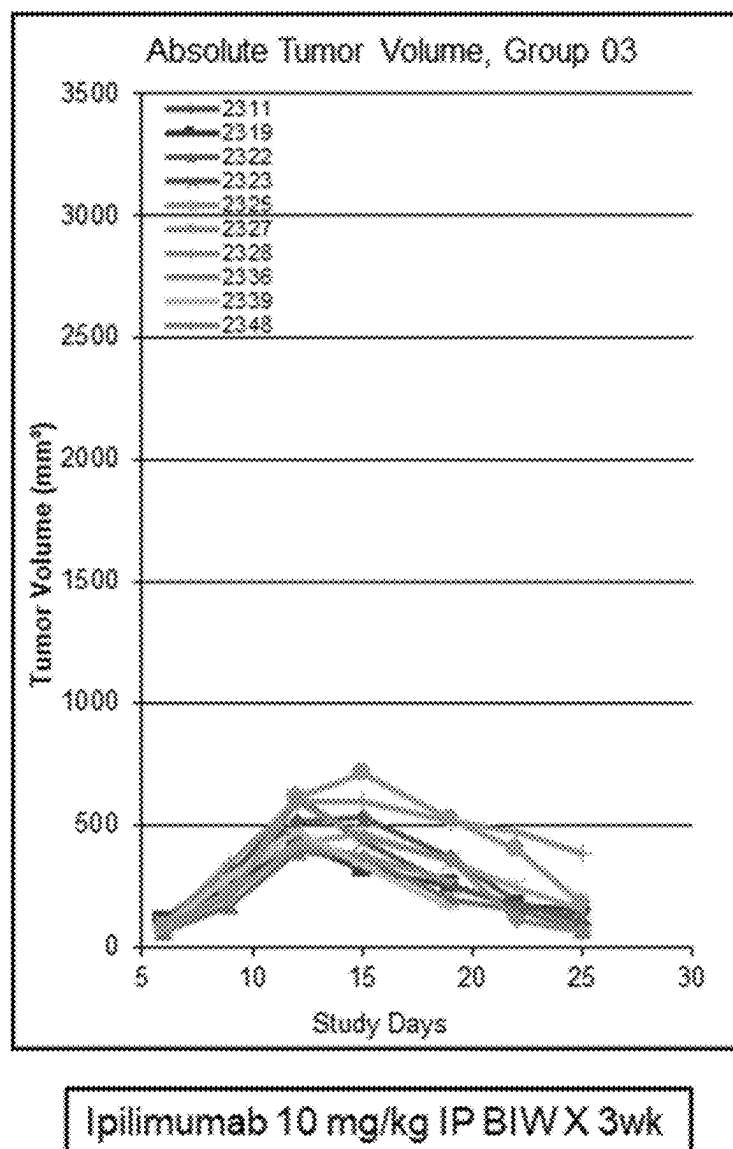


Fig. 26



Fig. 27

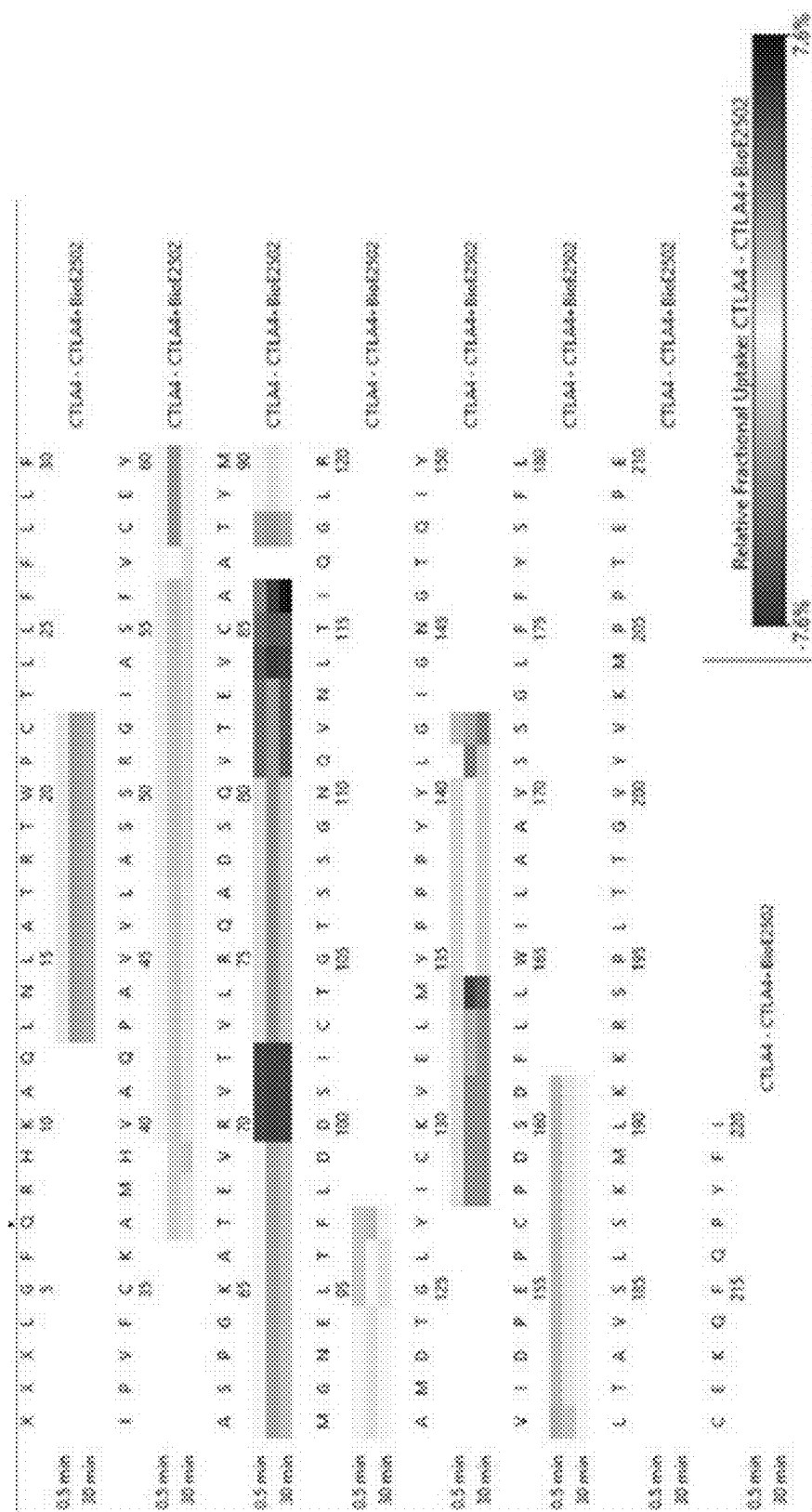


Fig. 28A

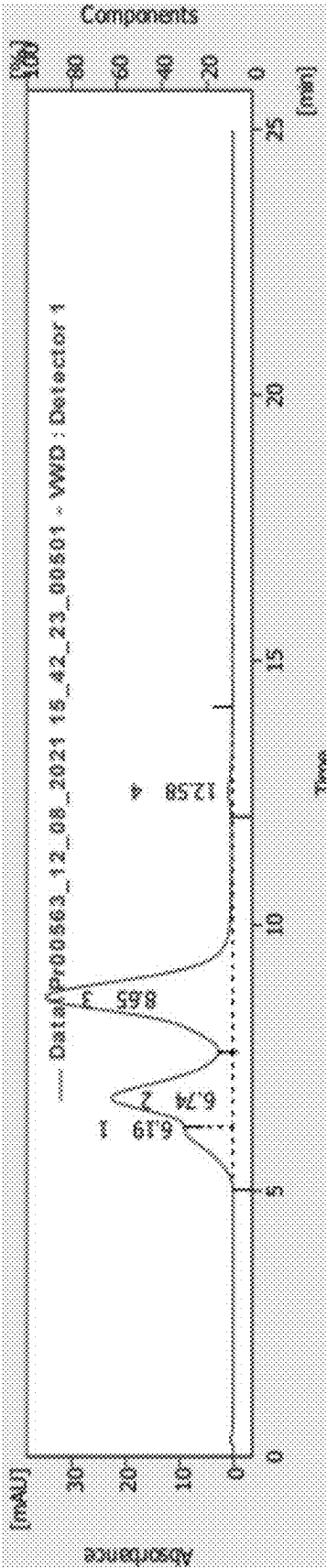


Fig. 28B

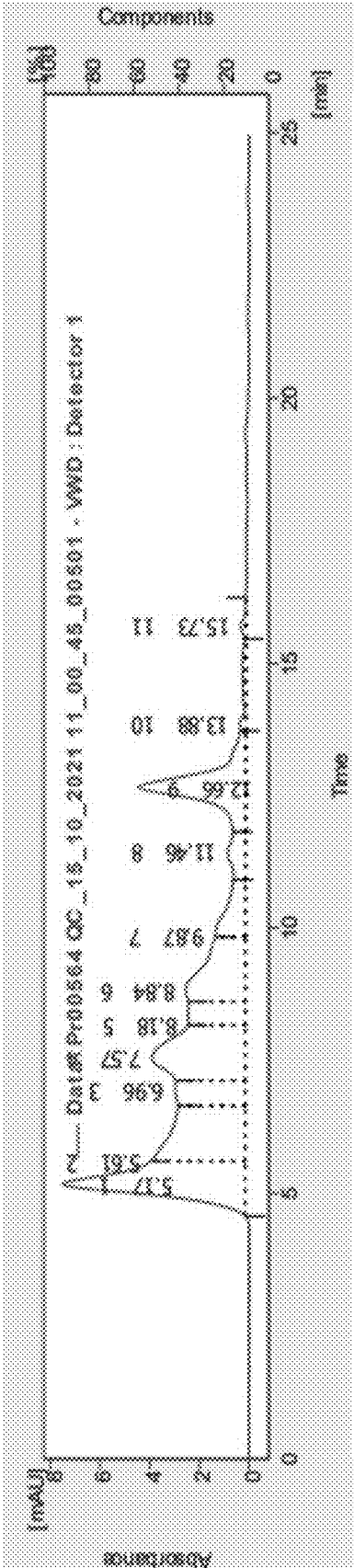


Fig. 29A

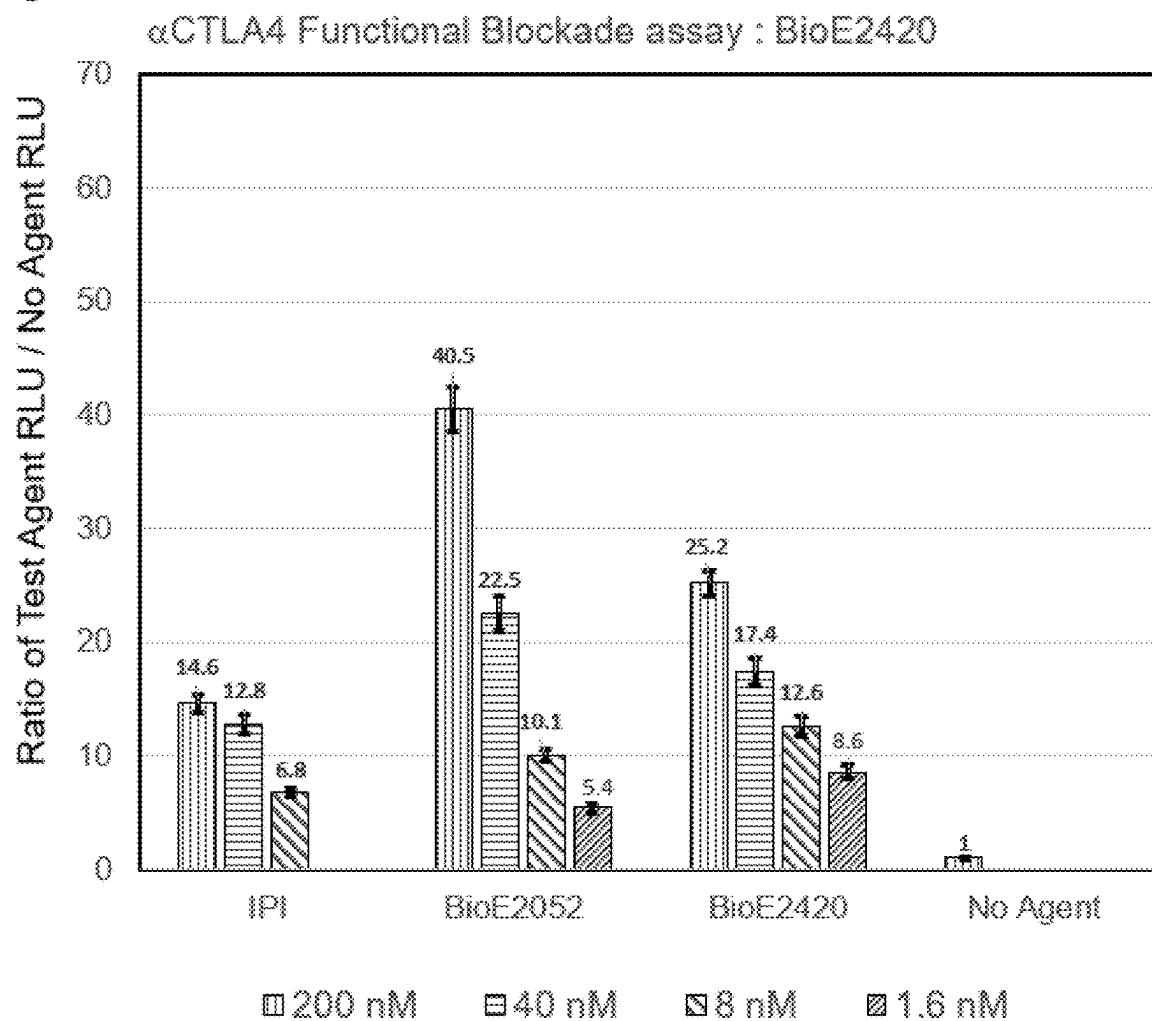
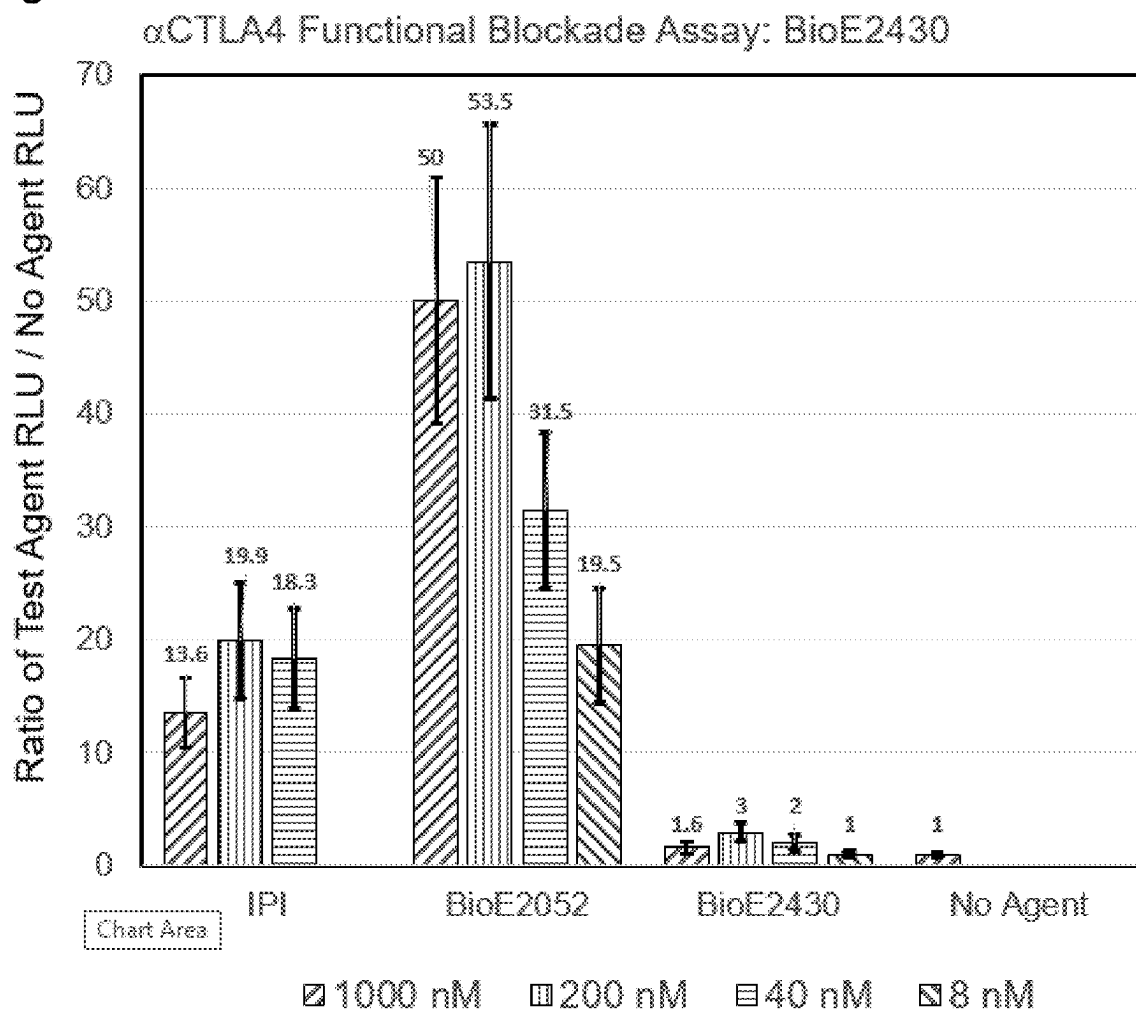


Fig. 29B



CTLA4-BINDING PROTEINS AND METHODS OF TREATING CANCER

RELATED APPLICATION

[0001] This application is the U.S. National Stage of International Patent Application No. PCT/US22/13120, filed Jan. 20, 2022, which claims the benefit of priority to U.S. Provisional Patent Application No. 63/139,510, filed on Jan. 20, 2021, the contents of both applications are fully incorporated by reference herein.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Feb. 5, 2024, is named BES-00125 SL.txt and is 212,712 bytes in size.

BACKGROUND

[0003] The striking clinical success of cancer immunotherapy with checkpoint blockade suggests it is likely to form the foundation of curative therapy for many malignancies (Reck et al. (2016) *N. Engl. J. Med.* 375:1823-1833; Hodi et al. (2010) *N. Engl. J. Med.* 363:711-723). The cytotoxic T-lymphocyte-associated antigen-4 (CTLA4) is an immune checkpoint that regulates T-cell proliferation at the early stage of naive T-cell activation, principally in the lymph nodes, thus providing a negative signal to T cells. Blockade of CTLA4 binding to its cognate ligand(s) induces an antitumor immune response by promoting the activation and proliferation of tumor-specific T cells.

[0004] Ipilimumab (Yervoy), a human monoclonal antibody that binds to human CTLA4 and blocks its interaction with ligands, demonstrated clinical efficacy in patients with melanoma, renal cell carcinoma, prostate cancer, urothelial carcinoma, and ovarian cancer. In 2011, ipilimumab was approved by the U.S. Food and Drug Administration (FDA) for the treatment of melanoma, and its 2020 sale is estimated at \$1.7 billion. However, ipilimumab has only a 22% long-term success rate in melanoma eradication, albeit effectively a cure. Additionally, ipilimumab is highly toxic with many side effects characteristic of autoimmune disease and TREG (Regulatory T cells) depletion, thereby limiting its broad use. Accordingly, a great need exists in the art for additional immunotherapeutic strategies with better efficacy and safety profiles.

SUMMARY

[0005] The present invention is based, at least in part, on the discovery that certain CTLA4-binding proteins show unexpected T cell activation properties. These CTLA4-binding proteins include those that bind multiple epitopes on CTLA4, e.g., Epitope 1 (¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1)) and Epitope 2 (⁶⁵SICT⁶⁸ (SEQ ID NO: 2)), as well as the entities that are devoid of the Fc region of the antibody (e.g., F(ab')₂ or diabodies). Surprisingly, the biparatopic entities presented herein showed synergistic effects on blocking CTLA4 ligand binding and T cell activation over the combination of monoparatopic counterparts (e.g., ipilimumab). In addition, the activity of the CTLA4-binding proteins devoid of the Fc domain surpassed the activity of ipilimumab or other CTLA4-binding antibodies with the Fc

domain. Such findings were surprising and unexpected as it is widely accepted that the interaction between the Fc region of a CTLA4 antibody with the Fc receptors is important for CTLA4-blocking activity (Bulliard et al. (2013) *J of Exp Medicine* 9:1685-1693; Waight et al. (2018) *Cancer Cell* 33:1033-1047; Ingram et al. (2018) *Proc Natl Acad Sci USA* 115:3912-3917; Vargas et al. (2018) *Cancer Cell* 33:1-15). Thus, the engineered antigen-binding proteins and pharmaceutical compositions comprising same promise clinical efficacy that surpasses that of ipilimumab.

[0006] In certain aspects, provided herein is an engineered antigen-binding protein that specifically binds to Epitope 1 defined by the residues ¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1) and Epitope 2 defined by the residues ⁶⁵SICT⁶⁸ (SEQ ID NO: 2) of CTLA4.

[0007] In certain aspects, provided herein is an engineered antigen-binding protein that specifically binds to CTLA4 and lacks the CH2 domain and/or a CH3 region of the constant region of an antibody. In some embodiments, the engineered antigen-binding protein is a diabody.

[0008] In certain aspects, provided herein is an isolated nucleic acid molecule that encodes an engineered antigen-binding protein of the present disclosure. Also provided herein is a vector comprising such nucleic acid. Further provided herein is a host cell which comprises the isolated nucleic acid, comprises a vector, or expresses an engineered antigen-binding protein of the present disclosure.

[0009] In certain aspects, provided herein is a pharmaceutical composition of an engineered antigen-binding protein of the present disclosure, the isolated nucleic acid, the vector, or the host cell. Further provided herein is a kit comprising at least one engineered antigen-binding protein of the present disclosure.

[0010] In certain aspects, provided herein is a method of producing at least one engineered antigen-binding protein of the present disclosure, wherein the method comprises the steps of: (i) culturing a host cell comprising a nucleic acid comprising a sequence encoding at least one engineered antigen-binding protein of the present disclosure under conditions suitable to allow expression of said engineered antigen-binding protein; and (ii) recovering the expressed engineered antigen-binding protein.

[0011] In certain aspects, provided herein is a method of preventing or treating a subject afflicted with cancer, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same.

[0012] In certain aspects, provided herein is a method of reducing proliferation of a cancer cell in a subject in need thereof, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same.

[0013] Numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in some embodiments, the at least one engineered antigen-binding protein or the pharmaceutical composition (a) reduces the number of proliferating cancer cells in the cancer; (b) reduces the volume or size of a tumor of the cancer; (c) increases the immune response against the cancer; and/or (d) activates the T cell.

[0014] In some embodiments, the engineered antigen-binding protein comprises an Fc domain that is heterologous

to the antigen-binding domain. In some embodiments, the heterologous Fc domain extends the half-life of the antigen-binding protein. In some embodiments, the Fc domain is an IgG1 Fc. In still other embodiments, the IgG1 Fc comprises a LALAPG mutation (Lo et al., (2017) *J. Biological Chem.*, 292:3900-08).

[0015] In some embodiments, the method further comprises administering to the subject an additional cancer therapy. In some embodiments, the additional cancer therapy is selected from the group consisting of immunotherapy, checkpoint blockade, cancer vaccines, chimeric antigen receptors, chemotherapy, radiation, target therapy, and surgery. In one embodiment, the additional cancer therapy is nivolumab.

[0016] In some embodiments, the cancer is selected from pancreatic cancer, lung cancer, non-small cell lung cancer (NSCLC), malignant pleural mesothelioma, small cell lung cancer (SCLC), renal cell carcinoma (RCC), breast cancer, liver cancer, hepatocellular carcinoma, kidney cancer, skin cancer, melanoma, thyroid cancer, gall bladder cancer, head-and-neck (squamous) cancer, stomach (gastric) cancer, head and neck cancer, bladder cancer, urothelial carcinoma, Merkel cell cancer, colon cancer, colorectal cancer, intestinal cancer, ovarian cancer, cervical cancer, testicular cancer, esophageal cancer, buccal cancer, brain cancer, blood cancers, lymphomas (B and T cell lymphomas), mesothelioma, cutaneous squamous cell cancer, Hodgkin's lymphoma, B-cell lymphoma, and a malignant or metastatic form thereof.

[0017] In some embodiments, the cancer is selected from melanoma (e.g., unresectable or metastatic melanoma), renal cell carcinoma (RCC), colorectal cancer, hepatocellular carcinoma, non-small cell lung cancer (NSCLC), malignant pleural mesothelioma, small cell lung cancer (SCLC), breast cancer, head and neck cancer, bladder cancer, urothelial carcinoma, Merkel cell cancer, cervical cancer, hepatocellular carcinoma, gastric cancer, cutaneous squamous cell cancer, Hodgkin's lymphoma, and B-cell lymphoma.

[0018] In certain aspects, provided herein is a method of increasing an immune response in a subject, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same.

[0019] In certain aspects, provided herein is a method of activating a T cell, the method comprising contacting the T cells with at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same.

[0020] In certain aspects, provided herein is a method of preventing or treating a disease or a condition characterized by aberrant expression or activity of a CTLA4 protein in a subject in need thereof, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same. In some embodiments, the disease or condition is a cancer, autoimmune disease, infection, or inflammatory disease.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 shows the CTLA4 blockade assay. The bioassay consists of two genetically engineered cell lines, CTLA4 Effector Cells (Jurkat) and aAPC/Raji Cells. When co-cultured, the CTLA4/CD80 and CD86 interaction inhibits the CD28 pathway activated luminescence (left panel).

The addition of anti-CTLA4 antibody blocks the CTLA4/CD80 and CD86 interaction, thereby re-establishing the CD28 pathway activated luminescence, which can be detected in a dose-dependent manner by addition of a luminescent agent (Glo) and quantitation with a luminometer (middle panel). When co-cultured with non-CTLA4-expressing Effector Cells (right panel), activation also induces luminescence by activation of the CD28 pathway but in a manner independent of anti-CTLA4 antibody.

[0022] FIG. 2A shows the CTLA4 blockade dose response to ipilimumab ("Ipi" or "IPI"), L3D10, or a combination of IPI and L3D10.

[0023] FIG. 2B shows the CTLA4 blockade dose response to ipilimumab ("Ipi" or "IPI") or L3D10 with the IL-2 level as an assay output.

[0024] FIG. 3 shows that anti-CTLA4 antibodies have the CTLA4 blocking activity that depends on the Fc receptor, CD32. The addition of an anti-CD32 antibody reduces the CTLA4 blockade mediated by various anti-CTLA4 antibodies. The antibodies tested herein include Ipi (ipilimumab), aCD32 (anti-CD32 antibody), L3D10 (BioLegend anti-CTLA4 antibody), and 26 HuIgG1.

[0025] FIG. 4 shows that anti-CTLA4 antibodies have the CTLA4 blocking activity that depends on the Fc receptor, CD32. The addition of an anti-CD32 antibody reduces the CTLA4 blockade mediated by various anti-CTLA4 antibodies, but not the antibody containing the LALA mutation that abrogates the interaction between the antibody's Fc region and CD32.

[0026] FIG. 5 shows that the 26 Fab' or 26 F(ab')₂ has an increased CTLA4 blocking activity as compared with the 26 HuIgG1 full-length antibody.

[0027] FIG. 6 shows that the increased CTLA4-blocking activity of 26 F(ab')₂ is independent of CD32.

[0028] FIG. 7 shows the CTLA4-blocking activity of ipilimumab, Ipi F(ab')₂, 121 HuIgG1, and 121 F(ab')₂.

[0029] FIG. 8 shows the CTLA4-blocking activity of ipilimumab, Ipi scFv, 121 HuIgG1, and 121 scFv.

[0030] FIGS. 9A and 9B show a schematic diagram of various antigen-binding proteins of the present disclosure. FIG. 9A shows the domain structures of various diabodies. The diagram is adapted from Kipriyanov et al. (1999) *J Mol Biol* 293:41-56, Volkel et al. (2001) *Protein Engineering* 14:815-823, Gall et al. (2004) *Protein Engineering, Design & Selection* 17:357-366, and Reusch et al. (2014) *mAbs* 6:727-738. FIG. 9B shows a schematic diagram of different types of antigen-binding proteins. FIG. 9A discloses "His6" as SEQ ID NO: 78.

[0031] FIGS. 10A-FIG. 10I show the domain structure and sequences of various diabodies. FIG. 10A shows the domain structure of exemplary anti-CTLA4 diabodies (BioE2051 and BioE2052). The diabodies comprise the VL and VH domains of ipilimumab and 121 antibody, and target two independent epitopes on CTLA4. FIG. 10A discloses "(G2S) 3" as SEQ ID NO: 75 and "His6" as SEQ ID NO: 78. FIG. 10B shows the diagram and sequence of BioE2052 (SEQ ID NO: 79). FIG. 10B discloses "His10" as SEQ ID NO: 92. FIG. 10C shows the diagram and sequence of BioE2051 (SEQ ID NO: 80). FIG. 10C discloses "His10" as SEQ ID NO: 92. FIG. 10D shows the diagram and sequence of BioE2022 (SEQ ID NO: 81) and BioE2023 (SEQ ID NO: 82) that form the Ipi F(ab')₂. FIG. 10E shows the diagram and sequence of BioE2201 (SEQ ID NO: 83) and BioE2023 (SEQ ID NO: 82) that form the Ipi Fab'. FIG. 10E discloses

“His10” as SEQ ID NO: 92. FIG. 10F shows the diagram and sequence of BioE2033 (SEQ ID NO: 84) and BioE2034 (SEQ ID NO: 88) that form the 121 F(ab')₂. FIG. 10G shows the diagram and sequence of BioE2202 (SEQ ID NO: 86) and BioE2034 (SEQ ID NO: 88) that form the 121 Fab'. FIG. 10H shows the comparison of the C-terminal sequences of BioE2022 (SEQ ID NO: 91) and BioE2201 (SEQ ID NO: 92). FIG. 10I shows the comparison of the C-terminal sequences of BioE2033 (SEQ ID NO: 93) and BioE2202 (SEQ ID NO: 93).

[0032] FIG. 11 shows that targeting two epitopes on CTLA4 increases the CTLA4-blocking activity. The anti-CTLA4 diabodies (BioE2051 and BioE2052) have an increased CTLA4-blocking activity as compared with ipilimumab or 121 HuIgG1.

[0033] FIG. 12 shows that targeting two epitopes on CTLA4 increases the CTLA4-blocking activity. The anti-CTLA4 diabody (BioE2052) or a combination of 121 scFv+ Ipi scFv has an increased CTLA4-blocking activity as compared with ipilimumab or 121 HuIgG1.

[0034] FIG. 13 shows that the anti-CTLA4 diabody (BioE2052) has an increased CTLA4-blocking activity as compared with ipilimumab or 121 HuIgG1.

[0035] FIG. 14 shows the increased CTLA4-blocking activity of the anti-CTLA4 diabody (BioE2052) as compared with ipilimumab.

[0036] FIG. 15 shows that the anti-CTLA4 diabody (BioE2052) has an increased CTLA4-blocking activity as compared with ipilimumab or 121 HuIgG1 (BioE2032).

[0037] FIG. 16 shows the CTLA4-blocking activity of ipilimumab, Ipi scFv, and Ipi F(ab')₂.

[0038] FIG. 17 shows the CTLA4-blocking activity of BioE2021 (Ipi scFv), BioE2031 (121 scFv), BioE2022 (Ipi F(ab')₂), BioE2033 (121 F(ab')₂), and ipilimumab.

[0039] FIG. 18 shows the CTLA4-blocking activity of the anti-CTLA4 diabody (BioE2052), a combination of BioE2021 (Ipi scFv)+BioE2031 (121 scFv), and a combination of BioE2022 (Ipi F(ab')₂)+BioE2033 (121 F(ab')₂).

[0040] FIG. 19 shows the CTLA4-blocking activity of the anti-CTLA4 diabody (BioE2052), BioE2201 (Ipi Fab'), BioE2202 (121 Fab'), a combination of BioE2201 (Ipi Fab')+BioE2202 (121 Fab'), and ipilimumab.

[0041] FIG. 20 shows the CTLA4-blocking activity of the anti-CTLA4 diabody (BioE2052 (06966)), BioE2022 (Ipi F(ab')₂), BioE2033 (121 F(ab')₂), and ipilimumab done in triplicate and data rendered in ratios.

[0042] FIG. 21 shows the CTLA4-blocking activity of the CTLA4-binding proteins (BioE2051, BioE2052, BioE2081, BioE2082, BioE2091, BioE2092, BioE2121, BioE2012, and ipilimumab).

[0043] FIG. 22 shows the CTLA4-blocking activity of the CTLA4-binding proteins (BioE2021 Ipi ScFv, BioE2022 Ipi F(ab')₂, BioE2052 Ipi 121-121Ipi, BioE2012, BioE2111 DART, and ipilimumab).

[0044] FIGS. 23A and 23B show the effects of BioE2052 and ipilimumab on mice body weight in the H22 model. FIG. 23A shows the effect of BioE2052 on body weight measured in grams. FIG. 23B shows the % change in body weight during treatment with BioE2052.

[0045] FIG. 24 shows the effects of administering BioE2052 and ipilimumab on tumor volume in the H22 Model

[0046] FIGS. 25A-25C show the changes in tumor volume in mice administered vehicle, BioE2052, or ipilimumab.

FIG. 25A shows the changes in tumor volume in mice administered vehicle. FIG. 25B shows the changes in tumor volume in mice administered BioE2052. FIG. 25C shows the changes in tumor volume in mice administered ipilimumab.

[0047] FIG. 26 shows the peptide coverage, as determined by deuterium exchange, for CTLA4 bound by BioE2052 (SEQ ID NO: 97).

[0048] FIG. 27 shows a heatmap of residues in CTLA4 that are susceptible to deuterium exchange in the presence of BioE2052 (SEQ ID NO: 97).

[0049] FIG. 28A is a chromatogram of size exclusion HPLC performed on a sample of monoclonal anti-CTLA4 antibody BioE2420.

[0050] FIG. 28B is a chromatogram of a size exclusion HPLC performed on a sample of monoclonal anti-CTLA4 antibody BioE2430.

[0051] FIG. 29A summarizes the results of a CTLA4 Functional Blockade assay of BioE2420.

[0052] FIG. 29B summarizes the results of a CTLA4 Functional Blockade assay of BioE2430.

DETAILED DESCRIPTION

[0053] Provided herein are engineered antigen-binding proteins that specifically bind to CTLA4 and show unexpected T cell activation properties. These proteins include those that bind to multiple epitopes on CTLA4, e.g., Epitope 1 (¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1)) and Epitope 2 (⁶⁵SICT⁶⁸ (SEQ ID NO: 2)), as well as F(ab')₂ or diabodies that are devoid of the Fc region of the antibody. The biparatopic entities presented herein showed unexpected synergistic effects on blocking CTLA4 ligand binding and T cell activation over the combination of monoparatopic counterparts (e.g., ipilimumab). In addition, the activity of the CTLA-binding proteins devoid of the Fc domain surpassed the activity of ipilimumab or other CTLA4-binding antibodies that comprise the Fc domain, which is contrary to the widely accepted belief that the interaction between the Fc region of a CTLA4 antibody with the Fc receptors is important for its CTLA4-blocking activity (Bulliard et al. (2013) *J of Exp Medicine* 9:1685-1693; Waight et al. (2018) *Cancer Cell* 33:1033-1047; Ingram et al. (2018) *Proc Natl Acad Sci USA* 115:3912-3917; Vargas et al. (2018) *Cancer Cell* 33:1-15). Thus, the engineered antigen-binding proteins and pharmaceutical compositions comprising same promise the clinical efficacy and safety that surpasses that of ipilimumab. Further provided herein are nucleic acids, vectors, host cells, conjugates, kits, pharmaceutical compositions, and methods of making the engineered antigen-binding proteins. Moreover, methods of treating a patient (e.g., those afflicted with a cancer or other diseases that are characterized by increased CTLA4 expression and/or activity), methods of increasing immune response, methods of reducing proliferation of a cancer cell, and methods of activating T cells are presented herein.

Definitions

[0054] The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element.

[0055] The term “antigen presenting cell” includes professional antigen presenting cells (e.g., B lymphocytes,

monocytes, dendritic cells, Langerhans cells) as well as other antigen presenting cells (e.g., keratinocytes, endothelial cells, astrocytes, fibroblasts, oligodendrocytes).

[0056] As used herein, the term “composite antibody” refers to an antibody which has variable regions comprising germline or non-germline immunoglobulin sequences from two or more unrelated variable regions. Additionally, the term “composite, human antibody” refers to an antibody which has constant regions derived from human germline or non-germline immunoglobulin sequences and variable regions comprising human germline or non-germline sequences from two or more unrelated human variable regions.

[0057] The terms “conjoint therapy” and “combination therapy,” as used herein, refer to the administration of two or more therapeutic substances. The different agents comprising the combination therapy may be administered concomitant with, prior to, or following the administration of one or more therapeutic agents.

[0058] By “detectable label” is meant a compound, substance, or composition that, when linked to a molecule of interest, renders the latter detectable, via spectroscopic, photochemical, biochemical, immunochemical, or chemical means. For example, useful labels include radioactive isotopes, magnetic beads, metallic beads, colloidal particles, fluorescent dyes, electron-dense reagents, enzymes (for example, as commonly used in an ELISA), biotin, digoxigenin, or haptens.

[0059] As used herein, the term “Fc region” is used to describe a C-terminal region of an immunoglobulin heavy chain, including native-sequence Fc regions and variant Fc regions. Although the boundaries of the Fc region of an immunoglobulin heavy chain might vary, the human IgG heavy-chain Fc region is usually defined to stretch from an amino acid residue at position Cys226, or from Pro230, to the carboxyl-terminus thereof. Suitable native-sequence Fc regions for use in the antibodies of the present invention include human IgG1, IgG2 (IgG2A, IgG2B), IgG3 and IgG4.

[0060] As used herein, “Fc receptor” or “FcR” describes a receptor that binds to the Fc region of an antibody. The preferred FcR is a native sequence human FcR. Moreover, a preferred FcR is one which binds an IgG antibody (a gamma receptor) and includes receptors of the FcγRI, FcγRII, and FcγRIII subclasses, including allelic variants and alternatively spliced forms of these receptors, FcγRII receptors include FcγRIIA (an “activating receptor”) and FcγRIIB (an “inhibiting receptor”), which have similar amino acid sequences that differ primarily in the cytoplasmic domains thereof. Activating receptor FcγRIIA contains an immunoreceptor tyrosine-based activation motif (ITAM) in its cytoplasmic domain. Inhibiting receptor FcγRIIB contains an immunoreceptor tyrosine-based inhibition motif (ITIM) in its cytoplasmic domain (see M. Daëron, *Annu. Rev. Immunol.* 15:203-234 (1997). FcRs are reviewed in Ravetch and Kinet, *Annu. Rev. Immunol.* 9:457-92 (1991); Capel et al., *Immunomethods* 4:25-34 (1994); and de Haas et al., *J. Lab. Clin. Med.* 126:330-41 (1995). Other FcRs, including those to be identified in the future, are encompassed by the term “FcR” herein.

[0061] Antibodies may also be “humanized”, which is intended to include antibodies made by a non-human cell having variable and constant regions which have been altered to more closely resemble antibodies that would be

made by a human cell. For example, by altering the non-human antibody amino acid sequence to incorporate amino acids found in human germline immunoglobulin sequences. The humanized antibodies of the present invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo), for example in the CDRs. The term “humanized antibody”, as used herein, also includes antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

[0062] “Interleukin-2 (IL-2)” is a cytokine signaling molecule that functions in the immune system. The IL-2 protein is produced primarily by activated T cells (CD4+ T cells); it regulates the activities of other T cells and B cells (increases growth and activity of these white blood cells) that are responsible for immunity. IL-2 is classified as a biologic response modifier that can modify the body’s response to cancer cells. The production of IL-2 exerts a wide spectrum of immunoregulatory effects on the immune system, e.g., increasing the proliferation and/or functional activity of other immune cells, such as tumor-infiltrating lymphocytes (TILs; T cells) and natural killer (NK) cells, enhancement of lymphocyte mitogenesis, lymphocyte cytotoxicity, induction of NK cells and lymphokine activated NK cells, and induction of interferon-γ production (S. L. Gaffen et al., 2004, *Cytokine*, 28 (3): 109-23). In T cells, IL-2 synthesis is tightly regulated at the mRNA level by signals from the T cell receptor (TCR) and CD28.

[0063] As used herein, the term “KD” is intended to refer to the dissociation equilibrium constant of a particular antibody-antigen interaction. The binding affinity of antibodies of the disclosed invention may be measured or determined by standard antibody-antigen assays, for example, competitive assays, saturation assays, or standard immunoassays such as ELISA or RIA.

[0064] A nucleic acid is “operably linked” when it is placed into a functional relationship with another nucleic acid sequence. For instance, a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence. With respect to transcription regulatory sequences, operably linked means that the DNA sequences being linked are contiguous and, where necessary to join two protein coding regions, contiguous and in reading frame. For switch sequences, operably linked indicates that the sequences are capable of effecting switch recombination.

[0065] The terms “prevent,” “preventing,” “prevention,” “prophylactic treatment,” and the like refer to reducing the probability of developing a disease, disorder, or condition in a subject, who does not have, but is at risk of or susceptible to developing a disease, disorder, or condition.

[0066] The term “remission” is art recognized, and refers to a condition in which the signs and symptoms of the cancer are reduced.

[0067] The term “selective” refers to a preferential action or function. The term “selective” can be quantified in terms of the preferential effect in a particular target of interest relative to other targets. For example, a measured variable (e.g., binding of the CTLA4-binding protein or the CTLA4-blocking activity) can be 10%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5-fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold,

7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, 11-fold, 12-fold, 13-fold, 14-fold, 15-fold, 16-fold, 17-fold, 18-fold, 19-fold, 20-fold, 25-fold, 30-fold, 35-fold, 40-fold, 45-fold, 50-fold, 55-fold, 60-fold, 70-fold, 80-fold, 90-fold, 100-fold, or greater or any range in between inclusive (e.g., 50% to 16-fold), different in a target of interest versus unintended or undesired targets. The same fold analysis can be used to confirm the magnitude of an effect in a given tissue, cell population, measured variable, measured effect, and the like.

[0068] By contrast, the term “specific” refers to an exclusionary action or function. For example, specific binding of an antibody or antigen-binding protein to a predetermined antigen refers to the ability of the antibody or antigen-binding protein to bind to the antigen of interest without binding to other antigens. Typically, the antibody binds with an affinity (K_D) of approximately less than 1×10^{-7} M, such as approximately less than 10^{-8} M, 10^{-9} M, 10^{-10} M, 10^{-11} M, or even lower to the predetermined antigen with an affinity that is at least 1.1-, 1.2-, 1.3-, 1.4-, 1.5-, 1.6-, 1.7-, 1.8-, 1.9-, 2.0-, 2.5-, 3.0-, 3.5-, 4.0-, 4.5-, 5.0-, 6.0-, 7.0-, 8.0-, 9.0-, or 10.0-fold or greater than its affinity for binding to a non-specific antigen (e.g., BSA, casein) other than the predetermined antigen or a closely-related antigen.

[0069] The term “sensitize” means to alter cells, such as cancer cells or tumor cells, in a way that allows for more effective treatment with a therapy (e.g., a CTLA4-binding protein). In some embodiments, normal cells are not affected to an extent that causes the normal cells to be unduly injured by the therapy (e.g., a CTLA4-binding protein). An increased sensitivity or a reduced sensitivity to a therapeutic treatment is measured according to a known method in the art for the particular treatment and methods described herein below, including, but not limited to, cell proliferative assays (Tanigawa N, Kern D H, Kikasa Y, Morton D L, Cancer Res 1982; 42:2159-2164), cell death assays (Weisenthal L M, Shoemaker R H, Marsden J A, Dill P L, Baker J A, Moran E M, Cancer Res 1984; 94:161-173; Weisenthal L M, Lippman M E, Cancer Treat Rep 1985; 69:615-632; Weisenthal L M, In: Kaspers G J L, Pieters R, Twentyman P R, Weisenthal L M, Veerman A J P, eds. Drug Resistance in Leukemia and Lymphoma. Langhorne, P A: Harwood Academic Publishers, 1993:415-432; Weisenthal L M, Contrib Gynecol Obstet 1994; 19:82-90). The sensitivity or resistance may also be measured in animal by measuring the tumor size reduction over a period of time, for example, 6 months for human and 4-6 weeks for mouse. A composition or a method sensitizes response to a therapeutic treatment if the increase in treatment sensitivity or the reduction in resistance is 5% or more, for example, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more, to 2-fold, 3-fold, 4-fold, 5-fold, 10-fold, 15-fold, 20-fold or more, compared to treatment sensitivity or resistance in the absence of such composition or method. The determination of sensitivity or resistance to a therapeutic treatment is routine in the art and within the skill of an ordinarily skilled clinician. It is to be understood that any method described herein for enhancing the efficacy of a CTLA4-binding protein, can be equally applied to methods for sensitizing hyperproliferative or otherwise cancerous cells (e.g., resistant cells) to the therapy.

[0070] As used herein, “subject” refers to any healthy animal, mammal or human, or any animal, mammal or human afflicted with a cancer. The term “subject” is interchangeable with “patient.”

[0071] The term “synergistic effect” refers to the combined effect of two or more therapeutic agents, such as two or more CTLA4 pathway modulators, either alone or in combination with another cancer therapy can be greater than the sum of the separate effects of individual agents alone. As used herein, the synergistic effect may also be used to refer to the effect of a single CTLA4-binding protein that comprises two or more binding moieties, wherein the effect (e.g., biological effect or therapeutic effect) is greater than the sum of the separate effects of the individual binding moieties.

[0072] Conventional T cells, also known as Tcons or Teffs, have effector functions (e.g., cytokine secretion, cytotoxic activity, anti-self-recognition, and the like) to increase immune responses by virtue of their expression of one or more T cell receptors. Tcons or Teffs are generally defined as any T cell population that is not a Treg and include, for example, naïve T cells, activated T cells, memory T cells, resting Tcons, or Tcons that have differentiated toward, for example, the Th1 or Th2 lineages. In some embodiments, Teffs are a subset of non-Treg T cells. In some embodiments, Teffs are CD4+Teffs or CD8+Teffs, such as CD4+ helper T lymphocytes (e.g., Th0, Th1, Th17, or Th17) and CD8+ cytotoxic T lymphocytes. As described further herein, cytotoxic T cells are CD8+T lymphocytes. “Naïve Tcons” are CD4+ T cells that have differentiated in bone marrow, and successfully underwent a positive and negative processes of central selection in a thymus, but have not yet been activated by exposure to an antigen. Naïve Tcons are commonly characterized by surface expression of L-selectin (CD62L), absence of activation markers such as CD25, CD44 or CD69, and absence of memory markers such as CD45RO. Naïve Tcons are therefore believed to be quiescent and non-dividing, requiring interleukin-7 (IL-7) and interleukin-15 (IL-15) for homeostatic survival (see, at least WO 2010/101870). The presence and activity of such cells are undesired in the context of suppressing immune responses. Unlike Tregs, Tcons are not anergic and can proliferate in response to antigen-based T cell receptor activation (Lechler et al. (2001) *Philos. Trans. R. Soc. Lond. Biol. Sci.* 356:625-637). In tumors, exhausted cells can present hallmarks of anergy.

[0073] The term “therapeutic effect” refers to a local or systemic effect in animals, particularly mammals, and more particularly humans, caused by a pharmacologically active substance. The term thus means any substance intended for use in the diagnosis, cure, mitigation, treatment or prevention of disease or in the enhancement of desirable physical or mental development and conditions in an animal or human.

[0074] The terms “therapeutically-effective amount” and “effective amount” as used herein means that amount of a compound, material, or composition comprising a compound encompassed by the present disclosure which is effective for producing some desired therapeutic effect in at least a sub-population of cells in an animal at a reasonable benefit/risk ratio applicable to any medical treatment. Toxicity and therapeutic efficacy of subject compounds may be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD_{50} and the ED_{50} . Compositions that exhibit large thera-

peutic indices are preferred. In some embodiments, the LD₅₀ (lethal dosage) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more reduced for the agent relative to no administration of the agent. Similarly, the ED₅₀ (i.e., the concentration which achieves a half-maximal inhibition of symptoms) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. Also, similarly, the IC₅₀ (i.e., the concentration which achieves half-maximal cytotoxic or cytostatic effect on cancer cells) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. In some embodiments, cancer cell growth in an assay can be inhibited by at least about 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100%. Cancer cell death can be promoted by at least about 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100%. In another embodiment, at least about a 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100% decrease in cancer cell numbers and/or a solid malignancy can be achieved.

CTLA4, Epitopes, and Ipilimumab

[0075] T cells, both CD4 (helper) and CD8 (cytotoxic), contribute to the adaptive immune response against pathogens and tumors, and activation and recruitment of specific T cells constitute a complex process. For a T cell to become fully activated (and subsequently proliferate and mediate effector function), at least 2 receptor-ligand interactions are required. The first of these occurs when the unique receptor of the T cell recognizes its cognate ligand, a short peptide presented in the context of a MHC molecule. This interaction is exquisitely specific, and if a good fit occurs, T-cell activation is initiated. However, full activation of a CD4 or CD8 T cell requires a second signal transmitted by costimulatory molecules present on the same antigen-presenting cell that expresses the peptide/MHC. This second signal is transmitted from costimulatory molecules (B7-1 (CD80) and/or B7-2 (CD86)) to a receptor on T cells known as CD28. Only when both signals are received and integrated does a specific T cell proliferate, acquire effector function, and migrate to sites of antigen expression.

[0076] CTLA4 is a homolog of CD28, suggesting that CTLA4 might serve, along with CD28, as a costimulatory molecule. However, several other studies provided opposing results, and for some time, it was not clear whether CTLA4 transmitted a stimulatory or inhibitory signal to T cells. The

generation of mice lacking CTLA4 provided a solution for this conundrum: Knockout mice developed a progressive accumulation of activated T cells and died of lymphoproliferative disease ~3 to 4 weeks after birth. These and other results suggested that blockade of CTLA4 with a monoclonal antibody could augment an adaptive immune response to an infectious agent or an evolving tumor. The subsequent study showed that CTLA4 blockade could attenuate the growth of several implanted murine tumors.

[0077] CTLA4 (also known as CD152) is a protein receptor that functions as an immune checkpoint and downregulates immune responses. CTLA4 is constitutively expressed in regulatory T cells but only upregulated in conventional T cells after activation—a phenomenon which is particularly notable in cancers. It acts as an “off” switch when bound to CD80 or CD86 on the surface of antigen-presenting cells. Thus, CTLA4 is a member of the immunoglobulin superfamily and encodes a protein which transmits an inhibitory signal to T cells.

[0078] The protein contains an extracellular V domain, a transmembrane domain, and a cytoplasmic tail. Alternate transcriptional splice variants, encoding different isoforms, have been characterized. The membrane-bound isoform functions as a homodimer interconnected by a disulfide bond, while the soluble isoform functions as a monomer. The intracellular domain is similar to that of CD28, in that it has no intrinsic catalytic activity and contains one YVKM motif (SEQ ID NO: 73) able to bind PI3K, PP2A and SHP-2 and one proline-rich motif able to bind SH3 containing proteins. The first role of CTLA4 in inhibiting T cell responses seem to be directly via SHP-2 and PP2A dephosphorylation of TCR-proximal signaling proteins such as CD3 and LAT. CTLA4 can also affect signaling indirectly via competing with CD28 for CD80/86 binding. CTLA4 can also bind PI3K, although the importance and results of this interaction are uncertain. CTLA4 is known to interact with various proteins such as CD80, CD86, CTXN3, MALL, PIK3R1, and TMEM218.

[0079] Mutations in CTLA4 have been associated with insulin-dependent diabetes mellitus, Graves’ disease, Hashimoto thyroiditis, celiac disease, systemic lupus erythematosus, thyroid-associated orbitopathy, and other autoimmune diseases.

[0080] The nucleic acids and polypeptide sequences of CTLA4 in humans and other organisms are well-known and include, for example, human CTLA4 (NM_001037631.3→NP_001032720.1 cytotoxic T-lymphocyte protein 4 isoform CTLA-4delTM; NM_005214.5→NP_005205.2 cytotoxic T-lymphocyte protein 4 isoform CTLA4-TM precursor), mouse CTLA4 (NM_001281976.1→NP_001268905.1 cytotoxic T-lymphocyte protein 4 isoform 2 precursor; NM_009843.4→NP_033973.2 cytotoxic T-lymphocyte protein 4 isoform 1 precursor), and rat (NM_031674.1→NP_113862.1 cytotoxic T-lymphocyte protein 4 precursor). Representative nucleic acid and polypeptide sequences are disclosed in Table 1 below.

TABLE 1

Exemplary sequence of CTLA4 and its epitopes	
SEQ ID NO: 1	
Epitope 1 of CTLA4 (amino acid residues 134-139 of CTLA4)	
1 mypppy	

TABLE 1-continued

Exemplary sequence of CTLA4 and its epitopes	
SEQ ID NO: 2	
Epitope 2 of CTLA4 (amino acid residues 65-68 of CTLA4)	
1 sict	
* The amino acid residues of 65-68 of ⁶⁵ SICT ⁶⁸ (SEQ ID NO: 2) correspond to the amino acid residues of the mature CTLA4 polypeptide. SEQ ID NO: 3 is the sequence of the CTLA4 polypeptide before maturation, thus the SICT epitope (SEQ ID NO: 2) corresponds to the amino acid residues 101-104 of SEQ ID NO: 3.	
SEQ ID NO: 3	
Human CTLA4 isoform CTLA4-TM amino acid sequence	
(NP_005205.2)	
1 maclgfqrhk aqlnlatrtw pctllffllf ipvfckamhv aqpavvlass rgiasfvcey	
61 aspgkatevr vtvrlrqadsq vtevcaatym mgneltfldd sictgtssgn qvnltiqglr	
121 amdtdglyick velmypppyy lgigngtqiy vidpepcpds dflwilaav ssgliffysfl	
181 ltavslskml kkrspittgv yvkmpptepe cekqfpyfi pin	
Human CTLA4 transcript variant 1 cDNA sequence	
SEQ ID NO: 4 Human CTLA4 transcript variant 1 cDNA sequence	
(NM_005214.5; CDS 173-844)	
1 gctttctatt caagtgcctt ctgtgtgtgc acatgtgttaa tacatatctg ggatcaaaagc	
61 tatctatata aagtccttga ttctgtgtgg gttcaaacac atttcaaagc ttcaggatcc	
121 tgaaggtttt tgctctactt cctgaagacc tgaacaccgc tccataaaag ccattggcttg	
181 ccttggtatt cagcggcaca aggtcagctt gaacctggct accaggacct ggccctgcac	
241 tctcctgttt ttctctctct tcatccctgt cttctgcaaa gcaatgcacg tggcccagcc	
301 tgctgtggta ctggccagca gccgagcat cgccagcttt gtgtgtgagt atgcatctcc	
361 aggcaaaagc actgaggtcc gggtagacgt gcttcggcag gctgacagcc aggtgactga	
421 agtctgtgcg gcaacctaca tgatggggaa tgagttgacc ttctagatg attccatctg	
481 cacgggcacc tccagtggaa atcaagtga cctcactatc caaggactga gggccatgga	
541 cacgggactc tacatctgca aggtggagct catgtaccca ccgccatact acctgggcat	
601 aggcaacgga acccagattt atgtaattga tccagaaccg tgcccagatt ctgacttcct	
661 cctctggatc ctgtcagcag ttagttcggg gtgtgttttt tatagcttcc tctcacagc	
721 tgtttctttg agcaaaatgc taaagaaaag aagccctctt acaacagggg tctatgtgaa	
781 aatgccccca acagagccag aatgtgaaaa gcaatttcag cttattttta ttcccatcaa	
841 ttgagaaacc attatgaaga agagagtcca tatttcaatt tccaagagct gaggcaattc	
901 taactttttt gctatccagc tatttttatt tgtttgtgca ttggggggga attcatctct	
961 cttaataata aagttggatg cggaacccaa attacgtgta ctacaattta aagcaaaagg	
1021 gtgaaaagac agagctggga tgtttctgtc acatcagctc cactttcagt gaaagcatca	
1081 cttgggatta atgtgggga gcagcattat gatgtgggtc aaggaattaa gttagggat	
1141 ggcacagccc aaagaaggaa aaggcaggga gcgagggaga agactatatt gtacacacct	
1201 tatatttacg tatgagacgt ttatagccga aatgatcttt tcaagttaaa ttttatgctt	
1261 ttattattct aaacaaatgt atgattacat caaggcttca aaaatactca catggctatg	
1321 ttttagccag tgatgctaaa ggttgatttg catatatata tatatatata tatatatata	
1381 tatatatata tatatatata tatatatata tatattttta ttgatagta ttgtgcatag	
1441 agccacgtat gtttttgtgt atttggtaat ggtttgaata taaacactat atggcagtgt	
1501 ctttccacct tgggtcccag ggaagttttg tggaggagct caggacacta ataccaccag	
1561 tagaacacaa ggtcatttgc taactagctt ggaacctgga tgaggtcata gcagtgtctg	
1621 attgcgtgtt ttctttctct gttgtgtgtt acatgtgctt tggggctttt acaccagttc	
1681 ctttcaatgg ttgcaaggaa agccacagct ggtggtatct gagttgactt gacagaacac	
1741 tgtcttgaag acaattggctt actccaggag acccaccaggt atgaccttct aggaagctcc	
1801 aspgkatevr vtvrlrqadsq vtevcaatym mgneltfldd sictgtssgn qvnltiqglr	
1861 aggtggcaga atgggggtgca tgaagggttc tgaataatga cactgctgtt gtttttaact	
1921 caatattttc catgaaaatg caacaacatg tataatattt ttaattaaat aaaaactctgt	
1981 ggtggtcgct ttccgga	
SEQ ID NO: 5	
Human CTLA4 isoform CTLA-4delTM amino acid sequence	
(NP_001032720.1)	
1 maclgfqrhk aqlnlatrtw pctllffllf ipvfckamhv aqpavvlass rgiasfvcey	
61 aspgkatevr vtvrlrqadsq vtevcaatym mgneltfldd sictgtssgn qvnltiqglr	
121 amdtdglyick velmypppyy lgigngtqiy viakekpsy nrglcnapn rarm	
SEQ ID NO: 6 Human CTLA4 transcript variant 2 cDNA sequence	
(NM_001037631.3; CDS 173-697)	
1 gctttctatt caagtgcctt ctgtgtgtgc acatgtgttaa tacatatctg ggatcaaaagc	
61 tatctatata aagtccttga ttctgtgtgg gttcaaacac atttcaaagc ttcaggatcc	
121 tgaaggtttt tgctctactt cctgaagacc tgaacaccgc tccataaaag ccattggcttg	
181 ccttggtatt cagcggcaca aggtcagctt gaacctggct accaggacct ggccctgcac	
241 tctcctgttt ttctctctct tcatccctgt cttctgcaaa gcaatgcacg tggcccagcc	
301 tgctgtggta ctggccagca gccgagcat cgccagcttt gtgtgtgagt atgcatctcc	
361 aggcaaaagc actgaggtcc gggtagacgt gcttcggcag gctgacagcc aggtgactga	
421 agtctgtgcy gcaacctaca tgatggggaa tgagttgacc ttctagatg attccatctg	
481 cacgggcacc tccagtggaa atcaagtga cctcactatc caaggactga gggccatgga	
541 cacgggactc tacatctgca aggtggagct catgtaccca ccgccatact acctgggcat	
601 aggcaacgga acccagattt atgtaattgc taaagaaaag aagccctctt acaacagggg	
661 tctatgtgaa aatgccccca acagagccag aatgtgaaaa gcaatttcag cttattttta	

TABLE 1-continued

Exemplary sequence of CTLA4 and its epitopes					
721	ttcccatcaa	ttgagaaacc	attatgaaga	agagagtcca	tatttcaatt
781	gaggcaattc	taactttttt	gctatccagc	tatttttatt	tgtttggtga
841	attcatctct	ctttaatata	aagttggatg	cggaacccaa	attacgtgta
901	aagcaaaaga	gtagaagac	agagctggga	tgtttctgtc	acatcagctc
961	gaaagcatca	cttgggatta	atatggggat	gcagcattat	gatgtgggtc
1021	gttagggaat	ggcacagccc	aaagaaggaa	aaggcagggg	gcgagggaga
1081	gtacacacct	tatatttacg	tatgagacgt	ttatagccga	aatgatcttt
1141	ttttatgcct	tttatttctt	aaacaaatgt	atgattacat	caaggcttca
1201	catggctatg	ttttagccag	tgatgctaaa	ggttgtattg	catatatata
1261	tatatatata	tatatatata	tatatatata	tatatatata	tattttttaa
1321	ttgtgcatag	agccacgtat	gtttttgtgt	atttgtaaat	ggtttgaata
1381	atggcagtg	ctttccacct	tgggtcccag	ggaagttttg	tggaggagct
1441	atacaccagg	tagaacacaa	ggctcattgc	taactagctt	ggaaactgga
1501	cgatgtcttg	attgtgtgga	attgtgctga	gttggtgttg	acatgtgctt
1561	acaccagttc	ctttcaatgg	tttgcaagga	agccacagct	ggtggtatct
1621	gacagaacac	tgtcttgaag	acaatggctt	actccaggag	acccacaggt
1681	aggaagctcc	agttcgtggg	gcccatttct	tacaaacatg	tggttaatgc
1741	agaaggcagc	aggtggcaga	atggggtgca	tgaaggtttc	tgaaaattaa
1801	gttttttaact	caatattttc	catgaaaatg	caacaacatg	tataatattt
1861	aaaaatctgt	ggtggctgctt	ttccgga		

SEQ ID NO: 7
 Mouse CTLA4 isoform 1 amino acid sequence (NP_033973.2)

1 macIglrryk aqlqlpsrtw pfvalltllf ipvfseaiqv tqpsvvllass hgvasfpcey
 61 spshntdevr vtvrlrqtn dq mtevcattft ekntvgfldy pfcsgrfnes rvnltiqglr
 121 avdtglylck velmypyppf vgmgngtgiy vidpepcpds dflilwilvav slglffysfl
 181 vtavslskml kkrspittgv yvkmpptepe cekqfipyfi pin

SEQ ID NO: 8
 Mouse CTLA4 transcript variant 1 cDNA sequence
 (NM_009843.4; CDS 147-818)

1 ctacacatat gtagcacgta ccttgatgca aagctgtcta tataaagtcc ccgagtctgt
 61 gtgggttcaa acacatctca aggtctctgg atcctgttgg gttttactct gtcacctgag
 121 gactctagca catttgcccc ccagccatgg ctgtgtctgg actccggagg tacaagctc
 181 aactgcagct gccttctagg acttggcctt ttgtagccct gctcactctt ctttccatcc
 241 cagtctcttc tgaagccata caggtgaccc aaccttcagt ggtgttggct agcagccatg
 301 gtgtgcccag ctttccatgt gaattattcac catcacacaa cactgatgag gtccgggtga
 361 ctgtgtctgc gcagacacaa gaccaaata ctgaggtctg tgccacgaca ttccacagaga
 421 agaatacagt gggcttccca gattaccctt tctgcagtgg tacctttaat gaaagcagag
 481 tgaacctcac catccaagga ctgagagctg ttgacacggg actgtacctc tgcaaggtgg
 541 aactcatgta ctttcaacaa ggggtctatg tgaaaatgcc ccccaacagag ccagaatgtg
 601 ttgatccaga accatgcccg gattctgact tctcctcttg gatccttctg gcagttagct
 661 tgggggttgt tttttacagt ttctcgtgca ctgctgtttc tttgagcaag atgctaaaga
 721 aaagaagtgc tottacaaca ggggtctatg tgaaaatgcc ccccaacagag ccagaatgtg
 781 aaaagcaatt tcagccttat tttattccca tcaactgaaa ggccgtttat gaagaagaag
 841 gagcatactt cagtctctaa aagctgaggg aatttcaact ttccctttct ctccagctat
 901 ttttacttgt ttgtatattt taaggagagt atgcctctct ttaatagaaa gctggatgca
 961 aaattccaat taagcatact acaattttaa gctaaggagc atgaacagag agctgggata
 1021 tttctgttgt gtcagaacca ttttactaaa agcatcactt ggaagcagca taaggatata
 1081 gcattatggt gtgggggtcaa ggggaacatta gggaaatggc cagcccaaaag aaaggaaggg
 1141 ggtgaaggaa gagattatat tgtacacatc ttgtatttac ctgagagatg tttatgactt
 1201 aaataatttt taaatttttc atgctgttat ttcttttaac aatgtataat tacacgaagg
 1261 ttttaacattt tattcacaga gctatgtgac atagccagtg gttccaaagg ttgtagtgtt
 1321 ccaagatgta tttttaagta atattgtaca tgggtgtttc atgtgctgtt gtgtatttgc
 1381 ttggtgtttg aatataaaca ctatgtatca gtgtcgtccc acagtgggtc ctggggagggt
 1441 ttggctgggg agcttaggac actaatccat caggttggac tcgaggtcct gcaccaactg
 1501 gcttggaaac tagatgagc tgtcacaggg ctgagttgca taaaccagat gtgatggagt
 1561 gtaaaactgg tctttacact cattttattt ttgtttctg cttttgtttt cttcaatgat
 1621 ttgcaaggaa accaaaagct ggcagtggtt gtatgaacct gacagaacac tgtcttcaag
 1681 gaaatgcctc attcctgaga ccagtaggtt tgttttttta ggaagtcca atactaggac
 1741 cccctacaag tactatggct cctcgaaaac acaaagttaa tgccacagga agcagcagat
 1801 ggtaggatgg gatgcacaag agttcctgaa aactaacact gttagtgttt tttttttaac
 1861 tcaatatttt ccatgaaaa gcaaccacat gtataatatt ttaattaaaa taaaagtttc
 1921 ttgtgattgt ttt

*Included in Table 1 are RNA nucleic acid molecules (e.g., thymidines replaced with uridines), nucleic acid molecules encoding orthologs of the encoded proteins, as well as DNA, cDNA, or RNA nucleic acid sequences comprising a nucleic acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with the nucleic acid sequence of any SEQ ID NO listed in Table 1, or a portion thereof. Such nucleic acid molecules can have a function of the full-length nucleic acid.

*Included in Table 1 are orthologs of the proteins, as well as polypeptide molecules comprising an amino acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with an amino acid sequence of any SEQ ID NO listed in Table 1, or a portion thereof. Such polypeptides can have a function of the full-length polypeptide.

*Included in Table 1 are other known CTLA4 nucleic acid and amino acid sequences.

[0081] Ipilimumab, sold under the brand name Yervoy, is a monoclonal antibody medication that works to activate the immune system by targeting CTLA4, a protein receptor that downregulates the immune system. Ipilimumab blocks the interaction between CTLA4 and its ligands. As described above, cytotoxic T lymphocytes (CTLs) can recognize and destroy cancer cells. However, an inhibitory mechanism interrupts this destruction. Ipilimumab turns off this inhibitory mechanism and boosts the body's immune response against cancer cells.

[0082] Ipilimumab binds to Epitope 1 (¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1)) of CTLA4. The MYPPPY motif (SEQ ID NO: 1), including Tyr139, is highly conserved across both CTLA4 and its immune stimulatory paralog CD28, and interfaces directly with both CD80 and with the CTLA4 inhibitor ipilimumab. Other antibodies, e.g., murine 26 antibody or humanized 121 antibody described herein and in U.S. Pat. No. 7,034,121 B2 bind to Epitope 2 (⁶⁵SICT⁶⁸ (SEQ ID NO: 2) of CTLA4. The significance of these epitopes were unknown prior to this disclosure.

[0083] Ipilimumab was approved by the US Food and Drug Administration (FDA) for treatment of melanoma (e.g., unresectable or metastatic melanoma in adults and pediatric patients), renal cell carcinoma (RCC), colorectal cancer, hepatocellular carcinoma, non-small cell lung cancer (NSCLC), and malignant pleural mesothelioma. Ipilimumab is also effective in combination with nivolumab that targets PD-1.

[0084] While effective, a major drawback of ipilimumab therapy is its association with severe and potentially fatal immunological adverse effects due to T cell activation and proliferation, occurring in ten to twenty percent of patients. Serious adverse effects include stomach pain, bloating, constipation, diarrhea, fever, trouble breathing, and urinating problems. Between 5.7 and 9.1% of individuals treated with ipilimumab develop checkpoint inhibitor induced colitis. Individual cases of severe neurologic disorders following ipilimumab have been observed, including acute inflammatory demyelination polyneuropathy and an ascending motor paralysis, and myasthenia gravis.

Antigen-Binding Proteins

[0085] Provided herein are antigen-binding proteins that bind to CTLA4. The antigen-binding proteins of the present disclosure can take any one of many forms of antigen-binding proteins known in the art. In various embodiments, the antigen-binding proteins of the present disclosure take the form of an antibody, or antigen-binding antibody fragment, or an antibody protein product.

[0086] In various embodiments of the present disclosure, the antigen-binding protein comprises, consists essentially of, or consists of an antibody or a fragment thereof. As used herein, the term "antibody" refers to a protein having a conventional immunoglobulin format, comprising heavy and light chains, and comprising variable and constant regions. For example, an antibody may be an IgG which is a "Y-shaped" structure of two identical pairs of polypeptide chains, each pair having one "light" (typically having a molecular weight of about 25 kDa) and one "heavy" chain (typically having a molecular weight of about 50-70 kDa). An antibody has a variable region and a constant region. In IgG formats, the variable region is generally about 100-110 or more amino acids, comprises three complementarity determining regions (CDRs), is primarily responsible for

antigen recognition, and substantially varies among other antibodies that bind to different antigens. Antibody-based antigen-binding proteins comprise the CDRs of the antibody, but not necessarily other regions (e.g., the constant region). The constant region allows the antibody to recruit cells and molecules of the immune system. The variable region is made of the N-terminal regions of each light chain and heavy chain, while the constant region is made of the C-terminal portions of each of the heavy and light chains. (Janeway et al., "Structure of the Antibody Molecule and the Immunoglobulin Genes", Immunobiology: The Immune System in Health and Disease, 4th ed. Elsevier Science Ltd./Garland Publishing, (1999)).

[0087] The general structure and properties of CDRs of antibodies have been described in the art. Briefly, in an antibody scaffold, the CDRs are embedded within a framework in the heavy and light chain variable region where they constitute the regions largely responsible for antigen binding and recognition. A variable region typically comprises at least three heavy or light chain CDRs (Kabat et al., 1991, Sequences of Proteins of Immunological Interest, Public Health Service N.I.H., Bethesda, Md.; see also Chothia and Lesk, 1987, J. Mol. Biol. 196:901-917; Chothia et al., 1989, Nature 342:877-883), within a framework region (designated framework regions 1-4, FR1, FR2, FR3, and FR4, by Kabat et al., 1991; see also Chothia and Lesk, 1987, supra).

[0088] CDR refers to a complementarity determining region (CDR) of which three make up the binding character of a light chain variable region (CDR-L1, CDR-L2 and CDR-L3) and three make up the binding character of a heavy chain variable region (CDR-H1, CDR-H2 and CDR-H3). CDRs contribute to the functional activity of an antibody molecule and are separated by amino acid sequences that comprise scaffolding or framework regions. The exact definitional CDR boundaries and lengths are subject to different classification and numbering systems. CDRs may therefore be referred to by Kabat, Chothia, contact or any other boundary definitions. Despite differing boundaries, each of these systems has some degree of overlap in what constitutes the so called "hypervariable regions" within the variable sequences. CDR definitions according to these systems may therefore differ in length and boundary areas with respect to the adjacent framework region. See for example Kabat, Chothia, and/or MacCallum et al., (Kabat et al., in "Sequences of Proteins of Immunological Interest," 5th Edition, U.S. Department of Health and Human Services, 1992; Chothia et al. (1987) J. Mol. Biol. 196, 901; and MacCallum et al., J. Mol. Biol. (1996) 262, 732, each of which is incorporated by reference in its entirety).

[0089] Antibodies can comprise any constant region known in the art. Human light chains are classified as kappa and lambda light chains. Heavy chains are classified as mu, delta, gamma, alpha, or epsilon, and define the antibody's isotype as IgM, IgD, IgG, IgA, and IgE, respectively. IgG has several subclasses, including, but not limited to IgG1, IgG2, IgG3, and IgG4. IgM has subclasses, including, but not limited to, IgM1 and IgM2. Embodiments of the present disclosure include all such classes or isotypes of antibodies. The light chain constant region can be, for example, a kappa- or lambda-type light chain constant region, e.g., a human kappa- or lambda-type light chain constant region. The heavy chain constant region can be, for example, an alpha-, delta-, epsilon-, gamma-, or mu-type heavy chain constant regions, e.g., a human alpha-, delta-, epsilon-, gamma-, or

mu-type heavy chain constant region. Accordingly, in various embodiments, the antibody is an antibody of isotype IgA, IgD, IgE, IgG, or IgM, including any one of IgG1, IgG2, IgG3 or IgG4. In various aspects, the antibody comprises a constant region comprising one or more amino acid modifications, relative to the naturally-occurring counterpart, in order to improve half-life/stability or to render the antibody more suitable for expression/manufacturability. In various instances, the antibody comprises a constant region wherein the C-terminal Lys residue that is present in the naturally-occurring counterpart is removed or clipped.

[0090] The antibody can be a monoclonal antibody. In some embodiments, the antibody comprises a sequence that is substantially similar to a naturally-occurring antibody produced by a mammal, e.g., mouse, rabbit, goat, horse, chicken, hamster, human, and the like. In this regard, the antibody can be considered as a mammalian antibody, e.g., a mouse antibody, rabbit antibody, goat antibody, horse antibody, chicken antibody, hamster antibody, human antibody, and the like. In certain aspects, the antigen-binding protein is an antibody, such as a human antibody. In certain aspects, the antigen-binding protein is a chimeric antibody or a humanized antibody. The term “chimeric antibody” refers to an antibody containing domains from two or more different antibodies. A chimeric antibody can, for example, contain the constant domains from one species and the variable domains from a second, or more generally, can contain stretches of amino acid sequence from at least two species. A chimeric antibody also can contain domains of two or more different antibodies within the same species. The term “humanized” when used in relation to antibodies refers to antibodies having at least CDR regions from a non-human source which are engineered to have a structure and immunological function more similar to true human antibodies than the original source antibodies. For example, humanizing can involve grafting a CDR from a non-human antibody, such as a mouse antibody, into a human antibody. Humanizing also can involve select amino acid substitutions to make a non-human sequence more similar to a human sequence. Information, including sequence information for human antibody heavy and light chain constant regions is publicly available through the Uniprot database as well as other databases well-known to those in the field of antibody engineering and production. For example, the IgG2 constant region is available from the Uniprot database as Uniprot number P01859, incorporated herein by reference.

[0091] An antibody can be cleaved into fragments by enzymes, such as, e.g., papain and pepsin. Papain cleaves an antibody to produce two Fab' fragments and a single Fc fragment. Pepsin cleaves an antibody to produce a F(ab')₂ fragment and a pFc' fragment. In various aspects of the present disclosure, the antigen-binding protein of the present disclosure is an antigen-binding fragment of an antibody (a.k.a., antigen-binding antibody fragment, antigen-binding fragment, antigen-binding portion). In various instances, the antigen-binding antibody fragment is a Fab' fragment or a F(ab')₂ fragment.

[0092] The architecture of antibodies has been exploited to create a growing range of alternative antibody formats that spans a molecular-weight range of at least about 12-150 kDa and has a valency (n) range from monomeric (n=1), to dimeric (n=2), to trimeric (n=3), to tetrameric (n=4), and potentially higher; such alternative antibody formats are referred to herein as “antibody protein products.” Antibody

protein products include those based on the full antibody structure and those that mimic antibody fragments which retain full antigen-binding capacity, e.g., scFvs, Fabs and VHH/VH (discussed below). The smallest antigen-binding fragment that retains its complete antigen binding site is the Fv fragment, which consists entirely of variable (V) regions. A soluble, flexible amino acid peptide linker is used to connect the V regions to a scFv (single chain fragment variable) fragment for stabilization of the molecule, or the constant (C) domains are added to the V regions to generate a Fab' fragment. Both scFv and Fab' fragments can be easily produced in host cells, e.g., prokaryotic host cells. Other antibody protein products include disulfide-bond stabilized scFv (ds-scFv), single chain Fab' (scFab'), as well as di- and multimeric antibody formats like dia-, tria- and tetra-bodies, or minibodies (miniAbs) that comprise different formats consisting of scFvs linked to oligomerization domains. The smallest fragments are VHH/VH of camelid heavy chain Abs as well as single domain Abs (sdAb). The building block that is most frequently used to create novel antibody formats is the single-chain variable (V)-domain antibody fragment (scFv), which comprises V domains from the heavy and light chain (VH and VL domain) linked by a peptide linker of ~15 amino acid residues. A peptibody or peptide-Fc fusion is yet another antibody protein product. The structure of a peptibody consists of a biologically active peptide grafted onto an Fc domain. Peptibodies are well-described in the art. See, e.g., Shimamoto et al., mAbs 4 (5): 586-591 (2012). Other antibody protein products include a single chain antibody (SCA); a diabody; a triabody; a tetrabody, and the like.

[0093] In various aspects, the antigen-binding protein of the present disclosure comprises, consists essentially of, or consists of any one of these antibody protein products. In various aspects, the antigen-binding protein of the present disclosure comprises, consists essentially of, or consists of any one of an scFv, Fab', F(ab')₂, VHH/VH, Fv fragment, ds-scFv, scFab', half antibody-scFv, heterodimeric Fab/scFv-Fc, heterodimeric scFv-Fc, heterodimeric IgG (CrossMab), tandem scFv, tandem biparatopic scFv, Fab/scFv-Fc, tandem Fab', single-chain diabody, dimeric antibody, multimeric antibody (e.g., a diabody, triabody, tetra-body), miniAb, peptibody VHH/VH of camelid heavy chain antibody, sdAb, diabody (single-chain diabody, homodimeric diabody, heterodimeric diabody, tandem diabody (TandAb), diabody that self-dimerizes), a triabody, a tetrabody. An ordinarily skilled artisan would understand that any bispecific antigen-binding protein formats can be used to generate biparatopic antigen-binding protein formats. In some embodiments, the antigen-binding protein is a dual-affinity re-targeting antibody (DART). In some embodiments, the antigen-binding protein is a bispecific T-cell engager (BiTE).

[0094] In various aspects, the antigen-binding protein of the present disclosure is linked to an agent. As described below, the agent may be any known in the art, including, but not limited to, chemotherapeutic agents, cytokines and growth factors, cytotoxic agents, detectable agent (e.g., fluorescein), and the like.

[0095] The antigen-binding proteins provided herein bind to CTLA4 in a non-covalent and reversible manner. In various embodiments, the binding strength of the antigen-binding protein to CTLA4 may be described in terms of its affinity, a measure of the strength of interaction between the

binding site of the antigen-binding protein and the epitope. In various aspects, the antigen-binding proteins provided herein have high-affinity for CTLA4 and thus will bind a greater amount of CTLA4 in a shorter period of time than low-affinity antigen-binding proteins. In various aspects, the antigen-binding protein has an equilibrium association constant, K_A , which is at least 105 mol^{-1} , at least 10^6 mol^{-1} , at least 10^7 mol^{-1} , at least 10^8 mol^{-1} , at least 10^9 mol^{-1} , or at least 10^{10} mol^{-1} . As understood by the artisan of ordinary skill, K_A can be influenced by factors including pH, temperature and buffer composition.

[0096] In various embodiments, the binding strength of the antigen-binding protein to CTLA4 may be described in terms of its sensitivity. K_D is the equilibrium dissociation constant, a ratio of k_{off}/k_{on} , between the antigen-binding protein and CTLA4. K_D and K_A are inversely related. The K_D value relates to the concentration of the antigen-binding protein (the amount of antigen-binding protein needed for a particular experiment) and so the lower the K_D value (lower concentration) the higher the affinity of the antigen-binding protein. In various aspects, the binding strength of the antigen-binding protein to CTLA4 may be described in terms of K_D . In various aspects, the K_D of the antigen-binding proteins provided herein is about 10^{-1} , about 10^{-2} , about 10^{-3} , about 10^{-4} , about 10^{-5} , about 10^{-6} , or less. In various aspects, the K_D of the antigen-binding proteins provided herein is micromolar, nanomolar, picomolar or femtomolar. In various aspects, the K_D of the antigen-binding proteins provided herein is within a range of about 10^{-4} to 10^{-6} or 10^{-7} to 10^{-9} or 10^{-10} to 10^{-12} or 10^{-13} to 10^{-15} . In various aspects, the K_D of the antigen-binding proteins provided herein is within a range of about $1.0 \times 10^{-12} \text{ M}$ to about $1.0 \times 10^{-8} \text{ M}$. In various aspects, the K_D of the antigen-binding proteins is within a range of about $1.0 \times 10^{-11} \text{ M}$ to about $1.0 \times 10^{-9} \text{ M}$.

[0097] In various aspects, the affinity of the antigen-binding proteins are measured or ranked using a flow cytometry- or Fluorescence-Activated Cell Sorting (FACS)-based assay. Flow cytometry-based binding assays are known in the art. See, e.g., Cedeno-Arias et al., *Sci Pharm* 79 (3): 569-581 (2011); Rathanaswami et al., *Analytical Biochem* 373:52-60 (2008); and Geuijen et al., *J Immunol Methods* 302 (1-2): 68-77 (2005). In various aspects, the affinity of the antigen-binding proteins are measured or ranked using a competition assay as described in Trikha et al., *Int J Cancer* 110:326-335 (2004) and Tam et al., *Circulation* 98 (11): 1085-1091 (1998), as well as below.

[0098] Avidity gives a measure of the overall strength of an antigen-binding protein-antigen complex. It is dependent on three major parameters: affinity of the antigen-binding protein for the epitope, valency of both the antigen-binding protein and CTLA4, and structural arrangement of the parts that interact. The greater an antigen-binding protein's valency (number of antigen binding sites), the greater the amount of antigen (CTLA4) it can bind. In various aspects, the antigen-binding proteins have a strong avidity for CTLA4. In various aspects, the antigen-binding proteins are multivalent. In various aspects, the antigen-binding proteins are bivalent. In various instances, the antigen antigen-binding proteins are monovalent.

[0099] In certain aspect, presented herein is an engineered antigen-binding protein that specifically binds to multiple epitopes on CTLA4.

[0100] In some embodiments, the engineered antigen-binding protein specifically binds Epitope 1 defined by the residues ¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1) and Epitope 2 defined by the residues ⁶⁵SICT⁶⁸ (SEQ ID NO: 2) of CTLA4.

[0101] In some embodiments, the engineered antigen-binding protein comprises: a) a heavy chain variable domain (VH) sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to a VH sequence selected from the group consisting of the VH sequences listed in Table 3; and/or b) a light chain variable domain (VL) sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to a VL sequence selected from the group consisting of the VL sequences listed in Table 3.

[0102] In some embodiments, the engineered antigen-binding protein comprises a sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to the sequence selected from SEQ ID NOS: 9, 10, 12, 14, 15, 17, 19, 20-24, 27, 28, and 29-54.

[0103] In some embodiments, the engineered antigen-binding protein comprises a sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to the sequence selected from: a) SEQ ID NOS: 9 and 10; b) SEQ ID NOS: 9 and 14; c) SEQ ID NOS: 9 and 17; d) SEQ ID NOS: 12 and 10; e) SEQ ID NOS: 12 and 14; f) SEQ ID NOS: 12 and 17; g) SEQ ID NOS: 15 and 10; h) SEQ ID NOS: 15 and 14; i) SEQ ID NOS: 15 and 17; j) SEQ ID NOS: 9, 10, 15, and 17; k) SEQ ID NOS: 12, 14, 15, and 17; l) SEQ ID NOS: 21 and 22; m) SEQ ID NOS: 27 and 28; n) SEQ ID NOS: 31, 32, and 33; o) SEQ ID NOS: 34, 35, and 36; p) SEQ ID NOS: 37, 38, and 39; q) SEQ ID NOS: 40, 41, and 42; r) SEQ ID NOS: 43 and 44; s) SEQ ID NOS: 45 and 46; t) SEQ ID NOS: 47, 48, and 49; u) SEQ ID NOS: 50, 51, and 52; and v) SEQ ID NOS: 53 and 54.

[0104] In some embodiments, the engineered antigen-binding protein comprises: a) a VH sequence selected from the group consisting of the VH sequences listed in Table 3; and/or b) a VL sequence selected from the group consisting of the VL sequences listed in Table 3.

[0105] In some embodiments, the engineered antigen-binding protein comprises a sequence selected from SEQ ID NOs: 9, 10, 12, 14, 15, 17, 19, 20-24, 27, 28, and 29-54.

[0106] In some embodiments, the engineered antigen-binding protein comprises a sequence selected from: a) SEQ ID NOs: 9 and 10; b) SEQ ID NOs: 9 and 14; c) SEQ ID NOs: 9 and 17; d) SEQ ID NOs: 12 and 10; e) SEQ ID NOs: 12 and 14; f) SEQ ID NOs: 12 and 17; g) SEQ ID NOs: 15 and 10; h) SEQ ID NOs: 15 and 14; i) SEQ ID NOs: 15 and 17; j) SEQ ID NOs: 9, 10, 15, and 17; k) SEQ ID NOs: 12, 14, 15, and 17; l) SEQ ID NOs: 21 and 22; m) SEQ ID NOs: 27 and 28; n) SEQ ID NOs: 31, 32, and 33; o) SEQ ID NOs: 34, 35, and 36; p) SEQ ID NOs: 37, 38, and 39; q) SEQ ID NOs: 40, 41, and 42; r) SEQ ID NOs: 43 and 44; s) SEQ ID NOs: 45 and 46; t) SEQ ID NOs: 47, 48, and 49; u) SEQ ID NOs: 50, 51, and 52; and v) SEQ ID NOs: 53 and 54.

[0107] In some embodiments, the engineered antigen-binding protein is selected from an antibody, Fv, F(ab')₂, Fab', dsFv, scFv, sc(Fv)₂, half antibody-scFv, tandem scFv, tandem biparatopic scFv, Fab/scFv-Fc, tandem Fab', single-chain diabody, tandem diabody (TandAb), Fab/scFv-Fc, heterodimeric Fab/scFv-Fc, heterodimeric scFv-Fc, heterodimeric IgG (CrossMab), DART, and diabody.

[0108] In certain embodiments, the engineered antigen-binding protein comprises an immunoglobulin heavy chain constant domain selected from the group consisting of IgG, IgG1, IgG2, IgG2A, IgG2B, IgG3, IgG4, IgA, IgM, IgD, and IgE constant domains.

[0109] In some embodiments, the engineered antigen-binding protein comprises an Fc domain. In some embodiments, the Fc domain is a functional or wild-type Fc domain that can bind to one or more Fc receptors. In some embodiments, the Fc domain may be nonfunctional, e.g., comprise a mutation, deletion, substitution, addition of one or more critical amino acids such that Fc domain (while structurally present) can no longer bind to one or more Fc receptors. In other embodiments, the engineered antigen-binding protein does not comprise (a) an Fc domain, or (b) the CH2 domain and/or CH3 domain of the constant region of an antibody. Accordingly, in some embodiments, the engineered antigen-binding protein does not bind to one or more Fc receptors, irrespective of whether the engineered antigen-binding protein comprises the Fc domain.

[0110] In certain aspects, provided herein is an engineered antigen-binding protein that specifically binds to CTLA4 and lacks the CH2 domain and/or a CH3 domain of the constant region of an antibody.

[0111] In some embodiments, the engineered antigen-binding protein comprises: a) a heavy chain variable domain (VH) sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to a VH sequence selected from the group consisting of the VH sequences listed in Table 3; and/or b) a light chain variable domain (VL) sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%,

91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to a VL sequence selected from the group consisting of the VL sequences listed in Table 3.

[0112] In some embodiments, the engineered antigen-binding protein comprises a sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to the sequence selected from SEQ ID NOs: 9, 10, 12, 14, 15, 17, 19, 20-24, 27, 28, and 29-54.

[0113] In some embodiments, the engineered antigen-binding protein comprises a sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to the sequence selected from: a) SEQ ID NOs: 9 and 10; b) SEQ ID NOs: 9 and 14; c) SEQ ID NOs: 9 and 17; d) SEQ ID NOs: 12 and 10; e) SEQ ID NOs: 12 and 14; f) SEQ ID NOs: 12 and 17; g) SEQ ID NOs: 15 and 10; h) SEQ ID NOs: 15 and 14; i) SEQ ID NOs: 15 and 17; j) SEQ ID NOs: 9, 10, 15, and 17; k) SEQ ID NOs: 12, 14, 15, and 17; l) SEQ ID NOs: 21 and 22; m) SEQ ID NOs: 27 and 28; n) SEQ ID NOs: 31, 32, and 33; o) SEQ ID NOs: 34, 35, and 36; p) SEQ ID NOs: 37, 38, and 39; q) SEQ ID NOs: 40, 41, and 42; r) SEQ ID NOs: 43 and 44; s) SEQ ID NOs: 45 and 46; t) SEQ ID NOs: 47, 48, and 49; u) SEQ ID NOs: 50, 51, and 52; and v) SEQ ID NOs: 53 and 54.

[0114] In some embodiments, the engineered antigen-binding protein comprises: a) a VH sequence selected from the group consisting of the VH sequences listed in Table 3; and/or b) a VL sequence selected from the group consisting of the VL sequences listed in Table 3.

[0115] In some embodiments, the engineered antigen-binding protein comprises a sequence selected from SEQ ID NOs: 9, 10, 12, 14, 15, 17, 19, 20-24, 27, 28, and 29-40.

[0116] In some embodiments, the engineered antigen-binding protein comprises a sequence selected from: a) SEQ ID NOs: 9 and 10; b) SEQ ID NOs: 9 and 14; c) SEQ ID NOs: 9 and 17; d) SEQ ID NOs: 12 and 10; e) SEQ ID NOs: 12 and 14; f) SEQ ID NOs: 12 and 17; g) SEQ ID NOs: 15 and 10; h) SEQ ID NOs: 15 and 14; i) SEQ ID NOs: 15 and 17; j) SEQ ID NOs: 9, 10, 15, and 17; k) SEQ ID NOs: 12, 14, 15, and 17; l) SEQ ID NOs: 21 and 22; m) SEQ ID NOs: 27 and 28; n) SEQ ID NOs: 31, 32, and 33; o) SEQ ID NOs: 34, 35, and 36; p) SEQ ID NOs: 37, 38, and 39; q) SEQ ID NOs: 40, 41, and 42; r) SEQ ID NOs: 43 and 44; s) SEQ ID NOs: 45 and 46; t) SEQ ID NOs: 47, 48, and 49; u) SEQ ID NOs: 50, 51, and 52; and v) SEQ ID NOs: 53 and 54.

[0117] In some embodiments, the engineered antigen-binding protein is selected from Fv, F(ab')₂, Fab', dsFv, scFv, sc(Fv)₂, half antibody-scFv, tandem scFv, tandem biparatopic scFv, Fab/scFv-Fc, tandem Fab', single-chain diabody, tandem diabody (TandAb), Fab/scFv-Fc, heterodimeric Fab/scFv-Fc, heterodimeric scFv-Fc, heterodimeric IgG (CrossMab), DART, and diabody.

[0118] In certain embodiments, the engineered antigen-binding protein is F(ab')₂. In some embodiments, the engineered antigen-binding protein comprises the sequence of SEQ ID NOs: 27 and 28. In some embodiments, the engineered antigen-binding protein comprises the sequence of SEQ ID NOs: 21 and 22.

[0119] In certain embodiments, the engineered antigen-binding protein binds two epitopes: Epitope 1 defined by the residues ¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1) and Epitope 2 defined by the residues ⁶⁵SICT⁶⁸ (SEQ ID NO: 2) of CTLA4.

[0120] In some embodiments, the engineered antigen-binding protein comprises at least two different VH domains and at least two different VL domains.

[0121] In some embodiments, the engineered antigen-binding protein is a DART or a diabody. In some embodiments, the diabody is a homodimeric diabody or a heterodimeric diabody. In some embodiments, the diabody is a single-chain diabody or a tandem diabody (TandAb). In some embodiments, the diabody is a tandem diabody.

[0122] In certain embodiments, the DART or the diabody comprises, in the N-terminal to C-terminal direction: a) a first VH; b) a first VL; c) a second VH; and d) a second VL; wherein the first VH and the first VL are linked by a Linker A; the first VL and the second VH are linked by a Linker B; and the second VH and the second VL are linked by a Linker C.

[0123] In some embodiments, the Linker A and/or Linker B comprise a peptide sequence of (GlyGlySer)_n, wherein n=1-20 (SEQ ID NO: 74). In some embodiments, the n=0, 1, 2, or 3. In some embodiments, the n=3.

[0124] In some embodiments, the Linker C comprises a peptide sequence of (GlyGlySer)_n, wherein n=1-20 (SEQ ID NO: 74). In some embodiments, the n is greater than 4. In some embodiments, the n=0, 1, 2, 3, or 4. In some embodiments, the n=3.

[0125] In some embodiments, each of the Linker A, Linker B, and Linker C comprises (GlyGlySer)_n, wherein n=3 (SEQ ID NO: 75).

[0126] In certain embodiments, the first VH and the second VL bind the Epitope 1 (SEQ ID NO: 1); and the first VL and second VH bind the Epitope 2 (SEQ ID NO: 2). In some embodiments, the first VH, first VL, second VH, second VL comprise the sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to SEQ ID NO: 15, SEQ ID NO: 14, SEQ ID NO: 12, and SEQ ID NO: 17, respectively. In some embodiments, the first VH, first VL, second VH, second VL comprise the sequence of SEQ ID NO: 15, SEQ ID NO: 14, SEQ ID NO: 12, and SEQ ID NO: 17, respectively.

[0127] In some embodiments, the engineered antigen-binding protein comprises the sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%,

99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to SEQ ID NO: 24 (BioE2052). In some embodiments, the engineered antigen-binding protein comprises the sequence of SEQ ID NO: 24 (BioE2052).

[0128] In certain embodiments, the first VH and the second VL bind the Epitope 2 (SEQ ID NO: 2); and the first VL and second VH bind the Epitope 1 (SEQ ID NO: 1). In some embodiments, the first VH, first VL, second VH, second VL comprise the sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to SEQ ID NO: 12, SEQ ID NO: 17, SEQ ID NO: 15, and SEQ ID NO: 14, respectively. In some embodiments, the first VH, first VL, second VH, second VL comprise the sequence as set forth in SEQ ID NO: 12, SEQ ID NO: 17, SEQ ID NO: 15, and SEQ ID NO: 14, respectively.

[0129] In some embodiments, the engineered antigen-binding protein comprises the sequence with at least or about 30%, 35%, 40%, 45%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100% identity to SEQ ID NO: 23 (BioE2051). In some embodiments, the engineered antigen-binding protein comprises the sequence of SEQ ID NO: 23 (BioE2051).

[0130] Numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in some embodiments, the CTLA4 is human CTLA4. In some embodiments, the engineered antigen-binding protein is chimeric, humanized, composite, murine, or human. In some embodiments, the engineered antigen-binding protein further comprises a peptide tag (e.g., Histidine tag). In some embodiments, the engineered antigen-binding protein further comprises a leader sequence. In some embodiments, the leader sequence comprises the sequence set forth in SEQ ID NO: 55. In some embodiments, the engineered antigen-binding protein is conjugated or detectably labeled, optionally wherein the engineered antigen-binding protein is PEGylated, e.g., to increase the half-life in circulation or for better solubility. In some embodiments, the engineered antigen-binding protein self-dimerizes. In some embodiments, the engineered antigen-binding protein: a) blocks the interaction between CTLA4 and its ligands (e.g., CD80 (B7-1) and CD86 (B7-2)); and/or b) increases the interleukin-2 (IL-2) expression by the T cells.

[0131] In certain aspects, provided herein is an isolated nucleic acid molecule that encodes the engineered antigen-binding protein of the present disclosure. Also provided herein is a vector comprising such isolated nucleic acid. Further provided herein is a host cell which comprises the said isolated nucleic acid, comprises the said vector, or expresses the engineered antigen-binding protein of the present disclosure.

[0132] In certain aspects, provided herein is a pharmaceutical composition comprising the engineered antigen-bind-

ing protein of the present disclosure, an isolated nucleic acid that encodes said engineered antigen-binding protein, a vector comprising said isolated nucleic acid, or a host cell comprising the said isolated nucleic acid, comprises the said vector, or expresses the engineered antigen-binding protein of the present disclosure.

[0133] In certain aspects, provided herein is a kit comprising at least one engineered antigen-binding protein of the present disclosure.

Sequence Identity/Homology

[0134] Function-conservative variants are those in which a given amino acid residue in a protein or enzyme has been changed without altering the overall conformation and function of the polypeptide, including, but not limited to, replacement of an amino acid with one having similar properties (such as, for example, polarity, hydrogen bonding potential, acidic, basic, hydrophobic, aromatic, and the like). Amino acids other than those indicated as conserved may differ in a protein so that the percent protein or amino acid sequence similarity between any two proteins of similar function may vary and may be, for example, from 70% to 99% as determined according to an alignment scheme such as by the Cluster Method, wherein similarity is based on the MEGALIGN algorithm. A function-conservative variant also includes a polypeptide which has at least 60% amino acid identity as determined by BLAST or FASTA algorithms, preferably at least 75%, more preferably at least 85%, still preferably at least 90%, and even more preferably at least 95%, and which has the same or substantially similar properties or functions as the native or parent protein to which it is compared.

[0135] The percent identity between two sequences is a function of the number of identical positions shared by the sequences (i.e., % identity=# of identical positions/total # of positions $\times$ 100), taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences. The comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm, as described in the non-limiting examples below.

[0136] The percent identity between two nucleotide sequences can be determined using the GAP program in the GCG software package (available on the World Wide Web at the GCG company website), using a NWSgapdna. CMP matrix and a gap weight of 40, 50, 60, 70, or 80 and a length weight of 1, 2, 3, 4, 5, or 6. The percent identity between two nucleotide or amino acid sequences can also be determined using the algorithm of E. Meyers and W. Miller (CABIOS, 4:11 17 (1989)) which has been incorporated into the

ALIGN program (version 2.0), using a PAM120 weight residue table, a gap length penalty of 12 and a gap penalty of 4. In addition, the percent identity between two amino acid sequences can be determined using the Needleman and Wunsch (J. Mol. Biol. (48): 444 453 (1970)) algorithm which has been incorporated into the GAP program in the GCG software package (available on the world wide web at the GCG company website), using either a Blossum 62 matrix or a PAM250 matrix, and a gap weight of 16, 14, 12, 10, 8, 6, or 4 and a length weight of 1, 2, 3, 4, 5, or 6.

[0137] The nucleic acid and protein sequences of the present invention can further be used as a "query sequence" to perform a search against public databases to, for example, identify related sequences. Such searches can be performed using the NBLAST and XBLAST programs (version 2.0) of Altschul, et al. (1990) J. Mol. Biol. 215:403 10. BLAST nucleotide searches can be performed with the NBLAST program, score=100, wordlength=12 to obtain nucleotide sequences homologous to the nucleic acid molecules of the present invention. BLAST protein searches can be performed with the XBLAST program, score=50, wordlength=3 to obtain amino acid sequences homologous to the protein molecules of the present invention. To obtain gapped alignments for comparison purposes, Gapped BLAST can be utilized as described in Altschul et al., (1997) Nucleic Acids Res. 25 (17): 3389 3402. When utilizing BLAST and Gapped BLAST programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used (available on the World Wide Web at the NCBI website).

Methods of Antibody Production and Related Methods

[0138] Suitable methods of making antigen-binding proteins (e.g., antibodies, antigen-binding antibody fragments, and antibody protein products) are known in the art. For instance, standard hybridoma methods for producing antibodies are described in, e.g., Harlow and Lane (eds.), *Antibodies: A Laboratory Manual*, CSH Press (1988), and CA. Janeway et al. (eds.), *Immunobiology*, 5th Ed., Garland Publishing, New York, NY (2001)).

[0139] Depending on the host species, various adjuvants can be used to increase the immunological response leading to greater antibody production by the host. Such adjuvants include but are not limited to Freund's, mineral gels such as aluminum hydroxide, and surface active substances such as lysolecithin, pluronic polyols, polyanions, peptides, oil emulsions, keyhole limpet hemocyanin, and dinitrophenol. BCG (bacilli Calmette-Guerin) and *Corynebacterium parvum* are potentially useful human adjuvants.

[0140] Other methods of antibody production are summarized in Table 2.

TABLE 2

Alternative Methods of Antibody Production	
Technique	Various references
EBV-hybridoma methods and Bacteriophage vector expression systems	Haskard and Archer, J. Immunol. Methods, 74(2), 361-67 (1984), Roder et al., Methods Enzymol., 121, 140-67 (1986), and Huse et al., Science, 246, 1275-81 (1989)).
methods of producing antibodies in non-human animals	U.S. Pat. Nos. 5,545,806, 5,569,825, and 5,714,352, and U.S. Patent Application Publication No. 2002/0197266

TABLE 2-continued

Alternative Methods of Antibody Production	
Technique	Various references
inducing in vivo production in the lymphocyte population or by screening recombinant immunoglobulin libraries or panels of highly specific binding reagents	Orlandi et al. (Proc Natl Acad Sci 86: 3833-3837; 1989), and Winter G and Milstein C (Nature 349: 293-299, 1991).
methods of producing recombinant proteins	Protein production and purification" Nat Methods 5(2): 135-146 (2008).
Phage display	Janeway et al., supra, Huse et al., supra, and U.S. Pat. No. 6,265,150). Related methods also are described in U.S. Pat. No. 5,403,484; U.S. Pat. No. 5,571,698; U.S. Pat. No. 5,837,500; U.S. Pat. No. 5,702,892. The techniques described in U.S. Pat. No. 5,780,279; U.S. Pat. No. 5,821,047; U.S. Pat. No. 5,824,520; U.S. Pat. No. 5,855,885; U.S. Pat. No. 5,858,657; U.S. Pat. No. 5,871,907; U.S. Pat. No. 5,969,108; U.S. Pat. No. 6,057,098; and U.S. Pat. No. 6,225,447
Antibodies can be produced by transgenic mice	U.S. Pat. Nos. 5,545,806 and 5,569,825, and Janeway et al., supra.

[0141] An antigen-binding protein can be engineered to increase or improve its pharmacokinetic (PK) properties (e.g., half-life). Numerous properties of an antigen-binding protein can influence pharmacokinetics including, but not limited to, molecular size, folding stability, solubility, target interaction, neonatal Fc binding capacity, and charge. Modifications to the antigen-binding protein include, but are not limited to antigen-binding domain conjugation to one or more carrier proteins, PEGylation, acylation (e.g., by conjugation to a fatty acid molecule), polysialylation, or glycosylation. Amino acid sequence modifications can be used to improve or optimize the PK properties of the protein, and conjugation to large, slowly metabolized macromolecules can also modify the PK properties of the protein. Macromolecules that can be conjugated to the antigen protein include, but are not limited to, proteins (e.g., albumin), polysaccharides (e.g., sepharose, agarose, cellulose, or cellulose beads), polymeric amino acids (polyglutamic acid or polylysine), amino acid copolymers, inactivated virus particles, inactivated bacterial toxins (e.g., leukotoxin or diphtheria, tetanus, or cholera toxins or molecules), inactivated bacteria, dendritic cells, thyroglobulin, polyamino acids (e.g., poly(D-lysine: D-glutamic acid)), VP6 polypeptides of rotaviruses, influenza virus hemagglutinin, influenza virus nucleoprotein, Keyhole Limpet Hemocyanin (KLH), and hepatitis B virus core protein and surface antigen (WO2021146436). Additional PK modulators known in the art include lipophiles, bile acids, steroids, phospholipid analogues, and vitamins, examples of which include, but are not limited to, cholesterol, fatty acids, cholic acid, lithocholic acid, dialkylglycerides, diacylglyceride, phospholipids, sphingolipids, naproxen, ibuprofen, vitamin E, and biotin (U.S. Pat. No. 9,322,018). Methods for producing modified antigen-binding proteins as described herein are known in the art.

[0142] In some embodiments, the antigen-binding protein receptor is fused or otherwise linked to a conventional fragment crystallizable region (Fc Region). For example, the Fc region can be an IgG1, IgG2, IgG3, or IgG4 Fc region.

In some embodiments, mutations in the Fc region of the antigen-binding protein can be engineered to modulate its interaction with the neonatal Fc receptor (FcRn), which is involved in receptor-mediated internalization and recycling of IgG occur via FcRn (Sokolosky and Szoka, Adv Drug Deliv Rev. (2015) 109-24), thereby improving its pharmacokinetic properties (US20210277092). For example, the Fc region can comprise a LALAPG mutation that inhibits binding of the antigen-binding protein to the neonatal Fc receptor. In some embodiments, the antigen-binding protein is fused or otherwise linked to an albumin-binding protein.

[0143] In some embodiments, the macromolecule is directly conjugated to the antigen-binding protein. In some embodiments, the macromolecule is fused to the antigen-binding peptide via a linker.

[0144] Modified antigen-binding proteins as described herein can have improved or optimized pharmacokinetic (PK) properties, for example, a plasma half-life in a human subject of greater than 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 9 hours, 10 hours, 12 hours, 18 hours, 24 hours, 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 10 days, 14 days, or 30 days.

[0145] Methods of testing the antigen-binding protein for the ability to bind to the epitope(s) of CTLA4 regardless of how the antigen-binding proteins are produced are known in the art and include any binding assay, such as, for example, radioimmunoassay (RIA), ELISA, Western blot, immunoprecipitation, SPR, and competitive inhibition assays (see, e.g., U.S. Patent Application Publication No. 2002/0197266).

Sequences

[0146] As used herein, coding region refers to regions of a nucleotide sequence comprising codons which are translated into amino acid residues, whereas noncoding region refers to regions of a nucleotide sequence that are not translated into amino acids (e.g., 5' and 3' untranslated regions).

[0147] Complement [to] or complementary refers to the broad concept of sequence complementarity between regions of two nucleic acid strands or between two regions of the same nucleic acid strand. It is known that an adenine residue of a first nucleic acid region is capable of forming specific hydrogen bonds (base pairing) with a residue of a second nucleic acid region which is antiparallel to the first region if the residue is thymine or uracil. Similarly, it is known that a cytosine residue of a first nucleic acid strand is capable of base pairing with a residue of a second nucleic acid strand which is antiparallel to the first strand if the residue is guanine. A first region of a nucleic acid is complementary to a second region of the same or a different nucleic acid if, when the two regions are arranged in an antiparallel fashion, at least one nucleotide residue of the first region is capable of base pairing with a residue of the second region. In some embodiments, the first region comprises a first portion and the second region comprises a second portion, whereby, when the first and second portions are arranged in an antiparallel fashion, at least or about 50%, and preferably at least or about 75%, at least or about 90%, or at least or about 95% of the nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion. In other embodiments, all nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion.

[0148] A nucleic acid is operably linked when it is placed into a functional relationship with another nucleic acid sequence. For instance, a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence. With respect to transcription regulatory sequences, operably linked means that the DNA sequences being linked are contiguous and, where necessary to join two protein coding regions, contiguous and in reading frame. For switch sequences, operably linked indicates that the sequences are capable of effecting switch recombination.

[0149] There is a known and definite correspondence between the amino acid sequence of a particular protein and the nucleotide sequences that can code for the protein, as defined by the genetic code (shown below). Likewise, there is a known and definite correspondence between the nucleotide sequence of a particular nucleic acid and the amino acid sequence encoded by that nucleic acid, as defined by the genetic code.

GENETIC CODE	
Alanine (Ala, A)	GCA, GCC, GCG, GCT
Arginine (Arg, R)	AGA, ACG, CGA, CGC, CGG, CGT
Asparagine (Asn, N)	AAC, AAT
Aspartic acid (Asp, D)	GAC, GAT
Cysteine (Cys, C)	TGC, TGT
Glutamic acid (Glu, E)	GAA, GAG
Glutamine (Gln, Q)	CAA, CAG
Glycine (Gly, G)	GGA, GGC, GGG, GGT
Histidine (His, H)	CAC, CAT
Isoleucine (Ile, I)	ATA, ATC, ATT
Leucine (Leu, L)	CTA, CTC, CTG, CTT, TTA, TTG
Lysine (Lys, K)	AAA, AAG
Methionine (Met, M)	ATG
Phenylalanine (Phe, F)	TTC, TTT
Proline (Pro, P)	CCA, CCC, CCG, CCT
Serine (Ser, S)	AGC, AGT, TCA, TCC, TCG, TCT

-continued

GENETIC CODE	
Threonine (Thr, T)	ACA, ACC, ACG, ACT
Tryptophan (Trp, W)	TGG
Tyrosine (Tyr, Y)	TAC, TAT
Valine (Val, V)	GTA, GTC, GTG, GTT
Termination signal (end)	TAA, TAG, TGA

[0150] An important and well-known feature of the genetic code is its redundancy, whereby, for most of the amino acids used to make proteins, more than one coding nucleotide triplet may be employed (illustrated above). Therefore, a number of different nucleotide sequences may code for a given amino acid sequence. Such nucleotide sequences are considered functionally equivalent since they result in the production of the same amino acid sequence in all organisms (although certain organisms may translate some sequences more efficiently than they do others). Moreover, occasionally, a methylated variant of a purine or pyrimidine may be found in a given nucleotide sequence. Such methylations do not affect the coding relationship between the trinucleotide codon and the corresponding amino acid.

[0151] In making the changes in the amino sequences of polypeptide, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biologic function on a protein is generally understood in the art. It is accepted that the relative hydropathic character of the amino acid contributes to the secondary structure of the resultant protein, which in turn defines the interaction of the protein with other molecules, for example, enzymes, substrates, receptors, DNA, antibodies, antigens, and the like. Each amino acid has been assigned a hydropathic index on the basis of their hydrophobicity and charge characteristics these are: isoleucine (+4.5); valine (+4.2); leucine (+3.8); phenylalanine (+2.8); cysteine/cystine (+2.5); methionine (+1.9); alanine (+1.8); glycine (−0.4); threonine (−0.7); serine (−0.8); tryptophane (−0.9); tyrosine (−1.3); proline (−1.6); histidine (−3.2); glutamate (−3.5); glutamine (−3.5); aspartate (<RTI 3.5); asparagine (−3.5); lysine (−3.9); and arginine (−4.5).

[0152] It is known in the art that certain amino acids may be substituted by other amino acids having a similar hydropathic index or score and still result in a protein with similar biological activity, i.e. still obtain a biological functionally equivalent protein.

[0153] As outlined above, amino acid substitutions are generally therefore based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. Exemplary substitutions which take various of the foregoing characteristics into consideration are well-known to those of skill in the art and include: arginine and lysine; glutamate and aspartate; serine and threonine; glutamine and asparagine; and valine, leucine and isoleucine.

[0154] In view of the foregoing, the nucleotide sequence of a DNA or RNA can be used to derive the polypeptide amino acid sequence, using the genetic code to translate the DNA or RNA into an amino acid sequence. Likewise, for polypeptide amino acid sequence, corresponding nucleotide sequences that can encode the polypeptide can be deduced from the genetic code (which, because of its redundancy,

will produce multiple nucleic acid sequences for any given amino acid sequence). Thus, description and/or disclosure herein of a nucleotide sequence which encodes a polypeptide should be considered to also include description and/or disclosure of the amino acid sequence encoded by the

nucleotide sequence. Similarly, description and/or disclosure of a polypeptide amino acid sequence herein should be considered to also include description and/or disclosure of all possible nucleotide sequences that can encode the amino acid sequence.

TABLE 3

Sequence of representative CTLA4-binding proteins or fragments thereof	
SEQ ID NO: 9	
Heavy chain variable domain (VH) of the 26 antibody-amino acid sequence	
1	MDVLVLFCL VAFPSCLVLSQ VQLKESGPGL VAPSQSLTIT CTVSGFSLTS YGVYVWRQPP
61	GKGLEWLVGI WAGGTTNYSN ALMSRLSISK DNSKSVFLK MSSLQTDITA MYCARGPPH
121	AMMKRGYAMD YWQGTSTVIV SS
SEQ ID NO: 10	
Light chain variable domain (VL) of the 26 antibody-amino acid sequence	
1	MDFQVQIFSF LLISASVILS RGQNVLTQSP AIMPASPGKE VTMTCSATSS ITYMSWYQQK
61	SGSSPRLLIY DTSNLSAGVP VRESGSGSGT SYSLTISRME AEDAATYYCQ QWSSYPLTFG
121	AGTKLELK
SEQ ID NO: 11	
Light chain variable domain (VL) of the 26 antibody-cDNA sequence	
1	atgggatttcc aagtcagat tttcagcttc ctgctaata gtgcctcagt catactgtcc
61	agaggacaaa atgttctcac ccagctctcca gcaatcatgc ctgcctctcc aggggagaag
121	gtcaccatga cctgcagtgc cactcaagt ataacttaca tgtcctggta ccagcagaag
181	tcaggatcct cccccagact cctgatttat gacacatcca acctggcttc tggagtccct
241	gttcgcttca gtggcagtg gtctgggacc tcttactctc tcacaatcag ccgaatggag
301	gctgaagatg ctgccactta ttactgccag cagtggagta gttaccgcgt cagcttcggt
361	gctgggacca agctggagct gaaa
SEQ ID NO: 12	
Heavy chain variable domain (VH) of the 121 antibody-amino acid sequence	
1	QVQLQESGPG LVKPSQTLST TCTVSGFSLT SYGVYVWRQP PGKGLEWLVGI IWAGGTTNYSN
61	SALMSRLTIS KDTSKNQVSL KLSSVTAADT AVYYCARGPP HAMMKRGYAM DYWGQGTILVT
121	VSS
SEQ ID NO: 13	
Heavy chain variable domain (VH) of the 121 antibody-cDNA	
1	caggtgcagc tgcaagagtc aggacctggc ctgggtgaagc cctcacagac actgtccttg
61	acttgcaactg tctctgggtt ttcattaacc tcatatgggt tatattgggt tgcaccagcct
121	ccaggaaagg gtctggagtg gctgggagta atatgggctg gtggtaccac aaattataat
181	tcggctctca tgtccagact gacaatcagc aaagacacat ccaagaacca agtttcctta
241	aaactcagca gtgtgactgc agcggacaca gccgtctact actgtgccc agggcccccg
301	cacgctatga tgaagagagg ctatgctatg gactactggg gacaaggaac ctagtcaca
361	gtctcctcag g
SEQ ID NO: 14	
Light chain variable domain (VL) of the 121 antibody-amino acid sequence	
1	DIQMTQSPSS LSASVGDRTV ITCSATSSIT YMSWYQQKPG KAPKLLIYDT SNLASGVPSR
61	FSGSGSGTDY TLTISSTLOPE DFATYYCQQW SSYPLTFGGG TKLEIK
SEQ ID NO: 15	
Heavy chain variable domain (VH) of ipilimumab-amino acid sequence	
1	QVQLVESGGG VVQPGRLRL SCAASGFTFS SYTMHWVRQA PGKGLEWVTF ISYDGNKYY
61	ADSVKGRFTI SRDNSKNLY LQMNSLRAED TAIYYCARTG WLGPFQYWGQ GTLVTVSS
SEQ ID NO: 16	
Heavy chain variable domain (VH) of ipilimumab-cDNA sequence	
1	caggtgcagc tgggtggagtc tgggggaggc gtgggtccagc ctgggaggtc cctgagactc
61	tcctgtgcag cctctggatt caccttcagt agctatacta tgcactgggt ccgccaggct
121	ccaggcaagg ggctggagtg ggtgacattt atatcatatg atggaaacaa taaatactac
181	gcagactccg tgaagggccg attcaccatc tccagagaca attccaagaa cagctgtgat
241	ctgcaaatga acagcctgag agctgaggac acggctatat attactgtgc gaggaccggc
301	tggctggggc cctttgacta ctggggccag ggaaccctgg tcaccgtctc ctgag
SEQ ID NO: 17	
Light chain variable domain (VL) of ipilimumab-amino acid sequence	
1	EIVLTQSPGT LSLSPGERAT LSCRASQSVG SSYLAWYQQK PGQAPRLLIY GAFSRATGIP
61	DRESGSGSGT DETLTISRLE PEDFAVYYCQ QYGSSPWTFG QGTVKVEIK

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof	
SEQ ID NO: 18 Light chain variable domain (VL) of ipilimumab-cDNA sequence	
1	gaaattgtgt tgacgcagtc tccaggcacc ctgtctttgt ctccagggga aagagccacc
61	ctctcctgcga gggccagtcga gagtggttggc agcagctact tagcctggta ccagcagaaa
121	cctggccagg ctcccaggct cctcatctat ggtgcattca gcagggccac tggcatccca
181	gacaggttca gtggcagtggt gtctggggaca gacttcactc tcaccatcag cagactggag
241	cctgaagatt ttgcagtgta ttactgtcag cagtatggta gctcaccgtg gacgttcggc
301	caagggacca aggtggaaat caaac
SEQ ID NO: 19	
BioE2021 (ipilimumab ScFv) amino acid sequence	
1	EIVLTQSPGT LSLSPGERAT LSCRASQSVG SSYLAWYQQK PGQAPRLLIY GAFSRATGIP
61	DRFSGSGSGT DFTLTISRLE PEDFAVYYCQ QYGSSPWTFG QGTKVEIKGG GSGGGSGGG
121	GGSQVQLVES GGGVVPGRS LRLSCAASGF TFSSYTMHWV RQAPGKGLEW VTFISYDGNM
181	KYYADSVKGR FTISRDNKSN TLYLQMNLSL AEDTAIYYCA RTGWLGPFDY WGQGLTVTVS
241	S
SEQ ID NO: 20	
BioE2031 (121 ScFv) amino acid sequence	
1	DIQMTQSPSS LSASVGRDVT ITCSATSSIT YMSWYQQKPG KAPKLLIYDT SNLASGVPSR
61	FSGSGSGTDY TLTISLQPE DFIATYCCQW SSYPLTFGGG TKVEIKGGGG SGGGGSGGG
121	SQVQLQESGP GLVKPSQTLT LTCTVSGFSL TSYGVYVWRQ PPGKGLEWLG VIWAGGTTNY
181	NSALMSRLTI SKDTSKNQVS LKLSSVTAAD TAVIYCARGP PHAMMKRGYA MDYWGQGLTV
241	TVSS
SEQ ID NO: 21	
121 VH and CH1 amino acid sequence	
1	QVQLQESGPG LVKPSQTLST TCTVSGFSLT SYGVYVWRP PGKGLEWLGW IWAGGTTNIN
61	SALMSRLTIS KDTSKNQVSL KLSVTAADT AVIYCARGPP HAMMKRGYAM DYWGQGLTVT
121	VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL
181	QSSGLYSLSS VVTVPSSSLG TQTYICNVNH KPSNTKVDKK V
*SEQ ID NO: 21 optionally comprises the sequence (EPKSCDKT) (SEQ ID NO: 76) at the C-terminal end, which corresponds to the hinge region for F(ab') ₂ or Fab', where the cysteine residue forms inter-chain bond with kappa light. Alternatively, F(ab') ₂ may comprise the sequence (e.g., proprietary sequence as indicated in FIG. 101) that forms inter-chain dimerization domain C-terminal to SEQ ID NO: 21.	
*See Table 3 and FIGS. 10F and 10I.	
SEQ ID NO: 22	
BioE2034 121 VL and CL amino acid sequence	
1	DIQMTQSPSS LSASVGRDVT ITCSATSSIT YMSWYQQKPG KAPKLLIYDT SNLASGVPSR
61	FSGSGSGTDY TLTISLQPE DEATYCCQW SSYPLTFGGG TKLEIKRTVA APSVFIFPPS
121	DEQLKSGTAS VVCLLNNEYR REAKVQWKVD NALQSGNSQE SVTEQDSKDS TYSLSSTLT
181	SKADYEKHKV YACEVTHQGL SSPVTKSENR GEC
SEQ ID NO: 23	
BioE2051 (anti-CTLA4 diabody => single-chain diabody; self-dimerizes; tandem diabody (TandAb)) amino acid sequence	
1	QVQLQESGPG LVKPSQTLST TCTVSGESLT SYGVYVWRQP PGKGLEWLGW IWAGGTTNIN
61	SALMSRLTIS KDTSKNQVSL KLSVTAADT AVIYCARGPP HAMMKRGYAM DYWGQGLTVT
121	VSSGGSGSGS GSEIVLTQSP GTLSLSPGER ATLSCRASQS VGSSYLAWYQ QKPGQAPRL
181	IYGAFSRATG IPDRESGSGS GTDETLTISR LEPEDFAVYY CQYGSPPWT FGQGTKVEIK
241	GGSGSGSGSQ VQLVESGGGV VQGRSLRLS CAASGFTFSS YTMHWVRQAP GKGLEWVTFI
301	SYDGNNKYA DSVKGRFTIS RDNSKNTLYL QMNSLRAEDT AIYYCARTGW LGPFYDWGQ
361	TLVTVSSGGS GSGGSDIQM TQSPSSLSAS VGDRVTITCS ATSSITYMSW YQKPKGKAP
421	LLIYDTSNLA SGVPSRESGS GSGTDYTLTI SSLQPEDFAT YYCQQWSSYP LTFGGGKTKVE
481	IK
SEQ ID NO: 24	
BioE2052 (anti-CTLA4 diabody => single-chain diabody; self-dimerizes; tandem diabody (TandAb)) amino acid sequence	
1	QVQLVESGGG VVQPGRLRL SCAASGETFS SYTMHWVRQA PGKGLEWVTF ISYDGNNKY
61	ADSVKGRFTI SRDNSKNTLY LQMNLSRAED TAIYYCARTG WLGPFYDWGQ GTLVTVSSG
121	SGSGGSGSDI QMQSPSSLSA SVGDRVTITC SATSSITYMS WYQKPKGKAP KLLIYDTSN
181	ASGVPSRFSG SSGGTDYTLT ISSLQPEDFA TYCQQWSSY PLTFGGGKTV EIKGSGSGS
241	GSQVQLQESG PGLVKPSQTL SLTCTVSGES LTSYGVYVWR QPPGKGLEWL GVIWAGGTTN
301	YNSALMSRLT ISKDTSKNQV SLKLSVTAAD TAVIYCARG PPHAMMKRGY AMDYWGQGL
361	VTVSSGSGSG GSGSEIVLTQ SPGLTSLSPG ERATLSRCS QSVGSSYLAW YQKPKGQAPR
421	LLIYGAFSRA TGIPDRESGS GSGTDFTLTI SRLPEPFAV YYCQQYGSPP WTFGGGKTKVE
481	IK

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof	
SEQ ID NO: 25	
BioE2051 (anti-CTLA4 diabody => single-chain diabody; self-dimerizes; tandem diabody (TandAb)) cDNA sequence	
1	CAAGTGCAGC TCCAAGAGTC CGGCCCCGGC CTCGTGAAAC CCAGCCAGAC ACTGTCTCTG
61	ACATGCACCG TGAGCGGCTT TTCTCTGACC AGCTACGGAG TGTATTGGGT GAGACAACCC
121	CCCGGCAAGG GACTGGAGTG GCTGGGAGTG ATTTGGGCGG GCGGCACCAC CAACTACAAT
181	AGCGCCCTCA TGTCTAGACT CACCATCTCC AAGGACACCA GCAAGAACCA AGTGTCCTC
241	AAGCTGTCCA GCGTCACAGC TGCCGACACC GCCGTGTACT ATGTGTCTAG AGGCCCCCCC
301	CATGCCATGA TGAAGAGAGG CTATGCCATG GATTACTGGG GCCAAGGCAC ACTGGTGACC
361	GTCAGCTCCG GAGGAAGCGG CGGCAGCGGA GGCTCCGAAA TTGTGCTCAC CCAGAGCCCC
421	GGCACACTGT CTCTGAGCCC CGGCGAAAGG GCCACACTGA GCTGCAGAGC CTCCCAATCC
481	GTGGGCAGCA GCTATCTGGC TTGGTATCAG CAGAAACCCG GCCAAGCCCC TAGACTGCTG
541	ATCTATGGAG CTTTTCCTAG AGCTACCGGC ATCCCCGACA GATTCTCCGG CAGCGGCAGC
601	GGCACAGACT TTACACTGAC AATTTCTAGA CTGGAACAG AGGATTTCGC CGTCTACTAC
661	TGCCAGCAGT ATGGAAGCAG CCCTTGGACC TTTGGCCAAG GCACCAAGGT GGAGATCAAG
721	GGAGGAAGCG GAGGCAGCGG AGGCAGCCAA GTGCAGCTCG TGGAAAGCGG AGGAGGCGTG
781	GTGCAGCCCC GCAGATCCCT CAGACTGAGC TGCGCCGCCA GCGGCTTCAC CTTCAGCTCC
841	TATACCATGC ACTGGGTGAG GCAAGCCCCG GCGAAAGGAC TGGAGTGGGT CACCTTCATC
901	AGCTACGACG GCAACAACAA GTACTACGCC GACAGCGTGA AGGGAAGGTT CACCATCTCT
961	AGAGACAACT CCAAGAACAC CCTCTACCTC CAGATGAAT CTCTGAGGGC CGAAGACACC
1021	GCCATCTACT ACTGCGCTAG AACCGGCTGG CTGGGACCCT TTGACTACTG GGGACAAGGC
1081	ACACTGGTCA CAGTGTCTTC CGGAGGAAGC GGAGGCTCCG GCGGCAGCGA TATCCAGATG
1141	ACCCAATCCC CTTCTCTCTT GAGCGCCTCC GTGGGAGATA GGGTCACCAT TACATGTAGC
1201	GCCACAAGCA GCATCACCTA CATGAGCTGG TACCAGAGA AACCCGGAAA GGCCCCCTAAG
1261	CTGCTCATCT ACGACACCTC CAATCTGGCC AGCGGCGTGC CTTCTAGATT TAGCGGCTCC
1321	GGAAGCGGCA CAGATTACAC ACTGACAATC AGCTCTCTGC AGCCAGAGGA CTTGCCACCC
1381	TACTACTGTC AGCAGTGGAG CAGTACCCT CTGACCTTG GCGGCGGCAC CAAGGTGGAA
1441	ATCAAACACC ACCATCACCA CCATCACCAT CATCAC
SEQ ID NO: 26	
BioE2052 (anti-CTLA4 diabody => single-chain diabody; self-dimerizes; tandem diabody (TandAb)) cDNA sequence	
1	CAAGTGCAGC TCGTGAATC CGGCGGAGGA GTCTGCAGC CCGGCAGAAG CCTCAGACTG
61	AGTGCGCCG CCAGCGGATT CACCTTCAGC AGCTACACCA TGCACTGGGT GAGGCAAGCC
121	CCCGGCAAGG GACTGGAGTG GGTCACTTC ATCTCTACG ATGGCAACAA CAAGTACTAC
181	GCGGACAGCG TGAAGGGAAG GTTTACCATC TCTAGAGATA ACTCCAAGAA CACCCTCTAC
241	CTCCAGATGA ACTCTCTGAG AGCTGAGGAC ACAGCCATCT ATTACTGCGC TAGAACCGGA
301	TGGCTGGGCC CTTTCGACTA CTGGGGACAA GGCACACTGG TGACAGTGTG CTCGGGAGGC
361	AGCGGAGGCA GCGGAGGCG CGACATCCAG ATGACCCAAT CCCCTAGCTC TCTGAGCGCC
421	AGCGTGGGCG ATAGAGTGAC CATTACATGC TCCGCCACCA GCAGCATCAC CTACATGAGC
481	TGGTACCAGC AAAAGCCCCG CAAAGCCCCC AAGCTGTGTA TCTACGATAC CAGCAATCTG
541	GCGAGCGGCG TGCCCTCTAG ATTTTCCGGC TCCGGAAGCG GCACCGATTA CACACTGACC
601	ATTTCTCTCT TGCAAGCCAGA GGATTTCCGC ACCTACTATT GCCAGCAGTG GTCCTCTTAC
661	CCTCTCACCT TTGGCGGCGG AACAAAGGTG GAGATCAAAG GCGGCAGCGG CGGCTCCGGA
721	GCCAGCCAAG TCCAGCTGCA AGAGTCCGGA CCCGACTGTT TGAACCTTC CCAGACACTG
781	TCTCTGACAT GCACAGTGAG CGGATTCTCC CTCACAAGCT ACGGCGTGTA CTGGGTGAGA
841	CAGCCTCCCG GCAAGGAGCT GGAGTGGCTG GCGCTCATCT GGGCTGGAGG CACCACCAAT
901	TACAACAGCG CTCTGATGTC TAGACTGACA ATCTCCAAGG ACACCAAGCA GAACCAAGTG
961	TACTCTGAAGC TGAGCTCCGT GACAGCCGCC GATACAGCCG TGTACTATTG TGCCAGAGGC
1021	CCCCCCCCACG CTATGATGAA GAGGGGCTAC GCCATGGACT ACTGGGGCCA AGGCACCCCTC
1081	GTCACAGTGA GTCCTGGAGG CAGCGGAGGA AGCGGCGGCA GCGAGATCGT GCTGACCCAG
1141	TCCCCCGGCA CACTGTCCCT CAGCCCCGGC GAAAGAGCCA CACTGAGCTG TAGAGCTTCC
1201	CAGAGCGTGG GCAGCAGCTA TCTGGCTTGG TATCAGCAGA AGCCCGGCCA AGCCCCAGA
1261	CTGCTCATCT ACGGAGCCTT CAGCAGAGCC ACCGGCATCC CCGACAGATT CAGCGGCAGC
1321	GGCAGCGGCA CCGATTTTAC CCTCACCATC TCCAGACTGG AGCCAGAGGA CTTCCGCTGT
1381	TATTACTGCC AACAGTACGG CAGCAGCCCT TGGACCTTG GACAAGGCAC CAAGGTGGAA
1441	ATCAAACACC ACCACCATCA CCATCATCAT CACCAC

SEQ ID NO: 27

Ipilimumab VH and CH1 amino acid sequence

1 QVQLVESGGG VVQPGKSLRL SCAASGFTES SYTMHWVRQA PGKGLEWVTF ISYDGNKKYY
 61 ADSVKGRFTI SRDNSKNTLY LOMNSLRAD TAIYYCARTG WLGPFDYWGQ GTLVTVSSAS
 121 TKGPSVFPLA PSSKSTSGGT AALGCLVKDY FPEPVTVSWN SGALTSGVHT FPAVLQSSGL
 181 YSLSSVVTVP SSSLTQTYI CNVNHKPSNT KVDKKV

*SEQ ID NO: 27 optionally comprises the sequence (EPKSCDKT) (SEQ ID NO: 76) at the C-terminal end, which corresponds to the hinge region for F(ab')₂ or Fab', where the cysteine residue forms inter-chain bond with kappa light. Alternatively, F(ab')₂ may comprise the sequence (e.g., proprietary sequence as indicated in FIG. 10H) that forms inter-chain dimerization domain C-terminal to SEQ ID NO: 27.

*See Table 3 and FIGS. 10D and 10H.

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
SEQ ID NO: 28 BioE2023 ipilimumab VL and CL amino acid sequence 1 EIVLTQSPGT LSLSPGERAT LSCRASQSVG SSYLAWYQQK PGQAPRLLIY GAFSRATGIP 61 DRFSGSGSGT DETLTISRLE PEDFAVYYCQ QYGSSPWTFG QGQTKVEIKRT VAAPSVEIFP 121 PSDQLKSGT ASVVCLLNNE YPREAKVQWK VDNLQSGNS QESVTEQDSK DSTYLSSTL 181 TLSKADYEKH KVIACEVTHQ GLSSPVTKSF NRGEC SEQ ID NO: 29 BioE2041 (anti-CTLA4 Tandem Biparatopic ScFv (TBS)) amino acid sequence VL IPI (G4S)3 VH IPI (KEA)6 VL 121 (G4S)3 VH 121 His tag (SEQ ID NOS 98, 102 and 98, respectively, in order of appearance) EIVLTQSPGTLSLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLTISRLEPEDFAVYYCQYQSSPWTFGQGTKEIKGGGGSGGGSGGGGQVQLVESGGGVVQPGRSLRLSCAASGFTESSYTMHWVRQAPGKGLEWVTFISYDGNMKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGGQTLVTVSSKEAKEAKEAKEAKEADIQMTQSPSSLSASVGDRTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLSAGVPSRFSGSGGTDTYTLTISLQPEDFATYYCQQWSSYPLTFGGGTKEIKGGGGSGGGSGGGGQVQLQESGPGLVKPSQTLSTCTVSGESLSDTSKNQVSLKLSSTVTAADTAIVYCARGPAPHAMMKRGYAMDYWGQGTLLVTVSS *BioE2041 self-dimerizes. *The underline represents the linker sequence. SEQ ID NO: 30 BioE2042 (anti-CTLA4 Tandem Biparatopic ScFv (TBS)) amino acid sequence VL 121 (G4S)3 VH 121 (KEA)6 VL IPI (G4S)3 VH IPI His tag (SEQ ID NOS 98, 102 and 98, respectively, in order of appearance) DIQMTQSPSSLSASVGDRTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLSAGVPSRESGSGSGTDTYTLTISLQPEDFATYYCQQWSSYPLTFGGGTKEIKGGGGSGGGSGGGGQVQLQESGPGLVKPSQTLSTCTVSGESLTSYGVYVWRQPPGKGLEWLGVIWAGGTTNYSALMSRLTISKDTSKNQVSLKLSSTVTAADTAIVYCARGPAPHAMMKRGYAMDYWGQGTLLVTVSSKEAKEAKEAKEAKEAIVLTQSPGTLSLSPGERATLSCRASQSVGSSYLAWYQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLTISRLEPEDFAVYYCQYQSSPWTFGQGTKEIKGGGGSGGGSGGGGQVQLVESGGGVVQPGRSLRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNMKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGGQGTLLVTVSS *BioE2042 self-dimerizes. *The underline represents the linker sequence. BioE2061 (anti-CTLA4 Tandem Fab) amino acid sequence VH 121--CH-(G4S)3--VH IPI-CH-His tag (VL 121--CL/VL IPI--CL) ("G4S)3" disclosed as SEQ ID NO: 98) BioE2061 comprises 3 polypeptides with the sequence of SEQ ID NOS: 31-33. See Wu et al. (2015) Mabs 7: 470-82 SEQ ID NO: 31 121-IPI Fab HC QVQLQESGPGSLVKPSQTLSTCTVSGFSLTSYGVYVWRQPPGKGLEWLGVIWAGGTTNYSALMSRLTISKDTSKNQVSLKLSSTVTAADTAIVYCARGPAPHAMMKRGYAMDYWGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTGGGSGGGSGGGGQVQLVESGGGVVQPGRSLRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNMKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHHHHHHHHHH SEQ ID NO: 32 121 LC DIQMTQSPSSLSASVGDRTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLSAGVPSRESGSGSGTDTYTLTISLQPEDFATYYCQQWSSYPLTFGGGTKEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALSQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVIACEVTHQGLSSPVTKSENREGEC SEQ ID NO: 33 IPI LC EIVLTQSPGTLSLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLTISRLEPEDFAVYYCQYQSSPWTFGQGTKEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALSQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVIACEVTHQGLSSPVTKSENREGEC BioE2062 (anti-CTLA4 Tandem Fab) amino acid sequence VH IPI-CH-(G4S)3-VH 121--CH-His tag (VL IPI--CL/VL 121--CL) ("G4S)3" disclosed as SEQ ID NO: 98) BioE2062 comprises 3 polypeptides with the sequence of SEQ ID NOS: 34-36. See Wu et al. (2015) Mabs 7: 470-82 SEQ ID NO: 34 IPI-121 Fab HC QVQLVESGGGVVQPGRSLRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNMKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKT

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
GGSGGGGGGGGGGQVQLQESGPGLVKPSQTLSTCTVSGFSLTSYGVYVWRQPPGKGLEWLGVIWAGGTTNYS ALMSRLTISKDTSKNQVSLKLSSTVAADTAVYYCARGPPHAMMKRGYAMDYWGQGLVTVSSASTKGPSVFPLAP SSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHNK PSNTKVDKKVEPKSCDKT SEQ ID NO: 35 IPI LC EIVLTQSPGTLSTLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLT ISRLEPEDFAVYYCQQYGGSPWTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK VDNALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENRGEC SEQ ID NO: 36 121 LC DIQMTQSPSSLSASVGRVTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLAGVPSRESGSGSGTDYTLTIS SLQPEDFATYYCQQWSSYPLTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD NALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENRGEC BioE2091 (anti-CTLA4 Tandem Fab) amino acid sequence VL IPI CL (G4S)3 VH 121 CH1 His tag (VH IPI CHI/VL 121 CL) BioE2091 comprises 3 polypeptides with the sequence of SEQ ID NOs: 37-39. See Wu et al. (2015) Mabs 7: 470-82 SEQ ID NO: 37 IPI LC-121 HC Fab EIVLTQSPGTLSTLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLT ISRLEPEDFAVYYCQQYGGSPWTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK VDNALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENRGECGGGGGGGGGG GGGGSQVQLQESGPGLVKPSQTLSTCTVSGFSLTSYGVYVWRQPPGKGLEWLGVIWAGGTTNYSALMSRLTIS KDTSKNQVSLKLSSTVAADTAVYYCARGPPHAMMKRGYAMDYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGT AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHNKPSNTKVDKK VEPKSCDKT SEQ ID NO: 38 IPI HC Fab QVQLVESGGGVVQPGSRSLRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDY FPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHNKPSNTKVDKKVEPKSCDKT SEQ ID NO: 39 121 LC DIQMTQSPSSLSASVGRVTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLAGVPSRESGSGSGTDYTLTIS SLQPEDFATYYCQQWSSYPLTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD NALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENRGEC BioE2092 (anti-CTLA4 Tandem Fab) amino acid sequence VL 121 CL (G4S)3 VH IPI CHI His tag (VH 121 CH1/VL IPI CL) BioE2092 comprises 3 polypeptides with the sequence of SEQ ID NOs: 40-42. See Wu et al. (2015) Mabs 7: 470-82 SEQ ID NO: 40 121 LC-IPI HC Fab DIQMTQSPSSLSASVGRVTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLAGVPSRFSGSGSGTDYTLTIS SLQPEDFATYYCQQWSSYPLTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD NALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENRGECGGGGGGGGGGGG GGGQVQLVESGGGVVQPGSRSLRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISR DNSKNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHNKPSNTKVDKKVEPKSCD KT SEQ ID NO: 41 121 HC Fab QVQLQESGPGLVKPSQTLSTCTVSGESLTSYGVYVWRQPPGKGLEWLGVIWAGGTTNYSALMSRLTISKDTSK NQVSLKLSSTVAADTAVYYCARGPPHAMMKRGYAMDYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGC LVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHNKPSNTKVDKKVEPKS CDKT SEQ ID NO: 42 IPI LC EIVLTQSPGTLSTLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLT ISRLEPEDFAVYYCQQYGGSPWTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK VDNALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENRGEC BioE2111 (anti-CTLA4 DART) amino acid sequence BioE2111 comprises 2 polypeptides with the sequence of SEQ ID NOs: 43-44.

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
SEQ ID NO: 43 VL IPI G3SG4 VH 121GGCGGGEVAALEKEVAALEKEVAALEKEVAALEK (SEQ ID NOS 101 and 99) EIVLTQSPGTLTSLSPGERATLSCRAQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLT ISRLPEPDAFVYCCQQYGSPPWTFGGGKVEIKGGGSGGGGQVQLQESGPGLVKPSQTLSTCTVSGESLTSYGV YWVRQPPGKGLEWLVGIWAGGTTNYSALMSRLTISKDTSKNQVSLKLSVTAADTAVVYCARGPPHAMMKRGYA MDYWGGQTLVTVSSGGCGGGEVAALEKEVAALEKEVAALEKEVAALEK *The underline represents the linker sequence. SEQ ID NO: 44 VL 121 G3SG4 VH IPI GGCGGKVAALKEKVAALKEKVAALKEKVAALKE (SEQ ID NOS 101 and 100) DIQMTQSPSSLSASVGDRTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLSGVPRESGSGSGTDYTLTIS SLQPEDFATYYCQQWSSYPLTFGGGKVEIKGGGSGGGGQVQLVESGGGVVQPGRSRLSCAASGFTESSTYTMHW VRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGGQ TLVTVSSGGCGGKVAALKEKVAALKEKVAALKEKVAALKE *The underline represents the linker sequence. BioE2012 (anti-CTLA4 Half antibody-scFv) BioE2012 comprises 2 polypeptides with the sequence of SEQ ID NOS: 45-46. IPI Fab Fc 121 ScFv [VL 121 (G4S)3 VH 121] *BioE2012 contains the Fc-null LALAPG mutation SEQ ID NO: 45 IPI 1/2Ab-121 scFv QVQLVESGGGVVQPGRSRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDN KNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGGQTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDY FPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTH TSPPSPAPEAAGGSPVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKENWYVDGVEVHNAKTKPREEQYNST YRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPIEKTIKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYP SDIAVEWESNGQPENNYKTTPPVLDSDGSFRLYSRLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGGG GSGGGGSDIQMTQSPSSLSASVGDRTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLSGVPRESGSGSGT DYTLLTISSLQPEDFATYYCQQWSSYPLTFGGGKVEIKGGGSGGGGSGGGGQVQLQESGPGLVKPSQTLSTCT TVSGFSLTSYGVYWRQPPGKGLEWLVGIWAGGTTNYSALMSRLTISKDTSKNQVSLKLSVTAADTAVVYCARG GPPHAMMKRGYAMDYWGGQTLVTVSS SEQ ID NO: 46 IPI LC EIVLTQSPGTLTSLSPGERATLSCRAQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLT ISRLPEPDAFVYCCQQYGSPPWTFGGGKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK VDNALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENREGC BioE2081 (anti-CTLA4 Heterodimeric Fab/scFv-Fc) BioE2081 comprises 3 polypeptides with the sequence of SEQ ID NOS: 47-49. 121 ScFv Fc IPI Fab Fc (Fc silent) KiH Fc *KiH stands for Knob in Hole. See e.g., Ridgway et al. (1996) <i>Protein Engineering</i> 9: 617-621 *BioE2081 contains the Fc-silent LALAPG mutation SEQ ID NO: 47 IPI Fab-Knob Fc QVQLVESGGGVVQPGRSRLSCAASGFTESSTYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDN KNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGGQTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDY FPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTH TCPPCPAPEAAGGSPVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKENWYVDGVEVHNAKTKPREEQYNST YRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPIEKTIKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYP SDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPG SEQ ID NO: 48 121 scFv-hole Fc DIQMTQSPSSLSASVGDRTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLSGVPRESGSGSGTDYTLTIS SLQPEDFATYYCQQWSSYPLTFGGGKVEIKGGGSGGGGSGGGGQVQLQESGPGLVKPSQTLSTCTVSGESL TSYGVYWRQPPGKGLEWLVGIWAGGTTNYSALMSRLTIKDTSKNQVSLKLSVTAADTAVVYCARGPPHAMM KRGYAMDYWGGQTLVTVSSGGGSEPKSQDKTHTCPPCPAPEAAGGSPVFLFPPKPKDTLMISRTPEVTCVVDV SHEDPEVKENWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPIEKTIKAKG QPREPQVCTLPSPRDELTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSR WQQGNVFCSCVMHEALHNHYTQKSLSLSPGK SEQ ID NO: 49 IPI LC EIVLTQSPGTLTSLSPGERATLSCRAQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRFSGSGSGTDETLT ISRLPEPDAFVYCCQQYGSPPWTFGGGKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK VDNALQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENREGC BioE2082 (anti-CTLA4 Heterodimeric Fab/scFv-Fc) BioE2082 comprises 3 polypeptides with the sequence of SEQ ID NOS: 50-52. 121 Fab Fc IPI ScFv Fc (Fc silent) KiH Fc

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
*KiH stands for Knob in Hole. See e.g., Ridgway et al. (1996) <i>Protein Engineering</i> 9: 617-621 *BioE2082 contains the Fc-silent LALAPG mutation SEQ ID NO: 50 121 Fab-knob Fc QVQLQESGPGLVKPSQTLSTCTVSGFSLTSYGVYVVRQPPGKGLEWLGVIWAGGTTNYSALMSRLTISKDTSK NQVSLKLSSTAAADVAVYICARGPPHAMMKRGYAMDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGC LVKDYFPEPVTVSWNSGALTSVHTFPAVLQSSGLYSLSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKS CDKTHTCPPCPAPEAAGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHEDPEVKENWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPI EKTISKAKGQPREPQVYTLPPCRDELTKNQVSLWCLV KGFYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFPLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLS PG SEQ ID NO: 51 IPI scFv-hole Fc EIVLTQSPGTLSTLSLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDDETLT ISRLEPEDFAVYYCQQYGS SPWTFQGQTKVEIKGGGGSGGGGGGGGSGVQLVSGGGGVVQPGRSRLRLSCAASGF TFSSYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAIYYCARTGWL GPFDYWGQGLTVTVSSGGGGSEPKSQDKTHTCPPCPAPEAAGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHED DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPI EKTISKAKGQPR EPQVCTLPSPRDELTKNQVSLCAVKGFPYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFPLYSKLTVDKSRWQQ GNVFCSCVMHEALHNHYTQKSLSLSPGK SEQ ID NO: 52 121 LC DIQMTQSPSSLSASVGDRVTITCSATSS ITYMSWYQQKPGKAPKLLIYDTSNLSAGVPSRESGSGSGTDYTLTIS SLOPEDFATYYCQQWSSYPLTFGGGTKEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWQKVD NALQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVTKSENRGEC BioE2121 (anti-CTLA4 Heterodimeric scFvs) BioE2121 comprises 2 polypeptides with the sequence of SEQ ID NOs: 53-54. IPI scFv-Fc/121 scFv-Fc (KiH) *KiH stands for Knob in Hole. See e.g., Ridgway et al. (1996) <i>Protein Engineering</i> 9: 617-621 *BioE2121 contains the Fc-silent LALAPG mutation SEQ ID NO: 53 IPI scFv-knob Fc EIVLTQSPGTLSTLSLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDFTLT ISRLEPEDFAVYYCQQYGS SPWTFQGQTKVEIKGGGGSGGGGGGGGSGVQLVSGGGGVVQPGRSRLRLSCAASGE TESSYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAIYYCARTGWL GPFDYWGQGLTVTVSSGGGGSEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDVSHED DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPI EKTISKAKGQPR EPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFPLYSKLTVDKSRWQQ GNVFCSCVMHEALHNHYTQKSLSLSPG SEQ ID NO: 54 121 scFv-hole Fc DIQMTQSPSSLSASVGDRVTITCSATSSITYMSWYQQKPGKAPKLLIYDTSNLSAGVPSRESGSGSGTDYTLTIS SLOPEDFATYYCQQWSSYPLTFGGGTKEIKGGGGSGGGGGGGGSGVQLQESGPGLVKPSQTLSTCTVSGESL TSYGVYVVRQPPGKGLEWLGVIWAGGTTNYSALMSRLTISKDTSKNQVSLKLSVTAADVAVYICARGPPHAMM KRGYAMDYWGQGLTVTVSSGGGGSEPKSQDKTHTCPPCPAPEAAGGPSVFLFPPPKPDTLMI SRTPEVTCVVVDV SHEDPEVKENWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPI EKTISKAKG QPREPQVCTLPSPRDELTKNQVSLCAVKGFPYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFPLYSKLTVDKSR WQQGNVFCSCVMHEALHNHYTQKSLSLSPGK SEQ ID NO: 59 BioE2410 amino acid sequence QVQLVESGGGVVQPGRSRLRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDN KNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGQGLTVTVSSGGGGGGGGSDIQMTQSPSSLSASVGDRVTITC SATSSITYMSWYQQKPGKAPKLLIYDTSNLSAGVPSRESGSGSGTDYTLTISLQPEDFATYYCQQWSSYPLTFG GGTKVEIKGGSGSGSGGQVQLQESGPGLVKPSQTLSTCTVSGFSLTSYGVYVVRQPPGKGLEWLGVIWAGGTTN YNSALMSRLTISKDTSKNQVSLKLSVTAADVAVYICARGPPHAMMKRGYAMDYWGQGLTVTVSSGGGGGGGGSE IVLTQSPGTLSTLSLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDDETLLI SRLEPEDFAVYYCQQYGS SPWTFQGQTKVEIKGGSGGDKTHTCPPCPAPELLGGPSVFLFPPPKPDTLMI SRTPE VTCVVVDVSHEDPEVKENWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE KTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFPYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFPLY SKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK SEQ ID NO: 60 BioE2410 nucleic acid sequence CAAGTGCAGCTGGTGGAGAGCGGTGGCGCGTGGTGCAGCCCGGTAGAACCTGAGACTGAGCTGCGCCGCAAGC GGCTTCACCTTCAGCAGCTACACCATGCCTGGGTGAGACAGCACC CGGCAAGGACTGGAATGGGTACGTTTC ATCAGCTACGACGGCAACACAGTACTACGCGACAGCGTGAAGGGCAGATTACCATCAGCCGAGACACAGC AAGAACACCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGACACCGCCATCTACTATTGCGCTAGAACGGGA

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
TGGCTGGGCCCCCTTCGATTATTGGGGGCAAGGCACGTTGGTGACCGTCTCAAGCGGCGGAAGCGGAGGCAGCGGA GGATCGGATATACAGATGACACAGTCGCTAGCTCACTGAGCGCAAGCGTGGGCGACAGAGTGACCATCACCTGC AGCGCCACAAAGCAGCATCACCTACATGAGCTGGTATCAGCAAAAACCGGCAAGGCCCCCAAGCTGCTGATCTAC GATACAAGCAACCTGGCAAGCGCGTGCTAGCAGATTCTCTGGCAGCGGCAGCGGCACCGACTATACTCTCACC ATAAGCAGTCTACAGCCTGAGGATTTCGCCACCTACTATTGTGACAGTGGTCGAGCTACCCCTGACCTTCGGG GGCGGCACCAAGGTAGAGATTAAAGGTGGGAGCGCGGGAGTGGTGGCAGCAGTCCAACCTGCAAGAGTCCGGA CCCGGTCTTGTGAAGCCTTCTCAGACTCTTAGCTTAACATGCACCGTGAGCGGCTTACGCTTACAAGCTACGGC GTCTACTGGGTACCTCAACCACTTGGTAAGGTTTAGAATGGCTAGGGGTGATCTGGGCCGGCGGCACCAAC TACAACAGCGCCTGATGAGCAGATGACGATCTCTAAGGACACTAGCAAAAACCAAGTGAGCCTCAAGCTGAGC AGCGTGACCGCGCCGACACCGCGCTATACTACTGTGCAAGAGGCCCCCCCCACGCGCATGATGAAGAGAGGCTAC GCCATGGATTACTGGGGGCAAGGCACCTCTAGTGACCGTGAGTAGTGGCGGGAGCGGTGGGAGCGCGGTTCGAA ATCGTGCTGACACAGAGCCTGGCACTCTGTCTCTGAGTCCCGCGGAGAGAGCTACCTGAGCTGCAGAGCATCT CAGAGCGTGGGCGAGCAGTACCTGGCCTGGTATCAGCAAAAGCCCGGCCAAGCCCCCTAGACTTCTGATCTACGGC GCATTACAGAGCGCACCGGCATCCCCGACAGATTACGTGGATCTGGCAGCGGAACGGACTTCAACCTAACAAATC AGCCGTCTGGAACCCGAAGACTTTGCGGTGTATTACTGTCAACAGTACGGTAGCAGCCCCCTGGACCTTCGCCAA GGCACAAGGTGGAGATCAAAGGCGGTTCCGGTGGCGACAAAACACACACCTGTCCCCCTGCCCCGCCCCCTGAG CTGTTAGGTGGTCTAGCGTGTCTCTGTTCCCCCAAGCCCAAGGACACCTGATGATCAGCAGAACCCCTGAG GTGACCTCGTGGTGGTGACCTGAGCCACGAGGACCTGAGGTGAAGTTCAACTGGTACGTGGACGGCGTGGAG GTGCACAACGCCAAGCAAGCCTAGAGAGGAGCAGTACAACAGCACCTACAGAGTGGTGAGCGTGTGACCGTG CTGCACCAAGACTGGCTGAACGGCAAGGAGTACAAGTGCAAGGTGAGCAACAAGGCCCTGCCGCCCTCAGAG AAGACCATCAGCAAGGCCAAGGGGAGCCTAGAGAGCCCCAAGTGTAACCTGCCCTAGCAGAGACGAGCTG ACCAAGAACCAAGTGTCTAACCTGTTTGTGAAGGGCTTCTACCTAGCGACATCGCCGTGGAGTGGGAGAGC AACGGGACGCTGAGAAACAACACAGACACCCCCCGTGTGGACAGCGACGGCAGCTTCTCTGTACAGC AAGCTGACCGTGGACAAGAGCAGATGGCAGCAAGGCAACGTGTTACGTGCAGCGTGATGCACGAGGCCCTGCAC AACCCTACACACAGAAGGCTGTCTGTGAGCCCCGGAAAA SEQ ID NO: 61 BioE2420 amino acid sequence QQLVESGGGVQPGRSRLRLSCAAGSFTFSYTMHWVRQAPGKGLEWVTFISYDGNKKYADSVKGRFTISRDN KNTLYLQMNLSRAEDTAIYYCARTGWLGPFDYWGQGLVTVSSGGSGSGSGSDIQMTQSPSSLSASVGRVITITC SATSSITYMSWYQKPKAPKLLIYDTSNLASGVPSRESGSGSDYTLTISSLPEDFATYYCQQWSSYPLTFG GGTKVEIKGSGSGSGSGVQLQESGPGLVKPSQTLSTCTVSGFSLTSYGVYVVRQPPGKLEWLGIWAGGTIN YNSALMSRLTISKDTSKNQVSLKLSVTAADTAVYYCARGPPHAMMKRGYAMDYWGQGLVTVSSGGSGSGSGSE IVLTQSPGTLSTLSPGERATLSCRASQSVGSSYLAWYQKPGQAPRLLIYGAFTSRATGIPDRESGSGSGTDFTLTI SRLEPEDFAVYYCQQYGSFPWTFGGGTVEIKGSGSGDKTHCTPPCPAPEAAGGPSVFLFPPKIDTLTMSIRTP VTCVVDVSHEDPEVKFNWYVDGVEVHNAKTPREEQYNSTYRVVSVLTVLHQDNLNGKEYKCKVSNKALGAPIE KTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYS KLTVDKSRWQQGNVFCSSVMHEALHNHYTQKLSLSLSPGK SEQ ID NO: 62 BioE2420 nucleic acid sequence CAAGTGACGCTGGTGGAGAGCGGTGGCGGCTGGTGACGCCGCTAGAACCTGAGACTGAGCTGCGCCGCAAGC GGCTTACCTTACAGCAGTACACCATGCACTGGGTGAGACAAGCACCCGGCAAGGGACTGGAATGGGTACAGTTC ATCAGCTACGACGGCAACAACAGTACTACGCCGACAGCGTGAAGGGCAGATTCAACATCAGCCGAGACAACAGC AAGAACACCTGTACTCTGAGATGAACAGCCTGAGAGCCGAGGACACCGCCATCTACTATTGCGCTAGAACGGGA TGGCTGGGCCCCCTTCGATTATTGGGGGCAAGGCACGTTGGTGACCGTCTCAAGCGGCGGAAGCGGAGGCAGCGGA GGATCGGATATACAGATGACACAGTCGCTAGCTCACTGAGCGCAAGCGTGGGCGACAGAGTGACCATCACCTGC AGCGCCACAAAGCAGCATCACCTACATGAGCTGGTATCAGCAAAAACCGGCAAGGCCCCCAAGCTGCTGATCTAC GATACAAGCAACCTGGCAAGCGCGTGCTAGCAGATTCTCTGGCAGCGGCAGCGGCACCGACTATACTCTCACC ATAAGCAGTCTACAGCCTGAGGATTTCGCCACCTACTATTGTGACAGTGGTCGAGCTACCCCTGACCTTCGGG GGCGGCACCAAGGTAGAGATTAAAGGTGGGAGCGCGGGAGTGGTGGCAGCAGTCCAACCTGCAAGAGTCCGGA CCCGGTCTTGTGAAGCCTTCTCAGACTCTTAGCTTAACATGCACCGTGAGCGGCTTACGCTTACAAGCTACGGC GTCTACTGGGTACGTCACCTCTGGTAAGGTTTAGAATGGCTAGGGGTGATCTGGGCCGGCGGCACCAAC TACAACAGCGCCTGATGAGCAGATGACGATCTCTAAGGACACTAGCAAAAACCAAGTGAGCCTCAAGCTGAGC AGCGTGACCGCGCCGACACCGCGTATACTACTGTGCAAGAGGCCCCCCCCACGCGCATGATGAAGAGAGGCTAC GCCATGGATTACTGGGGGCAAGGCACCTCTAGTGACCGTGAGTAGTGGCGGGAGCGGTGGGAGCGCGGTTCGAA ATCGTGCTGACACAGAGCCTGGCACTCTGTCTCTGAGTCCCGCGGAGAGAGCTACCTGAGCTGCAGAGCATCT CAGAGCGTGGGCGAGCAGTACCTGGCCTGGTATCAGCAAAAGCCCGGCCAAGCCCCCTAGACTTCTGATCTACGGC GCATTACAGAGCGCACCGGCATCCCCGACAGATTACGTGGATCTGGCAGCGGAACGGACTTCAACCTAACAAATC AGCCGTCTGGAACCCGAAGACTTTGCGGTGTATTACTGTCAACAGTACGGTAGCAGCCCCCTGGACCTTCGCCAA GGCACAAGGTGGAGATCAAAGGCGGTTCCGGTGGCGACAAAACACACACCTGTCCCCCTGCCCCGCCCCCTGAG CTGTCGGTGGTCTTACGCTGTTCTGTTCCTGTTCCCCCAAGCCCAAGGACACCTGATGATCAGCAGAACCCCTGAG GTGACCTCGTGGTGGTGACCTGAGCCACGAGGACCTGAGGTGAAGTTCAACTGGTACGTGGACGGCGTGGAG GTGCACAACGCCAAGCAAGCCTAGAGAGGAGCAGTACAACAGCACCTACAGAGTGGTGAGCGTGTGACCGTG CTGCACCAAGACTGGCTGAACGGCAAGGAGTACAAGTGCAAGGTGAGCAACAAGGCCCTGGCGCCCCCTCAGAG AAGACCATCAGCAAGGCCAAGGGGAGCCTAGAGAGCCCCAAGTGTAACCTGCCCTAGCAGAGACGAGCTG ACCAAGAACCAAGTGTCTAACCTGTTTGTGAAGGGCTTCTACCTAGCGACATCGCCGTGGAGTGGGAGAGC AACGGGACGCTGAGAAACAACACAGACACCCCCCGTGTGGACAGCGACGGCAGCTTCTCTGTACAGC AAGCTGACCGTGGACAAGAGCAGATGGCAGCAAGGCAACGTGTTACGTGCAGCGTGATGCACGAGGCCCTGCAC AACCCTACACACAGAAGGCTGTCTGTGAGCCCCGGAAAA SEQ ID NO: 63 BioE2430 amino acid sequence QQLVESGGGVQPGRSRLRLSCAAGSFTFSYTMHWVRQAPGKGLEWVTFISYDGNKKYADSVKGRFTISRDN KNTLYLQMNLSRAEDTAIYYCARTGWLGPFDYWGQGLVTVSSGGSGSGSGSDIQMTQSPSSLSASVGRVITITC

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
SATSSITYMSWYQKPGKAPKLLIYDTSNLASGVPSRFSGSGSDTYTLTISSLQPEDFATYYCQQWSSYPLTFG GGTKVEIKGSGSGSGSQVLQESGPGLVKPSQTLSTLCTVSGFSLTSYGVYVVRQPPGKGLEWLGVIWAGGTTN YNSALMSRLTISKDTSKNQVSLKLSVTAADTAVYYCARGPPHAMMKRGYAMDYNGQGLTVTVSSGSGSGSGSE IVLTQSPGTLISLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSGTDETLTI SRLEPEDFAVYYCQYQYSSPWFQGGTKVEIKGSGGGEAHKSEIAHRYNDLGEQHFKGLVLIAFSQYLQKCSYDE HAKLVQEVTDFAKTCVADESANCDKSLHTLFGDKLCAIPNLRENYGELADCTCKQEPERNECFLQHKDDNPSLP PFERPEAEAMCTSFKENPTTEMGHYLHEVARRHPYFYAPELLYYAEQYNEILTQCCAEADKESCLTPKLDGVKEK ALVSSVRQRMKSSMQKFGERAFKAWAVARLSQTFPNADFAEITKLATDLTKVNECCCHGDLLECAADRDELAKY MCENQATISLKLQTCCKPILLKKAHCLSEVEHDTMPADLPAIAADFVEDQEVCKNYAEAKDVELGTFLYEYSRRH PDYSVSLLLRLAKKYEATLEKCCAEANPPACYGTVLAEFQPLVEEPKLNVTNCDLYEKLGEYGFQNAILVRYTQ KAPQVSTPTLVEAARNLGRVGTGKCTLPEDQRLPCVEDYLSALNLRVCLLHEKTPVSEHVTKCCSGSLVERRPCE SALTVDETYVPKEFAETFTFHSIDCTLPEKEKQIKKQTALAEVLKHKPKATAEQLKTVMDFAQFLDTCCKAAD KDTCFSTEGPNLVTRCKDALAHHHHHHHHHH SEQ ID NO: 64 BioE2430 nucleic acid sequence CAAGTGCATTTGGTAGAGTCCGGCGGCGCGTGTGTGCAACCCGGCAGAAAGCCTTAGACTGAGCTGCGCGCGCAAGC GGCTTACCTTCAGCAGCTACACCTAGCACTGGGTGAGACAAAGCCCCGGCAAGGGCCTGGAATGGGTGACCTTC ATCAGCTACGACGGCAACAACAGTACTACGCTGATAGCGTGAAGGGCAGATTCACTATCAGCCGTGACAAACAGC AAGAACACCTGTACTGTCAGATGAACAGCCTGAGAGCCGAGGACACCGCCATCTACTACTGCGCAAGAACCGGC TGGCTGGGCCCCCTCGACTACTGGGGCCAAAGGACATTAGTTACTGTTAGCAGCGGCGGGAGCGGGGCAGCGGA GGATCGGATATACAGATGACACAGAGCCCTAGCAGCCTGAGCGCAAGCGTGGGCGACAGAGTGACCATCACCTGC AGCGCCACAGCAGCATCACCTACATGAGCTGGTATCAGCAAAAACCCGGCAAGGCCCAAGCTGCTGATCTAC GACACAAGCAACCTGGCAGCGCGCTGCTAGCAGATTCTCTGGCAGCGGCGAGCGGCACAGATTATCTCTACA ATCAGCAGTCTGCAGCCTGAGGATTTGCCACATACTACTGCCAACAGTGAGCAGCTACCCCTGACCTTCGGC GCGCGCATAAAGTAGAGATTAAAGGCGGCGAGCGGTGGAAGTGGCGGGAGCCCAAGTGCAACTGCAAGAGAGCGGT CCCGGCCTGGTGAAGCCTTCTCAGACCTTGAGCCTTACTGTCACCGTGAGCGGCTTACGCTGACAAGCTACGGC GTGTACTGGGTGCGGCAACCCCGGCAAGGCTTGGAATGGCTGGGGGTGATCTGGGCGGCGGCGACCAAC TACAACAGCGCCCTGATGAGCAGATGACCATTAGCAAGGACACAAGCAAGAACCAAGTGAGCCTGAAGCTATCT AGCGTGACCGCGCGACACCCCGCTCTATTACTGTGCAAGAGGCCCCCACGCATGATGAAGAGAGGCTAC GCCATGGATTATTGGGGCAAGGCACCTTGTTACCGTCTCTTCCGGGGTAGCGGCGGCGAGCGAGGTTCCGAG ATCGTGTAACTCAGACCGCTGCGACCTGTCTGCTGAGCCTGGTGAAGAGGCTACGCTAAGCTGAGAGCTGATCT CAGAGCGTGGGCGAGCAGTACCTGGCTGGTATCAGCAGAAACCCGGCCAGGCCCTAGACTGCTGATCTATGGT GCCTTCAGCAGAGCCACCGGCATCCCCGACAGATTACGCGGTTCCGGTAGTGGCACAGACTTCACTCTGACCATC AGCCGGCTAGAACCTGAGGATTTTGCCGTTTACTACTGTGACAGTATGGCAGCAGCCCTGGACCTTCCGCGCAA GGCACGAAGGTGGAGATAAAGGTGGAAGCGGCGGGGAAGCCCAAGAGCGAGATCGCCACAGATACAACGAC CTGGGCGAGCAGCACTTCAAGGGCTGGTGTGATCGCCTTCTCTCAGTACCTGCAGAGTGACGCTACGACGAG CACGCCAAGCTGGTGCAAGAGGTGACCGACTTCGCCAAGACCTGCGTGGCCGACGAGAGCGCGCCAACTGCGAC AAGAGCTGCACACCTGTTCGGCGACAGCTGTGCGCCATCCCAACCTGAGAGAGAACTACGGCGAGCTGGCC GACTGCTGCACCAAGCAAGAGCCGGAGAGAAATGAGTGCTTCCCTGCAGCACAAGGACGACAAACCTAGCCTGCCC CCTTCGAGAGACCTGAGGCCCGAGGCCATGTGCACTAGCTTCAAGGAGAACCCACACCTTCATGGGCGACTAC CTGACAGAGGTGGTAGAGCTCATCATATTTCTACGCCCTGAGCTGCTGTATGACAGAGAGTACAACGAG ATCCTGACCCCAATGTTGTGCCGAGGCGGACAAGAGAGCTGCCGACCCCAAGCTGGACGGCGTGAAGGAGAAG GCCCTGGTGAGCAGCGTGCAGCAGAGAATGAAGTGACGACGATGCAGAAAGTTGCGGGAGAGAGCTTTCAAGGCC TGGGCGGTGGCTAGACTGTCTCAGACCTTCCCAACGCGGACTTCGCGAGATCACCAAGCTGGCCACCGACCTG ACCAAGGTGAACAAGGAGTGCTGCCACGGCGACCTGCTGGAGTGCGCGAGCAGAGCCGAGCTGGCCAAGTAC ATGTGCGAGAACAAGCCTATCTCATCCAAATTACAGACCTGCTGCGACAAGCCCTGCTGAAGAAAGCTCAT TGCCCTCAGCGAGGTGGAGCATGACACCATGCCCGCGATCTGCGCGCATCGCCGCGACTTCGTGGAGGACCAA GAGGTGTGCAAGAATAACGCCGAGGCGAAGGATGTGTTCTGGGCGACCTTCTGTACGAGTACAGCCGAAGGCAC CCGCTACAGCGTGAGCTGTCTGTGAGACTGGCCAAAGATACGAGGCCACTTTGGAGAAATGCTGCGCTGAG GCAAACCCCTGCTGTCTACGGCACCTGTGCGCGAGTTTACGCCCTGGTAGAGAGCGCAAGAACCTGGTG AAGACCACTGCGACCTGTACGAGAAGCTGGGCGAGTACGGCTTTCAGAACGCCATCTGCTGAGATACACACAG AAGGCCCCCAAGTGAGACACCCACCTTGGTGGAGCGCAAGAACTGGGCGAGAGTGGGCAACCAATGTTGC ACGCTGCGGAGGACCAAGACTGCCCTGCGTGGAGGACTACCTGAGCGCCATCTGAACAGAGTGTGCTGCTG CACGAGAAGACCCCGTGAGCGAGCAGTGACCAAGTGCTGACGTGGCAGCCTGGTGAGAGAAGACCTGCTTC AGCGCCCTGACCGTGAGCAGACCTACGTGCCAAGGAGTTCAAGGCGAGACTTACCTTCCACAGCGACATC TGACTCTCCCTGAGAAGGAGAAGCAGATCAAGAAGCAGACCGCCCTGGCGAGCTGGTGAAGCACAAGCCCAAG GCCACCGCGAGCAGCTGAAGACCGTGATGGACGACTTCGCTCAGTTCTCGGACACCTGCTGCAAGGCCGCGAC AAGGACACCTGCTTACGACCGAGGGCCCAACCTGGTGACAAGATGAAGGACGCGCTGGCCCAACCAACACAC CACCACCACCAACCAAC SEQ ID NO: 65 BioE2440 VH amino acid sequence DVQLQESGPGLVKPSQSLSLTCSVTGYSITSGYVWNWIRQPPGNKLEWMGYISHDGNNTNYPNLSLKNRISITRDT KINQFLKLSVTTEDSATYYCTRNHYGGTMDYWGQTAVSVSSAKTTPSVYPLAPVCGDTTGSSTVLGCLVKG YFPEPVTLTWNSGSLSSGVHTPPAVLQSDLYTLSSSVTVTSSTWPSQSI TCNVHAPASSTKVKKI EPRGPPTIKP CPPCKCPAPNAAGGSPVFI PPPIKIDVLMISLSPIVTCVVVDVSEDDPDVQISWFMNVVEVHTAQTQTHREDYNS TLRVVSALPIHQDQNMVSGKEFKFKVNNKDLGAPI ERTISPKPGSVRAPQVYVLPPEEEMTKKQVTLTCMVTD PEDIYVEVTNNKGT ELNKNTEPVLDSGYSFMYSKLRLVEKKNWVERNSYS CSVVEGLHNHHTKFSRSTPGK SEQ ID NO: 66 BioE2440 VH nucleic acid sequence GACGTGACAGTGCAGAGAGCGGCCCCGGCTGGTGAAGCCGAGCCAATCTCTGAGCCTGACCTGCAGTGTGACG GGATACTCAATCAGCTCTGGTGTACGTGGAACTGGATCAGACAGTTCCCGGCAACAAGCTGGAGTGGATGGGC TACATCAGCCACGACGGCAACACCAACTACAACCTAGCCTGAAGAACAGAATCAGCATACAAGAGACACAAGC

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
AAGAATCAGTTCTTCTCTGAAGCTGAACAGCGTGACCACCGAGGACAGCGCCACCTACTACTGCACAAGAACTAC GGCTACGGCGGCGCACCATTGACTACTGGGGCCAAGGCACCGCGTGAGCGTAAGCAGCGCCAAAGACAACAGCCCT AGCGTGTACCCGCTGGCCCCCTTTGCGGCGACACCACCGGTAGCAGCGTGACCTGGGGTGCCTGGTGAAGGGC TACTTCCCTGAGCCCGTGACCTGACTTGGAAATAGCGGCAGCGCTGAGCAGCGGCGTGACACACCTTCCCCGCCGTG CTGCAGAGCGACCTGTACACCTGAGCGCCCATCGTGACCTGCGTGGTGGTGGACGTGAGCGAGGACGACCCCGACGTG TGCAACGTGGCCACCCCGCAAGCAGCACCAGGTGGACAAGAAGATCGAACCAAGAGGCCCCACTATTAAACCT TGCCCCCCTGCAAGTGTCCCGCACCCTAATGCGAGCGGCGCCCTAGCGTGTTTCTTCCCCCAGATCAAG GACGTGCTGATGATCAGCCTGAGCCCCATCGTGACCTGCGTGGTGGTGGACGTGAGCGAGGACGACCCCGACGTG CAGATCAGCTGGTTCGTGAACAACGTGGAGGTGCACACCGCTCAGACACAGACCCACAGAGAGGACTACAACAGC ACCCGTGAGAGTGGTGAAGCGCCCTGCCCCATTGACGACCAAGACTGGATGAGCGGCAAGGAGTTCAAGTGCAAGGTG AACAACAAGGACCTGGCGCCCCCATCGAGAGAACCATCAGCAAGCCCAAGGGCAGCGTGAGAGCCCCCAAGTG TACGTGCTGCCCCCCCTGAGGAGGAGATGACCAAGAAGCAAGTGACCTAACCTGTATGGTGACCGACTTCATG CCTGAGGACATCTACGTGGAGTGGACCAACAACGGCAAGACCGAGCTGAACACAGAACACCGAGCCCGTGCTG GACAGCGACGGCAGCTACTTTCATGTACAGCAAGCTGAGAGTGGAGAAGAAGAACTGGGTGGAGAGAAACAGCTAC AGCTGCAGCGTGGTGACGAGGCGCTGCACAACCACCACACCAAGAGCTTCAGCAGAACCCCGGCAAG SEQ ID NO: 67 BioE2440 VL amino acid sequence DIVMTQSQKFMSTSVGDRVSVTKASQNVGTYYVWYEQKLQSPKALIFASRYRTGVPDRFTGSGSGTDETLTI SNVQSEDLAEYFCQYQYDYSYPLTFGGGTGLEIKRADAAPTIVSIFPPSSSEQLTSGGASVVCFLNNFYPKDINVKWIKI DGSERQNGVLNSWTDQDSKDSYSSMSSTLTTLTKDEYERHNSYTCETHKSTSPIVKSENRENC SEQ ID NO: 68 BioE2440 VL nucleic acid sequence GACATCGTGATGACACAGTCTCAGAAGTTCATGAGCACAAGCGTGGGCGACAGAGTGAGCGTGACCTGCAAGGCA TCTCAGAACTGTGGGCGACCTTGGCTGGTACGAGCAGAAGCTGGGCGAGAGCCCAAGGCCCTGATCTTCAGC GCAAGCTACAGATATACCGGCGTGCCCGACAGATTACCGGCGAGCGGCGAGGCGCACCGACTTCACCTGACCATC AGCAACGTGACAGAGCGAGGACCTGGCCGAGTACTTCTGTACGAGTACGACAGCTACCCCTGACCTTCGGCGGC GGCACCAGCTCGAGATCAGCGCGCAGATGCTGCTCCTACCGTGAGCATCTTCCGCGCTCCAGCGAACCACTC ACTAGCGGAGGCGCGTCACTGGTCTGCTTCTTAACAATTTCTACCTTAAGGACATCAACGTCAGTGAAGATT GACGGATCGGAACGCCAGAACGGAGTGTGAACCTATGGAGTGTACAGGATTCCAAAGACTCGACTTACTCCATG TCCAGCACCTGACCTGACCAAGACGAGTACGAAAGGCACAAGCTCGTACAGTGCGAAGCCACCCACAAGACT TCCACCTCGCCCATCGTGAAGTCTTCAATCGCAATGAGTGC SEQ ID NO: 69 BioE2450 VH amino acid sequence QVQLVESGGGVQPGRLRLSCAASGFTFSSYTMHWVRQAPGKGLEWVTFISYDGNKKYYADSVKGRFTISRDN KNTLYLQMNSLRAEDTAIYYCARTGWLGPFDYWGQGLTVTVSSASTKGPSVFPPLAPSSKSTSGGTAALGCLVKDY FPEPVTWNSNGLALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKVEPKSKDKTH TCPPCPAPEAAGGSPVFLFPPPKDITLMSRIPEVTCTVVDVSHEDPEVKENWYVDGVEVHNAKTKPREEQYNST YRVVSVLTVLHQDWLNGKEYKCKVSNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYP SDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSLKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK SEQ ID NO: 70 BioE2450 VH nucleic acid sequence CAAGTGCAGCTAGTTGAGTCTGGAGGCGCGTGCTGTCAGCCCGGCAGAAGCCTGAGACTGAGCTGCGCCGCAAGC GGCTTCACCTTCAGCAGCTACACCTGCACTGGGTGAGACAAGCCCCCGCAAGGGCCTGGAGTGGGTGACCTTC ATCAGCTACGACGGCAACAACAGTACTACGCCGACAGCGTGAAGGGCAGATTACCATCAGCAGAGACAACAGC AAGAACACCCGTGATCTGCAGATGAACAGCCTGAGAGCCGAGGACACCGCCATCTACTACTGCGCTAGAACCGGC TGGCTGGGCCCTTCGATCTAGCGGCGCAAGGCACCTGGTAAACGTTAAGCAGCGCTAGTACCAAGGGCCTAGC GTGTTTCCTTTAGCCCCAGCAGCAAAAGCACAAGCGGAGGCACCGCGCCCTGGGCTGTCTGGTGAAGGATTAC TTTCTGTAACCCGTGACTGTATCATGGAATAGCGGGGCCCTGACTAGCGGAGTGATACCTTCCCCCGCGTGCTG CAGAGCAGCGGCTGTACAGCCTGAGCAGCGTGGTGACCGTGCTGACGAGCAGCCTGGGCGACAGACCTACATC TGCAACGTGAACCAAGCCTAGCAACACCAAGGTGGACAAGAAGGTGGAGCCCAAGAGCTGTGACAAAACGCAC ACCTGTCCGCGCTGCCCGCCCCCTGAGGCTGCCCGCGGCCCTAGCGTGTTCTCTGTTTCCCCCAAGCCCAAGGAC ACCCGTGATGATCAGCAGAACCCTGAGGTGACCTGCGTGGTGGTGGACGTGAGCCACGAGGACCCCTGAGGTGAAG TTCAACTGGTACGTGGACGGCGTGGAGGTGCACAACGCCAAGACCAAGCCTAGAGAGGAGCAGTACAACAGCACC TACAGAGTGGTGAAGTGTGACCGTGCTGCACCAAGACTGGCTGAACGGCAAGGAGTACAAGTGCAAGGTGAGC AACAAGGCCCTGGGCGCCCCATCGAGAAGACCATCAGCAAGGCCAAGGGCGAGCTAGAGAGCCCCAAGTGATC ACCCGTGCCCCCTAGCAGAGACGAGCTGACCAAGAACCAAGTGAGCCTGACCTGCTAGTGAAGGGTTTCTACCT AGCGACATCGCCGTGGATGGGAGAGCAACGGCGAGCCTGAGAACAACTACAAGACCACCCCCCGTGCTGGAC AGCAGCGCAGCTTTCTTCTGTACAGCAAGCTGACCGTGGACAAGAGCAGATGGCAGCAAGGCACACGTGTTTCA TGCAGCGTGATGCACGAGGCGCTGCACAACCACTACACACAGAAGAGCCTGAGCCTGAGCCCCGGCAAG SEQ ID NO: 71 BioE2450 VL amino acid sequence EIVLTQSPGTLSTLSPGERATLSCRASQSVGSSYLAWYQQKPGQAPRLLIYGAFSRATGIPDRESGSGSTDETLT ISRLEPEDFAVYFCQYGSFPWTFQGTQKVEIKRTVAAPSVFIIPPSSDEQLKSGTASVVCFLNNFYPREAVQWK VDNALQSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSENREGE SEQ ID NO: 72 BioE2450 VL nucleic acid sequence GAGATCGTGCTGACACAGAGCCCGGCACACTGTCTCTGAGCCCCGGCGAGAGGACACACTGAGCTGCAGAGCC AGCCAGAGCGTGGGCGAGCAGCTATCTGGCTGGTACCGAGAGAAGCCCGGCCAAGCCCTAGACTGCTGATTAC GGAGCCTTTAGCAGAGCCACCGGCATCCCCGACAGATTACGCGGAGCGGCGAGCGGCACCGATTTCACACTGACA

TABLE 3-continued

Sequence of representative CTLA4-binding proteins or fragments thereof
ATCTCTAGACTGGAGCCAGAGGACTTCGCCGTGTACTACTGCCAGCAGTACGGCAGCAGCCCTTGGACCTTTGGC CAAGGCACCAAGTGGAGATCAAGATCAAGAGAACTGTGGCCGCGCCGTAGTGTATCTTCCCTCCATCGGAT GAACAGCTTAAGTCCGGCAGCGCTCTGTGGTCTGCCTGCTCAATAACTTTTACCCTAGGGAAGCTAAAGTCCAA TGGAAAGTGGATAACGCCCTGCAGTCAGGAAACAGCCAGGAATCGGTTACCGAACAGGACAGCAAGGACAGCACT TACTCCTTGTGTCGACTCTTACTCTGAGCAAGGCCGATTACGAGAAGCAAGGTCTACGCTGCGAGGTCACC CATCAGGGACTCTCGTCCCGGTGACCAAATCCTTCAATAGAGGCGAATGC

*Included in Table 3 are RNA nucleic acid molecules (e.g., thymidines replaced with uridines), nucleic acid molecules encoding orthologs of the encoded proteins, as well as DNA, cDNA, or RNA nucleic acid sequences comprising a nucleic acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with the nucleic acid sequence of any SEQ ID NO listed in Table 3, or a portion thereof. Such nucleic acid molecules can have a function of the full-length nucleic acid.

*Included in Table 3 are proteins, as well as polypeptide molecules comprising an amino acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with an amino acid sequence of any SEQ ID NO listed in Table 3, or a portion thereof. Such polypeptides can have a function of the full-length polypeptide.

*The polypeptide molecules or proteins of the present disclosure may further comprise an optional tag (e.g., His tag, etc.) and/or a leader sequence. An exemplary leader sequence may comprise the following sequence: MDFQVQIFSLISASVILSRG (SEQ ID NO: 55)

TABLE 4

Exemplary antigen-binding proteins that bind CTLA4	
Name	Construct
BioE2012	Half antibody-ScFv IPI Fab Fc 121 ScFv [VL121 (G4S)3 VH 121]
BioE2021	Ipi scFv [VL Ipi]-(Gly4Ser)3-[VH Ipi]-His10 tag
BioE2022	Ipi F(ab') ₂ . BioE2022 also refers to the VH and CH1 domains of ipi that form ipi F(ab') ₂ with BioE2023. See FIGS. 10D and 10H.
BioE2023	Ipi VL and CL domains
BioE2031	121 scFv [VL 121]-(Gly4Ser)3-[VH 121]-His10 tag
BioE2032	121 HuIgG1
BioE2033	121 F(ab') ₂ . BioE2033 also refers to the VH and CH1 domains of 121 that form 121 F(ab') ₂ with BioE2034. See FIGS. 10F and 10I.
BioE2034	121 VL and CL domains
BioE2041	Tandem Biparatopic ScFv (TBS); self-dimerizes VL1PI (G4S)3 VHIPI (KEA)6 VL121 (G4S)3 VH121 His6 tag
BioE2042	Tandem Biparatopic ScFv (TBS); self-dimerizes VL121 (G4S)3 VH 121 (KEA)6 VL1PI (G4S)3 VH IPI His6 tag
BioE2051	Anti-CTLA4 diabody (single-chain diabody; self-dimerizes; Tandem Diabody (TandAb)) comprising: [VH 121]-(GlyGlySer)3-[VL Ipi]-(GlyGlySer)3-[VH Ipi]-(GlyGlySer)3-[VL121]-His10 Tag (See FIG. 10A and FIG. 10C)
BioE2052	Anti-CTLA4 diabody (single-chain diabody; self-dimerizes; Tandem Diabody (TandAb)) comprising: [VH Ipi]-(GlyGlySer)3-[VL 121]-(GlyGlySer)3-[VH 121]-(GlyGlySer)3-[VL Ipi]-His10 Tag (see FIG. 10A and FIG. 10B)
BioE2061	Tandem Fab VH121--CH -(G4S)3 --VH IPI-CH-His tag (VL 121--CL/VL IPI--CL) BioE2061 comprises 3 polypeptides with the sequence of SEQ ID NOs: 31-33. See Wu et al. (2015) <i>Mabs</i> 7: 470-82
BioE2062	Tandem Fab VH IPI-CH-(G4S)3-VH 121--CH-His tag (VL IPI--CL/VL 121--CL) BioE2062 comprises 3 polypeptides with the sequence of SEQ ID NOs: 34-36. See Wu et al. (2015) <i>Mabs</i> 7: 470-82
BioE2081	Heterodimeric Fab/ScFv-Fc 121ScFv Fc IPI Fab Fc (Fc silent) KiH Fc BioE2081 comprises 3 polypeptides with the sequence of SEQ ID NOs: 47-49. BioE2081 contains the Fc-silent LALAPG mutation. *KiH stands for Knob in Hole. See e.g., Ridgway et al. (1996) <i>Protein Engineering</i> 9: 617-621

TABLE 4-continued

Exemplary antigen-binding proteins that bind CTLA4	
Name	Construct
BioE2082	Heterodimeric Fab/ScFv-Fc 121 Fab Fc IPI ScFv Fc (Fc silent) KiH Fc BioE2082 comprises 3 polypeptides with the sequence of SEQ ID NOs: 50-52. BioE2082 contains the Fc-silent LALAPG mutation. *KiH stands for Knob in Hole. See e.g., Ridgway et al. (1996) <i>Protein Engineering</i> 9: 617-621
BioE2091	Tandem Fab VL IPI CL (G4S)3 VH 121 CH1 His tag (VH IPI CH1/VL 121 CL) BioE2091 comprises 3 polypeptides with the sequence of SEQ ID NOs: 37-39. See Wu et al. (2015) <i>MAbs</i> 7: 470-82
BioE2092	Tandem Fab VL 121 CL (G4S)3 VH IPI CH1 His tag (VH 121 CH1/VL IPI CL) BioE2092 comprises 3 polypeptides with the sequence of SEQ ID NOs: 40-42. See Wu et al. (2015) <i>MAbs</i> 7: 470-82
BioE2111	DART BioE2111 comprises 2 polypeptides with the sequence of SEQ ID NOs: 43-44.
BioE2201	Ipi Fab'. BioE2201 also refers to the VH and CH1 domains of ipi that form ipi Fab' with BioE2023. See FIGS. 10E and 10H.
BioE2202	121 Fab'. BioE2202 also refers to the VH and CH1 domains of 121 that form 121 Fab' with BioE2034. See FIGS. 10G and 10I.
BioE2121	Heterodimeric scFvs IPI scFv-Fc/121 scFv-Fc (KiH) BioE2121 comprises 2 polypeptides with the sequence of SEQ ID NOs: 53-54. BioE2121 contains the Fc-silent LALAPG mutation. *KiH stands for Knob in Hole. See e.g., Ridgway et al. (1996) <i>Protein Engineering</i> 9: 617-621

*Ipi: ipilimumab (see U.S. Pat. No. 6,984,720 B1)

*121: the humanized antibody disclosed in U.S. Pat. No. 7,034,121 B2

Nucleic Acids and Vectors

[0155] A further object of the invention relates to nucleic acid sequences encoding monoclonal antibodies and fragments thereof, immunoglobulins, and polypeptides of the present invention.

[0156] For example, in certain embodiments, the present invention relates, in part, to a nucleic acid sequence encoding the VH domain of mAbs 9A3, 14C3, 8H3, or 4A3; or the VL domain of mAbs 9A3, 14C3, 8H3, or 4A3.

[0157] Typically, said nucleic acid is a DNA or RNA molecule, which may be included in any suitable vector, such as a plasmid, cosmid, episome, artificial chromosome, phage or a viral vector.

[0158] The terms “vector”, “cloning vector” and “expression vector” mean the vehicle by which a DNA or RNA sequence (e.g. a foreign gene) can be introduced into a host cell, so as to transform the host and promote expression (e.g. transcription and translation) of the introduced sequence. Thus, a further object of the invention relates to a vector comprising a nucleic acid of the present invention.

[0159] Such vectors may comprise regulatory elements, such as a promoter, enhancer, terminator and the like, to cause or direct expression of said polypeptide upon administration to a subject. Examples of promoters and enhancers used in the expression vector for animal cell include early promoter and enhancer of SV40 (Mizukami T. et al. 1987), LTR promoter and enhancer of Moloney mouse leukemia virus (Kuwana Y et al. 1987), promoter (Mason J O et al. 1985) and enhancer (Gillies S D et al. 1983) of immunoglobulin H chain and the like.

[0160] Any expression vector for animal cell can be used. Examples of suitable vectors include pAGE107 (Miyaji H et al. 1990), pAGE103 (Mizukami T et al. 1987), pHSG274 (Brady G et al. 1984), pKCR (O'Hare K et al. 1981), pSG1 beta d2-4 (Miyaji H et al. 1990) and the like. Other representative examples of plasmids include replicating plasmids comprising an origin of replication, or integrative plasmids, such as for instance pUC, pcDNA, pBR, and the like. Representative examples of viral vector include adenoviral, retroviral, herpes virus and AAV vectors. Such recombinant viruses may be produced by techniques known in the art, such as by transfecting packaging cells or by transient transfection with helper plasmids or viruses. Typical examples of virus packaging cells include PA317 cells, PsiCRIP cells, GPenv-positive cells, 293 cells, etc. Detailed protocols for producing such replication-defective recombinant viruses may be found for instance in WO 95/14785, WO 96/22378, U.S. Pat. Nos. 5,882,877, 6,013,516, 4,861, 719, 5,278,056 and WO 94/19478.

[0161] Accordingly, the nucleic acids of the present disclosure in some aspects are incorporated into a vector. In this regard, the present disclosure provides vectors comprising any of the presently disclosed nucleic acids. In various aspects, the vector is a recombinant expression vector. For purposes herein, the term “recombinant expression vector” means a genetically-modified oligonucleotide or polynucleotide construct that permits the expression of an mRNA, protein, polypeptide, or peptide by a host cell, when the construct comprises a nucleotide sequence encoding the mRNA, protein, polypeptide, or peptide, and the vector is

contacted with the cell under conditions sufficient to have the mRNA, protein, polypeptide, or peptide expressed within the cell. The vectors of the present disclosure are not naturally-occurring as a whole. However, parts of the vectors can be naturally-occurring. The presently disclosed vectors can comprise any type of nucleotides, including, but not limited to DNA and RNA, which can be single-stranded or double-stranded, synthesized or obtained in part from natural sources, and which can contain natural, non-natural or altered nucleotides. The vectors can comprise naturally-occurring or non-naturally-occurring internucleotide linkages, or both types of linkages. In some aspects, the altered nucleotides or non-naturally occurring internucleotide linkages do not hinder the transcription or replication of the vector.

[0162] The vector of the present disclosure can be any suitable vector, and can be used to transduce, transform or transfect any suitable host. Suitable vectors include those designed for propagation and expansion or for expression or both, such as plasmids and viruses. The vector can be a plasmid based expression vector. In various aspects, the vector is selected from the group consisting of the pUC series (Fermentas Life Sciences), the pBluescript series (Stratagene, LaJolla, CA), the pET series (Novagen, Madison, WI), the pGEX series (Pharmacia Biotech, Uppsala, Sweden), and the pEX series (Clontech, Palo Alto, CA). Bacteriophage vectors, such as λ GT10, λ GTI 1, λ ZapII (Stratagene), λ EMBL4, and λ NMI 149, also can be used. Examples of plant expression vectors include pBIOL, pBI101.2, pBI101.3, pBI121 and pBIN19 (Clontech). Examples of animal expression vectors include pEUK-Cl, pMAM and pMAMneo (Clontech). In some aspects, the vector is a viral vector, e.g., a retroviral vector. In various aspects, the vector is an adenovirus vector, an adeno-associated virus (AAV) vector, a Herpes Simplex Virus (HSV) vector, a Vesicular stomatitis virus (VSV) vector, vaccinia virus vector, or lentivirus vector. See, e.g., Howarth et al., *Cell Biol. Toxicol.* 26 (1): 1-20 (2010). In various aspects, the vector is a baculovirus vector which infects arthropods, e.g., insects. In various aspects, the baculovirus vector is an Autographacalifornica multiple nuclear virus (AcMNPV) or a Bombyxmorinuclear polyhedrosis (BmNPV). See, e.g., Khan, *Adv Pharm Bull* 3 (2): 257-263 (2013); Miller, *Bioessays* 11 (4): 91-96 (1989); Atkinson et al., *Pestic Sci* 28:215-224 (1990).

[0163] The vectors of the present disclosure can be prepared using standard recombinant DNA techniques described in, for example, Sambrook et al., *supra*, and Ausubel et al., *supra*. Constructs of expression vectors, which are circular or linear, can be prepared to contain a replication system functional in a prokaryotic or eukaryotic host cell. Replication systems can be derived, e.g., from ColEI, 2 μ plasmid, λ , SV40, bovine papilloma virus, and the like. In some aspects, the vector comprises regulatory sequences, such as transcription and translation initiation and termination codons, which are specific to the type of host (e.g., bacterium, fungus, plant, or animal) into which the vector is to be introduced, as appropriate and taking into consideration whether the vector is DNA- or RNA-based.

[0164] The vector can include one or more marker genes, which allow for selection of transformed or transfected hosts. Marker genes include biocide resistance, e.g., resistance to antibiotics, heavy metals, etc., complementation in an auxotrophic host to provide prototrophy, and the like.

Suitable marker genes for the presently disclosed expression vectors include, for instance, neomycin/G418 resistance genes, hygromycin resistance genes, histidinol resistance genes, tetracycline resistance genes, and ampicillin resistance genes.

[0165] The vector can comprise a native or normative promoter operably linked to the nucleotide sequence encoding the polypeptide (including functional portions and functional variants thereof), or to the nucleotide sequence which is complementary to or which hybridizes to the nucleotide sequence encoding the polypeptide. The selection of promoters, e.g., strong, weak, inducible, tissue-specific and developmental-specific, is within the ordinary skill of the artisan. Similarly, the combining of a nucleotide sequence with a promoter is also within the skill of the artisan. The promoter can be a non-viral promoter or a viral promoter, e.g., a cytomegalovirus (CMV) promoter, an SV40 promoter, an RSV promoter, and a promoter found in the long-terminal repeat of the murine stem cell virus.

[0166] In another aspect, the present invention provides isolated nucleic acids that hybridize under selective hybridization conditions to a polynucleotide disclosed herein. Thus, the polynucleotides of this embodiment can be used for isolating, detecting, and/or quantifying nucleic acids comprising such polynucleotides. For example, polynucleotides of the present invention can be used to identify, isolate, or amplify partial or full-length clones in a deposited library. In some embodiments, the polynucleotides are genomic or cDNA sequences isolated, or otherwise complementary to, a cDNA from a human or mammalian nucleic acid library. Preferably, the cDNA library comprises at least 80% full-length sequences, preferably, at least 85% or 90% full-length sequences, and, preferably, at least 95% full-length sequences. The cDNA libraries can be normalized to increase the representation of rare sequences. Low or moderate stringency hybridization conditions are typically, but not exclusively, employed with sequences having a reduced sequence identity relative to complementary sequences. Moderate and high stringency conditions can optionally be employed for sequences of greater identity. Low stringency conditions allow selective hybridization of sequences having about 70% sequence identity and can be employed to identify orthologous or paralogous sequences. Optionally, polynucleotides of this invention will encode at least a portion of an antibody encoded by the polynucleotides described herein. The polynucleotides of this invention embrace nucleic acid sequences that can be employed for selective hybridization to a polynucleotide encoding an antibody of the present invention. See, e.g., Ausubel, *supra*; Colligan, *supra*, each entirely incorporated herein by reference.

Host Cells

[0167] Provided herein are host cells comprising a nucleic acid or vector of the present disclosure. A further object of the present invention relates to a cell which has been transfected, infected or transformed by a nucleic acid and/or a vector according to the invention. The term "transformation" means the introduction of a "foreign" (i.e. extrinsic or extracellular) gene, DNA or RNA sequence to a host cell, so that the host cell will express the introduced gene or sequence to produce a desired substance, typically a protein

or enzyme coded by the introduced gene or sequence. A host cell that receives and expresses introduced DNA or RNA has been “transformed.”

[0168] The nucleic acids of the present invention may be used to produce a recombinant polypeptide of the invention in a suitable expression system. The term “expression system” means a host cell and compatible vector under suitable conditions, e.g. for the expression of a protein coded for by foreign DNA carried by the vector and introduced to the host cell.

[0169] Common expression systems include *E. coli* host cells and plasmid vectors, insect host cells and Baculovirus vectors, and mammalian host cells and vectors. Other examples of host cells include, without limitation, prokaryotic cells (such as bacteria) and eukaryotic cells (such as yeast cells, mammalian cells, insect cells, plant cells, etc.). Specific examples include *E. coli*, *Kluyveromyces* or *Saccharomyces* yeasts, mammalian cell lines (e.g., Vero cells, CHO cells, 3T3 cells, COS cells, etc.) as well as primary or established mammalian cell cultures (e.g., produced from lymphoblasts, fibroblasts, embryonic cells, epithelial cells, nervous cells, adipocytes, etc.). Examples also include mouse SP2/0-Ag14 cell (ATCC CRL1581), mouse P3X63-Ag8.653 cell (ATCC CRL1580), CHO cell in which a dihydrofolate reductase gene (hereinafter referred to as “DHFR gene”) is defective (Urlaub G et al; 1980), rat YB2/3HL.P2.G11.16Ag.20 cell (ATCC CRL 1662, hereinafter referred to as “YB2/0 cell”), and the like. The YB2/0 cell is preferred, since ADCC activity of chimeric or humanized antibodies is enhanced when expressed in this cell.

[0170] The present invention also relates to a method of producing a recombinant host cell expressing an antibody or a polypeptide of the invention according to the invention, said method comprising the steps consisting of (i) introducing in vitro or ex vivo a recombinant nucleic acid or a vector as described herein into a competent host cell, (ii) culturing in vitro or ex vivo the recombinant host cell obtained and (iii), optionally, selecting the cells which express and/or secrete said antibody or polypeptide. Such recombinant host cells can be used for the production of antibodies and polypeptides of the invention.

[0171] As used herein, the term “host cell” refers to any type of cell that can contain the presently disclosed vector and is capable of producing an expression product encoded by the nucleic acid (e.g., mRNA, protein). The host cell in some aspects is an adherent cell or a suspended cell, i.e., a cell that grows in suspension. The host cell in various aspects is a cultured cell or a primary cell, i.e., isolated directly from an organism, e.g., a human. The host cell can be of any cell type, can originate from any type of tissue, and can be of any developmental stage.

[0172] In certain aspects, the antigen-binding protein is a glycosylated protein and the host cell is a glycosylation-competent cell. In various aspects, the glycosylation-competent cell is an eukaryotic cell, including, but not limited to, a yeast cell, filamentous fungi cell, protozoa cell, algae cell, insect cell, or mammalian cell. Such host cells are described in the art. See, e.g., Frenzel, et al., *Front Immunol* 4:217 (2013). In various aspects, the eukaryotic cells are mammalian cells. In various aspects, the mammalian cells are non-human mammalian cells. In some aspects, the cells are Chinese Hamster Ovary (CHO) cells and derivatives thereof (e.g., CHO-K1, CHO pro-3), mouse myeloma cells (e.g., NS0, GS-NS0, Sp2/0), cells engineered to be deficient in

dihydrofolatereductase (DHFR) activity (e.g., DUKX-X11, DG44), human embryonic kidney 293 (HEK293) cells or derivatives thereof (e.g., HEK293T, HEK293-EBNA), green African monkey kidney cells (e.g., COS cells, VERO cells), human cervical cancer cells (e.g., HeLa), human bone osteosarcoma epithelial cells U2-OS, adenocarcinoma human alveolar basal epithelial cells A549, human fibrosarcoma cells HT1080, mouse brain tumor cells CAD, embryonic carcinoma cells P19, mouse embryo fibroblast cells NIH 3T3, mouse fibroblast cells L929, mouse neuroblastoma cells N2a, human breast cancer cells MCF-7, retinoblastoma cells Y79, human retinoblastoma cells SO-Rb50, human liver cancer cells Hep G2, mouse B myeloma cells J558L, or baby hamster kidney (BHK) cells (Gaillet et al. 2007; Khan, *Adv Pharm Bull* 3 (2): 257-263 (2013)).

[0173] For purposes of amplifying or replicating the vector, the host cell is in some aspects is a prokaryotic cell, e.g., a bacterial cell.

[0174] Also provided by the present disclosure is a population of cells comprising at least one host cell described herein. The population of cells in some aspects is a heterogeneous population comprising the host cell comprising vectors described, in addition to at least one other cell, which does not comprise any of the vectors. Alternatively, in some aspects, the population of cells is a substantially homogeneous population, in which the population comprises mainly host cells (e.g., consisting essentially of) comprising the vector. The population in some aspects is a clonal population of cells, in which all cells of the population are clones of a single host cell comprising a vector, such that all cells of the population comprise the vector. In various embodiments of the present disclosure, the population of cells is a clonal population comprising host cells comprising a vector as described herein.

[0175] In certain aspects the host cell is a human cell that is autologous or allogeneic to the subject. In some embodiments, a nucleic acid of the present invention is transduced via a viral vector or transformed in other suitable methods (e.g., electroporation, etc.). Such host cells are transferred (e.g., grafted, implanted, etc.) to the subject for a prolonged treatment of the disease or condition, e.g., cancer.

Manufacturing Methods

[0176] Also provided herein are methods of producing an antigen-binding protein which binds to CTLA4. In various embodiments, the method comprises culturing a host cell comprising a nucleic acid comprising a nucleotide sequence encoding the antigen-binding protein as described herein in a cell culture medium and harvesting the antigen-binding protein from the cell culture medium. The host cell can be any of the host cells described herein. In various aspects, the host cell is selected from the group consisting of: CHO cells, NS0 cells, COS cells, VERO cells, and BHK cells. In various aspects, the step of culturing a host cell comprises culturing the host cell in a growth medium to support the growth and expansion of the host cell. In various aspects, the growth medium increases cell density, culture viability and productivity in a timely manner. In various aspects, the growth medium comprises amino acids, vitamins, inorganic salts, glucose, and serum as a source of growth factors, hormones, and attachment factors. In various aspects, the growth medium is a fully chemically defined media consisting of amino acids, vitamins, trace elements, inorganic salts, lipids and insulin or insulin-like growth factors. In

addition to nutrients, the growth medium also helps maintain pH and osmolality. Several growth media are commercially available and are described in the art. See, e.g., Arora, "Cell Culture Media: A Review" *Mater Methods* 3:175 (2013).

[0177] In various aspects, the method comprises culturing the host cell in a feed medium. In various aspects, the method comprises culturing in a feed medium in a fed-batch mode. Methods of recombinant protein production are known in the art. See, e.g., Li et al., "Cell culture processes for monoclonal antibody production" *MAbs* 2 (5): 466-477 (2010).

[0178] The method making an antigen-binding protein can comprise one or more steps for purifying the protein from a cell culture or the supernatant thereof and preferably recovering the purified protein. In various aspects, the method comprises one or more chromatography steps, e.g., affinity chromatography (e.g., protein A affinity chromatography, nickel resin for Histidine (His) tags), ion exchange chromatography, hydrophobic interaction chromatography. In various aspects, the method comprises purifying the protein using a Protein A affinity chromatography resin.

[0179] In various embodiments, the method further comprises steps for formulating the purified protein, etc., thereby obtaining a formulation comprising the purified protein. Such steps are described in *Formulation and Process Development Strategies for Manufacturing*, eds. Jameel and Hersenson, John Wiley & Sons, Inc. (Hoboken, NJ), 2010.

[0180] In various aspects, the antigen-binding protein linked to a polypeptide and the antigen-binding protein is part of a fusion protein. Thus, the present disclosure further provides methods of producing a fusion protein comprising an antigen-binding protein which binds to CTLA4. In various embodiments, the method comprises culturing a host cell comprising a nucleic acid comprising a nucleotide sequence encoding the fusion protein as described herein in a cell culture medium and harvesting the fusion protein from the cell culture medium.

[0181] Accordingly, the engineered antigen-binding protein of the present invention may be produced by any technique known in the art, such as, without limitation, any chemical, biological, genetic or enzymatic technique, either alone or in combination.

[0182] Knowing the amino acid sequence of the desired sequence, one skilled in the art can readily produce said antibodies or polypeptides, by standard techniques for production of polypeptides. For instance, they can be synthesized using well-known solid phase method, preferably using a commercially available peptide synthesis apparatus (such as that made by Applied Biosystems, Foster City, Calif.) and following the manufacturer's instructions. Alternatively, antibodies and other polypeptides of the present invention can be synthesized by recombinant DNA techniques as is well-known in the art. For example, these fragments can be obtained as DNA expression products after incorporation of DNA sequences encoding the desired (poly) peptide into expression vectors and introduction of such vectors into suitable eukaryotic or prokaryotic hosts that will express the desired polypeptide, from which they can be later isolated using well-known techniques.

[0183] In particular, the present invention further relates to a method of producing an antibody or a polypeptide of the invention, which method comprises the steps consisting of: (i) culturing a transformed host cell according to the inven-

tion under conditions suitable to allow expression of said antibody or polypeptide; and (ii) recovering the expressed antibody or polypeptide.

[0184] Antibodies and other polypeptides of the present invention are suitably separated from the culture medium by conventional immunoglobulin purification procedures such as, for example, protein A-Sepharose, hydroxylapatite chromatography, gel electrophoresis, dialysis, affinity chromatography, ammonium sulfate or ethanol precipitation, acid extraction, anion or cation exchange chromatography, phosphocellulose chromatography, hydrophobic interaction chromatography, hydroxylapatite chromatography and lectin chromatography. High performance liquid chromatography ("HPLC") can also be employed for purification. See, e.g., Colligan, *Current Protocols in Immunology*, or *Current Protocols in Protein Science*, John Wiley & Sons, NY, N.Y., (1997-2001), e.g., Chapters 1, 4, 6, 8, 9, 10, each entirely incorporated herein by reference.

[0185] Chimeric antibodies (e.g., mouse-human chimeras or non-rodent-human chimeras) of the present invention can be produced by obtaining nucleic sequences encoding VL and VH domains as previously described, constructing a human chimeric antibody expression vector by inserting them into an expression vector for animal cell having genes encoding human antibody CH and human antibody CL, and expressing the coding sequence by introducing the expression vector into an animal cell. The CH domain of a human chimeric antibody can be any region which belongs to human immunoglobulin, such as the IgG class or a subclass thereof, such as IgG1, IgG2, IgG3 and IgG4. Similarly, the CL of a human chimeric antibody can be any region which belongs to Ig, such as the kappa class or lambda class. The chimeric and humanized monoclonal antibodies, comprising both human and non-human portions, which can be made using standard recombinant DNA techniques, are within the scope of the invention. Such chimeric and humanized monoclonal antibodies can be produced by recombinant DNA techniques known in the art, for example using methods described in Robinson et al. International Patent Publication PCT/US86/02269; Akira et al. European Patent Application 184,187; Taniguchi, M. European Patent Application 171,496; Morrison et al. European Patent Application 173,494; Neuberger et al. PCT Application WO 86/01533; Cabilly et al. U.S. Pat. No. 4,816,567; Cabilly et al. European Patent Application 125,023; Better et al. (1988) *Science* 240:1041-1043; Liu et al. (1987) *Proc. Natl. Acad. Sci. USA* 84:3439-3443; Liu et al. (1987) *J. Immunol.* 139:3521-3526; Sun et al. (1987) *Proc. Natl. Acad. Sci.* 84:214-218; Nishimura et al. (1987) *Cancer Res.* 47:999-1005; Wood et al. (1985) *Nature* 314:446-449; Shaw et al. (1988) *J. Natl. Cancer Inst.* 80:1553-1559; Morrison, S. L. (1985) *Science* 229:1202-1207; Oi et al. (1986) *Biotechniques* 4:214; Winter U.S. Pat. No. 5,225,539; Jones et al. (1986) *Nature* 321:552-525; Verhoeven et al. (1988) *Science* 239:1534; and Beidler et al. (1988) *J. Immunol.* 141:4053-4060.

[0186] In addition, humanized antibodies can be made according to standard protocols such as those disclosed in U.S. Pat. No. 5,565,332. In another embodiment, antibody chains or specific binding pair members can be produced by recombination between vectors comprising nucleic acid molecules encoding a fusion of a polypeptide chain of a specific binding pair member and a component of a replicable generic display package and vectors containing nucleic acid molecules encoding a second polypeptide chain

of a single binding pair member using techniques known in the art, e.g., as described in U.S. Pat. Nos. 5,565,332, 5,871,907, or 5,733,743. Humanized antibodies of the present invention can be produced by obtaining nucleic acid sequences encoding CDR domains, as previously described, constructing a humanized antibody expression vector by inserting them into an expression vector for animal cell having genes encoding (i) a heavy chain constant region identical to that of a human antibody and (ii) a light chain constant region identical to that of a human antibody, and expressing the genes by introducing the expression vector into an animal cell.

[0187] The humanized antibody expression vector may be either of a type in which a gene encoding an antibody heavy chain and a gene encoding an antibody light chain exists on separate vectors or of a type in which both genes exist on the same vector (tandem type).

[0188] Methods for producing humanized antibodies based on conventional recombinant DNA and gene transfection techniques are well-known in the art (See, e.g., Riechmann L. et al. 1988; Neuberger M S. et al. 1985). Antibodies can be humanized using a variety of techniques known in the art including, for example, CDR-grafting (EP 239,400; PCT publication WO91/09967; U.S. Pat. Nos. 5,225,539; 5,530,101; and 5,585,089), veneering or resurfacing (EP 592,106; EP 519,596; Padlan E A (1991); Studnicka G M et al. (1994); Roguska M A. et al. (1994)), and chain shuffling (U.S. Pat. No. 5,565,332). The general recombinant DNA technology for preparation of such antibodies is also known (see European Patent Application EP 125023 and International Patent Application WO 96/02576).

[0189] In addition, methods for producing antibody fragments are well-known. For example, Fab fragments of the present invention can be obtained by treating an antibody which specifically reacts with a ganglioside with a protease such as papain. Also, Fabs can be produced by inserting DNA encoding Fabs of the antibody into a vector for prokaryotic expression system, or for eukaryotic expression system, and introducing the vector into a prokaryote or eukaryote (as appropriate) to express the Fabs.

[0190] Similarly, F(ab')₂ fragments of the present invention can be obtained treating an antibody which specifically reacts with a ganglioside with a protease, pepsin. Also, the F(ab')₂ fragment can be produced by binding Fab' described below via a thioether bond or a disulfide bond.

[0191] Fab' fragments of the present invention can be obtained treating F(ab')₂ which specifically reacts with a ganglioside with a reducing agent, dithiothreitol. Also, the Fab' fragments can be produced by inserting DNA encoding a Fab' fragment of the antibody into an expression vector for prokaryote, or an expression vector for eukaryote, and introducing the vector into a prokaryote or eukaryote (as appropriate) to perform its expression.

[0192] In addition, scFvs of the present invention can be produced by obtaining cDNA encoding the VH and VL domains as previously described, constructing DNA encoding scFv, inserting the DNA into an expression vector for prokaryote, or an expression vector for eukaryote, and then introducing the expression vector into a prokaryote or eukaryote (as appropriate) to express the scFv. To generate a humanized scFv fragment, a well-known technology called CDR grafting may be used, which involves selecting the complementary determining regions (CDRs) from a donor scFv fragment, and grafting them onto a human scFv

fragment framework of known three dimensional structure (see, e.g., WO98/45322; WO 87/02671; U.S. Pat. Nos. 5,859,205; 5,585,089; 4,816,567; EP0173494).

[0193] The diabody molecules of the present invention can be produced using a variety of methods well known in the art, including de novo protein synthesis and recombinant expression of nucleic acids encoding the binding proteins. The desired nucleic acid sequences can be produced by recombinant methods or by solid-phase DNA synthesis. Exemplary methods of producing a diabody are known in the art (see e.g., U.S. Pat. Nos. 5,637,481A, 9,017,687B1, US20180194840A1).

Modification of the Antigen-Binding Proteins

[0194] Amino acid sequence modification(s) of the antigen-binding proteins (e.g., antibody or fragments thereof, e.g., diabody, F(ab')₂), described herein are contemplated. For example, it may be desirable to improve the binding affinity and/or other biological properties of the (e.g., antibody or fragments thereof, e.g., diabody, F(ab')₂). It is known that when a humanized antibody is produced by simply grafting only CDRs in VH and VL of an antibody derived from a non-human animal in FRs of the VH and VL of a human antibody, the antigen binding activity is reduced in comparison with that of the original antibody derived from a non-human animal. It is considered that several amino acid residues of the VH and VL of the non-human antibody, not only in CDRs but also in FRs, are directly or indirectly associated with the antigen binding activity. Hence, substitution of these amino acid residues with different amino acid residues derived from FRs of the VH and VL of the human antibody would reduce binding activity and can be corrected by replacing the amino acids with amino acid residues of the original antibody derived from a non-human animal.

[0195] Modifications and changes may be made in the structure of the antibodies of the present invention, and in the DNA sequences encoding them, and still obtain a functional molecule that encodes an antibody and polypeptide with desirable characteristics. For example, certain amino acids may be substituted by other amino acids in a protein structure without appreciable loss of activity. Since the interactive capacity and nature of a protein define the protein's biological functional activity, certain amino acid substitutions can be made in a protein sequence, and, of course, in its DNA encoding sequence, while nevertheless obtaining a protein with like properties. It is thus contemplated that various changes may be made in the antibodies sequences of the invention, or corresponding DNA sequences that encode said polypeptides, without appreciable loss of their biological activity.

[0196] In one embodiment, amino acid changes may be achieved by changing codons in the DNA sequence to encode conservative substitutions based on conservation of the genetic code. Specifically, there is a known and definite correspondence between the amino acid sequence of a particular protein and the nucleotide sequences that can code for the protein, as defined by the genetic code (shown below). Likewise, there is a known and definite correspondence between the nucleotide sequence of a particular nucleic acid and the amino acid sequence encoded by that nucleic acid, as defined by the genetic code (see genetic code chart above).

[0197] As described above, an important and well-known feature of the genetic code is its redundancy, whereby, for most of the amino acids used to make proteins, more than one coding nucleotide triplet may be employed (illustrated above). Therefore, a number of different nucleotide sequences may code for a given amino acid sequence. Such nucleotide sequences are considered functionally equivalent since they result in the production of the same amino acid sequence in all organisms (although certain organisms may translate some sequences more efficiently than they do others). Moreover, occasionally, a methylated variant of a purine or pyrimidine may be found in a given nucleotide sequence. Such methylations do not affect the coding relationship between the trinucleotide codon and the corresponding amino acid.

[0198] In making the changes in the amino sequences of polypeptide, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biologic function on a protein is generally understood in the art. It is accepted that the relative hydropathic character of the amino acid contributes to the secondary structure of the resultant protein, which in turn defines the interaction of the protein with other molecules, for example, enzymes, substrates, receptors, DNA, antibodies, antigens, and the like. Each amino acid has been assigned a hydropathic index on the basis of their hydrophobicity and charge characteristics these are: isoleucine (+4.5); valine (+4.2); leucine (+3.8); phenylalanine (+2.8); cysteine/cystine (+2.5); methionine (+1.9); alanine (+1.8); glycine (−0.4); threonine (−0.7); serine (−0.8); tryptophane (−0.9); tyrosine (−1.3); proline (−1.6); histidine (−3.2); glutamate (−3.5); glutamine (−3.5); aspartate (<RTI 3.5); asparagine (−3.5); lysine (−3.9); and arginine (−4.5).

[0199] It is known in the art that certain amino acids may be substituted by other amino acids having a similar hydropathic index or score and still result in a protein with similar biological activity, i.e. still obtain a biological functionally equivalent protein.

[0200] As outlined above, amino acid substitutions are generally therefore based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. Exemplary substitutions which take various of the foregoing characteristics into consideration are well-known to those of skill in the art and include: arginine and lysine; glutamate and aspartate; serine and threonine; glutamine and asparagine; and valine, leucine and isoleucine.

[0201] Another type of amino acid modification of the antibody of the invention may be useful for altering the original glycosylation pattern of the antibody to, for example, increase stability. By “altering” is meant deleting one or more carbohydrate moieties found in the antibody, and/or adding one or more glycosylation sites that are not present in the antibody. Glycosylation of antibodies is typically N-linked. “N-linked” refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. The tripeptide sequences asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline, are the recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain. Thus, the presence of either of these tripeptide sequences in a polypeptide creates a potential glycosylation site. Addition of glycosylation sites to the antibody is conveniently accomplished by altering the amino acid

sequence such that it contains one or more of the above-described tripeptide sequences (for N-linked glycosylation sites). Another type of covalent modification involves chemically or enzymatically coupling glycosides to the antibody. These procedures are advantageous in that they do not require production of the antibody in a host cell that has glycosylation capabilities for N- or O-linked glycosylation. Depending on the coupling mode used, the sugar(s) may be attached to (a) arginine and histidine, (b) free carboxyl groups, (c) free sulfhydryl groups such as those of cysteine, (d) free hydroxyl groups such as those of serine, threonine, or hydroxyproline, (e) aromatic residues such as those of phenylalanine, tyrosine, or tryptophan, or (f) the amide group of glutamine. For example, such methods are described in WO87/05330.

[0202] Similarly, removal of any carbohydrate moieties present on the antibody may be accomplished chemically or enzymatically. Chemical deglycosylation requires exposure of the antibody to the compound trifluoromethanesulfonic acid, or an equivalent compound. This treatment results in the cleavage of most or all sugars except the linking sugar (N-acetylglucosamine or N-acetyl galactosamine), while leaving the antibody intact. Chemical deglycosylation is described by Sojahn H. et al. (1987) and by Edge, A S. et al. (1981). Enzymatic cleavage of carbohydrate moieties on antibodies can be achieved by the use of a variety of endo- and exo-glycosidases as described by Thotakura, N R. et al. (1987).

[0203] Other modifications can involve the formation of immunoconjugates. For example, in one type of covalent modification, antibodies or proteins are covalently linked to one of a variety of non proteinaceous polymers, e.g., polyethylene glycol, polypropylene glycol, or polyoxyalkylenes, in the manner set forth in U.S. Pat. Nos. 4,640,835; 4,496,689; 4,301,144; 4,670,417; 4,791,192 or 4,179,337.

[0204] Conjugation of antibodies or other proteins of the present invention with heterologous agents can be made using a variety of bifunctional protein coupling agents including but not limited to N-succinimidyl (2-pyridyldithio) propionate (SPDP), succinimidyl (N-maleimidomethyl)cyclohexane-1-carboxylate, iminothiolane (IT), bifunctional derivatives of imidesters (such as dimethyl adipimidate HCL), active esters (such as disuccinimidyl suberate), aldehydes (such as glutaraldehyde), bis-azido compounds (such as bis (p-azidobenzoyl) hexanediamine), bis-diazonium derivatives (such as bis-(p-diazoniumbenzoyl)-ethylenediamine), diisocyanates (such as toluene 2,6 diisocyanate), and bis-active fluorine compounds (such as 1,5-difluoro-2,4-dinitrobenzene). For example, carbon labeled 1-isothiocyanatobenzyl methyl-diethylene triamine-pentaacetic acid (MX-DTPA) is an exemplary chelating agent for conjugation of radionucleotide to the antibody (WO 94/11026).

[0205] In another aspect, the present invention features an antigen-binding protein (e.g., antibody or fragments thereof, e.g., diabody, F(ab')₂) that specifically bind CTLA4 conjugated to a moiety that allows detection in vivo or in vitro. Conjugated antigen-binding protein can be used to monitor its presence in blood or tissues as part of a clinical testing procedure. Examples of detectable moieties include various enzymes, prosthetic groups, fluorescent materials, luminescent materials, bioluminescent materials, and radioactive materials. Examples of suitable enzymes include horseradish peroxidase, alkaline phosphatase, β -galactosidase, or

acetylcholinesterase; examples of suitable prosthetic group complexes include streptavidin/biotin and avidin/biotin; examples of suitable fluorescent materials include umbelliferone, fluorescein, fluorescein isothiocyanate (FITC), rhodamine, dichlorotriazinylamine fluorescein, dansyl chloride or phycoerythrin (PE); an example of a luminescent material includes luminol; examples of bioluminescent materials include luciferase, luciferin, and aequorin, and examples of suitable radioactive material include ^{125}I , ^{131}I , ^{35}S , or ^3H . As used herein, the term “labeled”, with regard to the antibody, is intended to encompass direct labeling of the antibody by coupling (i.e., physically linking) a detectable substance, such as a radioactive agent or a fluorophore (e.g. fluorescein isothiocyanate (FITC) or phycoerythrin (PE) or indocyanine (Cy5)) to the antibody, as well as indirect labeling of the antibody by reactivity with a detectable substance. For example, an antibody may be labeled with a nucleic acid sequence that may be amplified and detected, or an antisense oligonucleotide to reduce expression of a particular gene, such that expression can then be detected and measured.

[0206] Techniques for conjugating such therapeutic moiety to an antigen-binding protein (e.g., antibody or fragments thereof) are well-known, see, e.g., Arnon et al., “Monoclonal Antibodies For Immunotargeting Of Drugs In Cancer Therapy”, in *Monoclonal Antibodies And Cancer Therapy*, Reisfeld et al. (eds.), pp. 243 56 (Alan R. Liss, Inc. 1985); Hellstrom et al., “Antibodies For Drug Delivery”, in *Controlled Drug Delivery* (2nd Ed.), Robinson et al. (eds.), pp. 623 53 (Marcel Dekker, Inc. 1987); Thorpe, “Antibody Carriers Of Cytotoxic Agents In Cancer Therapy: A Review”, in *Monoclonal Antibodies '84: Biological And Clinical Applications*, Pinchera et al. (eds.), pp. 475 506 (1985); “Analysis, Results, And Future Prospective Of The Therapeutic Use Of Radiolabeled Antibody In Cancer Therapy”, in *Monoclonal Antibodies For Cancer Detection And Therapy*, Baldwin et al. (eds.), pp. 303 16 (Academic Press 1985), and Thorpe et al., “The Preparation And Cytotoxic Properties Of Antibody-Toxin Conjugates”, *Immunol. Rev.*, 62:119 58 (1982).

Conjugates

[0207] The present disclosure also provides antigen-binding proteins attached, linked or conjugated to a second moiety (e.g., a heterologous moiety, a conjugate moiety). Accordingly, the present disclosure provides a conjugate comprising an antigen-binding protein and a heterologous moiety. As used herein, the term “heterologous moiety” is synonymous with “conjugate moiety” and refers to any molecule (chemical or biochemical, naturally-occurring or non-coded) which is different from the antigen-binding proteins of the present disclosure. Various heterologous moieties include, but are not limited to, a polymer, a carbohydrate, a lipid, a nucleic acid, an oligonucleotide, a DNA or RNA, an amino acid, peptide, polypeptide, protein, therapeutic agent, (e.g., a cytotoxic agent, cytokine), or a diagnostic agent.

[0208] In some embodiments, the heterologous moiety is a polymer. The polymer can be branched or unbranched. The polymer can be of any molecular weight. The polymer in some embodiments has an average molecular weight of between about 2 kDa to about 100 kDa (the term “about” indicating that in preparations of a water soluble polymer, some molecules will weigh more, some less, than the stated molecular weight). The average molecular weight of the

polymer is in some aspect between about 5 kDa and about 50 kDa, between about 12 kDa to about 40 kDa or between about 20 kDa to about 35 kDa.

[0209] In some embodiments, the polymer is modified to have a single reactive group, such as an active ester for acylation or an aldehyde for alkylation, so that the degree of polymerization can be controlled. The polymer in some embodiments is water soluble so that the protein to which it is attached does not precipitate in an aqueous environment, such as a physiological environment. In some embodiments, when, for example, the composition is used for therapeutic use, the polymer is pharmaceutically acceptable. Additionally, in some aspects, the polymer is a mixture of polymers, e.g., a co-polymer, a block co-polymer.

[0210] In some embodiments, the polymer is selected from the group consisting of: polyamides, polycarbonates, polyalkylenes and derivatives thereof including, polyalkylene glycols, polyalkylene oxides, polyalkylene terephthalates, polymers of acrylic and methacrylic esters, including poly(methyl methacrylate), poly(ethyl methacrylate), poly(butylmethacrylate), poly(isobutyl methacrylate), poly(hexylmethacrylate), poly(isodecyl methacrylate), poly(laurylmethacrylate), poly(phenyl methacrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), and poly(octadecyl acrylate), polyvinyl polymers including polyvinyl alcohols, polyvinyl ethers, polyvinyl esters, polyvinyl halides, poly(vinyl acetate), and polyvinylpyrrolidone, polyglycolides, polysiloxanes, polyurethanes and co-polymers thereof, celluloses including alkyl cellulose, hydroxyalkyl celluloses, cellulose ethers, cellulose esters, nitro celluloses, methyl cellulose, ethyl cellulose, hydroxypropyl cellulose, hydroxy-propyl methyl cellulose, hydroxybutyl methyl cellulose, cellulose acetate, cellulose propionate, cellulose acetate butyrate, cellulose acetate phthalate, carboxylethyl cellulose, cellulose triacetate, and cellulose sulphate sodium salt, polypropylene, polyethylenes including poly(ethylene glycol), poly(ethylene oxide), and poly(ethylene terephthalate), and polystyrene.

[0211] A particularly preferred water-soluble polymer for use herein is polyethylene glycol (PEG). As used herein, polyethylene glycol is meant to encompass any of the forms of PEG that can be used to derivatize other proteins, such as mono-(C1-C10) alkoxy- or aryloxy-polyethylene glycol. PEG is a linear or branched neutral polyether, available in a broad range of molecular weights, and is soluble in water and most organic solvents.

[0212] In some embodiments, the heterologous moiety is a carbohydrate. In some embodiments, the carbohydrate is a monosaccharide (e.g., glucose, galactose, fructose), a disaccharide (e.g., sucrose, lactose, maltose), an oligosaccharide (e.g., raffinose, stachyose), a polysaccharide (a starch, amylose, amylopectin, cellulose, chitin, callose, laminarin, xylan, mannan, fucoidan, galactomannan).

[0213] In some embodiments, the heterologous moiety is a lipid. The lipid, in some embodiments, is a fatty acid, eicosanoid, prostaglandin, leukotriene, thromboxane, N-acyl ethanolamine), glycerolipid (e.g., mono-, di-, tri-substituted glycerols), glycerophospholipid (e.g., phosphatidylcholine, phosphatidylinositol, phosphatidylethanolamine, phosphatidylserine), sphingolipid (e.g., sphingosine, ceramide), sterol lipid (e.g., steroid, cholesterol), prenol lipid, saccharolipid, or a polyketide, oil, wax, cholesterol, sterol, fat-soluble vitamin, monoglyceride, diglyceride, triglyceride, a phospholipid.

[0214] In some embodiments, the heterologous moiety is a therapeutic agent. The therapeutic agent can be any of those known in the art. Examples of therapeutic agents that are contemplated herein include, but are not limited to, natural enzymes, proteins derived from natural sources, recombinant proteins, natural peptides, synthetic peptides, cyclic peptides, antibodies, receptor agonists, cytotoxic agents, immunoglobulins, beta-adrenergic blocking agents, calcium channel blockers, coronary vasodilators, cardiac glycosides, antiarrhythmics, cardiac sympathomimetics, angiotensin converting enzyme (ACE) inhibitors, diuretics, inotropes, cholesterol and triglyceride reducers, bile acid sequestrants, fibrates, 3-hydroxy-3-methylglutaryl (HMG)-CoA reductase inhibitors, niacin derivatives, antiadrenergic agents, alpha-adrenergic blocking agents, centrally acting antiadrenergic agents, vasodilators, potassium-sparing agents, thiazides and related agents, angiotensin II receptor antagonists, peripheral vasodilators, antiandrogens, estrogens, antibiotics, retinoids, insulins and analogs, alpha-glucosidase inhibitors, biguanides, meglitinides, sulfonylureas, thiazolidinediones, androgens, progestogens, bone metabolism regulators, anterior pituitary hormones, hypothalamic hormones, posterior pituitary hormones, gonadotropins, gonadotropin-releasing hormone antagonists, ovulation stimulants, selective estrogen receptor modulators, antithyroid agents, thyroid hormones, bulk forming agents, laxatives, antiperistaltics, flora modifiers, intestinal adsorbents, intestinal anti-infectives, anorectic, anticachectic, antibulimics, appetite suppressants, antiobesity agents, antacids, upper gastrointestinal tract agents, anticholinergic agents, aminosalicic acid derivatives, biological response modifiers, corticosteroids, antispasmodics, 5-HT₄ partial agonists, antihistamines, cannabinoids, dopamine antagonists, serotonin antagonists, cytoprotectives, histamine H₂-receptor antagonists, mucosal protective agent, proton pump inhibitors, *H. pylori* eradication therapy, erythropoiesis stimulants, hematopoietic agents, anemia agents, heparins, antifibrinolytics, hemostatics, blood coagulation factors, adenosine diphosphate inhibitors, glycoprotein receptor inhibitors, fibrinogen-platelet binding inhibitors, thromboxane-A₂ inhibitors, plasminogen activators, antithrombotic agents, glucocorticoids, mineralcorticoids, corticosteroids, selective immunosuppressive agents, antifungals, drugs involved in prophylactic therapy, AIDS-associated infections, cytomegalovirus, non-nucleoside reverse transcriptase inhibitors, nucleoside analog reverse transcriptase inhibitors, protease inhibitors, anemia, Kaposi's sarcoma, aminoglycosides, carbapenems, cephalosporins, glycopoptides, lincosamides, macrolides, oxazolidinones, penicillins, streptogramins, sulfonamides, trimethoprim and derivatives, tetracyclines, anthelmintics, amebicides, biguanides, cinchona alkaloids, folic acid antagonists, quinoline derivatives, *Pneumocystis carinii* therapy, hydrazides, imidazoles, triazoles, nitroimidazoles, cyclic amines, neuraminidase inhibitors, nucleosides, phosphate binders, cholinesterase inhibitors, adjunctive therapy, barbiturates and derivatives, benzodiazepines, gamma aminobutyric acid derivatives, hydantoin derivatives, iminostilbene derivatives, succinimide derivatives, anticonvulsants, ergot alkaloids, antimigrane preparations, biological response modifiers, carbamic acid eaters, tricyclic derivatives, depolarizing agents, nondepolarizing agents, neuromuscular paralytic agents, CNS stimulants, dopaminergic reagents, monoamine oxidase inhibitors, COMT inhibitors, alkyl sulphonates,

ethylenimines, imidazotetrazines, nitrogen mustard analogs, nitrosoureas, platinum-containing compounds, antimetabolites, purine analogs, pyrimidine analogs, urea derivatives, antracyclines, actinomycins, camptothecin derivatives, epipodophyllotoxins, taxanes, *vinca* alkaloids and analogs, antiandrogens, antiestrogens, nonsteroidal aromatase inhibitors, protein kinase inhibitor antineoplastics, azaspirodecanedione derivatives, anxiolytics, stimulants, monoamine reuptake inhibitors, selective serotonin reuptake inhibitors, antidepressants, benzisoxazole derivatives, butyrophenone derivatives, dibenzodiazepine derivatives, dibenzothiazepine derivatives, diphenylbutylpiperidine derivatives, phenothiazines, thienobenzodiazepine derivatives, thioxanthene derivatives, allergenic extracts, nonsteroidal agents, leukotriene receptor antagonists, xanthines, endothelin receptor antagonist, prostaglandins, lung surfactants, mucolytics, antimitotics, uricosurics, xanthine oxidase inhibitors, phosphodiesterase inhibitors, methamine salts, nitrofurantoin derivatives, quinolones, smooth muscle relaxants, parasympathomimetic agents, halogenated hydrocarbons, esters of amino benzoic acid, amides (e.g. lidocaine, articaine hydrochloride, bupivacaine hydrochloride), antipyretics, hypnotics and sedatives, cyclopyrrolones, pyrazolopyrimidines, nonsteroidal anti-inflammatory drugs, opioids, para-aminophenol derivatives, alcohol dehydrogenase inhibitor, heparin antagonists, adsorbents, emetics, opioid antagonists, cholinesterase reactivators, nicotine replacement therapy, vitamin A analogs and antagonists, vitamin B analogs and antagonists, vitamin C analogs and antagonists, vitamin D analogs and antagonists, vitamin E analogs and antagonists, vitamin K analogs and antagonists.

[0215] The antigen-binding proteins of the present disclosure can be conjugated to one or more cytokines and growth factors that are effective in inhibiting tumor metastasis, and wherein the cytokine or growth factor has been shown to have an antiproliferative effect on at least one cell population. Such cytokines, lymphokines, growth factors, or other hematopoietic factors include, but are not limited to: M-CSF, GM-CSF, TNF, IL-1, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-11, IL-12, IL-13, IL-14, IL-15, IL-16, IL-17, IL-18, IFN, TNF α , TNF1, TNF2, G-CSF, Meg-CSF, GM-CSF, thrombopoietin, stem cell factor, and erythropoietin. Additional growth factors for use herein include angiogenin, bone morphogenic protein-1, bone morphogenic protein-2, bone morphogenic protein-3, bone morphogenic protein-4, bone morphogenic protein-5, bone morphogenic protein-6, bone morphogenic protein-7, bone morphogenic protein-8, bone morphogenic protein-9, bone morphogenic protein-10, bone morphogenic protein-11, bone morphogenic protein-12, bone morphogenic protein-13, bone morphogenic protein-14, bone morphogenic protein-15, bone morphogenic protein receptor 1A, bone morphogenic protein receptor 1B, brain derived neurotrophic factor, ciliary neurotrophic factor, ciliary neurotrophic factor receptor α , cytokine-induced neutrophil chemotactic factor 1, cytokine-induced neutrophil chemotactic factor 2 α , cytokine-induced neutrophil chemotactic factor 2 β , β endothelial cell growth factor, endothelin 1, epithelial-derived neutrophil attractant, glial cell line-derived neurotrophic factor receptor α 1, glial cell line-derived neurotrophic factor receptor α 2, growth related protein, growth related protein α , growth related protein β , growth related protein γ , heparin binding epidermal growth factor, hepatocyte growth factor, hepatocyte growth factor receptor, insulin-like growth factor

I, insulin-like growth factor receptor, insulin-like growth factor II, insulin-like growth factor binding protein, keratinocyte growth factor, leukemia inhibitory factor, leukemia inhibitory factor receptor α , nerve growth factor nerve growth factor receptor, neurotrophin-3, neurotrophin-4, pre-B cell growth stimulating factor, stem cell factor, stem cell factor receptor, transforming growth factor α , transforming growth factor β , transforming growth factor β 1, transforming growth factor β 1.2, transforming growth factor β 2, transforming growth factor β 3, transforming growth factor β 5, latent transforming growth factor β 1, transforming growth factor β binding protein I, transforming growth factor β binding protein II, transforming growth factor β binding protein III, tumor necrosis factor receptor type I, tumor necrosis factor receptor type II, urokinase-type plasminogen activator receptor, and chimeric proteins and biologically or immunologically active fragments thereof.

[0216] The present disclosure also provides conjugates comprising an antigen-binding protein of the present disclosure linked to a polypeptide, such that the conjugate is a fusion protein. Therefore, the present disclosure provides fusion proteins comprising an antigen-binding protein of the present disclosure linked to a polypeptide. In various embodiments, the polypeptide is a diagnostic label, e.g., a fluorescent protein, such as green fluorescent protein, or other tag, e.g., Myc tag. In various aspects, the polypeptide is one of the cytokines, lymphokines, growth factors, or other hematopoietic factors listed above.

Compositions, Pharmaceutical Compositions, and Formulations

[0217] Compositions comprising an antigen-binding protein, a nucleic acid, a vector, a host cell, or a conjugate as presently disclosed are provided herein. The compositions in some aspects comprise the antigen-binding proteins in isolated and/or purified form. In some aspects, the composition comprises a single type (e.g., structure) of an antigen-binding protein of the present disclosure or comprises a combination of two or more antigen-binding proteins of the present disclosure, wherein the combination comprises two or more antigen-binding proteins of different types (e.g., structures).

[0218] In some aspects, the composition comprises agents which enhance the chemico-physico features of the antigen-binding protein, e.g., via stabilizing the antigen-binding protein at certain temperatures, e.g., room temperature, increasing shelf life, reducing degradation, e.g., oxidation protease mediated degradation, increasing half-life of the antigen-binding protein, etc. In some aspects, the composition comprises any of the agents disclosed herein as a heterologous moiety or conjugate moiety, optionally in admixture with the antigen-binding proteins of the present disclosure or conjugated to the antigen-binding proteins.

[0219] In various aspects of the present disclosure, the composition additionally comprises a pharmaceutically acceptable carrier, diluents, or excipient. In some embodiments, the antigen-binding protein, a nucleic acid, a vector, a host cell, or a conjugate as presently disclosed (hereinafter referred to as "active agents") is formulated into a pharmaceutical composition comprising the active agent, along with a pharmaceutically acceptable carrier, diluent, or excipient. In this regard, the present disclosure further provides pharmaceutical compositions comprising an active agent which is intended for administration to a subject, e.g., a mammal.

[0220] In some embodiments, the active agent is present in the pharmaceutical composition at a purity level suitable for administration to a patient. In some embodiments, the active agent has a purity level of at least about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98% or about 99%, and a pharmaceutically acceptable diluent, carrier or excipient. In some embodiments, the compositions contain an active agent at a concentration of about 0.001 to about 30.0 mg/ml.

[0221] In various aspects, the pharmaceutical compositions comprise a pharmaceutically acceptable carrier. As used herein, the term "pharmaceutically acceptable carrier" includes any of the standard pharmaceutical carriers, such as a phosphate buffered saline solution, water, emulsions such as an oil/water or water/oil emulsion, and various types of wetting agents. The term also encompasses any of the agents approved by a regulatory agency of the US Federal government or listed in the US Pharmacopeia for use in animals, including humans.

[0222] The pharmaceutical composition can comprise any pharmaceutically acceptable ingredients, including, for example, acidifying agents, additives, adsorbents, aerosol propellants, air displacement agents, alkalizing agents, anti-caking agents, anticoagulants, antimicrobial preservatives, antioxidants, antiseptics, bases, binders, buffering agents, chelating agents, coating agents, coloring agents, desiccants, detergents, diluents, disinfectants, disintegrants, dispersing agents, dissolution enhancing agents, dyes, emollients, emulsifying agents, emulsion stabilizers, fillers, film forming agents, flavor enhancers, flavoring agents, flow enhancers, gelling agents, granulating agents, humectants, lubricants, mucoadhesives, ointment bases, ointments, oleaginous vehicles, organic bases, pastille bases, pigments, plasticizers, polishing agents, preservatives, sequestering agents, skin penetrants, solubilizing agents, solvents, stabilizing agents, suppository bases, surface active agents, surfactants, suspending agents, sweetening agents, therapeutic agents, thickening agents, tonicity agents, toxicity agents, viscosity-increasing agents, water-absorbing agents, water-miscible cosolvents, water softeners, or wetting agents. See, e.g., the *Handbook of Pharmaceutical Excipients*, Third Edition, A. H. Kibbe (Pharmaceutical Press, London, U K, 2000), which is incorporated by reference in its entirety. *Remington's Pharmaceutical Sciences*, Sixteenth Edition, E. W. Martin (Mack Publishing Co., Easton, Pa., 1980), which is incorporated by reference in its entirety.

[0223] In various aspects, the pharmaceutical composition comprises formulation materials that are nontoxic to recipients at the dosages and concentrations employed. In specific embodiments, pharmaceutical compositions comprising an active agent and one or more pharmaceutically acceptable salts; polyols; surfactants; osmotic balancing agents; tonicity agents; anti-oxidants; antibiotics; antimycotics; bulking agents; lyoprotectants; anti-foaming agents; chelating agents; preservatives; colorants; analgesics; or additional pharmaceutical agents. In various aspects, the pharmaceutical composition comprises one or more polyols and/or one or more surfactants, optionally, in addition to one or more excipients, including but not limited to, pharmaceutically acceptable salts; osmotic balancing agents (tonicity agents); anti-oxidants; antibiotics; antimycotics; bulking agents; lyoprotectants; anti-foaming agents; chelating agents; preservatives; colorants; and analgesics.

[0224] In certain embodiments, the pharmaceutical composition can contain formulation materials for modifying, maintaining or preserving, for example, the pH, osmolarity, viscosity, clarity, color, isotonicity, odor, sterility, stability, rate of dissolution or release, adsorption or penetration of the composition. In such embodiments, suitable formulation materials include, but are not limited to, amino acids (such as glycine, glutamine, asparagine, arginine or lysine); antimicrobials; antioxidants (such as ascorbic acid, sodium sulfite or sodium hydrogen-sulfite); buffers (such as borate, bicarbonate, Tris-HCl, citrates, phosphates or other organic acids); bulking agents (such as mannitol or glycine); chelating agents (such as ethylenediamine tetraacetic acid (EDTA)); complexing agents (such as caffeine, polyvinylpyrrolidone, beta-cyclodextrin or hydroxypropyl-beta-cyclodextrin); fillers; monosaccharides; disaccharides; and other carbohydrates (such as glucose, mannose or dextrans); proteins (such as serum albumin, gelatin or immunoglobulins); coloring, flavoring and diluting agents; emulsifying agents; hydrophilic polymers (such as polyvinylpyrrolidone); low molecular weight polypeptides; salt-forming counterions (such as sodium); preservatives (such as benzalkonium chloride, benzoic acid, salicylic acid, thimerosal, phenethyl alcohol, methylparaben, propylparaben, chlorhexidine, sorbic acid or hydrogen peroxide); solvents (such as glycerin, propylene glycol or polyethylene glycol); sugar alcohols (such as mannitol or sorbitol); suspending agents; surfactants or wetting agents (such as pluronics, PEG, sorbitan esters, polysorbates such as polysorbate 20, polysorbate, triton, tromethamine, lecithin, cholesterol, tyloxapal); stability enhancing agents (such as sucrose or sorbitol); tonicity enhancing agents (such as alkali metal halides, preferably sodium or potassium chloride, mannitol sorbitol); delivery vehicles; diluents; excipients and/or pharmaceutical adjuvants. See, *Remington's Pharmaceutical Sciences*, 18th Edition, (A. R. Genrmo, ed.), 1990, Mack Publishing Company.

[0225] The pharmaceutical compositions can be formulated to achieve a physiologically compatible pH. In some embodiments, the pH of the pharmaceutical composition can be for example between about 4 or about 5 and about 8.0 or about 4.5 and about 7.5 or about 5.0 to about 7.5. In various embodiments, the pH of the pharmaceutical composition is between 5.5 and 7.5.

[0226] The present disclosure provides methods of producing a pharmaceutical composition. In various aspects, the method comprises combining the antigen-binding protein, conjugate, fusion protein, nucleic acid, vector, host cell, or a combination thereof, with a pharmaceutically acceptable carrier, diluent, or excipient.

Administration

[0227] With regard to the present disclosure, the active agent, or pharmaceutical composition comprising the same, can be administered to the subject via any suitable route of administration. For example, the active agent can be administered to a subject via parenteral, nasal, oral, pulmonary, topical, vaginal, or rectal administration. The following discussion on routes of administration is merely provided to illustrate various embodiments and should not be construed as limiting the scope in any way.

[0228] Formulations suitable for parenteral administration include aqueous and non-aqueous, isotonic sterile injection solutions, which can contain anti-oxidants, buffers, bacte-

riostats, and solutes that render the formulation isotonic with the blood of the intended recipient, and aqueous and non-aqueous sterile suspensions that can include suspending agents, solubilizers, thickening agents, stabilizers, and preservatives. The term, "parenteral" means not through the alimentary canal but by some other route such as subcutaneous, intramuscular, intraspinal, or intravenous. The active agent of the present disclosure can be administered with a physiologically acceptable diluent in a pharmaceutical carrier, such as a sterile liquid or mixture of liquids, including water, saline, aqueous dextrose and related sugar solutions, an alcohol, such as ethanol or hexadecyl alcohol, a glycol, such as propylene glycol or polyethylene glycol, dimethylsulfoxide, glycerol, ketals such as 2,2-dimethyl-153-dioxolane-4-methanol, ethers, poly(ethyleneglycol) 400, oils, fatty acids, fatty acid esters or glycerides, or acetylated fatty acid glycerides with or without the addition of a pharmaceutically acceptable surfactant, such as a soap or a detergent, suspending agent, such as pectin, carbomers, methylcellulose, hydroxypropylmethylcellulose, or carboxymethylcellulose, or emulsifying agents and other pharmaceutical adjuvants.

[0229] Oils, which can be used in parenteral formulations include petroleum, animal, vegetable, or synthetic oils. Specific examples of oils include peanut, soybean, sesame, cottonseed, corn, olive, petrolatum, and mineral. Suitable fatty acids for use in parenteral formulations include oleic acid, stearic acid, and isostearic acid. Ethyl oleate and isopropyl myristate are examples of suitable fatty acid esters.

[0230] Suitable soaps for use in parenteral formulations include fatty alkali metal, ammonium, and triethanolamine salts, and suitable detergents include (a) cationic detergents such as, for example, dimethyl dialkyl ammonium halides, and alkyl pyridinium halides, (b) anionic detergents such as, for example, alkyl, aryl, and olefin sulfonates, alkyl, olefin, ether, and monoglyceride sulfates, and sulfosuccinates, (c) nonionic detergents such as, for example, fatty amine oxides, fatty acid alkanolamides, and polyoxyethylenepolypropylene copolymers, (d) amphoteric detergents such as, for example, alkyl- β -aminopropionates, and 2-alkyl-imidazole quaternary ammonium salts, and (e) mixtures thereof.

[0231] The parenteral formulations in some embodiments contain from about 0.5% to about 25% by weight of the active agent of the present disclosure in solution. Preservatives and buffers can be used. In order to minimize or eliminate irritation at the site of injection, such compositions can contain one or more nonionic surfactants having a hydrophile-lipophile balance (HLB) of from about 12 to about 17. The quantity of surfactant in such formulations will typically range from about 5% to about 15% by weight. Suitable surfactants include polyethylene glycol sorbitan fatty acid esters, such as sorbitan monooleate and the high molecular weight adducts of ethylene oxide with a hydrophobic base, formed by the condensation of propylene oxide with propylene glycol. The parenteral formulations in some aspects are presented in unit-dose or multi-dose sealed containers, such as ampoules and vials, and can be stored in a freeze-dried (lyophilized) condition requiring only the addition of the sterile liquid excipient, for example, water, for injections, immediately prior to use. Extemporaneous injection solutions and suspensions in some aspects are prepared from sterile powders, granules, and tablets of the kind previously described.

[0232] Injectable formulations are in accordance with the present disclosure. The requirements for effective pharmaceutical carriers for injectable compositions are well-known to those of ordinary skill in the art (see, e.g., *Pharmaceutics and Pharmacy Practice*, J. B. Lippincott Company, Philadelphia, PA, Banker and Chalmers, eds., pages 238-250 (1982), and *ASHP Handbook on Injectable Drugs*, Toissel, 4th ed., pages 622-630 (1986)).

Dosages

[0233] The active agents of the disclosure are believed to be useful in methods of inhibiting tumor growth, as well as other methods, as further described herein, including methods of treating or preventing cancer. For purposes of the disclosure, the amount or dose of the active agent administered should be sufficient to effect, e.g., a therapeutic or prophylactic response, in the subject or animal over a reasonable time frame. For example, the dose of the active agent of the present disclosure should be sufficient to treat cancer as described herein in a period of from about 1 to 4 minutes, 1 to 4 hours or 1 to 4 weeks or longer, e.g., 5 to 20 or more weeks, from the time of administration. In certain embodiments, the time period could be even longer. The dose will be determined by the efficacy of the particular active agent and the condition of the animal (e.g., human), as well as the body weight of the animal (e.g., human) to be treated.

[0234] Many assays for determining an administered dose are known in the art. For purposes herein, an assay, which comprises comparing the extent to which cancer is treated upon administration of a given dose of the active agent of the present disclosure to a mammal among a set of mammals, each set of which is given a different dose of the active agent, could be used to determine a starting dose to be administered to a mammal. The extent to which cancer is treated upon administration of a certain dose can be represented by, for example, the extent of tumor regression achieved with the active agent in a mouse xenograft model. Methods of assaying tumor regression are known in the art and described herein in the Examples.

[0235] The dose of the active agent of the present disclosure also will be determined by the existence, nature and extent of any adverse side effects that might accompany the administration of a particular active agent of the present disclosure. Typically, the attending physician will decide the dosage of the active agent of the present disclosure with which to treat each individual patient, taking into consideration a variety of factors, such as age, body weight, general health, diet, sex, active agent of the present disclosure to be administered, route of administration, and the severity of the condition being treated. By way of example and not intending to limit the present disclosure, the dose of the active agent of the present disclosure can be about 0.0001 to about 1 g/kg body weight of the subject being treated/day, from about 0.0001 to about 0.001 g/kg body weight/day, or about 0.01 mg to about 1 g/kg body weight/day.

Controlled Release Formulations

[0236] In some embodiments, the active agents described herein can be modified into a depot form, such that the manner in which the active agent of the present disclosure is released into the body to which it is administered is controlled with respect to time and location within the body

(see, for example, U.S. Pat. No. 4,450,150). Depot forms of active agents of the present disclosure can be, for example, an implantable composition comprising the active agents and a porous or non-porous material, such as a polymer, wherein the active agent is encapsulated by or diffused throughout the material and/or degradation of the non-porous material. The depot is then implanted into the desired location within the body of the subject and the active agent is released from the implant at a predetermined rate.

[0237] The pharmaceutical composition comprising the active agent in certain aspects is modified to have any type of in vivo release profile. In some aspects, the pharmaceutical composition is an immediate release, controlled release, sustained release, extended release, delayed release, or biphasic release formulation. Methods of formulating peptides for controlled release are known in the art. See, for example, Qian et al., *J Pharm* 374:46-52 (2009) and International Patent Application Publication Nos. WO 2008/130158; WO2004/033036; WO2000/032218; and WO 1999/040942.

[0238] The instant compositions can further comprise, for example, micelles or liposomes, or some other encapsulated form, or can be administered in an extended release form to provide a prolonged storage and/or delivery effect.

Method of Preventing or Treating a Disease

[0239] The antigen-binding proteins of the present disclosure are useful for inhibiting tumor growth. Without being bound to a particular theory, the inhibiting action of the antigen-binding proteins provided herein allow such entities to be useful in methods of treating cancer.

[0240] In certain aspects, provided herein is a method of preventing or treating a subject afflicted with cancer, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same.

[0241] In certain aspects, also provided herein is a method of reducing proliferation of a cancer cell in a subject in need thereof, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same.

[0242] Further provided herein are methods of inhibiting tumor growth in a subject and methods of reducing tumor size in a subject. In various embodiments, the methods comprise administering to the subject the pharmaceutical composition of the present disclosure in an amount effective for inhibiting tumor growth or reducing tumor size in the subject.

[0243] In certain embodiments, the therapeutically effective amount of an engineered antigen-binding protein or pharmaceutical composition is administered to a subject in need thereof. In other embodiments, the cells that are autologous or allogeneic to the subject are obtained and transduced (e.g., via a viral vector, such as AAV) or otherwise transformed with a nucleic acid (or a vector comprising same) that encodes any one of the engineered antigen-binding protein of the present disclosure. In preferred embodiments, such nucleic acid is stably integrated into the host cell genome. Upon confirming transformation of the nucleic acid, the cells are introduced to the subject (e.g., grafted or implanted) to supply a continued source of the antigen-binding proteins (i.e., expressed by the grafted cells and secreted into blood).

[0244] In various aspects, the growth of melanoma (e.g., unresectable or metastatic melanoma), renal cell carcinoma (RCC), colorectal cancer, hepatocellular carcinoma, non-small cell lung cancer (NSCLC), malignant pleural mesothelioma, small cell lung cancer (SCLC), breast cancer, head and neck cancer, bladder cancer, urothelial carcinoma, Merkel cell cancer, cervical cancer, hepatocellular carcinoma, gastric cancer, cutaneous squamous cell cancer, Hodgkin's lymphoma, or B-cell lymphoma is inhibited. In various aspects, the size of melanoma (e.g., unresectable or metastatic melanoma), renal cell carcinoma (RCC), colorectal cancer, hepatocellular carcinoma, non-small cell lung cancer (NSCLC), malignant pleural mesothelioma, small cell lung cancer (SCLC), breast cancer, head and neck cancer, bladder cancer, urothelial carcinoma, Merkel cell cancer, cervical cancer, hepatocellular carcinoma, gastric cancer, cutaneous squamous cell cancer, Hodgkin's lymphoma, or B-cell lymphoma is reduced.

[0245] As used herein, the term “inhibit” or “reduce” and words stemming therefrom may not be a 100% or complete inhibition or reduction. Rather, there are varying degrees of inhibition or reduction of which one of ordinary skill in the art recognizes as having a potential benefit or therapeutic effect. In this respect, the antigen-binding proteins of the present disclosure may inhibit tumor growth or reduce tumor size to any amount or level. In various embodiments, the inhibition provided by the methods of the present disclosure is at least or about a 10% inhibition (e.g., at least or about a 20% inhibition, at least or about a 30% inhibition, at least or about a 40% inhibition, at least or about a 50% inhibition, at least or about a 60% inhibition, at least or about a 70% inhibition, at least or about a 80% inhibition, at least or about a 90% inhibition, at least or about a 95% inhibition, at least or about a 98% inhibition). In various embodiments, the reduction provided by the methods of the present disclosure is at least or about a 10% reduction (e.g., at least or about a 20% reduction, at least or about a 30% reduction, at least or about a 40% reduction, at least or about a 50% reduction, at least or about a 60% reduction, at least or about a 70% reduction, at least or about a 80% reduction, at least or about a 90% reduction, at least or about a 95% reduction, at least or about a 98% reduction).

[0246] As used herein, the term “treat,” as well as words related thereto, do not necessarily imply 100% or complete treatment. Rather, there are varying degrees of treatment of which one of ordinary skill in the art recognizes as having a potential benefit or therapeutic effect. In this respect, the methods of treating cancer of the present disclosure can provide any amount or any level of treatment. Furthermore, the treatment provided by the method of the present disclosure can include treatment of one or more conditions or symptoms or signs of the cancer being treated. Also, the treatment provided by the methods of the present disclosure can encompass slowing the progression of the cancer. For example, the methods can treat cancer by virtue of enhancing the T cell activity or an immune response against the cancer, reducing tumor or cancer growth, reducing metastasis of tumor cells, increasing cell death of tumor or cancer cells, and the like. In various aspects, the methods treat by way of delaying the onset or recurrence of the cancer by at least 1 day, 2 days, 4 days, 6 days, 8 days, 10 days, 15 days, 30 days, two months, 3 months, 4 months, 6 months, 1 year, 2 years, 3 years, 4 years, or more. In various aspects, the methods treat by way of increasing the survival of the subject.

[0247] Numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in some embodiments, the at least one engineered antigen-binding protein or the pharmaceutical composition (a) reduces the number of proliferating cancer cells in the cancer; (b) reduces the volume or size of a tumor of the cancer; (c) increases the immune response against the cancer; and/or (d) activates the T cell.

[0248] In some embodiments, the method further comprises administering to the subject an additional cancer therapy. In some embodiments, the additional cancer therapy is selected from the group consisting of immunotherapy, checkpoint blockade, cancer vaccines, chimeric antigen receptors, chemotherapy, radiation, target therapy, and surgery, optionally wherein the additional cancer therapy is nivolumab.

[0249] In certain aspects, provided herein is a method of increasing an immune response in a subject, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same.

[0250] In certain aspects, provided herein is a method of activating a T cell, the method comprising contacting the T cells with at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same. Such method may be used in vivo, in vitro, or ex vivo.

[0251] In certain aspects, provided herein is a method of preventing or treating a disease or a condition characterized by aberrant expression or activity of a CTLA4 protein in a subject in need thereof, the method comprising administering to the subject at least one engineered antigen-binding protein of the present disclosure or a pharmaceutical composition comprising same. In some embodiments, the disease or condition is a cancer, autoimmune disease, infection, or inflammatory disease.

Cancer

[0252] Cancer, tumor, or hyperproliferative disorder refer to the presence of cells possessing characteristics typical of cancer-causing cells, such as uncontrolled proliferation, immortality, metastatic potential, rapid growth and proliferation rate, and certain characteristic morphological features. Cancer cells are often in the form of a tumor, but such cells may exist alone within an animal, or may be a non-tumorigenic cancer cell, such as a leukemia cell. Cancers include, but are not limited to, B cell cancer, e.g., multiple myeloma, Waldenström's macroglobulinemia, the heavy chain diseases, such as, for example, alpha chain disease, gamma chain disease, and mu chain disease, benign monoclonal gammopathy, and immunocytic amyloidosis, melanomas, breast cancer, lung cancer, bronchus cancer, colorectal cancer, prostate cancer, pancreatic cancer, stomach cancer, ovarian cancer, urinary bladder cancer, brain or central nervous system cancer, peripheral nervous system cancer, esophageal cancer, cervical cancer, uterine or endometrial cancer, cancer of the oral cavity or pharynx, liver cancer, kidney cancer, testicular cancer, biliary tract cancer, small bowel or appendix cancer, salivary gland cancer, thyroid gland cancer, adrenal gland cancer, osteosarcoma, chondrosarcoma, cancer of hematologic tissues, and the like. Other non-limiting examples of types of cancers applicable to the methods encompassed by the present invention

include human sarcomas and carcinomas, e.g., fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, osteogenic sarcoma, chordoma, angiosarcoma, endotheliosarcoma, lymphangiosarcoma, lymphangioendotheliosarcoma, synovoma, mesothelioma, Ewing's tumor, leiomyosarcoma, rhabdomyosarcoma, colon carcinoma, colorectal cancer, pancreatic cancer, breast cancer, ovarian cancer, prostate cancer, squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, sweat gland carcinoma, sebaceous gland carcinoma, papillary carcinoma, papillary adenocarcinomas, cystadenocarcinoma, medullary carcinoma, bronchogenic carcinoma, renal cell carcinoma, hepatoma, bile duct carcinoma, liver cancer, choriocarcinoma, seminoma, embryonal carcinoma, Wilms' tumor, cervical cancer, bone cancer, brain tumor, testicular cancer, lung carcinoma, small cell lung carcinoma (SCLC), bladder carcinoma, epithelial carcinoma, glioma, astrocytoma, medulloblastoma, cranio-pharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuroma, oligodendroglioma, meningioma, neuroblastoma, retinoblastoma; leukemias, e.g., acute lymphocytic leukemia and acute myelocytic leukemia (myeloblastic, promyelocytic, myelomonocytic, monocytic and erythroleukemia); chronic leukemia (chronic myelocytic (granulocytic) leukemia and chronic lymphocytic leukemia); and polycythemia vera, lymphoma (Hodgkin's disease and non-Hodgkin's disease), multiple myeloma, Waldenstrom's macroglobulinemia, and heavy chain disease. In some embodiments, cancers are epithelial in nature and include but are not limited to, bladder cancer, breast cancer, cervical cancer, colon cancer, gynecologic cancers, renal cancer, laryngeal cancer, lung cancer, oral cancer, head and neck cancer, ovarian cancer, pancreatic cancer, prostate cancer, or skin cancer. In other embodiments, the cancer is breast cancer, prostate cancer, lung cancer, or colon cancer. In still other embodiments, the epithelial cancer is non-small-cell lung cancer, nonpapillary renal cell carcinoma, cervical carcinoma, ovarian carcinoma (e.g., serous ovarian carcinoma), or breast carcinoma. The epithelial cancers may be characterized in various other ways including, but not limited to, serous, endometrioid, mucinous, clear cell, Brenner, or undifferentiated.

[0253] In some embodiments, the cancer is selected from pancreatic cancer, lung cancer, non-small cell lung cancer (NSCLC), malignant pleural mesothelioma, small cell lung cancer (SCLC), renal cell carcinoma (RCC), breast cancer, liver cancer, hepatocellular carcinoma, kidney cancer, skin cancer, melanoma, thyroid cancer, gall bladder cancer, head-and-neck (squamous) cancer, stomach (gastric) cancer, head and neck cancer, bladder cancer, urothelial carcinoma, Merkel cell cancer, colon cancer, colorectal cancer, intestinal cancer, ovarian cancer, cervical cancer, testicular cancer, esophageal cancer, buccal cancer, brain cancer, blood cancers, lymphomas (B and T cell lymphomas), mesothelioma, cutaneous squamous cell cancer, Hodgkin's lymphoma, B-cell lymphoma, and a malignant or metastatic form thereof.

Cancer Therapy

[0254] The therapeutic agents of the present invention can be used alone or can be administered in combination therapy with, e.g., chemotherapeutic agents, hormones, antiangiogens, radiolabelled, compounds, or with surgery, cryotherapy, immunotherapy, cancer vaccine, immune cell engineering (e.g., CAR-T), and/or radiotherapy. The preceding

treatment methods can be administered in conjunction with other forms of conventional therapy (e.g., standard-of-care treatments for cancer well-known to the skilled artisan), either consecutively with, pre- or post-conventional therapy. For example, agents of the present invention can be administered with a therapeutically effective dose of chemotherapeutic agent. In other embodiments, agents of the present invention are administered in conjunction with chemotherapy to enhance the activity and efficacy of the chemotherapeutic agent. The Physicians' Desk Reference (PDR) discloses dosages of chemotherapeutic agents that have been used in the treatment of various cancers. The dosing regimen and dosages of these aforementioned chemotherapeutic drugs that are therapeutically effective will depend on the particular cancer being treated, the extent of the disease and other factors familiar to the physician of skill in the art, and can be determined by the physician.

[0255] Immunotherapy is a targeted therapy that may comprise, for example, the use of cancer vaccines and/or sensitized antigen presenting cells. For example, an oncolytic virus is a virus that is able to infect and lyse cancer cells, while leaving normal cells unharmed, making them potentially useful in cancer therapy. Replication of oncolytic viruses both facilitates tumor cell destruction and also produces dose amplification at the tumor site. They may also act as vectors for anticancer genes, allowing them to be specifically delivered to the tumor site. The immunotherapy can involve passive immunity for short-term protection of a host, achieved by the administration of pre-formed antibody directed against a cancer antigen or disease antigen (e.g., administration of a monoclonal antibody, optionally linked to a chemotherapeutic agent or toxin, to a tumor antigen). For example, anti-VEGF is known to be effective in treating renal cell carcinoma. Immunotherapy can also focus on using the cytotoxic lymphocyte-recognized epitopes of cancer cell lines. Alternatively, antisense polynucleotides, ribozymes, RNA interference molecules, triple helix polynucleotides and the like, can be used to selectively modulate biomolecules that are linked to the initiation, progression, and/or pathology of a tumor or cancer.

[0256] Immunotherapy also encompasses immune checkpoint modulators. Immune checkpoints are a group of molecules on the cell surface of CD4+ and/or CD8+ T cells that fine-tune immune responses by down-modulating or inhibiting an anti-tumor immune response. Immune checkpoint proteins are well-known in the art and include, without limitation, CTLA4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, 2B4, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, HHLA2, TM6SF2, KIR3DL3, and A2aR (see, for example, WO 2012/177624). Inhibition of one or more immune checkpoint inhibitors can block or otherwise neutralize inhibitory signaling to thereby upregulate an immune response in order to more efficaciously treat cancer. In some embodiments, the cancer therapy one or more inhibitors of immune checkpoints (immune checkpoint inhibition therapy), such as PD1, PD-L1, and/or CD47 inhibitors. In some embodiments, the cancer therapy is nivolumab.

[0257] Adoptive cell-based immunotherapies can be combined with the therapies of the present invention. Well-known adoptive cell-based immunotherapeutic modalities, including, without limitation, irradiated autologous or allo-

geneic tumor cells, tumor lysates or apoptotic tumor cells, antigen-presenting cell-based immunotherapy, dendritic cell-based immunotherapy, adoptive T cell transfer, adoptive CAR T cell therapy, autologous immune enhancement therapy (AIET), cancer vaccines, and/or antigen presenting cells. Such cell-based immunotherapies can be further modified to express one or more gene products to further modulate immune responses, such as expressing cytokines like GM-CSF, and/or to express tumor-associated antigen (TAA) antigens, such as Mage-1, gp-100, and the like.

[0258] The term “chimeric antigen receptor” or “CAR” refers to engineered T cell receptors (TCR) having a desired antigen specificity. T lymphocytes recognize specific antigens through interaction of the T cell receptor (TCR) with short peptides presented by major histocompatibility complex (MHC) class I or II molecules. For initial activation and clonal expansion, naive T cells are dependent on professional antigen-presenting cells (APCs) that provide additional co-stimulatory signals. TCR activation in the absence of co-stimulation can result in unresponsiveness and clonal anergy. To bypass immunization, different approaches for the derivation of cytotoxic effector cells with grafted recognition specificity have been developed. CARs have been constructed that consist of binding domains derived from natural ligands or antibodies specific for cell-surface components of the TCR-associated CD3 complex. Upon antigen binding, such chimeric antigen receptors link to endogenous signaling pathways in the effector cell and generate activating signals similar to those initiated by the TCR complex. Since the first reports on chimeric antigen receptors, this concept has steadily been refined and the molecular design of chimeric receptors has been optimized and routinely use any number of well-known binding domains, such as scFV and another protein binding fragments described herein.

[0259] In other embodiments, immunotherapy comprises non-cell-based immunotherapies. In some embodiments, compositions comprising antigens with or without vaccine-enhancing adjuvants are used. Such compositions exist in many well-known forms, such as peptide compositions, oncolytic viruses, recombinant antigen comprising fusion proteins, and the like. In some embodiments, immunomodulatory cytokines, such as interferons, G-CSF, imiquimod, TNF α , and the like, as well as modulators thereof (e.g., blocking antibodies or more potent or longer lasting forms) are used. In some embodiments, immunomodulatory interleukins, such as IL-2, IL-6, IL-7, IL-12, IL-17, IL-23, and the like, as well as modulators thereof (e.g., blocking antibodies or more potent or longer lasting forms) are used. In some embodiments, immunomodulatory chemokines, such as CCL3, CCL26, and CXCL7, and the like, as well as modulators thereof (e.g., blocking antibodies or more potent or longer lasting forms) are used. In some embodiments, immunomodulatory molecules targeting immunosuppression, such as STAT3 signaling modulators, NF κ B signaling modulators, and immune checkpoint modulators, are used.

[0260] In still other embodiments, immunomodulatory drugs, such as immunocytostatic drugs, glucocorticoids, cytostatics, immunophilins and modulators thereof (e.g., rapamycin, a calcineurin inhibitor, tacrolimus, ciclosporin (cyclosporin), pimecrolimus, abetimus, gusperimus, ridaforolimus, everolimus, temsirolimus, zotarolimus, etc.), hydrocortisone (cortisol), cortisone acetate, prednisone, prednisolone, methylprednisolone, dexamethasone, betame-

thasone, triamcinolone, beclometasone, fludrocortisone acetate, deoxycorticosterone acetate (doca) aldosterone, a non-glucocorticoid steroid, a pyrimidine synthesis inhibitor, leflunomide, teriflunomide, a folic acid analog, methotrexate, anti-thymocyte globulin, anti-lymphocyte globulin, thalidomide, lenalidomide, pentoxifylline, bupropion, curcumin, catechin, an opioid, an IMPDH inhibitor, mycophenolic acid, myriocin, fingolimod, an NF- κ B inhibitor, raloxifene, drotrecogin alfa, denosumab, an NF- κ B signaling cascade inhibitor, disulfiram, olmesartan, dithiocarbamate, a proteasome inhibitor, bortezomib, MG132, Prol, NPI-0052, curcumin, genistein, resveratrol, parthenolide, thalidomide, lenalidomide, flavopiridol, non-steroidal anti-inflammatory drugs (NSAIDs), arsenic trioxide, dehydroxymethylepoxyquinomycin (DHMEQ), I3C (indole-3-carbinol)/DIM (di-indolmethane) (I3C/DIM), Bay 11-7082, luteolin, cell permeable peptide SN-50, IKK α -super repressor overexpression, NF κ B decoy oligodeoxynucleotide (ODN), or a derivative or analog of any thereof, are used. In yet other embodiments, immunomodulatory antibodies or protein are used. For example, antibodies that bind to CD40, Toll-like receptor (TLR), OX40, GITR, CD27, or to 4-1BB, T-cell bispecific antibodies, an anti-IL-2 receptor antibody, an anti-CD3 antibody, OKT3 (muromonab), orelizumab, teplizumab, visilizumab, an anti-CD4 antibody, clenoliximab, keliximab, zanolimumab, an anti-CD11a antibody, efalizumab, an anti-CD18 antibody, erlizumab, rovelizumab, an anti-CD20 antibody, afutuzumab, ocrelizumab, ofatumumab, pascolizumab, rituximab, an anti-CD23 antibody, lumiliximab, an anti-CD40 antibody, teneliximab, toralizumab, an anti-CD40L antibody, ruplizumab, an anti-CD62L antibody, aselizumab, an anti-CD80 antibody, galiximab, an anti-CD147 antibody, gavilimumab, a B-Lymphocyte stimulator (BLyS) inhibiting antibody, belimumab, an CTLA4-Ig fusion protein, abatacept, belatacept, an anti-CTLA4 antibody, ipilimumab, tremelimumab, an anti-eotaxin 1 antibody, bertilimumab, an anti- α 4-integrin antibody, natalizumab, an anti-IL-6R antibody, tocilizumab, an anti-LFA-1 antibody, odulimumab, an anti-CD25 antibody, basiliximab, daclizumab, inolimumab, an anti-CD5 antibody, zolimumab, an anti-CD2 antibody, siplizumab, nerelimomab, faralimumab, atlizumab, atorolimumab, cedelizumab, dorlimomab aritox, dorlixizumab, fontolizumab, gantenerumab, gomiliximab, lebrilizumab, maslimomab, morolimumab, pexelizumab, reslizumab, rovelizumab, talizumab, telimomab aritox, vopaliximab, vepalimumab, aflibercept, alefacept, rilonacept, an IL-1 receptor antagonist, anakinra, an anti-IL-5 antibody, mepolizumab, an IgE inhibitor, omalizumab, talizumab, an IL12 inhibitor, an IL23 inhibitor, ustekinumab, and the like.

[0261] Nutritional supplements that enhance immune responses, such as vitamin A, vitamin E, vitamin C, and the like, are well-known in the art (see, for example, U.S. Pat. Nos. 4,981,844 and 5,230,902 and PCT Publ. No. WO 2004/004483) can be used in the methods described herein.

[0262] Similarly, various agents or a combination thereof can be used to treat a cancer. For example, chemotherapy, radiation, epigenetic modifiers (e.g., histone deacetylase (HDAC) modifiers, methylation modifiers, phosphorylation modifiers, and the like), targeted therapy, and the like are well-known in the art.

[0263] In some embodiments, chemotherapy is used. Chemotherapy includes the administration of a chemotherapeutic agent. Such a chemotherapeutic agent may be, but is not

limited to, those selected from among the following groups of compounds: platinum compounds, cytotoxic antibiotics, antimetabolites, anti-mitotic agents, alkylating agents, arsenic compounds, DNA topoisomerase inhibitors, taxanes, nucleoside analogues, plant alkaloids, and toxins; and synthetic derivatives thereof. Exemplary compounds include, but are not limited to, alkylating agents: cisplatin, treosulfan, and trofosfamide; plant alkaloids: vinblastine, paclitaxel, docetaxol; DNA topoisomerase inhibitors: teniposide, crinatalol, and mitomycin; anti-folates: methotrexate, mycophenolic acid, and hydroxyurea; pyrimidine analogs: 5-fluorouracil, doxifluridine, and cytosine arabinoside; purine analogs: mercaptopurine and thioguanine; DNA antimetabolites: 2'-deoxy-5-fluorouridine, aphidicolin glycinate, and pyrazoloimidazole; and antimitotic agents: halichondrin, colchicine, and rhizoxin. Compositions comprising one or more chemotherapeutic agents (e.g., FLAG, CHOP) may also be used. FLAG comprises fludarabine, cytosine arabinoside (Ara-C) and G-CSF. CHOP comprises cyclophosphamide, vincristine, doxorubicin, and prednisone. In another embodiment, PARP (e.g., PARP-1 and/or PARP-2) inhibitors are used and such inhibitors are well-known in the art (e.g., Olaparib, ABT-888, BSI-201, BGP-15 (N-Gene Research Laboratories, Inc.); INO-1001 (Inotek Pharmaceuticals Inc.); PJ34 (Soriano et al., 2001; Pacher et al., 2002b); 3-aminobenzamide (Trevigen); 4-amino-1,8-naphthalimide; (Trevigen); 6 (5H)-phenanthridinone (Trevigen); benzamide (U.S. Pat. Re. 36,397); and NU1025 (Bowman et al.)). The mechanism of action is generally related to the ability of PARP inhibitors to bind PARP and decrease its activity. PARP catalyzes the conversion of beta-nicotinamide adenine dinucleotide (NAD⁺) into nicotinamide and poly-ADP-ribose (PAR). Both poly(ADP-ribose) and PARP have been linked to regulation of transcription, cell proliferation, genomic stability, and carcinogenesis (Bouchard V. J. et al. *Experimental Hematology*, Volume 31, Number 6, June 2003, pp. 446-454 (9); Herceg Z.; Wang Z.-Q. *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis*, Volume 477, Number 1, 2 Jun. 2001, pp. 97-110 (14)). Poly(ADP-ribose) polymerase 1 (PARP1) is a key molecule in the repair of DNA single-strand breaks (SSBs) (de Murcia J. et al. 1997. *Proc Natl Acad Sci USA* 94:7303-7307; Schreiber V, Dantzer F, Ame J C, de Murcia G (2006) *Nat Rev Mol Cell Biol* 7:517-528; Wang Z Q, et al. (1997) *Genes Dev* 11:2347-2358). Knockout of SSB repair by inhibition of PARP1 function induces DNA double-strand breaks (DSBs) that can trigger synthetic lethality in cancer cells with defective homology-directed DSB repair (Bryant H E, et al. (2005) *Nature* 434:913-917; Farmer H, et al. (2005) *Nature* 434:917-921). The foregoing examples of chemotherapeutic agents are illustrative, and are not intended to be limiting.

[0264] In other embodiments, radiation therapy is used. The radiation used in radiation therapy can be ionizing radiation. Radiation therapy can also be gamma rays, X-rays, or proton beams. Examples of radiation therapy include, but are not limited to, external-beam radiation therapy, interstitial implantation of radioisotopes (I-125, palladium, iridium), radioisotopes such as strontium-89, thoracic radiation therapy, intraperitoneal P-32 radiation therapy, and/or total abdominal and pelvic radiation therapy. For a general overview of radiation therapy, see Hellman, Chapter 16: Principles of Cancer Management: Radiation Therapy, 6th edition, 2001, DeVita et al., eds., J. B. Lip-

pencott Company, Philadelphia. The radiation therapy can be administered as external beam radiation or teletherapy wherein the radiation is directed from a remote source. The radiation treatment can also be administered as internal therapy or brachytherapy wherein a radioactive source is placed inside the body close to cancer cells or a tumor mass. Also encompassed is the use of photodynamic therapy comprising the administration of photosensitizers, such as hematoporphyrin and its derivatives, Vertoporphin (BPD-MA), phthalocyanine, photosensitizer Pc4, demethoxyhypocrellin A; and 2BA-2-DMHA.

[0265] In other embodiments, hormone therapy is used. Hormonal therapeutic treatments can comprise, for example, hormonal agonists, hormonal antagonists (e.g., flutamide, bicalutamide, tamoxifen, raloxifene, leuprolide acetate (LUPRON), LH-RH antagonists), inhibitors of hormone biosynthesis and processing, and steroids (e.g., dexamethasone, retinoids, deltoids, betamethasone, cortisol, cortisone, prednisone, dehydrotestosterone, glucocorticoids, mineralocorticoids, estrogen, testosterone, progestins), vitamin A derivatives (e.g., all-trans retinoic acid (ATRA)); vitamin D3 analogs; antigestagens (e.g., mifepristone, onapristone), or antiandrogens (e.g., cyproterone acetate).

[0266] In other embodiments, photodynamic therapy (also called PDT, photoradiation therapy, phototherapy, or photochemotherapy) is used for the treatment of some types of cancer. It is based on the discovery that certain chemicals known as photosensitizing agents can kill one-celled organisms when the organisms are exposed to a particular type of light.

[0267] In yet other embodiments, laser therapy is used to harness high-intensity light to destroy cancer cells. This technique is often used to relieve symptoms of cancer such as bleeding or obstruction, especially when the cancer cannot be cured by other treatments. It may also be used to treat cancer by shrinking or destroying tumors.

Inflammatory Disorders

[0268] The engineered antigen-binding proteins and/or pharmaceutical compositions described herein can be used, for example, for preventing or treating (reducing, partially or completely, the adverse effects of) an inflammatory disease, such as chronic inflammatory bowel disease, systemic lupus erythematosus, psoriasis, muckle-wells syndrome, rheumatoid arthritis, multiple sclerosis, or Hashimoto's disease, an allergic disease, asthma; an infectious disease; an inflammatory disease such as a TNF-mediated inflammatory disease (e.g., an inflammatory disease of the gastrointestinal tract, such as pouchitis, a cardiovascular inflammatory condition, such as atherosclerosis, or an inflammatory lung disease. The engineered antigen-binding proteins and/or pharmaceutical compositions can be used for suppressing rejection in organ transplantation or other situations in which tissue rejection might occur; for improving immune functions; or for suppressing the proliferation or function of immune cells.

[0269] In certain embodiments, the inflammatory disorders include inflammation of any tissue and organs of the body, including musculoskeletal inflammation, vascular inflammation, neural inflammation, digestive system inflammation, ocular inflammation, inflammation of the reproductive system, and other inflammation.

[0270] In some embodiments, the musculoskeletal inflammation include conditions affecting skeletal joints, including

joints of the hand, wrist, elbow, shoulder, jaw, spine, neck, hip, knee, ankle, and foot, and conditions affecting tissues connecting muscles to bones such as tendons. Examples of such immune disorders, which may be treated with the methods and compositions described herein include, but are not limited to, arthritis (including, for example, osteoarthritis, rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, acute and chronic infectious arthritis, arthritis associated with gout and pseudogout, and juvenile idiopathic arthritis), tendonitis, synovitis, tenosynovitis, bursitis, fibrositis (fibromyalgia), epicondylitis, myositis, and osteitis (including, for example, Paget's disease, osteitis pubis, and osteitis fibrosa cystica).

[0271] In some embodiments, the ocular immune disorders refers to an immune disorder that affects any structure of the eye, including the eye lids. Examples of ocular immune disorders which may be treated with the methods and compositions described herein include, but are not limited to, blepharitis, blepharochalasis, conjunctivitis, dacryoadenitis, keratitis, keratoconjunctivitis sicca (dry eye), scleritis, trichiasis, and uveitis

[0272] In some embodiments, the nervous system immune disorders which may be treated with the methods and compositions described herein include, but are not limited to, encephalitis, Guillain-Barre syndrome, meningitis, neuromyotonia, narcolepsy, multiple sclerosis, myelitis and schizophrenia. Examples of inflammation of the vasculature or lymphatic system which may be treated with the methods and compositions described herein include, but are not limited to, arthrosclerosis, arthritis, phlebitis, vasculitis, and lymphangitis.

[0273] In some embodiments, the digestive system immune disorders which may be treated with the methods and pharmaceutical compositions described herein include cholangitis, cholecystitis, enteritis, enterocolitis, gastritis, gastroenteritis, inflammatory bowel disease, ileitis, and proctitis. Inflammatory bowel diseases include, for example, certain art-recognized forms of a group of related conditions. Several major forms of inflammatory bowel diseases are known, with Crohn's disease (regional bowel disease, e.g., inactive and active forms) and ulcerative colitis (e.g., inactive and active forms) the most common of these disorders. In addition, the inflammatory bowel disease encompasses irritable bowel syndrome, microscopic colitis, lymphocytic-plasmocytic enteritis, coeliac disease, collagenous colitis, lymphocytic colitis and eosinophilic enterocolitis. Other less common forms of IBD include indeterminate colitis, pseudomembranous colitis (necrotizing colitis), ischemic inflammatory bowel disease, Behcet's disease, sarcoidosis, scleroderma, IBD-associated dysplasia, dysplasia associated masses or lesions, and primary sclerosing cholangitis.

[0274] In some embodiments, the reproductive system immune disorders which may be treated with the methods and pharmaceutical compositions described herein include, but are not limited to, cervicitis, chorioamnionitis, endometritis, epididymitis, omphalitis, oophoritis, orchitis, salpingitis, tubo-ovarian abscess, urethritis, vaginitis, vulvitis, and vulvodynia.

[0275] In some embodiments, the inflammatory disorders include acute disseminated alopecia universalis, Behcet's disease, Chagas' disease, chronic fatigue syndrome, dysautonomia, encephalomyelitis, ankylosing spondylitis, aplastic anemia, hidradenitis suppurativa, autoimmune hepatitis, autoimmune oophoritis, celiac disease, Crohn's disease,

diabetes mellitus type 1, type 2 diabetes, giant cell arteritis, Goodpasture's syndrome, Graves' disease, Guillain-Barre syndrome, Hashimoto's disease, Henoch-Schonlein purpura, Kawasaki's disease, lupus erythematosus, microscopic colitis, microscopic polyarteritis, mixed connective tissue disease, Muckle-Wells syndrome, multiple sclerosis, myasthenia gravis, opsoclonus myoclonus syndrome, optic neuritis, orchitis, thyroiditis, pemphigus, polyarteritis nodosa, polymyalgia, rheumatoid arthritis, Reiter's syndrome, Sjogren's syndrome, temporal arteritis, Wegener's granulomatosis, warm autoimmune haemolytic anemia, interstitial cystitis, Lyme disease, morphea, psoriasis, sarcoidosis, scleroderma, ulcerative colitis, and vitiligo.

[0276] The methods and compositions described herein may be used to treat T-cell mediated hypersensitivity diseases having an inflammatory component. Such conditions include contact hypersensitivity, contact dermatitis (including that due to poison ivy), urticaria, skin allergies, respiratory allergies (hay fever, allergic rhinitis, house dust mite allergy) and gluten-sensitive enteropathy (Celiac disease).

[0277] Other immune disorders which may be treated with the methods and pharmaceutical compositions include, for example, appendicitis, dermatitis, dermatomyositis, endocarditis, fibrositis, gingivitis, glossitis, hepatitis, hidradenitis suppurativa, iritis, laryngitis, mastitis, myocarditis, nephritis, otitis, pancreatitis, parotitis, pericarditis, peritonitis, pharyngitis, pleuritis, pneumonitis, prostatitis, pyelonephritis, and stomatitis, transplant rejection (involving organs such as kidney, liver, heart, lung, pancreas (e.g., islet cells), bone marrow, cornea, small bowel, skin allografts, skin homografts, and heart valve xenografts, serum sickness, and graft vs host disease), acute pancreatitis, chronic pancreatitis, acute respiratory distress syndrome, Sexary's syndrome, congenital adrenal hyperplasia, nonsuppurative thyroiditis, hypercalcemia associated with cancer, pemphigus, bullous dermatitis herpetiformis, severe erythema multiforme, exfoliative dermatitis, seborrheic dermatitis, seasonal or perennial allergic rhinitis, bronchial asthma, contact dermatitis, atopic dermatitis, drug hypersensitivity reactions, allergic conjunctivitis, keratitis, herpes zoster ophthalmicus, iritis and oiridocyclitis, chorioretinitis, optic neuritis, symptomatic sarcoidosis, fulminating or disseminated pulmonary tuberculosis chemotherapy, idiopathic thrombocytopenia purpura in adults, secondary thrombocytopenia in adults, acquired (autoimmune) haemolytic anemia, regional enteritis, autoimmune vasculitis, multiple sclerosis, chronic obstructive pulmonary disease, solid organ transplant rejection, sepsis. Preferred treatments include treatment of transplant rejection, rheumatoid arthritis, psoriatic arthritis, multiple sclerosis, Type 1 diabetes, asthma, inflammatory bowel disease, systemic lupus erythematosus, psoriasis, chronic obstructive pulmonary disease, and inflammation accompanying infectious conditions (e.g., sepsis).

Clinical Efficacy/Response to a Therapy

[0278] Clinical efficacy can be measured by any method known in the art. For example, the response to a therapy relates to any response of the cancer, e.g., a tumor, to the therapy, preferably to a change in tumor mass and/or volume after initiation of neoadjuvant or adjuvant chemotherapy. Tumor response may be assessed in a neoadjuvant or adjuvant situation where the size of a tumor after systemic intervention can be compared to the initial size and dimensions as measured by CT, PET, mammogram, ultrasound or

palpation and the cellularity of a tumor can be estimated histologically and compared to the cellularity of a tumor biopsy taken before initiation of treatment. Response may also be assessed by caliper measurement or pathological examination of the tumor after biopsy or surgical resection. Response may be recorded in a quantitative fashion like percentage change in tumor volume or cellularity or using a semi-quantitative scoring system such as residual cancer burden (Symmans et al., *J. Clin. Oncol.* (2007) 25:4414-4422) or Miller-Payne score (Ogston et al., (2003) *Breast* (Edinburgh, Scotland) 12:320-327) in a qualitative fashion like “pathological complete response” (pCR), “clinical complete remission” (cCR), “clinical partial remission” (cPR), “clinical stable disease” (cSD), “clinical progressive disease” (cPD) or other qualitative criteria. Assessment of tumor response may be performed early after the onset of neoadjuvant or adjuvant therapy, e.g., after a few hours, days, weeks or preferably after a few months. A typical endpoint for response assessment is upon termination of neoadjuvant chemotherapy or upon surgical removal of residual tumor cells and/or the tumor bed.

[0279] In some embodiments, clinical efficacy of the therapeutic treatments described herein may be determined by measuring the clinical benefit rate (CBR). The clinical benefit rate is measured by determining the sum of the percentage of patients who are in complete remission (CR), the number of patients who are in partial remission (PR) and the number of patients having stable disease (SD) at a time point at least 6 months out from the end of therapy. The shorthand for this formula is CBR=CR+PR+SD over 6 months. In some embodiments, the CBR for a particular anti-immune checkpoint therapeutic regimen is at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, or more.

[0280] Additional criteria for evaluating the response to a cancer therapy are related to “survival,” which includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); “recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g., time of diagnosis or start of treatment) and end point (e.g., death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence.

[0281] For example, in order to determine appropriate threshold values, a particular anti-cancer therapeutic regimen can be administered to a population of subjects and the outcome can be correlated to biomarker measurements that were determined prior to administration of any cancer therapy. The outcome measurement may be pathologic response to therapy given in the neoadjuvant setting. Alternatively, outcome measures, such as overall survival and disease-free survival can be monitored over a period of time for subjects following the cancer therapy for whom biomarker measurement values are known. In certain embodiments, the same doses of anti-cancer agents are administered to each subject. In related embodiments, the doses administered are standard doses known in the art for anti-cancer agents. The period of time for which subjects are monitored

can vary. For example, subjects may be monitored for at least 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40, 45, 50, 55, or 60 months. Biomarker measurement threshold values that correlate to outcome of a cancer therapy can be determined using methods such as those described in the Examples section.

Kits

[0282] In some embodiments, the antigen-binding proteins of the present disclosure are provided in a kit. In various aspects, the kit comprises the antigen-binding protein(s) as a unit dose. For purposes herein “unit dose” refers to a discrete amount dispersed in a suitable carrier. In various aspects, the unit dose is the amount sufficient to provide a subject with a desired effect, e.g., inhibition of tumor growth, reduction of tumor size, treatment of cancer. Accordingly, provided herein are kits comprising an antigen-binding protein of the present disclosure optionally provided in unit doses. In various aspects, the kit comprises several unit doses, e.g., a week or month supply of unit doses, optionally, each of which is individually packaged or otherwise separated from other unit doses. In some embodiments, the components of the kit/unit dose are packaged with instructions for administration to a patient. In some embodiments, the kit comprises one or more devices for administration to a patient, e.g., a needle and syringe, and the like. In some aspects, the antigen-binding protein of the present disclosure, a pharmaceutically acceptable salt thereof, a conjugate comprising the antigen-binding protein, or a multimer or dimer comprising the antigen-binding protein, is pre-packaged in a ready to use form, e.g., a syringe, an intravenous bag, etc. In some aspects, the kit further comprises other therapeutic or diagnostic agents or pharmaceutically acceptable carriers (e.g., solvents, buffers, diluents, etc.), including any of those described herein. In particular aspects, the kit comprises an antigen-binding protein of the present disclosure, along with an agent, e.g., a therapeutic agent, used in chemotherapy or radiation therapy.

[0283] The following examples are given merely to illustrate the present disclosure and not in any way to limit its scope.

EXAMPLES

Example 1: CTLA4 Blockade Assay

[0284] CTLA4 blockade assays described herein used the commercially available kit, CTLA4 blockade bioassay (cat. #JA3001 and JA3005; Promega Corporation, Madison, WI). CTLA4, also known as CD152, is an immune inhibitory receptor constitutively expressed on regulatory T cells (Tregs) and upregulated in activated T cells. CTLA4 plays a critical role in regulating immune responses to tumor antigens and autoantigens. CTLA4 is the counterpart of the co-stimulatory B7-CD28 pathway. When CTLA4 expression is upregulated on the surface of T cells, the T cells bind B7 with a higher avidity, and thus out-compete the positive co-stimulatory signal from CD28. In addition, engagement of CTLA4 by either of its ligands, CD80 (B7-1) or CD86 (B7-2) on an adjacent antigen presenting cell (APC) inhibits CD28 co-stimulation of T cell activation, cell proliferation and cytokine production.

[0285] The CTLA4 blockade assay involved two cell lines: Jurkat T cells (immortalized lymphocytic leukemia T cells, also referred to as the CTLA4 effector cells) and Raji cells (also referred to as the antigen-presenting cells (APC)). Jurkat T cells express human CTLA4 and a luciferase reporter driven by a native promoter which responds to TCR/CD28 activation. Raji cells express an engineered cell surface protein designed to activate cognate TCRs in an antigen-independent manner and endogenously expressing CTLA4 ligands CD80 (B7-1) and CD86 (B7-2), collectively called the B7 ligands. When the two cell types are co-cultured, CTLA4 competes with CD28 for their shared ligands, CD80 and CD86, and thus inhibits CD28 pathway activation and promoter-mediated luminescence. Addition of an anti-CTLA4 antibody blocks the interaction of CTLA4 with its ligands CD80 and CD86 and results in promoter-mediated luminescence (FIG. 1).

[0286] Jurkat cells can also be activated by anti-CD3, anti-CD28, or anti-CD3 with cell surface B7 ligands to express IL-2. Thus, in addition to the luminescence, the expression level of IL-2 protein is a metric of activation. Accordingly, when Jurkat cells are co-cultured with Raji cells that endogenously express B7 ligands and now engineered to express anti-CD3, the Jurkat cells are activated and express both IL-2 and luciferase. The expression levels of IL-2 and luciferase are linearly correlated with each other and with activation. However, the presence of CTLA4 extracellular domain, which binds the B7 ligands and blocks the Jurkat cell activation, abrogates both IL-2 and luciferase expression. Agents that block CTLA4 function, e.g., blocking the CTLA4's binding of B7 ligands, enable the Jurkat cell expression of IL-2 and luciferase. There is a linear correlation between IL-2 expression, luciferase expression/activity, and blocking efficacy of the anti-CTLA4 agent. The correlation between the CTLA4 blockade as measured by this assay and the primary T cell activation has been well documented, further validating the applicability of this assay (Waight et al. (2018) *Cancer Cell* 33:1033-1047).

[0287] While the activity curves in both assays are expected to have a similar profile, there is an effect from Fc receptors in T cell activation (Waight et al. (2018) *Cancer Cell* 33:1033-1047). Specifically, the anti-CTLA4 antibodies in certain in vitro T cell activation settings with antigen presenting cells show a biological effect dependent on Fc receptor interactions. Raji cells express FcγRII (CD32), and blockade of anti-CTLA4 antibody Fc interactions with FcγRII through either an anti-CD32 antibody or through deletion of Fc domains leads to an attenuation of an anti-CTLA4 antibody's activity as well as a decrease in the absolute level of response. This attenuation of an anti-CTLA4 antibody's activity has been reported to extend to antigen stimulation of primary T cell activation, where the FcγRIIIA (CD16) receptor was essential and not FcγRI, FcγRIIA/B, or FcγRIIB (Waight et al. (2018) *Cancer Cell* 33:1033-1047). Raji cells used in the Promega assay do not express FcγRIIIA (CD16) and no negative activity dampening effect is seen with the anti-CD32 antibodies although the anti-CD32 antibodies are strong attenuators of anti-CTLA4 biological action. Thus, it is widely accepted that the interaction between the Fc region of a CTLA4 antibody with the Fc receptors is important for its CTLA4-blocking activity (Bulliard et al. (2013) *J of Exp Medicine* 9:1685-1693; Waight et al. (2018) *Cancer Cell* 33:1033-1047; Ingram et

al. (2018) *Proc Natl Acad Sci USA* 115:3912-3917; Vargas et al. (2018) *Cancer Cell* 33:1-15).

[0288] The CTLA4 blockade assay described herein involved 120 samples on 2 plates of the 96-well plate. The assay was performed with samples in duplicate unless otherwise noted, in a titration curve of the protein concentration or the concentration of binding valency (number of antigen-binding sites). The standard control anti-CTLA4 antibodies were also assayed in duplicate in titration and within the test sample assay plate. For exceptional situations, the control anti-CTLA4 antibody titration curves from another plate, but within in the same assay kit, were used. The Promega CTLA4 Blockade Bioassay is designed and fitted for a Relative Luciferase Activity (RLU) readout. The activities of samples were reported herein as the ratio of Test Agent Relative Luciferase Units/No Test Agent Relative Luciferase Units (background luciferase units representative of basal levels of T cell activation). The assays were incubated from 12 to 16 hours. While variation in Relative Luciferase Units occurred among experiments, but the ratios of the CTLA4-blocking activity remained constant. In some experiments, rather than measuring luciferase activity, the assay well media was assayed for human IL-2 levels. Conventional Human IL-2 ELISA assays were used. Generally, the entire contents of the assay well were analyzed directly for IL-2.

[0289] The control anti-Human CTLA4 antibodies used were: Cat. No. JA1020 from Promega, ipilimumab from SelleckChem, and L3D10 from BioLegends.

Example 2: Importance of Fc Receptor in CTLA4 Blocking Activity of the Anti-CTLA4 Antibodies

[0290] In order to develop novel CTLA4-binding proteins with increased potency and safety profiles, it was hypothesized that there are sites spanning the extracellular domain of CTLA4 that are amenable to binding proteins which can allosterically block CTLA4 ligand binding and that affect the biological signaling of the CTLA4 receptor. It was also hypothesized that such signaling can be the basis of novel potent immune onco-therapeutics.

[0291] Experimental antibodies and their derivatives gene sequences, sequence confirmation, expression, purification and analytical protein analysis were produced by either Proteos (Kalamazoo, MI) or Absolute Antibody™ (Oxford, UK). The antibodies' and derivatives' gene sequences were synthesized, inserted into proprietary expression vectors, expressed in 293 cells by standard transient transfection protocols and were purified by either protein A columns, or Ni columns when a His6 tag (SEQ ID NO: 78) was used. All experimental antibodies had less than 0.1 EU per mg.

[0292] FIG. 2A and FIG. 2B show the Promega CTLA4 Blockade Bioassay of titration curve dose response of ipilimumab (IPI) (Selleckchem, Cat. No. A2001) and L3D10 (BioLegends) executed on 2 separate assays where one assay set was readout in the standard luciferase activity assay (FIG. 2A) and the other was readout as levels of IL-2 protein expressed (FIG. 2B). The RLU ratios (10^{-15}) were consistent with those reported in the Promega technical manual and consistent with a similar activity titration curve for Promega's anti-CTLA4 antibody, JA1020 (data not shown). RLU ratios of 30+over background have been observed when Jurkat effector cells alone are incubated with anti-CD3, anti-CD28 and an anti-Fc crosslinking antibody. The titration profiles of ipilimumab and L3D10 each showed

similar half max activity concentrations of approximately 50 nM for either luciferase or IL-2 readouts.

[0293] Ipilimumab and L3D10 (May et al. (2005), Du et al. (2018)) both bind close to the ligand binding site of CTLA4 for B7s, ¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1), however both ipilimumab and L3D10 have different biochemical and biological properties (Du et al. (2018a), Du et al. (2018b)) that likely arise from their respective unique sequence structure. As shown in FIG. 2B, the blockade activities of both ipilimumab and L3D10 are likely mechanistically similar as their activities are not additive when the antibody titration series are added together. Concentrations of ipilimumab as high as 5000 nM and also concentrations of ipilimumab and L3D10 admixture of 2000 nM each have been tested and their titration curves appear to plateau in the range of 1000 nM and the activity ratios stayed in the 10⁻¹⁵ range.

[0294] It is identified herein that the binding sites along CTLA4 distal to the CTLA4's ligand binding sites have novel properties. One such site is ⁶⁵SICT⁶⁸ (SEQ ID NO: 2). An antibody (hereinafter "antibody 26" or "26 antibody"), an anti-human CTLA4 mouse antibody described in patent U.S. Pat. No. 7,034,121B2 (which is incorporated herein by reference) binds this site and allosterically blocks CTLA4 binding to B7s. A chimeric antibody was generated by fusing the VH and VL domains of the antibody 26 to a human IgG1 scaffold, essentially the same IgG1 framework of ipilimumab. This chimeric antibody is referred to as 26 HuIgG1. As shown in FIG. 3, 26 HuIgG1 was active in the CTLA4 blockade assay, however the activity maximum observed was only half that of ipilimumab or L3D10. Additionally, the half maximal activity for 26 HuIgG1 was about 100 nM compared to approximately 50 nM for ipilimumab. When the anti-CD32 antibody was present at 10 µg per ml and blocked FcγRII binding, the entire activity titration curve for ipilimumab, L3D10, and 26 HuIgG1 was displaced about 50% to a lower RLU ratio (FIG. 3). The action of anti-CD32 antibodies demonstrated that the Fc region of the anti-CTLA4 antibodies is essential to activity although it may not be as simple as Fc binding to receptor.

[0295] FIG. 4 shows a comparison of the activities of 26 HuIgG1 with 26 HuIgG1 with a LALA mutation, which abolishes the FcγR and C1q interaction (Schlothauer et al. (2016) *Protein Engineering, Design & Selection* 10:457-466). Within the experimental resolution of this assay, there is not substantial effect for eliminating Fc receptor binding by this mutation nor does adding an anti-CD32 antibody have any detectable effect. Notably for ipilimumab, the addition of anti-CD32 reduced its activity by almost 50%, consistent with the results observed in FIG. 3, and by Promega and Waight et al. (2018) *Cancer Cell* 33:1033-1047. Thus, the interaction between FcγR and the anti-CTLA4-antibody, which occurs via the Fc region of the antibody, is required for full CTLA4-blocking activity.

Example 3: Surprising and Unexpected Activity of the CTLA4-Binding Proteins that Lack the Fc Region

[0296] 26 antibody derivatives with no C_H2 C_H3 domains and thereby lacking any means to bind to FcγR. 26 F(ab')₂ (synthetic) were engineered by using a fungal protein dimerization domain (a proprietary technology of Absolute Antibody™) to synthetically construct a dimer of two 26 Fabs, where each 26 Fab' is a molecularly engineered 26 V_H

C_H1 and 26 V_LC_L. The experiment of engineered antibody derivatives of 26 HuIgG1 without the Fc region, specifically 26 F(ab')₂ (syn) and 26 Fab', unexpectedly showed the activity curves that were more active (FIG. 5) than either 26 HuIgG1 or 26 HuIgG1+anti-CD32. Strikingly, FIG. 5 showed that the F(ab')₂ form of 26 has its activity titration curve shifted to a higher level of activity, more than 2× of 26 HuIgG1 and about 10+% more active than L3D10. However, the displacement of 26 F(ab')₂ to a higher activity was not uniform through the titration but is evident at high concentrations. The addition of the anti-CD32 antibody at 10 µg per ml to the titrations of 26 F(ab')₂ (syn) had no effect on the activity titration profile (FIG. 6). 26 Fab' had an activity titration curve similar to 26 HuIgG1. Also notable is that 26 Fab' is a monovalent binder while 26HuIgG1 is a bivalent and the activity titration curve of 26 Fab' is upshifted compared to 26 HuIgG1.

[0297] The structural effect observed for 26 Fv and Fc domains was not case specific only for the derivatives of the 26 antibody. As demonstrated herein, the ipilimumab activity in the CTLA4 blockade assay was decreased by anti-CD32 antibodies. Thus, binding to FcγRII is essential for the full ipilimumab activity. Nonetheless, the Ipi F(ab')₂ and Ipi ScFv constructs showed that eliminating the Fc region was not equivalent to blocking its interaction with FcγRII via anti-CD32 antibody. FIG. 7 shows the activity titration curves of Ipi F(ab')₂ (constructed as a synthetic dimer of Ipi Fabs with Absolute Antibody™) and a conventional scFv of the Ipi Fv regions in Ipi V_L (Gly₄Ser)₃ Ipi V_H ("(Gly₄Ser)₃" disclosed as SEQ ID NO: 98). As shown in FIG. 7, the activity titration curve for Ipi F(ab')₂ (syn) was twice the magnitude of ipilimumab for the high concentrations of the curve, but less so, albeit still more active at the lower concentrations. At the highest concentration of 1000 nM, the activity of Ipi ScFv was equivalent to ipilimumab but less at the lower concentrations. The activities of Ipi ScFv at the 200, 40, and 8 nM concentrations were about 50% of that seen for the ipilimumab control in this assay, similar to that observed for the combination of ipilimumab+anti-CD32 antibody.

[0298] The addition of an Fc region to an anti-CTLA4 binding domain may "bridge" through Fc receptor binding to accessory immune cells (Waight et al. (2018) *Cancer Cell* 33:1033-1047), in this assay Raji cells, which can affect the resulting biology (T cell activation) of blocking CTLA4 ligand binding. Waight et al. (2018) have shown that this bridging is a structural process, since mutations that degrade or enhance Fc binding to the receptor consequently affect the activity titration curve seen for an anti-CTLA4 antibody; reducing Fc binding affinity downshifts the entire titration curve. However, in therapeutic applications of anti-CTLA4 biologics, the addition of an Fc region could be adding positive or negative modulation of T cell activation and other inadvertent activities such as Fc-directed depletion of CTLA4 expressing cells, e.g., TREG cells. These biological activities are additional to the basic action of blocking CTLA4 from binding its ligand, but a more functionally focused anti-CTLA4. As shown in FIG. 7, Ipi F(ab')₂ (syn) can be almost twice as active as ipilimumab in this CTLA4 blockade assay. Note that the activity titration curves start at concentrations as high as 5000 nM, and ipilimumab plateaus at 1000 nM; the addition of an Fc region affects and limits a conventional anti-CTLA4 antibody's blockade potency.

[0299] For the F(ab')₂, Fab', and ScFv forms of 26 antibody and ipilimumab, there appears to be a concentration dependent effect on activity, evident on close examination of their activity titration curves. In comparing the activity curves of ipilimumab and 26 HuIgG1 with their sub-antibody derivatives F(ab')₂, Fab', and ScFv forms, there is a subtle inflection, a decrease in activity generally below 200 nM.

[0300] 121 Hu IgG1 is the human form of mouse antibody 26. Table 3 shows the human V_k light chain for the humanized form of antibody 26. Table 3 also shows the human V_H heavy chain of the humanized antibody 26. The 121 HuIgG1 is the assembly of these into a full antibody. Exclusive of the V_L and V_H sequences are the identical sequences for IgG1 (C_L C_{H1}, C_{H2}, C_{H3}) found in ipilimumab. Thus, ipilimumab and 121 HuIgG1 differ only in the V_L and V_H variable domains. FIG. 7 and FIG. 8 show the activity titration curve for 121 HuIgG1, which is displaced to a lower activity curve compared to ipilimumab and consistent for it being a humanized form of 26 HuIgG1.

[0301] 121 ScFv is a single chain form containing 121 VL and 121 VH: 121VL-(Gly₄Ser)₃-121VH-His₆ ("(Gly4Ser) 3" disclosed as SEQ ID NO: 98).

[0302] FIG. 7 and FIG. 8 show the activity titration curve for 121 ScFv and 121 HuIgG1.

[0303] FIG. 16 shows the ratios of RLU for ipilimumab, Ipi ScFv, and Ipi F(ab')₂. Ipi ScFv showed the lower activity, but F(ab')₂ showed higher activity. As described above, this was contrary to the lowered CTLA4-blocking activity observed when the anti-CD32 antibody was added to ipilimumab to block the FcγRII receptor from interacting with the ipilimumab Fc region. As also described above, this was contrary to the published work by Waight et al. (2018) and others listed above as well as Promega. Thus, this increased activity for F(ab')₂ was surprising and unexpected.

[0304] FIG. 17 shows the CTLA4-blocking activity of BioE2021 (Ipi scFv), BioE2031 (121 scFv), BioE2022 (Ipi F(ab')₂), BioE2033 (121 F(ab')₂), and ipilimumab. The experiment was done in triplicate, except the ipilimumab control which is only assayed in duplicate. As described above, the assay is formatted such that only 60 samples can be assay per plate and therefore limits the number of samples formally assayed under similar experimental conditions. Each pre-packed Promega assay allows for 120 samples, 2 plates of 60. BioE2021=Ipi ScFv, BioE2031=121 ScFv, BioE2022=Ipi F(ab')₂, BioE2033=121 F(ab')₂.

Example 4: Biparatopic CTLA4-Binding Proteins that Modulate CTLA4 and Activate T Cells

[0305] The observations described herein suggested that engineering better avidity, thereby increasing the local molecular concentration of an anti-CTLA4 binding protein, could lead to much greater potency. In addition, with the antibody designs that do not include the Fc region, a more pristine CTLA4 therapeutic may be possible without the consequences of Fc-driven immune accessory cell activity, notably TREG depletion. With this aim, biparatopic anti-CTLA4 constructs that could bind 2 different epitopes on CTLA4 were designed.

[0306] FIG. 9A shows a schematic diagram of the exemplary diabody designs (see Kipriyanov et al. (1999), Volk et al. (2001), Gall et al. (2004), and Reusch et al. (2014)). The basic diabody comprises just the VL and VH domains arrange to fold either upon itself, monomeric form, or

dimerize with itself, homodimeric diabody. The linkers between the VL and VH domains in conjunction with the amino acid sequence unique to each VL VH domains determine whether the diabody is monomeric or dimeric. FIG. 9A shows 4 forms with exemplary linker variations. These forms are a mere guide for construction and the optimal constructs cannot be predicted.

[0307] FIG. 10A shows a schematic diagram of the exemplary 2 constructs implemented; BioE2051 and BioE2052. BioE2051 and BioE2052 gene constructs were synthesized by conventional means, inserted in proprietary mammalian expression vectors of Absolute Antibody™, expressed in 293 cells. Protein product was purified by Ni column and analyzed by size exclusion chromatography (SEC). BioE2051 production yield was low, approximately 0.5 mg per 250 liter 293 cells, and revealed 2 peaks on SEC, labelled as BioE2051 P1, BioE2051 P2. BioE2052's final yield was 4 mg per 250 ml 293 cells and revealed a single peak on SEC. BioE2051 P1, BioE2051 P2 and BioE2052 had endotoxin levels of <0.1 Endotoxin Unit per mg.

[0308] BioE2051 P1, BioE2052 P2, and BioE2052 were assayed in the Promega CTLA4 blockade assay and all three diabody constructs were significantly more active throughout an activity titration curve than either ipilimumab or 121 HuIgG1 (FIG. 11). BioE2052 was further analyzed and compared to a mixture of Ipi ScFv and 121ScFv (FIG. 12). Ipi ScFv and 121 ScFv activity curves were additive but considerably less active than their covalent combination in a diabody form. BioE2052's activity is plateaued as, the lower concentration portion of the curve, after the first dilution point, showed activities that are 3 plus times greater than the corresponding concentration mixture of Ipi ScFv and 121 ScFv.

[0309] FIG. 13 depicts the results of the Promega CTLA4 Blockade bioassay done in triplicate and the read out is shown in Relative Luciferase Units, measured on a Glow-Max Navigator. Error bars shown are standard deviations. FIG. 14 depicts the readings rendered into ratios for ipilimumab and BioE2052. The activity titration curve for BioE2052 is displaced to 3× higher levels throughout the titration curve. The half-max activity concentration is between 8 and 40 nM. Arranging the V_L and V_H domains into a diabody configuration has led to both an unpredicted increase in the absolute activity in the Promega CTLA4 blockade assay and a decrease in the half max concentration. Such significant increase in the potency and specific activity was unpredicted, and exceeds the effect of mere avidity.

[0310] The covalent arrangement of Ipi and 121 VH and VL domains resulted in a diabody with unpredicted and unprecedented biological activity in the CTLA4 blockade assay. As described above, FIG. 14 shows the results expressed as the ratio of the relative luciferase units divided by the background no test agent relative luciferase unit. This ratio measurement allowed comparisons between and within the assays. The error bars are standard deviations. The assays were done in triplicate.

[0311] FIG. 15 is same as FIG. 13 but expressed as the ratios with the error bars. This is similar to FIG. 14, but with BioE2032 (BioE2032=121 HuIgG1) included.

[0312] FIG. 18 shows the CTLA4-blocking activity of the anti-CTLA4 diabody (BioE2052), a combination of BioE2021 (Ipi scFv)+BioE2031 (121 scFv), and a combination of BioE2022 (Ipi F(ab')₂)+BioE2033 (121 F(ab')₂). The two samples on the right of the graph are: BioE2021+

BioE2031 (mixture of Ipi ScFv+121 ScFv, each at 1000 nM, 200 nM, 40 nM and 8 nM) and BioE2022+BioE2033 (mixture of Ipi F(ab')₂+121 F(ab')₂). This assay was done in triplicate. The three BioE2052 samples on the left are

(referred to herein as the H22 mouse model or H22 model). The mice were administered BioE2052, ipilimumab, or vehicle (i.e., saline) i.p. according to the treatment schedule shown in Table 5.

TABLE 5

Treatment Groups and Schedule							
Group	No.	Treatment	Dose level (mg/kg)	Dosing Solution (mg/ml)	Dosing Volume (μL/g)	ROA	Dosing Frequency & Duration
1	10	Vehicle	—	—	40 μL/mouse	i.p.	QD × 20 days
2	10	BioE2052	80 μg/mouse	2	40 μL/mouse	i.p.	QD × 20 days
3	10	Ipilimumab	10	1	10	i.p.	BIW × 3 weeks
Unassigned	2	BioE2052	80 μg/mouse	2	40 μL/mouse	i.p.	Single dose

BioE2052 (07409), which is a new lot of BioE2052, expressed and purified by Absolute Antibody™. It is from the Absolute Antibody™ project 07409. FT BioE2052 is a frozen thawed sample from the 07409 lot, demonstrating that freezing the recombinant protein at -80° C. and then thawing had marginal effects on its activity. Sample BioE2052 (06966) is the original material (Absolute Antibody™ project 06966). This has been stored at 4° C. and has maintained activity for at least 4 months. This material was assayed in duplicate.

[0313] FIG. 19 shows the CTLA4-blocking activity of the anti-CTLA4 diabody (BioE2052), BioE2201 (Ipi Fab'), BioE2202 (121 Fab'), a combination of BioE2201 (Ipi Fab')+BioE2202 (121 Fab'), and ipilimumab. Note the inferior level of activity of the mixture, which at best were additive of each component's activity as compared with the diabody BioE2052, which demonstrated synergism in activity.

[0314] In summary, activity is BioE2052>Ipi F(ab')₂+121 F(ab')₂>Ipi F(ab')₂+ipilimumab~Ipi ScFv+121 ScFv>>Ipi Fab'+121 Fab' (except at 1000 nM)>Ipi ScFv, 121 ScFv>Ipi Fab', 121 Fab'.

[0315] FIG. 20 shows the CTLA4-blocking activity of the anti-CTLA4 diabody (BioE2052 (06966)), BioE2022 (Ipi F(ab')₂), BioE2033 (121 F(ab')₂), and ipilimumab done in triplicate and data rendered in ratios.

[0316] The CTLA4-blocking activities of the CTLA4-binding proteins BioE2051, BioE2052, BioE2081, BioE2082, BioE2091, BioE2092, BioE2121, BioE2012, and ipilimumab are shown in FIG. 21.

[0317] The CTLA4-blocking activity of the CTLA4-binding proteins BioE2021 Ipi ScFv, BioE2022 Ipi F(ab')₂, BioE2052 Ipi121-121Ipi, BioE2012, BioE2111 DART, and ipilimumab are shown in FIG. 22.

Example 5: In Vivo Efficacy of BioE2052

[0318] To investigate the in vivo efficacy of BioE2052, a mouse model expressing a humanized CTLA4 and having a xenograft of liver tumor cells was dosed with ipilimumab or BioE2052. Specifically, xenografts of H22 mouse liver tumor cells were implanted into the right front flank of 6-9 week old female PD1/CTLA4 HuGEMM BALB/c mice

Cell Culture

[0319] The H22 tumor cell lines were maintained in vitro with RPMI1640 medium supplemented with 10% fetal bovine serum at 37° C., 5% CO₂. The tumor cells were subcultured twice weekly. The cells in an exponential growth phase were harvested and counted for tumor inoculation.

Tumor Inoculation

[0320] Each mouse was inoculated subcutaneously in the right front flank region with H22 tumor cells (1×10⁶) in 0.1 ml of PBS for tumor development. The date of tumor cell inoculation was denoted as Day 0.

Randomization

[0321] Two days before the randomization, two mice with a tumor volume (TV)>100 mm³ were injected with BioE2052 and observed daily before randomization. The two mice were observed and measured along with other mice in three groups after randomization.

[0322] The randomization started when the mean tumor size reached approximately 85 mm³ (Day 6). A total of 30 mice were enrolled in the study and allocated into 3 groups shown in Table 5, with 10 mice per group. Randomization was performed based on "Matched distribution" method (Study Director™ software, version 3.1.399.19).

Observation and Data Collection

[0323] After tumor inoculation, the animals were checked daily for morbidity and mortality. During routine monitoring, the animals were checked for any effects of tumor growth and treatments on behavior such as mobility, food and water consumption, body weight gain/loss (Body weights would be measured twice per week after randomization), eye/hair matting and any other abnormalities. Mortality and observed clinical signs were recorded for individual animals in detail.

[0324] Tumor volumes were measured twice per week after randomization in two dimensions using a caliper, and the volume was expressed in mm³ using the formula: V=(L×W×W)/2, where V is tumor volume, L is tumor length (the longest tumor dimension) and W is tumor width (the longest tumor dimension perpendicular to L). Dosing as well as

tumor and body weight measurements were conducted in a Laminar Flow Cabinet for 20 days (Day 6 to Day 25).

[0325] The body weights and tumor volumes were measured by using Study Director™ software (version 3.1.399.19).

Statistical Analysis

[0326] To compare tumor volumes of different groups at a pre-specified day, Bartlett's test was used to validate the assumption of homogeneity of variance across all groups. When the p-value of Bartlett's test is >0.05 , one-way ANOVA was performed to test overall equality of means across all groups. If the p-value of the one-way ANOVA is <0.05 , post hoc testing was performed by running Tukey's HSD (honest significant difference) tests for all pairwise comparisons and Dunnett's tests for comparing each treatment group with the vehicle group. When the p-value of Bartlett's test was <0.05 , the Kruskal-Wallis test was performed to test overall equality of medians among all groups. If the p-value the Kruskal-Wallis test was <0.05 , post hoc testing was performed by running Conover's non-parametric test for all pairwise comparisons or for comparing each treatment group with the vehicle group, both with single-step p-value adjustment. All statistical analyses were done in R-a language and environment for statistical computing and graphics (version 3.3.1). All tests were two-sided unless otherwise specified, and p-values of <0.05 were regarded as statistically significant.

[0327] For survival analysis, the survival time was analyzed by the Kaplan-Meier method. The survival time was defined as the time from the day of randomization until animal death or ethical endpoint. For each group, the median survival time (MST) and the increased in life-span (ILS) were calculated. The Kaplan-Meier curves were also constructed for each group and the log-rank test was used to compare survival curves between groups.

Results

[0328] The results of body weights and body weight changes at different time points are shown in FIGS. 23A and 23B, respectively.

[0329] Tumor volume was also assessed during the study. Mean tumor growth curves at different time points are shown in FIG. 24. The mean tumor growth curve stopped at the first animal sacrifice due to tumor size in each group. During this study, 3 mice were sacrificed due to a TV >3000 mm³. One mouse that was not assigned was also sacrificed due to a TV >3000 mm³ (not shown).

[0330] Tumor growth inhibition observed in BioE2052-treated and ipilimumab-treated mice is summarized in Table 6 below. The data was used on the specific day when the first mouse was sacrificed due to tumor size.

TABLE 6

Antitumor Activity of BioE2052 and Ipilimumab in H22 Model					
Group	Treatment Description	Tumor Size (mm ³) ^a on Day 22	T/C (%) on Day 22	TGI (%) on Day 22	P value ^b
1	Vehicle, 40 μ L/mouse, i.p., QD	1562.45 $\pm$ 240.26	—	—	—

TABLE 6-continued

Antitumor Activity of BioE2052 and Ipilimumab in H22 Model					
Group	Treatment Description	Tumor Size (mm ³) ^a on Day 22	T/C (%) on Day 22	TGI (%) on Day 22	P value ^b
2	BioE2052, 80 μ g/mouse, i.p., QD	1436.65 $\pm$ 359.16	92	8	0.568
3	Ipilimumab, 10 mg/kg, i.p., BIW	236.93 $\pm$ 45.47	15	85	<0.001

Note:

^a Mean $\pm$ SEM;

^b compared with group 1 tumor volume on Day 22.

In this study, the mean tumor size of the vehicle control mice reached 1562.45 mm³ on Day 22 post randomization.

[0331] BioE2052 was administered at 80 μ g/mouse, QD, which produced slight anti-tumor efficacy in the H22 model, with a TGI value of 8% on Day 22. However, no statistically significant difference ($P>0.05$) was observed when compared with vehicle control group. Importantly, a therapeutic effect was observed in the BioE2052-treated group as three of the ten mice in this group exhibited tumor regression or stabilization (FIGS. 25A-25C).

[0332] Ipilimumab administered at 10 mg/kg, BIW, produced significant anti-tumor efficacy in the H22 model, with TGI value of 85% on Day 22, yielding a statistically significant difference ($p<0.05$) when compared with vehicle control group. Two ipilimumab-treated animals died, but this mortality rate is often observed in ipilimumab-treated subjects.

[0333] The β half-life of BioE2052 was about 30 minutes after a rapid drop (short a phase) upon i.p. injection. After 24 hours, only about 1% of the initial material was found in the mouse circulation.

Example 6: BioE2052-CTLA4 Interaction

[0334] To better understand the binding of BioE2502 to CTLA4, hydrogen-deuterium exchange mass spectrometry (HDX) was performed. HDX identifies binding regions between to interacting peptides as the bound regions of both are protected from deuterium exchange. CTLA4 and the antibody were combined at 1:1 ratio without dilution in order to drive occupancy (the more antibody present the better). Data was acquired for 0, 10 sec, 5 min, and 30 min. HDX generates a heat map in which the degree of H to D exchange is indicated by the color scheme. In this case, the stronger the blue hue, the more the protection in the presence of antibody. The epitopes on CTLA4 that are bound by BioE2502 were identified (FIGS. 26 and 27).

TABLE 7

Summary of Hydrogen-Deuterium Exchange Experiments		
Complex	Protection increases with antibody present	Protection decreases with antibody present
CTLA4 + BioE2502	Leu13 to Cys22 Gln80 to Ala86 Ile128 to Met134 weakly Ala61 to Ser80	Weakly Val151 to Asp162

Example 7: BioE2052-IgG1 Fc Fusion Proteins

[0335] In an effort to increase half-life and stability of BioE2052, constructs were generated that encode BioE2052 fused to a conventional IgG1 Fc or to an IgG1 Fc with an LALAPG mutation. This mutation blocks BioE2052 binding to the Fcγ receptor, thus eliminating antibody-dependent cellular cytotoxicity (ADCC). This overcomes a significant shortfall of ipilimumab, which may be exclusively a Treg depletory antibody rather than a CTLA4-blocking antibody. **[0336]** The addition of an albumin-binding protein to BioE2052 was also examined. Specifically, the carboxy end of BioE2052 was fused to the amino end of mouse albumin. The SEC profile of BioE2052 fused to mouse albumin showed 2 peaks. The two peaks were combined in these preliminary experiments but can be collected and individually analyzed.

Example 8: Antigen-Binding Proteins Fused to an IgG Fc Peptide or an Albumin-Binding Proteins Display Increases Stability

[0337] Two monoclonal anti-CTLA4 antigen-binding proteins (BioE2420 and BioE2430 (SEQ ID NOS: 61 and 63, respectively)) were generated using commercially available methods (absoluteantibody.com/our-technology/our-recombinant-technology/). BioE2420 comprises the amino acid sequence of BioE2052 (SEQ ID NO: 24) with an Fc peptide fused to the carboxy terminal of the BioE2052 peptide. BioE2430 comprises the amino acid sequence of BioE2052 (SEQ ID NO: 24) with an albumin peptide fused to the carboxy terminal of the BioE2052 peptide.

[0338] Monomers were purified by size exclusion HPLC (SEC-HPLC) (FIGS. 28A and 28B). The activity of these purified monomers was assessed using the CTLA4 Functional Blockade Assay (Promega; see example 7). As shown in FIG. 29A, BioE2420 has less anti-CTLA4 activity relative to BioE2052, but more than Ipi, which acts in a manner other than by blocking CTLA4. BioE2430 has very little activity, roughly equivalent to the negative control (FIG. 29B). These results show that antigen-binding proteins modified to have another protein domain fused to the carboxy terminus can have anti-CTLA4 activity.

Example 9

[0339] To improve or modify the pharmacokinetic (PK) properties of antigen-binding proteins, antigen-binding

domains (e.g., BioE2052) are modified (e.g., fused) to comprise a PK modulator. The modified antigen-binding protein is administered to H22 model mice according to the schedule shown in Table 6. Blood samples are drawn at regular intervals and assayed for the presence of the modified antigen-binding protein to determine the bioavailability of the modified antigen-binding protein. Results are compared to those observed in H22 mice that are administered vehicle (e.g., saline), those that are administered ipilimumab, and those that are administered only the antigen-binding protein (i.e., no Fc domain). Results show an improvement in the bioavailability of the modified antigen-binding protein relative to the normal control group and the group that are administered only the antigen binding protein. Additionally, survivability is improved in mice administered the modified antigen-binding protein relative to the normal control group and the group that are administered only the antigen binding protein.

[0340] Tumor volume and growth are measured in the H22 mice that are administered the modified antigen-binding protein and are compared to tumor volume and growth observed in H22 mice that are administered only vehicle (e.g., saline), H22 mice that are administered ipilimumab, and H22 mice that are administered only the antigen-binding protein. Results show an improvement in tumor volume and growth in mice receiving the modified antigen-binding protein relative to the normal control group and the group administered only the antigen binding protein.

INCORPORATION BY REFERENCE

[0341] All publications, patents, and patent applications mentioned herein are hereby incorporated by reference in their entirety as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated by reference. In case of conflict, the present application, including any definitions herein, will control.

EQUIVALENTS

[0342] Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the present invention described herein. Such equivalents are intended to be encompassed by the following claims.

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gaggcaattc taactttttt gctatccagc tatttttatt tgtttgtgca tttgggggga	840
attcatctct ctttaatata aagttggatg cggaacccaa attacgtgta ctacaattta	900
aagcaaagga gtgaaaagac agagctggga tgtttctgtc acatcagctc cactttcagt	960
gaaagcatca cttgggatta atatggggat gcagcattat gatgtgggtc aaggaattaa	1020
gttaggggaat ggcacagccc aaagaaggaa aaggcaggga gcgagggaga agactatatt	1080
gtacacacct tatatttacg tatgagacgt ttatagccga aatgatcttt tcaagttaaa	1140
ttttatgcct tttatttctt aaacaaatgt atgattacat caaggcttca aaaatactca	1200
catggctatg ttttagccag tgatgctaaa ggttgatttg catatatata tatatatata	1260
tatatatata tatatatata tatatatata tatatatata tatattttaa tttgatagta	1320
ttgtgcatag agccacgtat gtttttgtgt atttgttaat ggtttgaata taaacactat	1380
atggcagtgct ctttccacct tgggtcccag ggaagttttg tggaggagct caggacacta	1440

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atacaccagg tagaacacaa ggtcatttgc taactagctt ggaaactgga tgaggtcata 1500
gcagtgttg attgctgga attgtgctga gttggtgttg acatgtgctt tggggctttt 1560
acaccagttc ctttcaatgg tttgcaagga agccacagct ggtgggtatct gagttgactt 1620
gacagaacac tgtcttgaag acaatggctt actccaggag acccacaggt atgaccttct 1680
aggaagctcc agttcgatgg gcccaattct tacaacatg tggttaatgc catggacaga 1740
agaaggcagc aggtggcaga atgggggtgca tgaagggttc tgaaaattaa cactgcttgt 1800
gtttttaact caatatattc catgaaaatg caacaacatg tataatatatt ttaattaaat 1860
aaaaatctgt ggtgggtcgtt ttccgga 1887

```

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<210> SEQ ID NO 7
<211> LENGTH: 223
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

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<400> SEQUENCE: 7

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```

Met Ala Cys Leu Gly Leu Arg Arg Tyr Lys Ala Gln Leu Gln Leu Pro
 1          5          10          15
Ser Arg Thr Trp Pro Phe Val Ala Leu Leu Thr Leu Leu Phe Ile Pro
 20          25          30
Val Phe Ser Glu Ala Ile Gln Val Thr Gln Pro Ser Val Val Leu Ala
 35          40          45
Ser Ser His Gly Val Ala Ser Phe Pro Cys Glu Tyr Ser Pro Ser His
 50          55          60
Asn Thr Asp Glu Val Arg Val Thr Val Leu Arg Gln Thr Asn Asp Gln
 65          70          75          80
Met Thr Glu Val Cys Ala Thr Thr Phe Thr Glu Lys Asn Thr Val Gly
 85          90          95
Phe Leu Asp Tyr Pro Phe Cys Ser Gly Thr Phe Asn Glu Ser Arg Val
100          105          110
Asn Leu Thr Ile Gln Gly Leu Arg Ala Val Asp Thr Gly Leu Tyr Leu
115          120          125
Cys Lys Val Glu Leu Met Tyr Pro Pro Pro Tyr Phe Val Gly Met Gly
130          135          140
Asn Gly Thr Gln Ile Tyr Val Ile Asp Pro Glu Pro Cys Pro Asp Ser
145          150          155          160
Asp Phe Leu Leu Trp Ile Leu Val Ala Val Ser Leu Gly Leu Phe Phe
165          170          175
Tyr Ser Phe Leu Val Thr Ala Val Ser Leu Ser Lys Met Leu Lys Lys
180          185          190
Arg Ser Pro Leu Thr Thr Gly Val Tyr Val Lys Met Pro Pro Thr Glu
195          200          205
Pro Glu Cys Glu Lys Gln Phe Gln Pro Tyr Phe Ile Pro Ile Asn
210          215          220

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<210> SEQ ID NO 8
<211> LENGTH: 1933
<212> TYPE: DNA
<213> ORGANISM: Mus musculus

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<400> SEQUENCE: 8

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```

ctacacatat gtagcacgta ccttgatca aagctgtcta tataaagtcc ccgagtctgt 60

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gtgggttcaa acacatctca aggcctcttg atcctgttgg gttttactct gtcacctgag	120
gacctcagca catttgcccc ccagccatgg cttgtcttgg actccggagg tacaaagctc	180
aactgcagct gccttctagg acttggcctt ttgtagccct gtcactctt cttttcatcc	240
cagtcttctc tgaagccata cagggtaccc aaccttcagt ggtgttggct agcagccatg	300
gtgtcgccag ctttccatgt gaattttcac catcacacaa cactgatgag gtccgggtga	360
ctgtgctgcg gcagacaaat gaccaaata ctgaggtctg tgccacgaca ttcacagaga	420
agaatacagt gggcttccta gattaccctt tctgcagtgg tacctttaat gaaagcagag	480
tgaacctcac catccaagga ctgagagctg ttgacacggg actgtacctc tgcaagggtg	540
aactcatgta cccaccgcca tactttgtgg gcatgggcaa cgggacgcag atttatgtca	600
ttgatccaga accatgcccg gattctgact tcttcctttg gatccttgte gcagttagct	660
tgggtgtgtt tttttacagt ttcctgggtc ctgctgttct tttgagcaag atgctaaaga	720
aaagaagtcc tcttacaaca ggggtctatg tgaaaaatgcc cccaacagag ccagaatgtg	780
aaaagcaatt tcagccttat tttattccca tcaactgaaa ggccgtttat gaagaagaag	840
gagcatactt cagtctctaa aagctgaggc aatttcaact ttccttttct ctccagctat	900
ttttacctgt ttgtatatat taaggagagt atgcctctct ttaatagaaa gctggatgca	960
aaattccaat taagcatact acaatttaaa gctaaggagc atgaacagag agctgggata	1020
tttctgttgt gtcagaacca ttttactaaa agcatcactt ggaagcagca taaggatata	1080
gcattatggt gtgggggtcaa gggaacatta gggaatggca cagcccaaag aaaggaagg	1140
ggtgaaggaa gagattatat tgtacacatc ttgtatttac ctgagagatg tttatgactt	1200
aaataathtt taaatttttc atgctgttat tttctttaac aatgtataat tacacgaagg	1260
tttaaacatt tattcacaga gctatgtgac atagccagtg gttccaaagg ttgtagtggt	1320
ccaagatgta tttttaagta atattgtaca tgggtgttct atgtgctgtt gtgtatttgc	1380
tgggtgtttg aatataaaca ctatgtatca gtgtcgtccc acagtgggtc ctggggaggt	1440
ttggctgggg agcttaggac actaatccat cagggttgga cagaggtcct gcaccaactg	1500
gcttggaaac tagatgaggc tgtcacaggg ctcagttgca taaaccgatg gtgatggagt	1560
gtaaacaggg tctttacact cattttatht tttgtttctg cttttgtttt ctccaatgat	1620
ttgcaaggaa accaaaagct ggcagtgttt gtatgaacct gacagaacac tgtcttcaag	1680
gaaatgcctc attcctgaga ccagtaggtt tgttttttta ggaagttcca atactaggac	1740
cccctacaag tactatggct cctcgaaaac acaaagttaa tgccacagga agcagcagat	1800
ggtaggatgg gatgcacaag agttcctgaa aactaacact gttagtgttt tttttttaac	1860
tcaatathtt ccatgaaaat gcaaccacat gtataatatt ttaattaaa taaaagtttc	1920
ttgtgattgt ttt	1933

<210> SEQ ID NO 9

<211> LENGTH: 142

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 9

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```

Met Asp Val Leu Val Leu Phe Leu Cys Leu Val Ala Phe Pro Ser Cys
1          5          10          15
Val Leu Ser Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Ala
20          25          30
Pro Ser Gln Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu
35          40          45
Thr Ser Tyr Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu
50          55          60
Glu Trp Leu Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser
65          70          75          80
Ala Leu Met Ser Arg Leu Ser Ile Ser Lys Asp Asn Ser Lys Ser Gln
85          90          95
Val Phe Leu Lys Met Ser Ser Leu Gln Thr Asp Asp Thr Ala Met Tyr
100         105         110
Tyr Cys Ala Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala
115        120        125
Met Asp Tyr Trp Gly Gln Gly Thr Ser Val Ile Val Ser Ser
130        135        140

```

```

<210> SEQ ID NO 10
<211> LENGTH: 128
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

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<400> SEQUENCE: 10

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```

Met Asp Phe Gln Val Gln Ile Phe Ser Phe Leu Leu Ile Ser Ala Ser
1          5          10          15
Val Ile Leu Ser Arg Gly Gln Asn Val Leu Thr Gln Ser Pro Ala Ile
20          25          30
Met Pro Ala Ser Pro Gly Glu Lys Val Thr Met Thr Cys Ser Ala Thr
35          40          45
Ser Ser Ile Thr Tyr Met Ser Trp Tyr Gln Gln Lys Ser Gly Ser Ser
50          55          60
Pro Arg Leu Leu Ile Tyr Asp Thr Ser Asn Leu Ala Ser Gly Val Pro
65          70          75          80
Val Arg Phe Ser Gly Ser Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile
85          90          95
Ser Arg Met Glu Ala Glu Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp
100         105         110
Ser Ser Tyr Pro Leu Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
115        120        125

```

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<210> SEQ ID NO 11
<211> LENGTH: 384
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

```

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<400> SEQUENCE: 11

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```

atggattttc aagtgcagat tttcagcttc ctgctaataca gtgcctcagt catactgtcc      60
agaggacaaa atgttctcac ccagttctcca gcaatcatgc ctgcattctcc aggggagaag      120

```

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gtcaccatga cctgcagtgc cacctcaagt ataacttaca tgtcctggta ccagcagaag   180
tcaggatcct cccccagact cctgatttat gacacatcca acctggcttc tggagtcct   240
gttcgcttca gtggcagtgg gtctgggacc tcttactctc tcacaatcag ccgaatggag   300
gctgaagatg ctgccactta ttactgccag cagtggagta gttaccgcgt cacgttcggt   360
gctgggacca agctggagct gaaa                                           384

```

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<210> SEQ ID NO 12
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

```

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<400> SEQUENCE: 12

```

```

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
1          5          10         15
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
          20         25         30
Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
          35         40         45
Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala Leu Met
          50         55         60
Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val Ser Leu
          65         70         75         80
Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
          85         90         95
Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met Asp Tyr
          100        105        110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
          115        120

```

```

<210> SEQ ID NO 13
<211> LENGTH: 371
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

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<400> SEQUENCE: 13

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```

cagggtgcagc tgcaagagtc aggacctggc ctggtgaagc cctcacagac actgtccttg   60
acttgcactg tctctggggt ttcattaacc tcatatgggt tatattgggt tcgccagcct   120
ccaggaaaagg gtctggagtg gctgggagta atatgggctg gtggtaccac aaattataat   180
tcggctctca tgtccagact gacaatcagc aaagacacat ccaagaacca agtttcctta   240
aaactcagca gtgtgactgc agcggacaca gccgtctact actgtgcccc aggccccccg   300
cacgctatga tgaagagagg ctatgctatg gactactggg gacaaggaac cctagtcaca   360
gtctcctcag g                                           371

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<210> SEQ ID NO 14
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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-continued

<220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 14

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met
 20 25 30

Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
 35 40 45

Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
 50 55 60

Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu
 65 70 75 80

Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr
 85 90 95

Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 15

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 15

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30

Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95

Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 16

<211> LENGTH: 355

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 16

caggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgagactc 60

tcctgtgcag cctctggatt caccttcagt agctatacta tgcactgggt ccgccaggct 120

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ccaggcaagg ggctggagtg ggtgacattt atatcatatg atggaaacaa taaatactac    180
gcagactccg tgaagggccg attcaccatc tccagagaca attccaagaa cacgctgtat    240
ctgcaaatga acagcctgag agctgaggac acggctatat attactgtgc gaggaccggc    300
tggtgggggc cctttgacta ctggggccag ggaaccctgg tcaccgtctc ctacg          355

```

```

<210> SEQ ID NO 17
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

```

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<400> SEQUENCE: 17

```

```

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
                20           25           30
Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
                35           40           45
Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
                50           55           60
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
65           70           75           80
Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
                85           90           95
Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
                100          105

```

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<210> SEQ ID NO 18
<211> LENGTH: 325
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polynucleotide

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<400> SEQUENCE: 18

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```

gaaattgtgt tgacgcagtc tccaggcacc ctgtctttgt ctccagggga aagagccacc    60
ctctcctgca gggccagtcg gagtgttggc agcagctact tagcctggta ccagcagaaa    120
cctggccagg ctcccaggct cctcatctat ggtgcattca gcagggccac tggcatccca    180
gacaggttca gtggcagtggt gtctggggaca gacttcactc tcaccatcag cagactggag    240
cctgaagatt ttgcagtgtg ttactgtcag cagtatggta gctcaccgtg gacgttcggc    300
caagggaaca aggtggaat caaac          325

```

```

<210> SEQ ID NO 19
<211> LENGTH: 241
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

```

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<400> SEQUENCE: 19

```

```

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly

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1	5	10	15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser	20	25	30
Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu	35	40	45
Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser	50	55	60
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu	65	70	80
Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro	85	90	95
Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Gly	100	105	110
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Val	115	120	125
Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg Ser Leu Arg Leu Ser	130	135	140
Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr Thr Met His Trp Val	145	150	160
Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Thr Phe Ile Ser Tyr	165	170	175
Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr	180	185	190
Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn Ser	195	200	205
Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala Arg Thr Gly Trp	210	215	220
Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser	225	230	235
			240

Ser

<210> SEQ ID NO 20
 <211> LENGTH: 244
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 20

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly	1	5	10	15
Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met	20	25	30	
Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr	35	40	45	
Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser	50	55	60	
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu	65	70	75	80
Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr	85	90	95	
Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Gly Ser Gly				

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100				105				110							
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser
	115					120						125			
Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln	Thr	Leu	Ser	Leu	Thr	Cys	Thr
	130					135					140				
Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr	Gly	Val	Tyr	Trp	Val	Arg	Gln
145					150					155					160
Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Ile	Trp	Ala	Gly	Gly
			165						170					175	
Thr	Thr	Asn	Tyr	Asn	Ser	Ala	Leu	Met	Ser	Arg	Leu	Thr	Ile	Ser	Lys
		180						185					190		
Asp	Thr	Ser	Lys	Asn	Gln	Val	Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala
	195						200					205			
Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Gly	Pro	Pro	His	Ala	Met
	210					215					220				
Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val
225					230					235				240	
Thr	Val	Ser	Ser												

<210> SEQ ID NO 21

<211> LENGTH: 221

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 21

Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln
1			5					10					15		
Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr
			20				25					30			
Gly	Val	Tyr	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu
	35					40						45			
Gly	Val	Ile	Trp	Ala	Gly	Gly	Thr	Thr	Asn	Tyr	Asn	Ser	Ala	Leu	Met
	50				55						60				
Ser	Arg	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val	Ser	Leu
65				70					75					80	
Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala
			85					90						95	
Arg	Gly	Pro	Pro	His	Ala	Met	Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr
		100						105					110		
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly
	115					120						125			
Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly
	130					135					140				
Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val
145					150					155					160
Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe
			165						170					175	
Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val
		180						185					190		
Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val

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195	200	205
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Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val
 210 215 220

<210> SEQ ID NO 22
 <211> LENGTH: 213
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 22

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met
20 25 30
Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
35 40 45
Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu
65 70 75 80
Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr
85 90 95
Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala Pro
100 105 110
Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr
115 120 125
Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys
130 135 140
Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu
145 150 155 160
Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
165 170 175
Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
180 185 190
Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
195 200 205
Asn Arg Gly Glu Cys
210

<210> SEQ ID NO 23
 <211> LENGTH: 482
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 23

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
1 5 10 15
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
20 25 30
Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu

35					40					45					
Gly 50	Val	Ile	Trp	Ala	Gly 55	Gly	Thr	Thr	Asn	Tyr	Asn 60	Ser	Ala	Leu	Met
Ser 65	Arg	Leu	Thr	Ile 70	Ser	Lys	Asp	Thr	Ser	Lys 75	Asn	Gln	Val	Ser	Leu 80
Lys	Leu	Ser	Ser	Val 85	Thr	Ala	Ala	Asp	Thr 90	Ala	Val	Tyr	Tyr	Cys 95	Ala
Arg	Gly	Pro	Pro	His 100	Ala	Met	Met	Lys 105	Arg	Gly	Tyr	Ala	Met 110	Asp	Tyr
Trp	Gly	Gln	Gly	Thr 115	Leu	Val	Thr 120	Val	Ser	Ser	Gly	Gly 125	Ser	Gly	Gly
Ser	Gly 130	Gly	Ser	Glu	Ile	Val 135	Leu	Thr	Gln	Ser	Pro 140	Gly	Thr	Leu	Ser
Leu 145	Ser	Pro	Gly	Glu	Arg 150	Ala	Thr	Leu	Ser	Cys 155	Arg	Ala	Ser	Gln	Ser 160
Val	Gly	Ser	Ser	Tyr 165	Leu	Ala	Trp	Tyr	Gln 170	Gln	Lys	Pro	Gly	Gln 175	Ala
Pro	Arg	Leu	Leu	Ile 180	Tyr	Gly	Ala	Phe 185	Ser	Arg	Ala	Thr	Gly 190	Ile	Pro
Asp	Arg	Phe 195	Ser	Gly	Ser	Gly	Ser 200	Gly	Thr	Asp	Phe	Thr 205	Leu	Thr	Ile
Ser	Arg 210	Leu	Glu	Pro	Glu	Asp 215	Phe	Ala	Val	Tyr	Tyr 220	Cys	Gln	Gln	Tyr
Gly 225	Ser	Ser	Pro	Trp	Thr 230	Phe	Gly	Gln	Gly	Thr 235	Lys	Val	Glu	Ile	Lys 240
Gly	Gly	Ser	Gly	Gly 245	Ser	Gly	Gly	Ser	Gln 250	Val	Gln	Leu	Val	Glu 255	Ser
Gly	Gly	Gly	Val 260	Val	Gln	Pro	Gly	Arg 265	Ser	Leu	Arg	Leu	Ser 270	Cys	Ala
Ala	Ser 275	Gly	Phe	Thr	Phe	Ser	Ser 280	Tyr	Thr	Met	His	Trp 285	Val	Arg	Gln
Ala	Pro 290	Gly	Lys	Gly	Leu	Glu 295	Trp	Val	Thr	Phe	Ile 300	Ser	Tyr	Asp	Gly
Asn 305	Asn	Lys	Tyr	Tyr	Ala 310	Asp	Ser	Val	Lys	Gly 315	Arg	Phe	Thr	Ile	Ser 320
Arg	Asp	Asn	Ser	Lys 325	Asn	Thr	Leu	Tyr	Leu 330	Gln	Met	Asn	Ser	Leu 335	Arg
Ala	Glu	Asp	Thr 340	Ala	Ile	Tyr	Tyr	Cys 345	Ala	Arg	Thr	Gly	Trp 350	Leu	Gly
Pro	Phe 355	Asp	Tyr	Trp	Gly	Gln	Gly 360	Thr	Leu	Val	Thr	Val 365	Ser	Ser	Gly
Gly 370	Ser	Gly	Gly	Ser	Gly	Gly 375	Ser	Asp	Ile	Gln	Met	Thr 380	Gln	Ser	Pro
Ser 385	Ser	Leu	Ser	Ala	Ser	Val 390	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser 400
Ala	Thr	Ser	Ser	Ile 405	Thr	Tyr	Met	Ser	Trp 410	Tyr	Gln	Gln	Lys	Pro	Gly
Lys	Ala	Pro	Lys 420	Leu	Leu	Ile	Tyr	Asp 425	Thr	Ser	Asn	Leu	Ala	Ser	Gly
Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly 440	Ser	Gly	Thr	Asp 445	Tyr	Thr	Leu

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Thr Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln
450 455 460

Gln Trp Ser Ser Tyr Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu
465 470 475 480

Ile Lys

<210> SEQ ID NO 24

<211> LENGTH: 482

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 24

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30

Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
85 90 95

Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
100 105 110

Leu Val Thr Val Ser Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Asp
115 120 125

Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly Asp
130 135 140

Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met Ser
145 150 155 160

Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr Asp
165 170 175

Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser Gly
180 185 190

Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp
195 200 205

Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr Phe
210 215 220

Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Ser Gly Gly Ser Gly
225 230 235 240

Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro
245 250 255

Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr
260 265 270

Ser Tyr Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu
275 280 285

Trp Leu Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala
290 295 300

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Leu Met Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val
305 310 315 320

Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
325 330 335

Cys Ala Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met
340 345 350

Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly Ser
355 360 365

Gly Gly Ser Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr
370 375 380

Leu Ser Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser
385 390 395 400

Gln Ser Val Gly Ser Ser Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly
405 410 415

Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly
420 425 430

Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
435 440 445

Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln
450 455 460

Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu
465 470 475 480

Ile Lys

<210> SEQ ID NO 25

<211> LENGTH: 1476

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 25

```

caagtgcagc tccaagagtc cggccccggc ctcgtgaaac ccagccagac actgtctctg      60
acatgcaccg tgagcggcgtt ttctctgacc agctacggag tgtattgggt gagacaaccc      120
cccggaagg gactggagtg gctgggagtg atttgggccg gcggcaccac caactacaat      180
agcgccctca tgtctagact caccatctcc aaggacacca gcaagaacca agtgtccctc      240
aagctgtcca gcgtcacagc tgccgacacc gccgtgtact attgtgctag agggccccc      300
catgccatga tgaagagagg ctatgccatg gattactggg gccaaaggac actggtgacc      360
gtcagctccg gaggaagcgg cggcagcgga ggctccgaaa ttgtgctcac ccagagcccc      420
ggcacactgt ctctgagccc cggcgaaagg gccacactga gctgcagagc ctcccaatcc      480
gtgggcagca gctatctggc ttggtatcag cagaaacccg gccaaagccc tagactgctg      540
atctatggag ccttttctag agctaccggc atccccgaca gattctccgg cagcggcagc      600
ggcacagact ttacactgac aatttctaga ctggaaccag aggatttcgc cgtctactac      660
tgccagcagt atggaagcag cccttggaac ttggccaag gcaccaaggt ggagatcaag      720
ggaggaagcg gaggcagcgg aggcagccaa gtgcagctcg tggaaagcgg aggaggcgtg      780
gtgcagcccc gcagatccct cagactgagc tgcgccgcca gcggcttcac cttcagctcc      840
tataccatgc actgggtgag gcaagcccc ggcaaaggac tggagtgggt caccttcac      900

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agctacgacg gcaacaacaa gtactacgcc gacagcgtga agggaagggt caccatctct 960
agagacaact ccaagaacac cctctacctc cagatgaact ctctgagggc cgaagacacc 1020
gccatctact actgcgctag aaccggctgg ctgggaccct ttgactactg gggacaaggc 1080
acactggtea cagtgtctc cggaggaagc ggaggctccg gcgcagcga tatccagatg 1140
acccaatccc cttcctctct gagcgctcc gtgggagata gggtcacat tacatgtagc 1200
gccacaagca gcatcaccta catgagctgg taccagcaga aaccggaaa ggcccctaag 1260
ctgctcatct acgacacctc caatctggcc agcggcgtgc cttctagatt tagcggctcc 1320
ggaagcggca cagattacac actgacaatc agctctctgc agccagagga cttcgccacc 1380
tactactgtc agcagtggag cagctaccct ctgaccttg gcggcggcac caaggtggaa 1440
atcaaacacc accatcacca ccatcaccac catcac 1476

```

<210> SEQ ID NO 26

<211> LENGTH: 1476

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 26

```

caagtgcagc tctggaatc cggcggagga gtcgtgcagc ccggcagaag cctcagactg 60
agctgcgccc ccagcggatt caccttcagc agctacacca tgcactgggt gaggaagcc 120
cccgcaaaag gactggagtg ggtcacattc atctcctacg atggcaacaa caagtactac 180
gccgacagcg tgaagggaag gtttaccatc tctagagata actccaagaa caccctctac 240
ctccagatga actctctgag agctgaggac acagccatct attactgcgc tagaaccgga 300
tggtggggcc ctttcgacta ctggggacaa ggcacactgg tgacagtgtc ctccggaggc 360
agcggaggga cgggaggcag cgacatccag atgacccaat cccctagctc tctgagcgcc 420
agcgtggggc atagagtgc cattacatgc tccgccacca gcagcatcac ctacatgagc 480
tggtaccagc aaaagcccg caaagccccc aagctgctga tctacgatac cagcaatctg 540
gccagcggcg tgccttctag attttccggc tccggaagcg gcaccgatta cacactgacc 600
atttcctctc tgcagccaga ggatttcgcc acctactatt gccagcagtg gtcctctac 660
cctctcacct ttggcggcgg aacaaagggt gagatcaaag gcgcagcgg cggtccgga 720
ggcagccaag tccagctgca agagtccgga cccggactgg tgaaaccttc ccagacactg 780
tctctgacat gcacagttag cggattctcc ctcacaagct acggcgtgta ctgggtgaga 840
cagcctcccg gcaaaggact ggagtggctg ggcgtcatct gggctggagg caccaccaat 900
tacaacagcg ctctgatgtc tagactgaca atctccaagg acaccagcaa gaaccaagtg 960
tctctgaagc tgagctccgt gacagcggcc gatacagccg tgtactattg tgccagaggc 1020
cccccccacg ctatgatgaa gaggggctac gccatggact actggggcca aggcacctc 1080
gtcacagtga gctccggagg cagcggagga agcggcggca gcgagatcgt gctgaccag 1140
tcccccgga cactgtccct cagccccggc gaaagagcca cactgagctg tagagcttcc 1200
cagagcgtgg gcagcageta tctggcttgg tatcagcaga agcccgcca agccccaga 1260
ctgctcatct acggagcctt cagcagagcc accggcatcc ccgacagatt cagcggcagc 1320

```

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```

ggcagcggca ccgattttac cctcaccatc tccagactgg agccagagga cttegccgtg 1380
tattactgcc aacagtacgg cagcagccct tggacctttg gacaaggcac caagtgga 1440
atcaaacacc accaaccatca ccatacatcat caccac 1476

```

```

<210> SEQ ID NO 27
<211> LENGTH: 216
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 27

```

```

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5              10              15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25              30
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40              45
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
50          55              60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70              75              80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
85          90              95
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
100         105         110
Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
115         120         125
Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly
130         135         140
Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
145         150         155         160
Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
165         170         175
Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser
180         185         190
Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser
195         200         205
Asn Thr Lys Val Asp Lys Lys Val
210         215

```

```

<210> SEQ ID NO 28
<211> LENGTH: 215
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 28

```

```

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1          5              10              15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
20          25              30

```

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```

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
   35                               40                               45

Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
   50                               55                               60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
   65                               70                               75                               80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
   85                               90                               95

Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala
  100                               105                               110

Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser
  115                               120                               125

Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu
  130                               135                               140

Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser
  145                               150                               155                               160

Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu
  165                               170                               175

Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val
  180                               185                               190

Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys
  195                               200                               205

Ser Phe Asn Arg Gly Glu Cys
  210                               215

```

<210> SEQ ID NO 29

<211> LENGTH: 503

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 29

```

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1           5           10           15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
 20           25           30

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35           40           45

Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50           55           60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65           70           75           80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
 85           90           95

Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Gly
100           105           110

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Val
115           120           125

Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg Ser Leu Arg Leu Ser
130           135           140

Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr Thr Met His Trp Val
145           150           155           160

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Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val	Thr	Phe	Ile	Ser	Tyr	165	170	175
Asp	Gly	Asn	Asn	Lys	Tyr	Tyr	Ala	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	180	185	190
Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	195	200	205
Leu	Arg	Ala	Glu	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys	Ala	Arg	Thr	Gly	Trp	210	215	220
Leu	Gly	Pro	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	225	230	235
Ser	Lys	Glu	Ala	Lys	Glu	Ala	Lys	Glu	Ala	Lys	Glu	Ala	Lys	Glu	Ala	245	250	255
Lys	Glu	Ala	Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	260	265	270
Ser	Val	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Ala	Thr	Ser	Ser	Ile	275	280	285
Thr	Tyr	Met	Ser	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	290	295	300
Leu	Ile	Tyr	Asp	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	305	310	315
Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	325	330	335
Gln	Pro	Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Tyr	340	345	350
Pro	Leu	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly	355	360	365
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	370	375	380
Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln	Thr	Leu	Ser	Leu	385	390	395
Thr	Cys	Thr	Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr	Gly	Val	Tyr	Trp	405	410	415
Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Ile	Trp	420	425	430
Ala	Gly	Gly	Thr	Thr	Asn	Tyr	Asn	Ser	Ala	Leu	Met	Ser	Arg	Leu	Thr	435	440	445
Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val	Ser	Leu	Lys	Leu	Ser	Ser	450	455	460
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Gly	Pro	Pro	465	470	475
His	Ala	Met	Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	485	490	495
Thr	Leu	Val	Thr	Val	Ser	Ser										500		

<210> SEQ ID NO 30

<211> LENGTH: 503

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 30

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1			5					10					15		
Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Ala	Thr	Ser	Ser	Ile	Thr	Tyr	Met
	20						25					30			
Ser	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr
	35					40					45				
Asp	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser
	50					55				60					
Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu
65				70					75					80	
Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Tyr	Pro	Leu	Thr
			85						90					95	
Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly
	100						105						110		
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser
	115						120					125			
Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln	Thr	Leu	Ser	Leu	Thr	Cys	Thr
	130					135					140				
Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr	Gly	Val	Tyr	Trp	Val	Arg	Gln
145					150				155					160	
Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Ile	Trp	Ala	Gly	Gly
			165					170						175	
Thr	Thr	Asn	Tyr	Asn	Ser	Ala	Leu	Met	Ser	Arg	Leu	Thr	Ile	Ser	Lys
		180						185					190		
Asp	Thr	Ser	Lys	Asn	Gln	Val	Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala
	195					200						205			
Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Gly	Pro	Pro	His	Ala	Met
	210					215					220				
Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val
225					230					235				240	
Thr	Val	Ser	Ser	Lys	Glu	Ala	Lys	Glu	Ala	Lys	Glu	Ala	Lys	Glu	Ala
				245					250					255	
Lys	Glu	Ala	Lys	Glu	Ala	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr
	260						265						270		
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser
	275						280					285			
Gln	Ser	Val	Gly	Ser	Ser	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly
	290					295					300				
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Phe	Ser	Arg	Ala	Thr	Gly
305				310					315					320	
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu
			325						330					335	
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln
			340					345					350		
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu
	355						360						365		
Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
	370					375					380				
Ser	Gln	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Val	Val	Gln	Pro	Gly
385				390						395				400	

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Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser
 405 410 415

Tyr Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
 420 425 430

Val Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser
 435 440 445

Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu
 450 455 460

Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr
 465 470 475 480

Cys Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly
 485 490 495

Thr Leu Val Thr Val Ser Ser
 500

<210> SEQ ID NO 31
 <211> LENGTH: 478
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 31

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 1 5 10 15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30

Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45

Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala Leu Met
 50 55 60

Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val Ser Leu
 65 70 75 80

Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met Asp Tyr
 100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
 115 120 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
 130 135 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
 145 150 155 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
 165 170 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
 180 185 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
 195 200 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
 210 215 220

Ser Cys Asp Lys Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly

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225	230	235	240
Gly Gly Gly Ser Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val	245	250	255
Gln Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr	260	265	270
Phe Ser Ser Tyr Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly	275	280	285
Leu Glu Trp Val Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr	290	295	300
Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys	305	310	315
Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala	325	330	335
Ile Tyr Tyr Cys Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp	340	345	350
Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro	355	360	365
Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr	370	375	380
Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr	385	390	395
Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro	405	410	415
Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr	420	425	430
Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn	435	440	445
His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser	450	455	460
Cys Asp Lys Thr His His His His His His His His His	465	470	475

<210> SEQ ID NO 32

<211> LENGTH: 213

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 32

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly	1	5	10	15
Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met	20	25	30	
Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr	35	40	45	
Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser	50	55	60	
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu	65	70	75	80
Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr	85	90	95	

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<210> SEQ ID NO 34
<211> LENGTH: 468
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 34

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
      20          25          30
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
      35          40          45
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
      50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
      65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
      85          90          95
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
      100          105          110
Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
      115          120          125
Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly
      130          135          140
Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
      145          150          155          160
Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
      165          170          175
Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser
      180          185          190
Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser
      195          200          205
Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr
      210          215          220
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
      225          230          235          240
Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln Thr
      245          250          255
Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr Gly
      260          265          270
Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu Gly
      275          280          285
Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala Leu Met Ser
      290          295          300
Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val Ser Leu Lys
      305          310          315          320
Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala Arg
      325          330          335

Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met Asp Tyr Trp

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340	345	350
Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro		
355	360	365
Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr		
370	375	380
Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr		
385	390	395
Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro		
405	410	415
Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr		
420	425	430
Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn		
435	440	445
His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser		
450	455	460
Cys Asp Lys Thr		
465		

<210> SEQ ID NO 35

<211> LENGTH: 215

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 35

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly		
1	5	10
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser		
20	25	30
Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu		
35	40	45
Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser		
50	55	60
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu		
65	70	75
Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro		
85	90	95
Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala		
100	105	110
Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser		
115	120	125
Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu		
130	135	140
Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser		
145	150	155
Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu		
165	170	175
Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val		
180	185	190
Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys		
195	200	205

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Ser Phe Asn Arg Gly Glu Cys
210 215

<210> SEQ ID NO 36
 <211> LENGTH: 213
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 36

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met
 20 25 30
 Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
 35 40 45
 Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
 50 55 60
 Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu
 65 70 75 80
 Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr
 85 90 95
 Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro
 100 105 110
 Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr
 115 120 125
 Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys
 130 135 140
 Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu
 145 150 155 160
 Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
 165 170 175
 Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
 180 185 190
 Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
 195 200 205
 Asn Arg Gly Glu Cys
 210

<210> SEQ ID NO 37
 <211> LENGTH: 459
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 37

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
 20 25 30
 Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45

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Ile	Tyr	Gly	Ala	Phe	Ser	Arg	Ala	Thr	Gly	Ile	Pro	Asp	Arg	Phe	Ser
50					55					60					
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu
65					70				75					80	
Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro
			85					90					95		
Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Arg	Thr	Val	Ala
		100					105						110		
Ala	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser
	115						120					125			
Gly	Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu
130						135					140				
Ala	Lys	Val	Gln	Trp	Lys	Val	Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser
145					150					155					160
Gln	Glu	Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu
			165						170					175	
Ser	Ser	Thr	Leu	Thr	Leu	Ser	Lys	Ala	Asp	Tyr	Glu	Lys	His	Lys	Val
		180						185					190		
Tyr	Ala	Cys	Glu	Val	Thr	His	Gln	Gly	Leu	Ser	Ser	Pro	Val	Thr	Lys
	195						200					205			
Ser	Phe	Asn	Arg	Gly	Glu	Cys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
210						215					220				
Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly
225					230					235					240
Leu	Val	Lys	Pro	Ser	Gln	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly
			245						250					255	
Phe	Ser	Leu	Thr	Ser	Tyr	Gly	Val	Tyr	Trp	Val	Arg	Gln	Pro	Pro	Gly
		260						265					270		
Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Ile	Trp	Ala	Gly	Gly	Thr	Thr	Asn
	275						280					285			
Tyr	Asn	Ser	Ala	Leu	Met	Ser	Arg	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser
290						295					300				
Lys	Asn	Gln	Val	Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr
305					310					315					320
Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Gly	Pro	Pro	His	Ala	Met	Met	Lys	Arg
			325					330						335	
Gly	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser
		340						345					350		
Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser
	355						360					365			
Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp
370						375					380				
Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr
385					390					395					400
Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr
			405						410					415	
Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln
			420					425					430		
Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp
	435						440					445			
Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr					

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450
455
<210> SEQ ID NO 38
<211> LENGTH: 224
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 38

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
      20          25          30
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
      35          40          45
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
      50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
      65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
      85          90          95
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
      100          105          110
Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
      115          120          125
Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly
      130          135          140
Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
      145          150          155          160
Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
      165          170          175
Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser
      180          185          190
Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser
      195          200          205
Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr
      210          215          220

<210> SEQ ID NO 39
<211> LENGTH: 213
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 39

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1          5          10          15
Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met
      20          25          30
Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
      35          40          45
Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser

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<210> SEQ ID NO 40
<211> LENGTH: 452
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
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<400> SEQUENCE: 40

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Ala	Thr	Ser	Ser	Ile	Thr	Tyr	Met
			20					25					30		
Ser	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr
		35					40					45			
Asp	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser
	50					55					60				
Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu
65					70					75					80
Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Tyr	Pro	Leu	Thr
				85					90					95	
Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Arg	Thr	Val	Ala	Ala	Pro
			100					105					110		
Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser	Gly	Thr
		115					120					125			
Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu	Ala	Lys
	130					135					140				
Val	Gln	Trp	Lys	Val	Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser	Gln	Glu
145					150					155					160
Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu	Ser	Ser
			165						170					175	

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Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
 180 185 190
 Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
 195 200 205
 Asn Arg Gly Glu Cys Gly Gly Gly Ser Gly Gly Gly Ser Gly
 210 215 220
 Gly Gly Gly Ser Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val
 225 230 235 240
 Gln Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr
 245 250 255
 Phe Ser Ser Tyr Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly
 260 265 270
 Leu Glu Trp Val Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr
 275 280 285
 Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys
 290 295 300
 Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala
 305 310 315 320
 Ile Tyr Tyr Cys Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp
 325 330 335
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
 340 345 350
 Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr
 355 360 365
 Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
 370 375 380
 Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro
 385 390 395 400
 Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr
 405 410 415
 Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn
 420 425 430
 His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser
 435 440 445
 Cys Asp Lys Thr
 450

<210> SEQ ID NO 41

<211> LENGTH: 229

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 41

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30
 Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45
 Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala Leu Met
 50 55 60

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Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val Ser Leu
 65 70 75 80
 Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95
 Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met Asp Tyr
 100 105 110
 Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
 115 120 125
 Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
 130 135 140
 Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
 145 150 155 160
 Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
 165 170 175
 Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
 180 185 190
 Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
 195 200 205
 Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
 210 215 220
 Ser Cys Asp Lys Thr
 225

<210> SEQ ID NO 42
 <211> LENGTH: 215
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 42

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
 20 25 30
 Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80
 Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
 85 90 95
 Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala
 100 105 110
 Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser
 115 120 125
 Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu
 130 135 140
 Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser
 145 150 155 160
 Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu

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<210> SEQ ID NO 43
<211> LENGTH: 273
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

Glu 1	Ile	Val	Leu	Thr 5	Gln	Ser	Pro	Gly	Thr 10	Leu	Ser	Leu	Ser	Pro 15	Gly
Glu	Arg	Ala	Thr 20	Leu	Ser	Cys	Arg	Ala 25	Ser	Gln	Ser	Val	Gly 30	Ser	Ser
Tyr	Leu 35	Ala	Trp	Tyr	Gln	Gln	Lys 40	Pro	Gly	Gln	Ala	Pro 45	Arg	Leu	Leu
Ile 50	Tyr	Gly	Ala	Phe	Ser	Arg 55	Ala	Thr	Gly	Ile	Pro 60	Asp	Arg	Phe	Ser
Gly 65	Ser	Gly	Ser	Gly 70	Thr	Asp	Phe	Thr	Leu	Thr 75	Ile	Ser	Arg	Leu	Glu 80
Pro	Glu	Asp	Phe 85	Ala	Val	Tyr	Tyr	Cys 90	Gln	Gln	Tyr	Gly	Ser	Ser 95	Pro
Trp	Thr	Phe 100	Gly	Gln	Gly	Thr	Lys 105	Val	Glu	Ile	Lys	Gly 110	Gly	Gly	Ser
Gly	Gly 115	Gly	Gly	Gln	Val	Gln	Leu 120	Gln	Glu	Ser	Gly	Pro 125	Gly	Leu	Val
Lys 130	Pro	Ser	Gln	Thr	Leu	Ser 135	Leu	Thr	Cys	Thr	Val 140	Ser	Gly	Phe	Ser
Leu 145	Thr	Ser	Tyr	Gly 150	Val	Tyr	Trp	Val	Arg	Gln 155	Pro	Pro	Gly	Lys	Gly 160
Leu	Glu	Trp 165	Leu	Gly	Val	Ile	Trp	Ala	Gly 170	Gly	Thr	Thr	Asn	Tyr	Asn
Ser	Ala	Leu 180	Met	Ser	Arg	Leu	Thr	Ile 185	Ser	Lys	Asp	Thr 190	Ser	Lys	Asn
Gln	Val 195	Ser	Leu	Lys	Leu	Ser	Ser 200	Val	Thr	Ala	Ala	Asp 205	Thr	Ala	Val
Tyr 210	Tyr	Cys	Ala	Arg	Gly 215	Pro	Pro	His	Ala	Met	Met 220	Lys	Arg	Gly	Tyr
Ala 225	Met	Asp	Tyr	Trp 230	Gly	Gln	Gly	Thr	Leu	Val 235	Thr	Val	Ser	Ser	Gly 240
Gly	Cys	Gly 245	Gly	Gly	Glu	Val	Ala	Ala 250	Glu	Glu	Lys	Glu	Val	Ala 255	Ala
Leu	Glu	Lys 260	Glu	Val	Ala	Ala	Leu	Glu 265	Lys	Glu	Val	Ala	Ala	Leu	Glu

Lys

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<210> SEQ ID NO 44
 <211> LENGTH: 266
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 44

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15
Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met
20           25           30
Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
35           40           45
Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
50           55           60
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu
65           70           75           80
Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr
85           90           95
Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Ser Gly Gly
100          105          110
Gly Gly Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro
115          120          125
Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser
130          135          140
Ser Tyr Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu
145          150          155          160
Trp Val Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp
165          170          175
Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr
180          185          190
Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr
195          200          205
Tyr Cys Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln
210          215          220
Gly Thr Leu Val Thr Val Ser Ser Gly Gly Cys Gly Gly Gly Lys Val
225          230          235          240
Ala Ala Leu Lys Glu Lys Val Ala Ala Leu Lys Glu Lys Val Ala Ala
245          250          255
Leu Lys Glu Lys Val Ala Ala Leu Lys Glu
260          265

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<210> SEQ ID NO 45
 <211> LENGTH: 701
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 45

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Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr

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20							25				30				
Thr	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
		35					40					45			
Thr	Phe	Ile	Ser	Tyr	Asp	Gly	Asn	Asn	Lys	Tyr	Tyr	Ala	Asp	Ser	Val
	50					55					60				
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr
65					70					75				80	
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys
				85					90					95	
Ala	Arg	Thr	Gly	Trp	Leu	Gly	Pro	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr
			100					105					110		
Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro
		115					120					125			
Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly
	130					135					140				
Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn
145					150					155					160
Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln
				165					170					175	
Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser
			180					185					190		
Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser
	195						200					205			
Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr
	210					215					220				
His	Thr	Ser	Pro	Pro	Ser	Pro	Ala	Pro	Glu	Ala	Ala	Gly	Gly	Pro	Ser
225					230					235				240	
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
				245					250					255	
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro
			260					265					270		
Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
		275					280					285			
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val
	290					295					300				
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
305					310					315				320	
Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Gly	Ala	Pro	Ile	Glu	Lys	Thr
				325					330					335	
Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu
		340						345					350		
Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys
		355					360						365		
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser
	370					375					380				
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp
385					390					395				400	
Ser	Asp	Gly	Ser	Phe	Arg	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp	Lys	Ser
				405					410					415	
Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala
			420					425					430		

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Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Gly
 435 440 445
 Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp Ile Gln Met Thr Gln Ser
 450 455 460
 Pro Ser Ser Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys
 465 470 475 480
 Ser Ala Thr Ser Ser Ile Thr Tyr Met Ser Trp Tyr Gln Gln Lys Pro
 485 490 495
 Gly Lys Ala Pro Lys Leu Leu Ile Tyr Asp Thr Ser Asn Leu Ala Ser
 500 505 510
 Gly Val Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Tyr Thr
 515 520 525
 Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys
 530 535 540
 Gln Gln Trp Ser Ser Tyr Pro Leu Thr Phe Gly Gly Gly Thr Lys Val
 545 550 555 560
 Glu Ile Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 565 570 575
 Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro
 580 585 590
 Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr
 595 600 605
 Ser Tyr Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu
 610 615 620
 Trp Leu Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala
 625 630 635 640
 Leu Met Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val
 645 650 655
 Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
 660 665 670
 Cys Ala Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met
 675 680 685
 Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 690 695 700

<210> SEQ ID NO 46

<211> LENGTH: 215

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 46

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
 20 25 30
 Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu

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65	70	75	80
Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro	85	90	95
Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala	100	105	110
Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser	115	120	125
Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu	130	135	140
Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser	145	150	155
Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu	165	170	175
Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val	180	185	190
Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys	195	200	205
Ser Phe Asn Arg Gly Glu Cys	210	215	

<210> SEQ ID NO 47

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 47

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg	1	5	10	15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr	20	25	30	
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val	35	40	45	
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val	50	55	60	
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr	65	70	75	80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys	85	90	95	
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr	100	105	110	
Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro	115	120	125	
Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly	130	135	140	
Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn	145	150	155	160
Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln	165	170	175	
Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser	180	185	190	

-continued

Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser
	195						200					205			
Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr
	210					215					220				
His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Ala	Ala	Gly	Gly	Pro	Ser
	225				230					235					240
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
			245						250					255	
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro
			260						265				270		
Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
		275					280					285			
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val
	290					295					300				
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
	305				310					315					320
Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Gly	Ala	Pro	Ile	Glu	Lys	Thr
			325						330				335		
Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu
			340					345					350		
Pro	Pro	Cys	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Trp	Cys
		355					360					365			
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser
	370					375					380				
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp
	385				390					395					400
Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser
			405					410					415		
Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala
			420					425				430			
Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	
		435				440					445				

<210> SEQ ID NO 48

<211> LENGTH: 481

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 48

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1			5					10					15		
Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Ala	Thr	Ser	Ser	Ile	Thr	Tyr	Met
	20					25						30			
Ser	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr
	35				40						45				
Asp	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser
	50				55					60					
Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu
	65			70					75					80	
Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Tyr	Pro	Leu	Thr
			85					90						95	

Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly
			100					105						110	
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser
		115					120					125			
Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln	Thr	Leu	Ser	Leu	Thr	Cys	Thr
		130				135					140				
Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr	Gly	Val	Tyr	Trp	Val	Arg	Gln
					150					155					160
Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Ile	Trp	Ala	Gly	Gly
				165					170					175	
Thr	Thr	Asn	Tyr	Asn	Ser	Ala	Leu	Met	Ser	Arg	Leu	Thr	Ile	Ser	Lys
			180					185					190		
Asp	Thr	Ser	Lys	Asn	Gln	Val	Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala
		195					200					205			
Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Gly	Pro	Pro	His	Ala	Met
		210				215					220				
Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val
				230						235					240
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Pro	Lys	Ser	Gln	Asp	Lys
				245					250					255	
Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Ala	Ala	Gly	Gly	Pro
			260					265					270		
Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser
		275					280					285			
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp
		290				295					300				
Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn
				310						315					320
Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val
				325					330					335	
Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu
			340					345				350			
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Gly	Ala	Pro	Ile	Glu	Lys
		355					360					365			
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Cys	Thr
						375					380				
Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Ser
				390						395					400
Cys	Ala	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu
				405					410					415	
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu
			420					425					430		
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Val	Ser	Lys	Leu	Thr	Val	Asp	Lys
		435					440					445			
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser						

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<210> SEQ ID NO 49
 <211> LENGTH: 215
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 49

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
 20 25 30
 Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80
 Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
 85 90 95
 Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala
 100 105 110
 Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser
 115 120 125
 Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu
 130 135 140
 Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser
 145 150 155 160
 Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu
 165 170 175
 Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val
 180 185 190
 Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys
 195 200 205
 Ser Phe Asn Arg Gly Glu Cys
 210 215

<210> SEQ ID NO 50
 <211> LENGTH: 452
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 50

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30
 Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45
 Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala Leu Met
 50 55 60
 Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val Ser Leu

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65	70	75	80
Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala	85	90	95
Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met Asp Tyr	100	105	110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly	115	120	125
Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly	130	135	140
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val	145	150	155
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe	165	170	175
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val	180	185	190
Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val	195	200	205
Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys	210	215	220
Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Ala	225	230	235
Ala Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr	245	250	255
Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val	260	265	270
Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val	275	280	285
Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser	290	295	300
Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu	305	310	315
Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Gly Ala	325	330	335
Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro	340	345	350
Gln Val Tyr Thr Leu Pro Pro Cys Arg Asp Glu Leu Thr Lys Asn Gln	355	360	365
Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala	370	375	380
Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr	385	390	395
Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu	405	410	415
Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser	420	425	430
Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser	435	440	445
Leu Ser Pro Gly	450		

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<211> LENGTH: 478
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 51

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1             5             10            15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
      20            25            30

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
      35            40            45

Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
      50            55            60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
65            70            75            80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
      85            90            95

Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Gly
      100           105           110

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Val
      115           120           125

Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg Ser Leu Arg Leu Ser
      130           135           140

Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr Thr Met His Trp Val
145           150           155           160

Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Thr Phe Ile Ser Tyr
      165           170           175

Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr
      180           185           190

Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn Ser
      195           200           205

Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys Ala Arg Thr Gly Trp
      210           215           220

Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser
225           230           235           240

Ser Gly Gly Gly Gly Ser Glu Pro Lys Ser Gln Asp Lys Thr His Thr
      245           250           255

Cys Pro Pro Cys Pro Ala Pro Glu Ala Ala Gly Gly Pro Ser Val Phe
      260           265           270

Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro
      275           280           285

Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val
      290           295           300

Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr
305           310           315           320

Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val
      325           330           335

Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys
      340           345           350

Lys Val Ser Asn Lys Ala Leu Gly Ala Pro Ile Glu Lys Thr Ile Ser

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355	360	365
Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Cys Thr Leu Pro Pro		
370	375	380
Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val		
385	390	395 400
Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly		
405	410	415
Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp		
420	425	430
Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp		
435	440	445
Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His		
450	455	460
Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys		
465	470	475

<210> SEQ ID NO 52
 <211> LENGTH: 213
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 52

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met
20 25 30
Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
35 40 45
Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu
65 70 75 80
Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr
85 90 95
Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro
100 105 110
Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr
115 120 125
Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys
130 135 140
Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu
145 150 155 160
Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
165 170 175
Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
180 185 190
Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
195 200 205
Asn Arg Gly Glu Cys
210

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<210> SEQ ID NO 53
 <211> LENGTH: 477
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 53

Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	1	5	10	15
Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Val	Gly	Ser	Ser	20	25	30	
Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	35	40	45	
Ile	Tyr	Gly	Ala	Phe	Ser	Arg	Ala	Thr	Gly	Ile	Pro	Asp	Arg	Phe	Ser	50	55	60	
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	65	70	75	80
Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	85	90	95	
Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly	Gly	100	105	110	
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Val	115	120	125	
Glu	Ser	Gly	Gly	Gly	Val	Val	Gln	Pro	Gly	Arg	Ser	Leu	Arg	Leu	Ser	130	135	140	
Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Tyr	Thr	Met	His	Trp	Val	145	150	155	160
Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val	Thr	Phe	Ile	Ser	Tyr	165	170	175	
Asp	Gly	Asn	Asn	Lys	Tyr	Tyr	Ala	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	180	185	190	
Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	195	200	205	
Leu	Arg	Ala	Glu	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys	Ala	Arg	Thr	Gly	Trp	210	215	220	
Leu	Gly	Pro	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	225	230	235	240
Ser	Gly	Gly	Gly	Gly	Ser	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr	His	Thr	245	250	255	
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Ala	Ala	Gly	Gly	Pro	Ser	Val	Phe	260	265	270	
Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	275	280	285	
Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	290	295	300	
Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	305	310	315	320
Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	325	330	335	
Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	340	345	350	

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Lys Val Ser Asn Lys Ala Leu Gly Ala Pro Ile Glu Lys Thr Ile Ser
355 360 365

Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro
370 375 380

Cys Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Trp Cys Leu Val
385 390 395 400

Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly
405 410 415

Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp
420 425 430

Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp
435 440 445

Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His
450 455 460

Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
465 470 475

<210> SEQ ID NO 54

<211> LENGTH: 481

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 54

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met
20 25 30

Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
35 40 45

Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
50 55 60

Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu
65 70 75 80

Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr
85 90 95

Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Gly Ser Gly
100 105 110

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser
115 120 125

Gly Pro Gly Leu Val Lys Pro Ser Gln Thr Leu Ser Leu Thr Cys Thr
130 135 140

Val Ser Gly Phe Ser Leu Thr Ser Tyr Gly Val Tyr Trp Val Arg Gln
145 150 155 160

Pro Pro Gly Lys Gly Leu Glu Trp Leu Gly Val Ile Trp Ala Gly Gly
165 170 175

Thr Thr Asn Tyr Asn Ser Ala Leu Met Ser Arg Leu Thr Ile Ser Lys
180 185 190

Asp Thr Ser Lys Asn Gln Val Ser Leu Lys Leu Ser Ser Val Thr Ala
195 200 205

Ala Asp Thr Ala Val Tyr Tyr Cys Ala Arg Gly Pro Pro His Ala Met

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210					215					220					
Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val
225					230					235					240
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Pro	Lys	Ser	Gln	Asp	Lys
				245					250					255	
Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Ala	Ala	Gly	Gly	Pro
			260					265					270		
Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser
		275					280					285			
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp
	290					295					300				
Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn
305					310					315					320
Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val
				325					330					335	
Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu
			340					345					350		
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Gly	Ala	Pro	Ile	Glu	Lys
		355					360					365			
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Cys	Thr
	370					375					380				
Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Ser
385					390					395					400
Cys	Ala	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu
				405					410					415	
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu
		420						425					430		
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Val	Ser	Lys	Leu	Thr	Val	Asp	Lys
		435					440					445			
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu
	450					455					460				
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly
465				470						475				480	

Lys

<210> SEQ ID NO 55
 <211> LENGTH: 22
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 55

Met Asp Phe Gln Val Gln Ile Phe Ser Phe Leu Leu Ile Ser Ala Ser
 1 5 10 15

Val Ile Leu Ser Arg Gly
 20

<210> SEQ ID NO 56

<400> SEQUENCE: 56

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<210> SEQ ID NO 57

<400> SEQUENCE: 57

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<210> SEQ ID NO 58

<400> SEQUENCE: 58

000

<210> SEQ ID NO 59

<211> LENGTH: 714

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 59

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
85 90 95
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
100 105 110
Leu Val Thr Val Ser Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Asp
115 120 125
Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly Asp
130 135 140
Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met Ser
145 150 155 160
Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr Asp
165 170 175
Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser Gly
180 185 190
Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp
195 200 205
Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr Phe
210 215 220
Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Ser Gly Gly Ser Gly
225 230 235 240
Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro
245 250 255
Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr
260 265 270

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Ser	Tyr	Gly	Val	Tyr	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu	Glu	275	280	285
Trp	Leu	Gly	Val	Ile	Trp	Ala	Gly	Gly	Thr	Thr	Asn	Tyr	Asn	Ser	Ala	290	295	300
Leu	Met	Ser	Arg	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val	305	310	315
Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	325	330	335
Cys	Ala	Arg	Gly	Pro	Pro	His	Ala	Met	Met	Lys	Arg	Gly	Tyr	Ala	Met	340	345	350
Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Ser	355	360	365
Gly	Gly	Ser	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	370	375	380
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	385	390	395
Gln	Ser	Val	Gly	Ser	Ser	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	405	410	415
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Phe	Ser	Arg	Ala	Thr	Gly	420	425	430
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	435	440	445
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	450	455	460
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	465	470	475
Ile	Lys	Gly	Gly	Ser	Gly	Gly	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	485	490	495
Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	500	505	510
Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	515	520	525
Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	530	535	540
Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	545	550	555
Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	565	570	575
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	580	585	590
Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	595	600	605
Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	610	615	620
Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	625	630	635
Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	645	650	655
Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	660	665	670
Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn			

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675	680	685
Val Phe Ser Cys Ser Val Met His Glu Ala Leu	His Asn His Tyr Thr	
690	695	700
Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys		
705	710	

<210> SEQ ID NO 60
 <211> LENGTH: 2142
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 60

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caagtgcagc tggtagagag cggtagcgcc gtggtgcagc ccgtagaag cctgagactg    60
agctgcgccg caagcggtct caccttcagc agctacacca tgcactgggt gagacaagca    120
cccgcaagg gactggaatg ggtcacgttc atcagctacg acggcaacaa caagtactac    180
gccgacagcg tgaaggcgag attcaccatc agccgagaca acagcaagaa caccctgtac    240
ctgcagatga acagcctgag agccgaggac accgccatct actattgcgc tagaacggga    300
tggctggggc ccttcgatta ttgggggcaa ggcacgttgg tgaccgtctc aagcggcgga    360
agcggaggga gccgaggatc ggatatacag atgacacagt cgctagctc actgagcgca    420
agcgtggggc acagagtgc catcacctgc agcgccacaa gcagcatcac ctacatgagc    480
tggatatcag aaaaaccggg caaggccccc aagctgtgta tctacgatac aagcaacctg    540
gcaagcggcg tgcctagcag attctctggc agcggcagcg gcaccgacta tactctcacc    600
ataagcagtc tacagcctga ggatttcgcc acctactatt gtcagcagtg gtcgagctac    660
ccccctgacct tcgggggcgg caccaaggta gagattaaag gtgggagcgg cgggagtggt    720
ggcagcccaag tccaactgca agagtccgga cccggtcttg tgaagccttc tcagactctt    780
agcttaacat gcaccgtgag cggtctcagc cttacaagct acggcgtcta ctgggtacgt    840
caaccacctg gtaagggttt agaattggta ggggtgatct gggccggcgg caccaccaac    900
tacaacagcg ccctgatgag cagactgacg atctctaagg aactagcaa aaaccaagtg    960
agcctcaagc tgagcagcgt gaccgcccgc gacaccgccc tatactactg tgcaagaggc   1020
ccccccacg ccctgatgaa gagaggctac gccatggatt actgggggca aggcactcta   1080
gtgaccgtga gtagtggcgg gagcgggtgg agcggcggtt ccgaaatcgt gctgacacag   1140
agccctggca ctctgtctct gagtccggc gagagagcta ccctgagctg cagagcatct   1200
cagagcgtgg gcagcagcta cctggcctgg tatcagcaaa agcccgcca agcccctaga   1260
cttctgatct acggcgcatc cagcagagcc accggcatcc ccgacagatt cagtggatct   1320
ggcagcggaa cggacttcac cctaacaatc agcgtctctg aaccgaaga ctttgcggtg   1380
tattactgtc aacagtacgg tagcagcccc tggaccttcg gccaaaggc aaaggtggag   1440
atcaaaggcg gttccggtgg cgacaaaaca cacacctgtc cccctgccc cgcccctgag   1500
ctgttaggtg gtcctagcgt gttcctgttc cccccaagc ccaaggacac cctgatgatc   1560
agcagaaccc ctgaggtgac ctgcgtggtg gtggacgtga gccacgagga ccctgaggtg   1620
aagttcaact ggtacgtgga cggcgtggag gtgcacaacg ccaagaccaa gcctagagag   1680
gagcagtaca acagcaccta cagagtgtg agcgtgctga ccgtgctgca ccaagactgg   1740

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ctgaacggca aggagtacaa gtgcaaggtg agcaacaagg cctgcccgc ccccatcgag 1800
aagaccatca gcaaggccaa ggggcagcct agagagcccc aagtgtacac cctgcccct 1860
agcagagacg agctgaccaa gaaccaagtg tcgctaacct gtttggtgaa gggcttctac 1920
cctagcgaca tcgccgtgga gtgggagagc aacgggcagc ctgagaacaa ctacaagacc 1980
accccccccg tgctggacag cgacggcagc ttcttctgt acagcaagct gaccgtggac 2040
aagagcagat ggcagcaagg caacgtgttc agctgcagcg tgatgcacga ggccctgcac 2100
aaccactaca cacagaagag cctgtcgctg agccccgaa aa 2142

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<210> SEQ ID NO 61
<211> LENGTH: 714
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 61

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Gln Val Gln Leu Val Glu Ser Gly Gly Val Val Gln Pro Gly Arg
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20     25     30
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
85     90     95
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
100    105    110
Leu Val Thr Val Ser Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Asp
115    120    125
Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly Asp
130    135    140
Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met Ser
145    150    155    160
Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr Asp
165    170    175
Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser Gly
180    185    190
Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp
195    200    205
Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr Phe
210    215    220
Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Ser Gly Gly Ser Gly
225    230    235    240
Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro
245    250    255
Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr
260    265    270

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Ser	Tyr	Gly	Val	Tyr	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu	Glu
		275					280					285			
Trp	Leu	Gly	Val	Ile	Trp	Ala	Gly	Gly	Thr	Thr	Asn	Tyr	Asn	Ser	Ala
	290					295					300				
Leu	Met	Ser	Arg	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val
305					310					315					320
Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr
				325					330					335	
Cys	Ala	Arg	Gly	Pro	Pro	His	Ala	Met	Met	Lys	Arg	Gly	Tyr	Ala	Met
			340					345					350		
Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Ser
		355					360					365			
Gly	Gly	Ser	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr
	370					375					380				
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser
385					390					395					400
Gln	Ser	Val	Gly	Ser	Ser	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly
			405						410						415
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Phe	Ser	Arg	Ala	Thr	Gly
			420					425					430		
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu
	435					440						445			
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln
	450					455					460				
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu
465					470					475					480
Ile	Lys	Gly	Gly	Ser	Gly	Gly	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys
			485					490						495	
Pro	Ala	Pro	Glu	Ala	Ala	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro
			500					505					510		
Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys
		515				520						525			
Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp
	530					535					540				
Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu
545					550					555					560
Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu
				565				570						575	
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn
			580					585					590		
Lys	Ala	Leu	Gly	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly
		595				600						605			
Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu
	610					615					620				
Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr
625					630					635					640
Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn
			645					650						655	
Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe
			660					665					670		

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Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn
675 680 685

Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
690 695 700

Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
705 710

<210> SEQ ID NO 62

<211> LENGTH: 2142

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 62

```

caagtgcagc tgggtggagag cgggtggcggc gtgggtgcagc ccggtagaag cctgagactg      60
agctgcgcgc caagcggtt cacccttcagc agctacacca tgcactgggt gagacaagca      120
cccggaagg gactggaatg ggtcacgttc atcagctacg acggcaacaa caagtactac      180
gccgacagcg tgaagggcag attcaccatc agccgagaca acagcaagaa caccctgtac      240
ctgcagatga acagcctgag agccgaggac accgccatct actattgcgc tagaacggga      300
tggctgggcc ccttcgatta ttggggggcaa ggcacgttgg tgaccgtctc aagcggcgga      360
agcggaggga gcggaggatc ggatatacag atgacacagt cgcctagctc actgagcgca      420
agcgtgggcg acagagtgc catcacctgc agcgccacaa gcagcatcac ctacatgagc      480
tggtatcagc aaaaaccggg caaggccccc aagctgctga tctacgatac aagcaacctg      540
gcaagcggcg tgcctagcag attctctggc agcggcagcg gcaccgacta tactctcacc      600
ataagcagtc tacagcctga ggatttcgcc acctactatt gtcagcagtg gtcgagctac      660
ccctgacct tcggggggcg caccaaggta gagattaaag gtgggagcgg cgggagtggg      720
ggcagccaag tccaactgca agagtccgga cccggtcttg tgaagccttc tcagactctt      780
agcttaacat gcaccgtgag cggcttcagc cttacaagct acggcgtcta ctgggtacgt      840
caaccacctg gtaagggttt agaattggta ggggtgatct gggccggcgg caccaccaac      900
tacaacagcg ccctgatgag cagactgacg atctctaagg aactagcaa aaaccaagtg      960
agcctcaagc tgagcagcgt gaccgccgcc gacaccgccg tatactactg tgcaagaggc     1020
ccccccacg ccatgatgaa gagaggttac gccatggatt actgggggca aggcactcta     1080
gtgaccgtga gtagtgccg gagcgggtgg agcggcgggt ccgaaatcgt gctgacacag     1140
agccctggca ctctgtctct gagtcccgcc gagagagcta ccctgagctg cagagcatct     1200
cagagcgtgg gcagcagcta cctggcctgg tatcagcaaa agcccggcc aagccctaga     1260
cttctgatct acggcgcatc cagcagagcc accggcatcc ccgacagatt cagtggatct     1320
ggcagcggaa cggacttcac cctaacaatc agccgtctgg aaccgaaga ctttgcggtg     1380
tattactgtc aacagtacgg tagcagcccc tggaccttcg gccaaaggcac aaagggtggag     1440
atcaaaggcg gttccgggtg cgacaaaaca cacacctgtc cccctgccc cgccctgag     1500
gtgcccgtg gtcctagcgt gttcctgttc cccccaagc ccaaggacac cctgatgatc     1560
agcagaaccc ctgaggtgac ctgcgtggtg gtggacgtga gccacgagga ccctgaggtg     1620
aagttcaact ggtacgtgga cggcgtggag gtgcacaacg ccaagaccaa gcctagagag     1680

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gagcagtaca acagcaccta cagagtgggtg agcgtgctga ccgtgctgca ccaagactgg 1740
ctgaacggca aggagtacaa gtgcaagggtg agcaacaagg ccctggggcgc ccccatcgag 1800
aagaccatca gcaaggccaa ggggcagcct agagagcccc aagtgtacac cctgccccct 1860
agcagagacg agctgaccaa gaaccaagtg tcgctaacct gtttggtgaa gggtttctac 1920
cctagcgaca tcgccgtgga gtgggagagc aacgggcagc ctgagaacaa ctacaagacc 1980
accccccccg tgctggacag cgacggcagc ttcttctgt acagcaagct gaccgtggac 2040
aagagcagat ggcagcaagg caacgtgttc agctgcagcg tgatgcacga ggccctgcac 2100
aaccactaca cacagaagag cctgtcgctg agccccgaa aa 2142

```

<210> SEQ ID NO 63

<211> LENGTH: 1081

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 63

```

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20          25          30
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
85          90          95
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
100          105          110
Leu Val Thr Val Ser Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Asp
115          120          125
Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly Asp
130          135          140
Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met Ser
145          150          155          160
Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr Asp
165          170          175
Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser Gly
180          185          190
Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp
195          200          205
Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr Phe
210          215          220
Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Ser Gly Gly Ser Gly
225          230          235          240
Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro
245          250          255
Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr

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260						265						270					
Ser	Tyr	Gly	Val	Tyr	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu	Glu		
		275					280					285					
Trp	Leu	Gly	Val	Ile	Trp	Ala	Gly	Gly	Thr	Thr	Asn	Tyr	Asn	Ser	Ala		
	290					295					300						
Leu	Met	Ser	Arg	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val		
305					310					315					320		
Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr		
			325						330					335			
Cys	Ala	Arg	Gly	Pro	Pro	His	Ala	Met	Met	Lys	Arg	Gly	Tyr	Ala	Met		
		340						345					350				
Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Ser		
	355						360					365					
Gly	Gly	Ser	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr		
	370					375					380						
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser		
385					390					395					400		
Gln	Ser	Val	Gly	Ser	Ser	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly		
			405						410					415			
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Phe	Ser	Arg	Ala	Thr	Gly		
		420						425					430				
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu		
	435					440						445					
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln		
	450					455					460						
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu		
465					470					475					480		
Ile	Lys	Gly	Gly	Ser	Gly	Gly	Glu	Ala	His	Lys	Ser	Glu	Ile	Ala	His		
			485						490					495			
Arg	Tyr	Asn	Asp	Leu	Gly	Glu	Gln	His	Phe	Lys	Gly	Leu	Val	Leu	Ile		
		500						505					510				
Ala	Phe	Ser	Gln	Tyr	Leu	Gln	Lys	Cys	Ser	Tyr	Asp	Glu	His	Ala	Lys		
		515					520					525					
Leu	Val	Gln	Glu	Val	Thr	Asp	Phe	Ala	Lys	Thr	Cys	Val	Ala	Asp	Glu		
	530					535					540						
Ser	Ala	Ala	Asn	Cys	Asp	Lys	Ser	Leu	His	Thr	Leu	Phe	Gly	Asp	Lys		
545					550					555					560		
Leu	Cys	Ala	Ile	Pro	Asn	Leu	Arg	Glu	Asn	Tyr	Gly	Glu	Leu	Ala	Asp		
			565						570					575			
Cys	Cys	Thr	Lys	Gln	Glu	Pro	Glu	Arg	Asn	Glu	Cys	Phe	Leu	Gln	His		
		580						585					590				
Lys	Asp	Asp	Asn	Pro	Ser	Leu	Pro	Pro	Phe	Glu	Arg	Pro	Glu	Ala	Glu		
	595					600						605					
Ala	Met	Cys	Thr	Ser	Phe	Lys	Glu	Asn	Pro	Thr	Thr	Phe	Met	Gly	His		
	610					615						620					
Tyr	Leu	His	Glu	Val	Ala	Arg	Arg	His	Pro	Tyr	Phe	Tyr	Ala	Pro	Glu		
625					630					635					640		
Leu	Leu	Tyr	Tyr	Ala	Glu	Gln	Tyr	Asn	Glu	Ile	Leu	Thr	Gln	Cys	Cys		
			645					650						655			
Ala	Glu	Ala	Asp	Lys	Glu	Ser	Cys	Leu	Thr	Pro	Lys	Leu	Asp	Gly	Val		
		660						665					670				

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Lys	Glu	Lys	Ala	Leu	Val	Ser	Ser	Val	Arg	Gln	Arg	Met	Lys	Cys	Ser	675	680	685
Ser	Met	Gln	Lys	Phe	Gly	Glu	Arg	Ala	Phe	Lys	Ala	Trp	Ala	Val	Ala	690	695	700
Arg	Leu	Ser	Gln	Thr	Phe	Pro	Asn	Ala	Asp	Phe	Ala	Glu	Ile	Thr	Lys	705	710	715
Leu	Ala	Thr	Asp	Leu	Thr	Lys	Val	Asn	Lys	Glu	Cys	Cys	His	Gly	Asp	725	730	735
Leu	Leu	Glu	Cys	Ala	Asp	Asp	Arg	Ala	Glu	Leu	Ala	Lys	Tyr	Met	Cys	740	745	750
Glu	Asn	Gln	Ala	Thr	Ile	Ser	Ser	Lys	Leu	Gln	Thr	Cys	Cys	Asp	Lys	755	760	765
Pro	Leu	Leu	Lys	Lys	Ala	His	Cys	Leu	Ser	Glu	Val	Glu	His	Asp	Thr	770	775	780
Met	Pro	Ala	Asp	Leu	Pro	Ala	Ile	Ala	Ala	Asp	Phe	Val	Glu	Asp	Gln	785	790	795
Glu	Val	Cys	Lys	Asn	Tyr	Ala	Glu	Ala	Lys	Asp	Val	Phe	Leu	Gly	Thr	805	810	815
Phe	Leu	Tyr	Glu	Tyr	Ser	Arg	Arg	His	Pro	Asp	Tyr	Ser	Val	Ser	Leu	820	825	830
Leu	Leu	Arg	Leu	Ala	Lys	Lys	Tyr	Glu	Ala	Thr	Leu	Glu	Lys	Cys	Cys	835	840	845
Ala	Glu	Ala	Asn	Pro	Pro	Ala	Cys	Tyr	Gly	Thr	Val	Leu	Ala	Glu	Phe	850	855	860
Gln	Pro	Leu	Val	Glu	Glu	Pro	Lys	Asn	Leu	Val	Lys	Thr	Asn	Cys	Asp	865	870	875
Leu	Tyr	Glu	Lys	Leu	Gly	Glu	Tyr	Gly	Phe	Gln	Asn	Ala	Ile	Leu	Val	885	890	895
Arg	Tyr	Thr	Gln	Lys	Ala	Pro	Gln	Val	Ser	Thr	Pro	Thr	Leu	Val	Glu	900	905	910
Ala	Ala	Arg	Asn	Leu	Gly	Arg	Val	Gly	Thr	Lys	Cys	Cys	Thr	Leu	Pro	915	920	925
Glu	Asp	Gln	Arg	Leu	Pro	Cys	Val	Glu	Asp	Tyr	Leu	Ser	Ala	Ile	Leu	930	935	940
Asn	Arg	Val	Cys	Leu	Leu	His	Glu	Lys	Thr	Pro	Val	Ser	Glu	His	Val	945	950	955
Thr	Lys	Cys	Cys	Ser	Gly	Ser	Leu	Val	Glu	Arg	Arg	Pro	Cys	Phe	Ser	965	970	975
Ala	Leu	Thr	Val	Asp	Glu	Thr	Tyr	Val	Pro	Lys	Glu	Phe	Lys	Ala	Glu	980	985	990
Thr	Phe	Thr	Phe	His	Ser	Asp	Ile	Cys	Thr	Leu	Pro	Glu	Lys	Glu	Lys	995	1000	1005
Gln	Ile	Lys	Lys	Gln	Thr	Ala	Leu	Ala	Glu	Leu	Val	Lys	His	Lys		1010	1015	1020
Pro	Lys	Ala	Thr	Ala	Glu	Gln	Leu	Lys	Thr	Val	Met	Asp	Asp	Phe		1025	1030	1035
Ala	Gln	Phe	Leu	Asp	Thr	Cys	Cys	Lys	Ala	Ala	Asp	Lys	Asp	Thr		1040	1045	1050
Cys	Phe	Ser	Thr	Glu	Gly	Pro	Asn	Leu	Val	Thr	Arg	Cys	Lys	Asp		1055	1060	1065

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Ala Leu Ala His His His His His His His His His His
1070 1075 1080

<210> SEQ ID NO 64
<211> LENGTH: 3243
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 64

caagtgcaat tggtagagtc cggcggcggc gttgtgcaac ccggcagaag ccttagactg 60
agctgcgccg caagcggett caccttcagc agctacacca tgcactgggt gagacaagcc 120
cccggaagg gcctggaatg ggtgaccttc atcagctacg acggcaacaa caagtactac 180
gctgatagcg tgaagggcag attcactatc agccgtgaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccatct actactgccc aagaaccggc 300
tggttggggc ccttcgacta ctggggccaa gggacattag ttactgttag cagcgcgggg 360
agcgggggca gcggaggatc ggatatacag atgacacaga gccctagcag cctgagcgca 420
agcgtggggc acagagtgc catcacctgc agcggccaaa gcagcatcac ctacatgagc 480
tggtatcagc aaaaaccggc caaggccccc aagctgctga tctacgacac aagcaacctg 540
gcaagcggcg tgcctagcag attctctggc agcggcagcg gcacagatta tactctcaca 600
atcagcagtc tgcagcctga ggatttcgcc acatactact gccaacagtg gagcagctac 660
cccttgacct tcggcgcgcg cactaaagta gagattaaag gcggcagcgg tggaagtggc 720
gggagccaag tgcaactgca agagagcggc cccggcctgg tgaagccttc tcagacctg 780
agccttacct gcacogtgag cggcttcagc ctgacaagct acggcgtgta ctgggtgagg 840
caaccccccg gcaaaggctt ggaatggctg ggggtgatct gggcggcgcg caccaccaac 900
tacaacacgc ccctgatgag cagactgacc attagcaagg acacaagcaa gaaccaagt 960
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ccccccacg ccatgatgaa gagaggctac gccatggatt attggggcca aggcacctg 1080
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agccctggca ccctgtcgct gagccctggt gaaagggcta cgctaagctg cagagcatct 1200
cagagcgtgg gcagcageta cctggcctgg tatcagcaga aaccgggcca agcccctaga 1260
ctgctgatct atggtgcctt cagcagagcc accggcatcc ccgacagatt cagcggttcc 1320
ggtagtggca cagacttcac tctgaccatc agccggctag aacctgagga ttttgcggtt 1380
tactactgtc agcagtatgg cagcagcccc tggaccttcg gccaaaggac gaagggtggag 1440
ataaaagggt gaagcggcgg ggaagccac aagagcgaga tcgcccacag atacaacgac 1500
ctgggcgagc agcacttcaa gggcctgggt ctgatgcct tctctcagta cctgcagaag 1560
tgcagctacg acgagcacgc caagctgggt caagagggtga ccgacttcgc caagacctgc 1620
gtggccgacg agagcgccgc caactgcgac aagagcctgc acacctgtt cggcgacaag 1680
ctgtgcgcca tccccaaact gagagagaac tacggcgagc tggccgactg ctgcaccaag 1740
caagagccgg agagaaatga gtgcttctct cagcacaagg acgacaaccc tagcctgccc 1800
cccttcgaga gacctgaggc cgaggccatg tgcactagct tcaaggagaa ccccaccacc 1860

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ttcatgggcc actacctgca cgaggtggct agacgtcatc catatttcta cgccctgag 1920
ctgctgtact atgcagagca gtacaacgag atcctgaccc aatgttgtgc cgaggcggac 1980
aaagagagct gcctgacccc caagctggac ggcgtgaagg agaaggccct ggtgagcagc 2040
gtcagacaga gaatgaagtg cagcagcatg cagaagtctg gggagagagc tttcaaggcc 2100
tgggccgtgg ctagactgtc tcagaccttc cccaacgccg acttcgccga gatcaccaag 2160
ctggccaccg acctgaccaa ggtgaacaag gagtgtgcc acggcgacct gctggagtgc 2220
gccgacgaca gagccgagct ggccaagtac atgtgcgaga accaagccac tatctcatcc 2280
aaattacaga cctgtgcga caagccctg ctgaagaaag ctcatcgct cagcgaggtg 2340
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tacagccgaa ggcaccccg ctacacgctg agcctgctgc tgagactggc caagaagtac 2520
gaggccactt tggagaaatg ctgcgctgag gcaaaccccc ctgcctgcta cggcaccgtg 2580
ctggccgagt ttcagccctt ggtagaggag ccgaagaacc tgggtgaagac caactgcgac 2640
ctgtacgaga agctgggcga gtacggcttt cagaacgcca tcctggtgag atacacacag 2700
aaggcccccc aagtgcgcac ccccacctt gtggaagccg caagaaacct gggcagagtg 2760
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tgtactctcc ctgagaagga gaagcagatc aagaagcaga ccgccctggc cgagctgggtg 3060
aagcacaaag ccaaggccac cgccgagcag ctgaagaccg tgatggacga cttcgctcag 3120
ttcctggaca cctgtgcaa ggcgcccgc aaggacacct gcttcagcac cgagggcccc 3180
aacctggtga caagatgcaa ggacgcccctg gccccaccac accaccacca ccaccaccac 3240
cac

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<210> SEQ ID NO 65

<211> LENGTH: 449

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 65

```

Asp Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
1           5           10           15
Ser Leu Ser Leu Thr Cys Ser Val Thr Gly Tyr Ser Ile Thr Ser Gly
20          25          30
Tyr Val Trp Asn Trp Ile Arg Gln Phe Pro Gly Asn Lys Leu Glu Trp
35          40          45
Met Gly Tyr Ile Ser His Asp Gly Asn Thr Asn Tyr Asn Pro Ser Leu
50          55          60
Lys Asn Arg Ile Ser Ile Thr Arg Asp Thr Ser Lys Asn Gln Phe Phe
65          70          75          80
Leu Lys Leu Asn Ser Val Thr Thr Glu Asp Ser Ala Thr Tyr Tyr Cys
85          90          95

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Lys

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<210> SEQ ID NO 66
<211> LENGTH: 1347
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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polynucleotide

<400> SEQUENCE: 66

gacgtgcagc tgcaagagag cggccccggc ctggtgaagc cgagccaatc tctgagcctg	60
acctgcagtg tgacgggata ctcaatcacc tctggttacg tgtggaactg gatcagacag	120
ttccccggca acaagctgga gtggatgggc tacatcagcc acgacggcaa caccaactac	180
aaccctagcc tgaagaacag aatcagcata acaagagaca caagcaagaa tcagttcttc	240
ctgaagctga acagcgtgac caccgaggac agcgccacct actactgcac aagaaactac	300
ggctacggcg gcaccatgga ctactggggc caaggcaccg ccgtgagcgt aagcagcgcc	360
aagacaacag cccctagcgt gtacccgctg gcccccgttt gcggcgacac caccggtagc	420
agcgtgaccc tgggggtgct ggtgaagggc tacttccctg agcccgtagc cctgacttgg	480
aatagcggca gcctgagcag cggcgtgcac accttccccg ccgtgctgca gagcgacctg	540
tacacctga gcagcagcgt gacagtgcg agcagcacct ggctagcca aagtatcacc	600
tgcaacgtgg cccaccccg c aagcagcacc aaggtggaca agaagatcga accaagaggc	660
cccactatta aaccttgccc cccctgcaag tgtcccgcac ccaatgcagc aggcggccct	720
agcgtgttca tcttcccccc caagatcaag gacgtgtgta tgatcagcct gagccccatc	780
gtgacctgcg tgggtggtga cgtgagcgag gacgaccccg acgtgcagat cagctgggtc	840
gtgaacaacg tggaggtgca caccgctcag acacagaccc acagagagga ctacaacagc	900
accttgagag tggtagcgcg cctgcccatt cagcaccaag actggatgag cggcaaggag	960
ttcaagtgca aggtgaacaa caaggacctg ggcgccccca tcgagagaac catcagcaag	1020
cccaagggca gcgtgagagc cccccaagtg tacgtgctgc cccccctga ggaggagatg	1080
accaagaagc aagtgacctt aacctgtatg gtgaccgact tcatgcctga ggacatctac	1140
gtggagtgga ccaacaacgg caagaccgag ctgaactaca agaacaccga gcccgtagctg	1200
gacagcgacg gcagctactt catgtacagc aagctgagag tggagaagaa gaactgggtg	1260
gagagaaaca gctacagctg cagcgtggtg cacgaggggc tgcacaacca ccacaccacc	1320
aagagcttca gcagaacccc cggcaag	1347

<210> SEQ ID NO 67
 <211> LENGTH: 214
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 67

Asp Ile Val Met Thr Gln Ser Gln Lys Phe Met Ser Thr Ser Val Gly	
1 5 10 15	
Asp Arg Val Ser Val Thr Cys Lys Ala Ser Gln Asn Val Gly Thr Tyr	
20 25 30	
Val Ala Trp Tyr Glu Gln Lys Leu Gly Gln Ser Pro Lys Ala Leu Ile	
35 40 45	
Phe Ser Ala Ser Tyr Arg Tyr Thr Gly Val Pro Asp Arg Phe Thr Gly	
50 55 60	
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asn Val Gln Ser	
65 70 75 80	

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Glu Asp Leu Ala Glu Tyr Phe Cys Gln Gln Tyr Asp Ser Tyr Pro Leu
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Ala Asp Ala Ala
100 105 110

Pro Thr Val Ser Ile Phe Pro Pro Ser Ser Glu Gln Leu Thr Ser Gly
115 120 125

Gly Ala Ser Val Val Cys Phe Leu Asn Asn Phe Tyr Pro Lys Asp Ile
130 135 140

Asn Val Lys Trp Lys Ile Asp Gly Ser Glu Arg Gln Asn Gly Val Leu
145 150 155 160

Asn Ser Trp Thr Asp Gln Asp Ser Lys Asp Ser Thr Tyr Ser Met Ser
165 170 175

Ser Thr Leu Thr Leu Thr Lys Asp Glu Tyr Glu Arg His Asn Ser Tyr
180 185 190

Thr Cys Glu Ala Thr His Lys Thr Ser Thr Ser Pro Ile Val Lys Ser
195 200 205

Phe Asn Arg Asn Glu Cys
210

<210> SEQ ID NO 68
 <211> LENGTH: 642
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 68

```

gacatcgtga tgacacagtc tcagaagttc atgagcacaa gcgtgggcca cagagtgagc      60
gtgacctgca aggcattctca gaacgtgggc acctacgtgg cctgggtacga gcagaagctg    120
gggcagagcc ccaaggccct gatcttcagc gcaagctaca gatataccgg cgtgcccagc      180
agattcaccg gcagcggcag cggcaccgac ttcaccctga ccatcagcaa cgtgcagagc      240
gaggacctgg ccgagtactt ctgtcagcag tacgacagct accccctgac ctteggcggc      300
ggcaccaagc tcgagatcaa gcgcgcagat gctgctccta ccgtgagcat ctcccgccg      360
tccagcgaac aactcactag cggaggcgcg tcagtgggtc gcttccttaa caattttctac    420
cctaaggaca tcaacgtcaa gtggaagatt gacggatcgg aacgccagaa cggagtgtctg    480
aactcatgga ctgatcagga ttccaaagac tcgacttact ccatgtccag caccctgacc    540
ctgaccaaag acgagtacga aaggcacaac tcgtacacgt gcgaagccac ccacaagact    600
tccacctcgc ccatcgtgaa gtcttcaat cgcaatgagt gc                          642

```

<210> SEQ ID NO 69
 <211> LENGTH: 448
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 69

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30

Thr	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
35 40 45															
Thr	Phe	Ile	Ser	Tyr	Asp	Gly	Asn	Asn	Lys	Tyr	Tyr	Ala	Asp	Ser	Val
50 55 60															
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr
65 70 75 80															
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys
85 90 95															
Ala	Arg	Thr	Gly	Trp	Leu	Gly	Pro	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr
100 105 110															
Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro
115 120 125															
Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly
130 135 140															
Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn
145 150 155 160															
Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln
165 170 175															
Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser
180 185 190															
Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser
195 200 205															
Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr
210 215 220															
His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Ala	Ala	Gly	Gly	Pro	Ser
225 230 235															
Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg
240 245 250 255															
Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro
260 265 270															
Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala
275 280 285															
Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val
290 295 300															
Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr
305 310 315 320															
Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Gly	Ala	Pro	Ile	Glu	Lys	Thr
325 330 335															
Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu
340 345 350															
Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys
355 360 365															
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser
370 375 380															
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp
385 390 395 400															
Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser
405 410 415															
Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala
420 425 430															

-continued

Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 435 440 445

<210> SEQ ID NO 70
 <211> LENGTH: 1344
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 70

```

caagtgcagc tagttgagtc tggaggcggc gtggtgcagc ccggcagaag cctgagactg    60
agctgcgccc caagcggett caccttcagc agctacacca tgcactgggt gagacaagcc    120
cccggaagg gcctggagtg ggtgacctc atcagctacg acggcaacaa caagtactac    180
gccgacagcg tgaagggcag attcaccatc agcagagaca acagcaagaa caccctgtac    240
ctgcagatga acagcctgag agccgaggac accgccatct actactgcgc tagaaccggc    300
tggtggggcc ccttcgacta ctggggccaa ggcaccctgg taacggtaag cagcgctagt    360
accaaagggc ctagcgtgtt tccttttagc cctagcagca aaagcacaag cggaggcacc    420
gccgcctcgg gctgtctggt gaaggattac ttctctgaac ccgtgactgt atcatggaat    480
agcggggccc tgactagcgg agtgcatacc ttcccccgcc tgctgcagag cagcggcctg    540
tacagcctga gcagcgtggt gaccgtgcct agcagcagcc tgggcacaca gacctacatc    600
tgcaacgtga accacaagcc tagcaacacc aagggtggaca agaaggtgga gcccagagc    660
tgtgacaaaa cgcacacctg tccgccgtgc cccgcccctg aggctgccgg cggccttagc    720
gtgttctctg tcccccccaa gcccaaggac accctgatga tcagcagaac cctgaggtg    780
acctgcgtgg tggtggacgt gagccacgag gaccctgagg tgaagttcaa ctggtacgtg    840
gacggcgtgg aggtgcacaa cgccaagacc aagcctagag aggagcagta caacagcacc    900
tacagagtgg tgagcgtgct gaccgtgctg caccaagact ggctgaacgg caaggagtac    960
aagtgcagg tgagcaacaa ggcctggggc gcccccatcg agaagaccat cagcaaggcc   1020
aaggggcagc ctagagagcc ccaagtgtac accctgcccc ctagcagaga cgagctgacc   1080
aagaaccaag tgagcctgac ctgcctagtg aagggtttct accctagcga catcgccgtg   1140
gagtgggaga gcaacgggca gcctgagaac aactacaaga ccaccccccc cgtgctggac   1200
agcgacggca gcttttctct gtacagcaag ctgaccgtgg acaagagcag atggcagcaa   1260
ggcaacgtgt tcagctgcag cgtgatgcac gaggccctgc acaaccacta cacacagaag   1320
agcctgagcc tgagccccgg caag                                     1344

```

<210> SEQ ID NO 71
 <211> LENGTH: 215
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 71

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
 20 25 30

-continued

Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80
 Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
 85 90 95
 Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala
 100 105 110
 Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser
 115 120 125
 Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu
 130 135 140
 Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser
 145 150 155 160
 Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu
 165 170 175
 Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val
 180 185 190
 Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys
 195 200 205
 Ser Phe Asn Arg Gly Glu Cys
 210 215

<210> SEQ ID NO 72
 <211> LENGTH: 651
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 72

```

gagatcgtgc tgacacagag ccccggcaca ctgtctctga gcccggcgga gagagccaca      60
ctgagctgca gagccagcca gagcgtgggc agcagctatc tggcttggtg ccagcagaag      120
cccggccaaag cccctagact gctgatttac ggagccttta gcagagccac cggcateccc      180
gacagattca gcggaagcgg cagcggcacc gatttcacac tgacaatctc tagactggag      240
ccagaggact tcgccgtgta ctactgccag cagtacggca gcagcccttg gacctttggc      300
caaggcacca aggtggagat caagatcaag agaactgtgg ccgcgccgtc agtgtttatc      360
ttccctccat cggatgaaca gcttaagtcc ggcacggcgt ctgtggtctg cctgctcaat      420
aacttttacc ctagggaagc taaagtccaa tggaaagtgg ataacgccct gcagtcagga      480
aacagccagg aatcggttac cgaacaggac agcaaggaca gcacttactc cttgtcgtcg      540
actcttactc tgagcaaggc cgattacgag aagcacaagg tctacgcctg cgaggtcacc      600
catcagggac tctcgtcccc ggtgacccaa tccttcaata gaggcgaatg c          651
  
```

<210> SEQ ID NO 73
 <211> LENGTH: 4
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 73

Tyr Val Lys Met
1

<210> SEQ ID NO 74

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<220> FEATURE:

<221> NAME/KEY: SITE

<222> LOCATION: (1) .. (60)

<223> OTHER INFORMATION: This sequence may encompass 1-20 'Gly Gly Ser' repeating units

<400> SEQUENCE: 74

Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly
1 5 10 15

Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly
20 25 30

Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser
35 40 45

Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser
50 55 60

<210> SEQ ID NO 75

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 75

Gly Gly Ser Gly Gly Ser Gly Gly Ser
1 5

<210> SEQ ID NO 76

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 76

Glu Pro Lys Ser Cys Asp Lys Thr
1 5

<210> SEQ ID NO 77

<211> LENGTH: 34

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 77

Gly Gly Cys Gly Gly Gly Glu Val Ala Ala Leu Glu Lys Glu Val Ala

-continued

1	5	10	15
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Ala Leu Glu Lys Glu Val Ala Ala Leu Glu Lys Glu Val Ala Ala Leu
 20 25 30

Glu Lys

<210> SEQ ID NO 78
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 78

His His His His His His
 1 5

<210> SEQ ID NO 79
 <211> LENGTH: 492
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 79

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg	
1 5 10 15	
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr	
20 25 30	
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val	
35 40 45	
Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val	
50 55 60	
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr	
65 70 75 80	
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys	
85 90 95	
Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr	
100 105 110	
Leu Val Thr Val Ser Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Asp	
115 120 125	
Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly Asp	
130 135 140	
Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met Ser	
145 150 155 160	
Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr Asp	
165 170 175	
Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser Gly	
180 185 190	
Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp	
195 200 205	
Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr Phe	
210 215 220	
Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Ser Gly Gly Ser Gly	
225 230 235 240	

<400> SEQUENCE: 80

Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln
1				5					10					15	
Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr
			20					25					30		
Gly	Val	Tyr	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu
		35					40					45			
Gly	Val	Ile	Trp	Ala	Gly	Gly	Thr	Thr	Asn	Tyr	Asn	Ser	Ala	Leu	Met
	50					55					60				
Ser	Arg	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val	Ser	Leu
65					70					75				80	
Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala

-continued

85								90				95			
Arg	Gly	Pro	Pro	His	Ala	Met	Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr
			100					105				110			
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Ser	Gly	Gly
			115					120				125			
Ser	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser
			130					135				140			
Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser
			145					150				155			
Val	Gly	Ser	Ser	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala
			165					170				175			
Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Phe	Ser	Arg	Ala	Thr	Gly	Ile	Pro
			180					185				190			
Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile
			195					200				205			
Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr
			210					215				220			
Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys
			225					230				235			
Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gln	Val	Gln	Leu	Val	Glu	Ser
			245					250				255			
Gly	Gly	Gly	Val	Val	Gln	Pro	Gly	Arg	Ser	Leu	Arg	Leu	Ser	Cys	Ala
			260					265				270			
Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Tyr	Thr	Met	His	Trp	Val	Arg	Gln
			275					280				285			
Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val	Thr	Phe	Ile	Ser	Tyr	Asp	Gly
			290					295				300			
Asn	Asn	Lys	Tyr	Tyr	Ala	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser
			305					310				315			
Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg
			325					330				335			
Ala	Glu	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys	Ala	Arg	Thr	Gly	Trp	Leu	Gly
			340					345				350			
Pro	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly
			355					360				365			
Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro
			370					375				380			
Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser
			385					390				395			
Ala	Thr	Ser	Ser	Ile	Thr	Tyr	Met	Ser	Trp	Tyr	Gln	Gln	Lys	Pro	Gly
			405					410				415			
Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr	Asp	Thr	Ser	Asn	Leu	Ala	Ser	Gly
			420					425				430			
Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu
			435					440				445			
Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln
			450					455				460			
Gln	Trp	Ser	Ser	Tyr	Pro	Leu	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu
			465					470				475			
Ile	Lys	His	His	His	His	His	His	His	His	His	His	His	His	His	His
			485					490				495			

-continued

<210> SEQ ID NO 81
 <211> LENGTH: 226
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 81

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95
 Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
 115 120 125
 Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly
 130 135 140
 Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
 145 150 155 160
 Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
 165 170 175
 Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser
 180 185 190
 Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser
 195 200 205
 Asn Thr Lys Val Asp Lys Lys Val Gly Gly Gly Gly Ser Gly Gly Gly
 210 215 220
 Gly Ser
 225

<210> SEQ ID NO 82
 <211> LENGTH: 217
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 82

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Gly Ser Ser
 20 25 30
 Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45

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Ile Tyr Gly Ala Phe Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50          55          60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
65          70          75          80

Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
          85          90          95

Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Ile Lys Arg Thr
          100          105          110

Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu
          115          120          125

Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro
          130          135          140

Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly
          145          150          155          160

Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr
          165          170          175

Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His
          180          185          190

Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val
          195          200          205

Thr Lys Ser Phe Asn Arg Gly Glu Cys
          210          215

```

<210> SEQ ID NO 83

<211> LENGTH: 234

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 83

```

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
          20          25          30

Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45

Thr Phe Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Ala Asp Ser Val
          50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
          65          70          75          80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys
          85          90          95

Ala Arg Thr Gly Trp Leu Gly Pro Phe Asp Tyr Trp Gly Gln Gly Thr
          100          105          110

Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
          115          120          125

Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly
          130          135          140

Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
          145          150          155          160

Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln

```

165										170					175				
Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser				
			180				185						190						
Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser				
			195				200						205						
Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr				
			210				215						220						
His	His	His	His	His	His	His	His	His	His										
			225				230												
<210> SEQ ID NO 84																			
<211> LENGTH: 231																			
<212> TYPE: PRT																			
<213> ORGANISM: Artificial Sequence																			
<220> FEATURE:																			
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide																			
<400> SEQUENCE: 84																			
Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln				
1				5				10						15					
Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr				
			20				25						30						
Gly	Val	Tyr	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu				
			35				40						45						
Gly	Val	Ile	Trp	Ala	Gly	Gly	Thr	Thr	Asn	Tyr	Asn	Ser	Ala	Leu	Met				
			50				55						60						
Ser	Arg	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	Asn	Gln	Val	Ser	Leu				
			65				70						75			80			
Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala				
			85						90						95				
Arg	Gly	Pro	Pro	His	Ala	Met	Met	Lys	Arg	Gly	Tyr	Ala	Met	Asp	Tyr				
			100						105						110				
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly				
			115						120						125				
Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly				
			130			135						140							
Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val				
			145			150						155			160				
Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe				
			165						170						175				
Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val				
			180						185						190				
Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val				
			195						200						205				
Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Gly	Gly	Gly				
			210			215						220							
Gly	Ser	Gly	Gly	Gly	Gly	Ser													
			225			230													
<210> SEQ ID NO 85																			
<400> SEQUENCE: 85																			
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<210> SEQ ID NO 86
<211> LENGTH: 239
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 86

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
1 5 10 15
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
20 25 30
Gly Val Tyr Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
35 40 45
Gly Val Ile Trp Ala Gly Gly Thr Thr Asn Tyr Asn Ser Ala Leu Met
50 55 60
Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val Ser Leu
65 70 75 80
Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95
Arg Gly Pro Pro His Ala Met Met Lys Arg Gly Tyr Ala Met Asp Tyr
100 105 110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
115 120 125
Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
130 135 140
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145 150 155 160
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
165 170 175
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
180 185 190
Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
195 200 205
Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
210 215 220
Ser Cys Asp Lys Thr His His His His His His His His His His
225 230 235

<210> SEQ ID NO 87

<400> SEQUENCE: 87

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<210> SEQ ID NO 88
<211> LENGTH: 213
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 88

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly

-continued

1	5	10	15
Asp Arg Val Thr Ile Thr Cys Ser Ala Thr Ser Ser Ile Thr Tyr Met			
	20	25	30
Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr			
	35	40	45
Asp Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser			
	50	55	60
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu			
	65	70	75
Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Tyr Pro Leu Thr			
	85	90	95
Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala Pro			
	100	105	110
Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr			
	115	120	125
Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys			
	130	135	140
Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu			
	145	150	155
Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser			
	165	170	175
Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala			
	180	185	190
Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe			
	195	200	205
Asn Arg Gly Glu Cys			
	210		

<210> SEQ ID NO 89

<400> SEQUENCE: 89

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<210> SEQ ID NO 90

<400> SEQUENCE: 90

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<210> SEQ ID NO 91

<211> LENGTH: 66

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 91

Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln			
1	5	10	15
Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser			
	20	25	30
Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser			
	35	40	45
Asn Thr Lys Val Asp Lys Lys Val Gly Gly Gly Gly Ser Gly Gly Gly			

-continued

50	55	60
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Gly Ser
65

<210> SEQ ID NO 92
<211> LENGTH: 74
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 92

Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln
1				5					10					15	
Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser
		20					25						30		
Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser
		35					40					45			
Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr
	50					55					60				
His	His	His	His	His	His	His	His	His	His	His					
65							70								

<210> SEQ ID NO 93
<211> LENGTH: 70
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 93

Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu
1				5						10				15	
Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr
		20					25						30		
Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val
		35					40					45			
Asp	Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr	His	His	His	His
	50					55					60				
His	His	His	His	His	His										
65							70								

<210> SEQ ID NO 94
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 94

His	His	His	His	His	His	His	His	His	His
1				5					10

<210> SEQ ID NO 95
<211> LENGTH: 62
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 95

Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu
1 5 10 15Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr
20 25 30Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val
35 40 45Asp Lys Lys Val Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
50 55 60

<210> SEQ ID NO 96

<400> SEQUENCE: 96

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<210> SEQ ID NO 97

<211> LENGTH: 220

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<220> FEATURE:

<221> NAME/KEY: MOD_RES

<222> LOCATION: (1)..(3)

<223> OTHER INFORMATION: Any amino acid

<400> SEQUENCE: 97

Xaa Xaa Xaa Leu Gly Phe Gln Arg His Lys Ala Gln Leu Asn Leu Ala
1 5 10 15Thr Arg Thr Trp Pro Cys Thr Leu Leu Phe Phe Leu Leu Phe Ile Pro
20 25 30Val Phe Cys Lys Ala Met His Val Ala Gln Pro Ala Val Val Leu Ala
35 40 45Ser Ser Arg Gly Ile Ala Ser Phe Val Cys Glu Tyr Ala Ser Pro Gly
50 55 60Lys Ala Thr Glu Val Arg Val Thr Val Leu Arg Gln Ala Asp Ser Gln
65 70 75 80Val Thr Glu Val Cys Ala Ala Thr Tyr Met Met Gly Asn Glu Leu Thr
85 90 95Phe Leu Asp Asp Ser Ile Cys Thr Gly Thr Ser Ser Gly Asn Gln Val
100 105 110Asn Leu Thr Ile Gln Gly Leu Arg Ala Met Asp Thr Gly Leu Tyr Ile
115 120 125Cys Lys Val Glu Leu Met Tyr Pro Pro Pro Tyr Tyr Leu Gly Ile Gly
130 135 140Asn Gly Thr Gln Ile Tyr Val Ile Asp Pro Glu Pro Cys Pro Asp Ser
145 150 155 160Asp Phe Leu Leu Trp Ile Leu Ala Ala Val Ser Ser Gly Leu Phe Phe
165 170 175Tyr Ser Phe Leu Leu Thr Ala Val Ser Leu Ser Lys Met Leu Lys Lys
180 185 190

-continued

Arg	Ser	Pro	Leu	Thr	Thr	Gly	Val	Tyr	Val	Lys	Met	Pro	Pro	Thr	Glu
		195					200					205			

Pro	Glu	Cys	Glu	Lys	Gln	Phe	Gln	Pro	Tyr	Phe	Ile
	210					215					220

<210> SEQ ID NO 98
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 98

Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
1				5					10					15

<210> SEQ ID NO 99
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 99

Gly	Gly	Cys	Gly	Gly	Gly	Glu	Val	Ala	Ala	Leu	Glu	Lys	Glu	Val	Ala
1				5					10					15	

Ala	Leu	Glu	Lys	Glu	Val	Ala	Ala	Leu	Glu	Lys	Glu	Val	Ala	Ala	Leu
		20						25					30		

Glu Lys

<210> SEQ ID NO 100
 <211> LENGTH: 34
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 100

Gly	Gly	Cys	Gly	Gly	Gly	Lys	Val	Ala	Ala	Leu	Lys	Glu	Lys	Val	Ala
1				5					10					15	

Ala	Leu	Lys	Glu	Lys	Val	Ala	Ala	Leu	Lys	Glu	Lys	Val	Ala	Ala	Leu
		20						25					30		

Lys Glu

<210> SEQ ID NO 101
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 101

Gly	Gly	Gly	Ser	Gly	Gly	Gly
1				5		

<210> SEQ ID NO 102
 <211> LENGTH: 18
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 102

Lys Glu Ala Lys Glu Ala Lys Glu Ala Lys Glu Ala Lys Glu Ala Lys
 1 5 10 15

Glu Ala

1. An engineered antigen-binding protein comprising antigen-binding domains that specifically binds to Epitope 1 defined by the residues ¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1) and Epitope 2 defined by the residues ⁶⁵SICT⁶⁸ (SEQ ID NO: 2) of cytotoxic T-lymphocyte-associated antigen-4 (CTLA4), or

an engineered antigen-binding protein that specifically binds to CTLA4 and lacks the CH2 domain and/or a CH3 domain of the constant region of an antibody.

2. The engineered antigen-binding protein of claim 1, wherein the engineered antigen-binding protein comprises:

- a) a heavy chain variable domain (VH) sequence with at least or about 85% identity to a VH sequence selected from the group consisting of the VH sequences listed in Table 3; and/or
- b) a light chain variable domain (VL) sequence with at least or about 85% identity to a VL sequence selected from the group consisting of the VL sequences listed in Table 3.

3. The engineered antigen-binding protein of claim 1, wherein the engineered antigen-binding protein comprises a sequence with at least or about 85% identity to the sequence selected from SEQ ID NOs: 9, 10, 12, 14, 15, 17, 19, 20-24, 27, 28, 29-54, and 59, 61, 63, 65, 67, and 71, optionally wherein the engineered antigen-binding protein comprises a sequence with at least or about 85% identity to the sequence selected from:

- a) SEQ ID NOs: 9 and 10;
- b) SEQ ID NOs: 9 and 14;
- c) SEQ ID NOs: 9 and 17;
- d) SEQ ID NOs: 12 and 10;
- e) SEQ ID NOs: 12 and 14;
- f) SEQ ID NOs: 12 and 17;
- g) SEQ ID NOs: 15 and 10;
- h) SEQ ID NOs: 15 and 14;
- i) SEQ ID NOs: 15 and 17;
- j) SEQ ID NOs: 9, 10, 15, and 17;
- k) SEQ ID NOs: 12, 14, 15, and 17;
- l) SEQ ID NOs: 21 and 22;
- m) SEQ ID NOs: 27 and 28;
- n) SEQ ID NOs: 31, 32, and 33;
- o) SEQ ID NOs: 34, 35, and 36;
- p) SEQ ID NOs: 37, 38, and 39;
- q) SEQ ID NOs: 40, 41, and 42;
- r) SEQ ID NOs: 43 and 44;
- s) SEQ ID NOs: 45 and 46;
- t) SEQ ID NOs: 47, 48, and 49;
- u) SEQ ID NOs: 50, 51, and 52;
- v) SEQ ID NOs: 53 and 54;
- w) SEQ ID NO: 59;
- x) SEQ ID NO: 61;

y) SEQ ID NO: 63;

z) SEQ ID NOs: 65 and 67; and

aa) SEQ ID NOs: 69 and 71.

4-7. (canceled)

8. The engineered antigen-binding protein of claim 1, wherein

- (a) the engineered antigen-binding protein is selected from an antibody, Fv, F(ab')₂, Fab', dsFv, scFv, sc(Fv)₂, half antibody-scFv, tandem scFv, tandem biparatopic scFv, Fab/scFv-Fc, tandem Fab', single-chain diabody, tandem diabody (TandAb), Fab/scFv-Fc, heterodimeric Fab/scFv-Fc, heterodimeric scFv-Fc, heterodimeric IgG (CrossMab), DART, and diabody;
- (b) the engineered antigen-binding protein comprises an immunoglobulin heavy chain constant domain selected from the group consisting of IgG, IgG1, IgG2, IgG2A, IgG2B, IgG3, IgG4, IgA, IgM, IgD, and IgE constant domains;
- (c) the engineered antigen-binding protein comprises an Fc domain, optionally wherein
 - (i) the Fc domain is heterologous relative to the antigen-binding domain,
 - (ii) the Fc domain is an IgG1 Fc,
 - (iii) the Fc domain does not bind to one or more Fc receptors, optionally wherein the Fc receptor is a neonatal Fc receptor;
 - (iv) the Fc domain comprises a mutation that inhibits binding of the antigen-binding protein to the neonatal Fc receptor, optionally wherein the mutation is a LALAPG mutation; or
- (d) the engineered antigen-binding protein does not comprise (i) an Fc domain, or (ii) the CH2 domain and/or CH3 domain of the constant region of an antibody.

9-23. (canceled)

24. The engineered antigen-binding protein of claim 1, wherein engineered antigen-binding protein lacks the CH2 domain and/or a CH3 domain of the constant region of an antibody, and:

- (a) the engineered antigen-binding protein is F(ab')₂;
- (b) the engineered antigen-binding protein comprises the sequence of SEQ ID NOs: 21 and 22;
- (c) the engineered antigen-binding protein comprises the sequence of SEQ ID NOs: 27 and 28; or
- (d) the engineered antigen-binding protein binds two epitopes: Epitope 1 defined by the residues ¹³⁴MYPPPY¹³⁹ (SEQ ID NO: 1) and Epitope 2 defined by the residues ⁶⁵SICT⁶⁸ (SEQ ID NO: 2) of CTLA4.

25-27. (canceled)

28. The engineered antigen-binding protein of claim 24, wherein the engineered antigen-binding protein comprises at least two different VH domains and at least two different VL

domains, optionally wherein the engineered antigen-binding protein is a DART or a diabody.

29. (canceled)

30. The engineered antigen-binding protein of claim **28**, wherein the engineered antigen-binding protein is a DART or a diabody, and the DART or the diabody comprises, in the N-terminal to C-terminal direction:

- a) a first VH;
- b) a first VL;
- c) a second VH; and
- d) a second VL;

wherein the first VH and the first VL are linked by a Linker A; the first VL and the second VH are linked by a Linker B; and the second VH and the second VL are linked by a Linker C.

31. The engineered antigen-binding protein of claim **30**, wherein

- (a) the Linker A and/or Linker B comprise a peptide sequence of (GlyGlySer)_n, wherein n=1-20 (SEQ ID NO: 74), optionally wherein the n=0, 1, 2, or 3, optionally wherein the n=3;
- (b) the Linker C comprises a peptide sequence of (GlyGlySer)_n, wherein n=1-20 (SEQ ID NO: 74), optionally wherein
 - (i) the n is greater than 4;
 - (ii) n=0, 1, 2, 3, or 4; or
 - (iii) n=3;
- (c) each of the Linker A, Linker B, and Linker C comprises (GlyGlySer)_n, wherein n=3 (SEQ ID NO: 75); or
- (d) the first VH and the second VL bind the Epitope 1 (SEQ ID NO: 1); and the first VL and second VH bind the Epitope 2 (SEQ ID NO: 2).

32-39. (canceled)

40. The engineered antigen-binding protein of claim **30**, wherein

- (a) the first VH, first VL, second VH, and second VL comprise the sequence with at least or about 85% identity to SEQ ID NO: 15, SEQ ID NO: 14, SEQ ID NO: 12, and SEQ ID NO: 17, respectively;
- (b) the engineered antigen-binding protein comprises the sequence with at least or about 85% identity to SEQ ID NO: 24 (BioE2052);
- (c) the first VH and the second VL bind the Epitope 2 (SEQ ID NO: 2); and the first VL and second VH bind the Epitope 1 (SEQ ID NO: 1);
- (d) the first VH, first VL, second VH, second VL comprise the sequence with at least or about 85% identity to SEQ ID NO: 12, SEQ ID NO: 17, SEQ ID NO: 15, and SEQ ID NO: 14, respectively; or
- (e) the engineered antigen-binding protein comprises the sequence with at least or about 85% identity to SEQ ID NO: 23 (BioE2051).

41-48. (canceled)

49. The engineered antigen-binding protein of claim **1**, wherein

- (a) the CTLA4 is a human CTLA4;
- (b) the engineered antigen-binding protein is chimeric, humanized, composite, murine, or human;
- (c) the engineered antigen-binding protein further comprises a peptide tag (e.g., His6 tag) (SEQ ID NO: 78) and/or a leader sequence;
- (d) the engineered antigen-binding protein is conjugated or detectably labeled, optionally wherein the engineered antigen-binding protein is PEGylated;

(e) the engineered antigen-binding protein self-dimerizes; or

(f) the engineered antigen-binding protein:

- (i) blocks the interaction between CTLA4 and its ligands (e.g., CD80 (B7-1) and CD86 (B7-2)); and/or
- (ii) increases the interleukin-2 (IL-2) expression by the T cells.

50-54. (canceled)

55. An isolated nucleic acid molecule that encodes the engineered antigen-binding protein of claim **1**, or a vector comprising the isolated nucleic acid.

56. (canceled)

57. A host cell which

- (a) comprises an isolated nucleic acid encoding the engineered antigen-binding protein of claim **1**;
- (b) comprises a vector comprising the isolated nucleic acid of (a); or
- (c) expresses the engineered antigen-binding protein of claim **1**.

58. A pharmaceutical composition comprising:

- (a) the engineered antigen-binding protein of claim **1**;
- (b) an isolated nucleic acid encoding the engineered antigen-binding protein of claim **1**;
- (c) a vector comprising the isolated nucleic acid of (b); or
- (d) a host cell comprising any one of (a)-(c).

59. A kit comprising at least one engineered antigen-binding protein of claim **1**.

60. A method of producing at least one engineered antigen-binding protein of claim **1**, wherein the method comprises the steps of: (i) culturing a host cell comprising a nucleic acid comprising a sequence encoding at least one engineered antigen-binding protein of claim **1** under conditions suitable to allow expression of said engineered antigen-binding protein; and (ii) recovering the expressed engineered antigen-binding protein.

61. A method of preventing or treating a subject afflicted with cancer or a method of reducing proliferation of a cancer cell in a subject in need thereof, the method comprising administering to the subject at least one engineered antigen-binding protein of claim **1** or a pharmaceutical composition comprising the at least one engineered antigen-binding protein.

62. (canceled)

63. The method of claim **61**, wherein

- (i) the at least one engineered antigen-binding protein or the pharmaceutical composition (a) reduces the number of proliferating cancer cells in the cancer; (b) reduces the volume or size of a tumor of the cancer; (c) increases the immune response against the cancer; and/or (d) activates the T cell;
- (ii) the method further comprises administering to the subject an additional cancer therapy, optionally wherein the additional cancer therapy is selected from the group consisting of immunotherapy, checkpoint blockade, cancer vaccines, chimeric antigen receptors, chemotherapy, radiation, target therapy, and surgery, optionally wherein the additional cancer therapy is nivolumab; or
- (iii) the cancer is selected from pancreatic cancer, lung cancer, non-small cell lung cancer (NSCLC), malignant pleural mesothelioma, small cell lung cancer (SCLC), renal cell carcinoma (RCC), breast cancer, liver cancer, hepatocellular carcinoma, kidney cancer, skin cancer,

melanoma (e.g., unresectable or metastatic melanoma), thyroid cancer, gall bladder cancer, head and neck cancer, stomach cancer, head and neck cancer, bladder cancer, urothelial carcinoma, Merkel cell cancer, colon cancer, colorectal cancer, intestinal cancer, ovarian cancer, cervical cancer, testicular cancer, esophageal cancer, buccal cancer, brain cancer, blood cancers, lymphomas (B and T cell lymphomas), mesothelioma, cutaneous squamous cell cancer, Hodgkin's lymphoma, B-cell lymphoma, and a malignant or metastatic form thereof.

64-67. (canceled)

68. A method of increasing an immune response in a subject, or a method of activating a T-cell in a subject, or a method of preventing or treating a disease of a condition characterized by aberrant expression or activity of a CTLA4 protein in a subject in need thereof, the method comprising administering to the subject at least one engineered antigen-binding protein of claim **1** or a pharmaceutical composition comprising the at least one engineered antigen-binding protein.

69-70. (canceled)

71. The method of claim **68**, wherein the subject is afflicted with a cancer, autoimmune disease, infection, or inflammatory disease.

* * * * *